

Frequency-Dependent Streaming Potentials

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An experimental apparatus and data acquisition system was constructed to measure the streaming potential coupling coefficients as a function of frequency. The purpose of the experiments was to measure, for the first time, the real and imaginary portion of streaming potentials. In addition, the measured frequency range was extended beyond any previous measurements. Frequency-dependent streaming potential experiments were conducted on one glass capillary and two porous glass filters. The sample pore diameters ranged from 1 mm to 34 μm . Two frequency-dependent models (Packard and Pride) were compared to the data. Both Pride's and Packard's models have a good fit to the experimental data in the low- and intermediate-frequency regime. In the high-frequency regime, the data fit the theory after being corrected for capacitance effects of the experimental setup. Pride's generalized model appears to have the ability to more accurately estimate pore sizes in the porous medium samples. Packard's model has one unknown model parameter while Pride's model has four unknown model parameters, two of which can be independently determined experimentally. Pride's additional parameters may allow for a determination of permeability. © 2001 Academic Press

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INTRODUCTION

The phenomenon of streaming potentials has been studied for many years. This has led to applications in such diverse fields as chemistry, biology, and geophysics (1–5). The overwhelming majority of this work has been in the area of dc streaming potentials, while very little work has been done in the area of frequency-dependent streaming potentials. Past theoretical and experimental work dealing with the frequency response of streaming potentials has dealt with single-frequency, low-frequency, and frequency response measurements. Single-frequency streaming potential measurements are made at one frequency while varying the pressure in order to get plots of streaming potentials versus pressure without using a flow-through apparatus (6). Low-frequency measurements are used to

extrapolate the frequency-dependent streaming potential value to the dc limit. This is done for the purposes of determining the zeta or surface potentials (7, 8). Low-frequency measurements are also used to determine the effective pore size and hydraulic permeability (9). Frequency-response measurements examine the frequency-dependent behavior of streaming potentials (10, 11). Packard (10) proposed a frequency-dependent streaming potential theory for capillaries where the streaming potential coupling coefficient remains constant at its dc value until the critical frequency is approached by the sinusoidal driving force. At frequencies higher than the critical frequency, the streaming potential coupling coefficient decays with increasing frequency. Packard's experiments were performed on a limited number of capillary samples of large radii (2.083–0.589 mm) and with no changes in solution chemistry. Packard was able to achieve a maximum measuring frequency of 200 Hz. Cooke (11) attempted to duplicate Packard's work, but with limited success; he could not match his capillary data to the theoretical curves of Packard. Cooke's data for porous glass filters appears to have the expected trend predicted by Packard's theory. On closer examination, however, Packard's theory cannot be satisfactorily fitted to Cooke's data. No one to date has satisfactorily fitted frequency-dependent streaming potential data to theoretical curves for porous media of any pore diameter or for capillaries with diameters less than 155 μm .

In 1994, Pride (12) proposed a generalized theory for frequency-dependent streaming potentials in porous media. This theory relates the transport properties and pore-geometry parameters to the samples' streaming potential frequency-response behavior. No experimental work has been performed to validate Pride's theory.

The understanding of frequency-dependent streaming potentials is vital to Earth scientists who study the electromagnetic signals generated by seismic/acoustic waves propagating in the earth. Earth scientists believe that these electromagnetic signals are generated by oscillatory fluid flow in rocks relative to the mineral matrix (13). This relative flow induces a streaming current, which oscillates as an electric dipole at the same frequency as the seismic/acoustic wave exciting the medium. This frequency-dependent streaming current has a counter-current that flows through the conductive part of the rock (bulk fluid) to develop a frequency-dependent streaming potential.

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Frequency-dependent streaming potentials induced by a seismic wave are often referred to as the seismoelectric effect (14, 15). Some Earth scientists believe that the streaming potential frequency response may be useful in determining *in situ* permeability (16, 17).

In this paper, the real and imaginary streaming potential frequency response for a capillary is developed based on Packard's model, which is then compared to Pride's model. Then a discussion of the experimental system and methodology is provided. This is followed by a presentation of the first data where the complex frequency response for one glass capillary and two porous glass filters is presented. These experiments cover a frequency range of 1–500 Hz.

DC STREAMING POTENTIALS

Streaming potentials are a subset of electrokinetic phenomena, which includes electroosmosis, electrophoresis, and sedimentation potentials. Electrokinetic phenomena are a consequence of a mobile space charge region that exists at the interfacial boundary of two different phases. This region is commonly referred to as the electrical double layer (EDL). The most simplified approximations of the EDL can be represented by a parallel-plate capacitor (Helmholtz model) or a charge distribution that decays exponentially away from the surface, Gouy–Chapman model (18). A more accurate model takes into account the finite size of ions by combining the Helmholtz and Gouy–Chapman models where a fixed layer exists at the surface (Stern layer) and a diffuse layer extends from the fixed layer into the bulk of the liquid phase. This model is often referred to as the Stern model of the electrical double layer. More specifically the interfacial region, shown schematically in Fig. 1, consists of the inner Helmholtz plane (IH) where ions are adsorbed to the surface and the outer Helmholtz plane (OH) where ions are rigidly held by electrostatic forces and cannot move. The closest plane to the surface at which fluid motion can take place is called the slipping plane. The slipping plane has a potential defined as the zeta potential (ζ), which is a characteristic of the solid and liquid that constitute the interface. The diffuse layer extends from the OH into the bulk of the liquid phase. The distance at which the diffuse layer potential (ψ_0) has been reduced to ψ_0/e is referred to as the Debye length. This is often used as a measure of how far the diffuse layer extends into the bulk fluid.

Streaming potentials occur when relative motion between the two phases displaces ions tangentially along the slipping plane by viscous effects in the liquid. This displacement of ions generates a convection current (I_{conv}) and has properties similar to an ideal current source. For a capillary, I_{conv} is defined by

$$I_{\text{conv}}(r) = \int v(r) \rho_c(r) dr, \quad [1]$$

in which $v(r)$ is the fluid particle velocity, dr is an infinitesimal part of the cross section, and $\rho_c(r)$ is the charge distribution

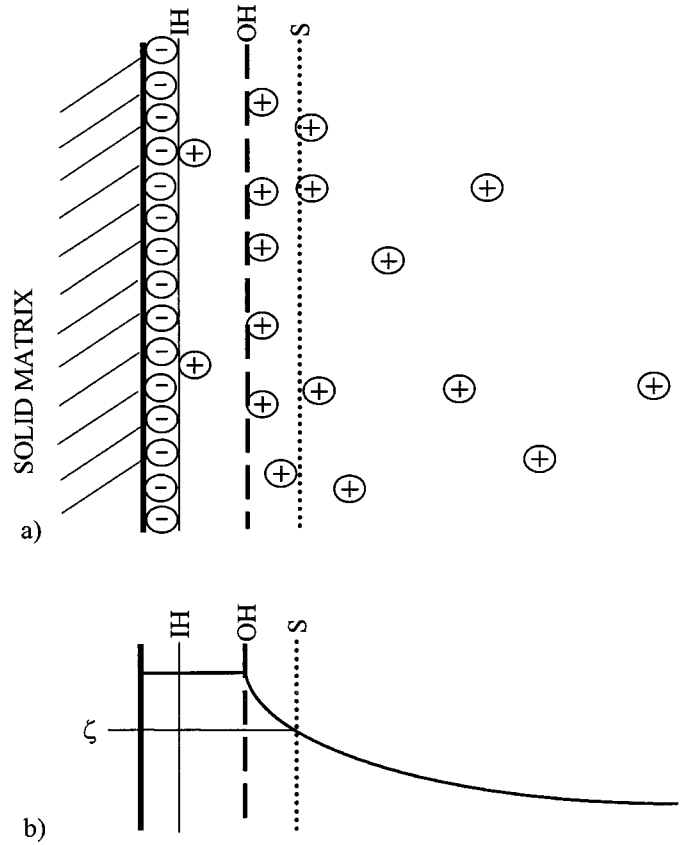


FIG. 1. (a) The Stern model of the electrical double layer. (b) One of the possible potential distributions of the Stern model. This model assumes that in the Stern layer the potential varies linearly.

in the capillary. The solution to this integral can be found in a variety of texts and is given by

$$I_{\text{conv}}(r) = \frac{\pi \varepsilon a^2 \zeta \Delta P}{\eta l}, \quad [2]$$

where ε is the permittivity of the fluid, a is the radius of the capillary, l is the length of the capillary, ΔP is the pressure across the sample, η is the viscosity of the fluid, and ζ is the zeta potential.

In steady-state equilibrium the convection current must be balanced by a conduction current (I_{cond}) (18). Hence by ohms law,

$$I_{\text{cond}}(r) = \frac{\pi \sigma a^2}{l} \Delta V, \quad [3]$$

where σ is the fluid conductivity and ΔV is the voltage measured across the sample. Equating the convection and conduction currents, which must be equal at equilibrium, leads to the Helmholtz–Smoluchowski equation (18)

$$\Delta V = \frac{\varepsilon \zeta}{\eta \sigma} \Delta P. \quad [4]$$

It should be noted when viewing Eq. [4] that it does not include geometry terms for the specimen. The ratio $\Delta V/\Delta P$ is referred to as the cross-coupling coefficient or simply the coupling coefficient in the rest of the paper. When the capillary or pore-space thickness approach the dimensions of the diffuse layer ($0.1 \mu\text{m}$ for 0.001 M solution), surface effects must be considered (2, 4). The radii of the samples used in our experiments are much larger than the Debye length; hence, these effects are not be addressed.

AC STREAMING POTENTIALS

The basic principles of ac streaming potentials are presented following the methodology of Packard (10). To better understand the physics of ac streaming potentials, appropriate comparisons are made to the frequency-dependent hydraulic problem previously solved by others (19).

The basic electrokinetic principles of dc and ac streaming potentials are the same. The difference is in the hydrodynamic part of the solution where the constant pressure of the dc case is replaced by a sinusoidal pressure in the ac case. Therefore, the starting point for the derivation must go back to the hydrodynamic problem where the Navier–Stokes equation,

$$\rho \frac{\partial \bar{v}(r, \omega)}{\partial t} \exp(-i\omega t) = -\nabla P(\omega) \exp(-i\omega t) + \eta \nabla^2 \bar{v}(r, \omega) \exp(-i\omega t), \quad [5]$$

is shown with a sinusoidal driving pressure applied across the sample. Equation [5] can be rearranged into the form

$$\left[\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + k^2 \right] v(r, \omega) = \frac{\Delta P(\omega)}{\eta l}, \quad [6]$$

where the gradient of the pressure has been replaced by $\Delta P(\omega)/l$ and

$$k^2 = \frac{-i\omega\rho}{\eta}. \quad [7]$$

It is apparent that the solution can be expressed using Bessel functions, with the general solution to Eq. [6] expressed as

$$\bar{v}(r, \omega) = -\frac{\Delta P(\omega)}{k^2 l \eta} + C_1 J_0(kr) + C_2 Y_0(kr). \quad [8]$$

The applied boundary conditions are $v(r, \omega) = 0$, when $r = a$; $v(r, \omega) = \text{finite}$, when $r = 0$, and also noting $Y_0(0) = \infty$, which then requires that $C_2 = 0$. This leads to

$$C_1 = -\frac{\Delta P(\omega)}{\eta l k^2} \frac{1}{J_0(ka)}. \quad [9]$$

Substituting Eq. [9] into Eq. [8] gives

$$\bar{v}(r, \omega) = \frac{\Delta P(\omega)}{\eta l k^2} \left[\frac{J_0(kr)}{J_0(ka)} - 1 \right] \quad [10]$$

as the frequency-dependent velocity inside a capillary. Integrating Eq. [10] over the cross-sectional area of the capillary gives

$$v(\omega) = \frac{\Delta P(\omega)}{\eta l k^2} \left[\frac{2}{ka} \frac{J_1(ka)}{J_0(ka)} - 1 \right] \quad [11]$$

as the average fluid velocity inside a capillary, which is the frequency-dependent hydraulic solution for a capillary.

Now that the frequency-dependent fluid flow inside a capillary has been presented, Eq. [10] can be substituted into Eq. [1] to give

$$I_{\text{conv}}(\omega) = \int 2\pi r \rho_c(r) \bar{v}(r, \omega) dr, \quad [12]$$

which is the frequency-dependent convection current inside a capillary, where the charge density in the capillary in cylindrical coordinates is given by

$$\rho_c(r) = -\varepsilon \nabla^2 \psi_0(r) = -\frac{\varepsilon}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \psi_0(r)}{\partial r} \right). \quad [13]$$

Integrating Eq. [12] from the center of the capillary to the slipping plane under the assumption that the capillary radius is large compared to the Debye length of the EDL (10, 12) gives

$$I_{\text{conv}}(\omega) = -\frac{2\pi \varepsilon a \zeta \Delta P(\omega)}{\eta l k} \frac{J_1(ka)}{J_0(ka)} \quad [14]$$

as the frequency-dependent convection current for a capillary.

As in the DC case, at equilibrium, the convection current must be balanced by a conduction current. The conduction current is determined using Ohm's law, where the conduction current is given by

$$I_{\text{cond}}(\omega) = \Delta V(\omega) \pi a^2 \frac{\sigma}{l}. \quad [15]$$

Setting the convection current equal to the conduction current gives

$$C(\omega) = \frac{\Delta V(\omega)}{\Delta P(\omega)} = \left[\frac{\varepsilon \zeta}{\sigma \eta} \right] \frac{-2}{ka} \frac{J_1(ka)}{J_0(ka)}. \quad [16]$$

This is the ac Helmholtz–Smoluchowski equation in a form similar to that presented by Packard (10), where $C(\omega)$ is the frequency-dependent cross-coupling coefficient. The ac Helmholtz–Smoluchowski equation reduces to the dc form in the limit as ω goes to zero, which can be demonstrated using Bessel function recursive relations or by taking the low-frequency approximation of Eq. [16]. Figure 2 shows the real and imaginary

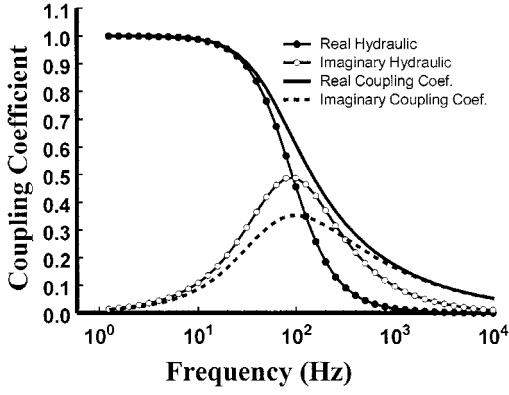


FIG. 2. Theoretical comparison of the frequency-dependent hydraulic solution, Eq. [11] and the frequency-dependent streaming potential solution. The real and imaginary portions of both solutions are shown. The coupling coefficients are normalized.

portions of the ac coupling coefficient, Eq. [16], and a plot of the frequency-dependent hydraulic equation, Eq. [11]. Figure 3 shows the phase part of Eqs. [11] and [16]. It can be seen in these two figures that the general behavior of the two equations is similar. In particular, the low-frequency behavior is identical but the high-frequency behavior diverges as one goes to higher frequencies.

At first examination of frequency-dependent streaming potentials and frequency-dependent hydraulics, it might be expected that the two phenomena would have identical behavior, since the frequency-dependent streaming potential behavior is governed by the frequency-dependent fluid flow. The different responses of the ac Helmholtz–Smoluchowski equation, Eq. [16], and the frequency-dependent hydraulic equation, Eq. [11], are easier to understand by looking at the series and asymptotic approximations for the low- and high-frequency cases, respectively. Following the methodology of Crandall (20) (which was developed for the acoustic case), the low-frequency and high-frequency approximations are made. The two separate equations are then combined to form a single equation.

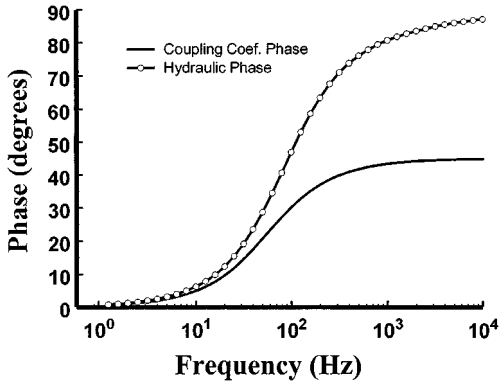


FIG. 3. Phase comparison of the frequency-dependent coupling coefficients and frequency-dependent hydraulics, where the phase is 0° for both curves at 0 Hz.

Starting with the ac Helmholtz–Smoluchowski expression, Eq. [16], and substituting the low-frequency approximations of the Bessel functions (20) J_0 and J_1 , with

$$J_0(x) = \sum_{n=0}^{\infty} \frac{(-1/4x^2)^n}{n! \Gamma(n+1)} \quad [17]$$

and

$$J_1(x) = \frac{x}{2} \sum_{n=0}^{\infty} \frac{(-1/4x^2)^n}{n! \Gamma(n+2)}, \quad [18]$$

where x represents ka and $ka < 1$. Substituting these two approximations into Eq. [16] and taking the $\lim n \rightarrow \infty$ gives

$$\begin{aligned} \frac{\Delta V(\omega)}{\Delta P(\omega)} &= \lim_{n \rightarrow \infty} \left[\frac{\varepsilon \zeta}{\sigma \eta} \right] \left[\frac{2}{ka} \left\{ \frac{\frac{x}{2} \sum_{n=0}^{\infty} \frac{(-1/4x^2)^n}{n! \Gamma(n+2)}}{\sum_{n=0}^{\infty} \frac{(-1/4x^2)^n}{n! \Gamma(n+1)}} \right\} \right] \\ &= \left[\frac{\varepsilon \zeta}{\sigma \eta} \right] \left[\frac{2}{ka} \frac{x}{2} \right]. \end{aligned} \quad [19]$$

Since $x = ka$, the low-frequency approximation reduces to

$$C_{ka < 1}(\omega) = \frac{\Delta V(\omega)}{\Delta P(\omega)} = \left[\frac{\varepsilon \zeta}{\sigma \eta} \right] = \frac{\Delta V}{\Delta P}. \quad [20]$$

The high-frequency approximation used for the Bessel functions J_0 and J_1 are given by

$$\frac{J_1(x\sqrt{-i})}{J_0(x\sqrt{-i})} = -i, \quad [21]$$

which can be found in Crandall (20) or which can be easily proven using the asymptotic approximations of Abramowitz and Stegun (21). In Eq. [16],

$$ka = a \sqrt{-\frac{i\rho\omega}{\eta}}, \quad [22]$$

from which the following substitution is made in Eq. [22]:

$$x\sqrt{-i} = a \sqrt{-\frac{i\rho\omega}{\eta}} = a \sqrt{\frac{\rho\omega}{\eta}} \sqrt{-i}. \quad [23]$$

Then substituting Eq. [21] and Eq. [23] into Eq. [16] gives

$$C_{ka > 10}(\omega) = \left[\frac{\varepsilon \zeta}{\sigma \eta} \right] \left[\frac{-2i}{ka} \right] = \left[\frac{\varepsilon \zeta}{\sigma \eta} \right] \left[\frac{-2i}{a \sqrt{\frac{\rho\omega}{\eta}} \sqrt{-i}} \right]. \quad [24]$$

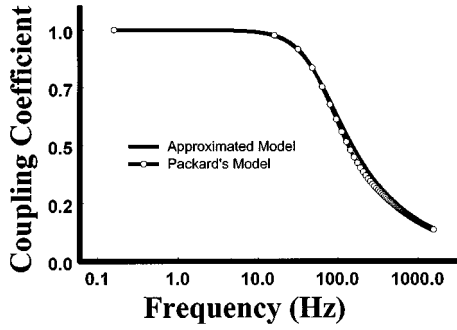


FIG. 4. Normalized comparison of the Bessel function solution, Eq. [16], to the approximated Bessel function solution, Eq. [26].

Equation [24] can then be modified to

$$C(\omega) = \left[\frac{\varepsilon \zeta}{\sigma \eta} \right] \left[\frac{-2}{a} \sqrt{\frac{\eta}{\omega \rho}} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} i \right) \right]. \quad [25]$$

Combining the low-frequency and high-frequency approximations gives

$$CA(\omega) = \left[\frac{\varepsilon \zeta}{\sigma \eta} \right] \left[1 - \frac{2}{a} \sqrt{\frac{\eta}{\omega \rho}} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} i \right) \right], \quad [26]$$

where $CA(\omega)$ represents the approximated cross-coupling coefficient. In Fig. 4, the solution using Bessel function approximations, Eq. [26], is plotted against the Bessel function solution of Eq. [16]. It can be seen in Fig. 4 that the two solutions are nearly identical. There is a slight divergence in the intermediate-frequency range, $1 > ka > 10$, which was not accounted for in the approximation. However, the error is smaller than measurements can detect.

It can be seen in Eq. [26] and Figs. 2, 3, and 4 that as the frequency is increased inertial effects start to retard the motion of the fluid within the pore space. This occurs as the fluid makes a transition from viscous dominated flow to inertial dominated flow. At the higher frequencies the flow becomes inefficient, requiring more energy to move the same amount of liquid the same distance. Therefore at frequencies higher than the transition from viscous to inertial flow, more pressure is required to shear the same quantity of ions from the diffuse zone than when strictly in the viscous flow regime. It can also be seen in Eq. [26] and Fig. 2 that at higher frequencies the real and imaginary parts of the coupling coefficient solution are decreasing at the same rate. This explains the 45° phase angle found at high frequencies in Fig. 3.

One might ask why the frequency-dependent hydraulics and frequency-dependent streaming potentials do not have the same behavior. By comparing the form of the frequency-dependent streaming potential (FSP) approximate solution to the frequency-dependent hydraulic (FDH) solution, valuable insight into the physics that causes the difference in behavior between FDH and FSP solutions can be obtained. This comparison

is made using Eqs. [20], [25], and [26] of the FSP approximation and Eq. [11] of the (FDH) solution. The low-frequency approximation to Eq. [11] is (20)

$$H(\omega) = \left(\frac{8}{a^2} + \frac{4}{3} i \rho \omega \right)^{-1}, \quad [27]$$

where $H(\omega)$ represents the frequency-dependent hydraulic solution. When $ka < 1$, the real term in Eq. [27] dominates over the imaginary term. This implies that at low frequencies the flow is real (no vorticity present) and as the frequency is increased an inertial component starts to develop in the flow (vorticity present). However, when $ka < 1$ the inertial term remains insignificant and the flow is essentially viscous. Therefore, both low-frequency FDH and FSP solution approximations approach the dc limit at low frequencies.

The high-frequency approximation of the FDH equation is given by (20)

$$H(\omega) = \left(i \rho \omega + \frac{1}{r} 2\eta \sqrt{\frac{\rho \omega}{2\eta}} (1 + i) \right)^{-1}, \quad [28]$$

where $(2\eta/\rho\omega)^{1/2}$ is the viscous skin depth. When examining Eq. [28], it can be seen that the bulk of the fluid is governed by the imaginary terms and thus inertial flow exists. However, when the viscous skin depth is sufficiently small, a second-order effect starts to dominate and the imaginary term starts to decrease at the same rate as the real term, giving rise to a diminished propagation velocity. It becomes apparent from the approximation analysis that the high-frequency solution causes the difference between the frequency-dependent hydraulics and the frequency-dependent streaming potential behavior. When the integral of Eq. [12] is evaluated, most of the contribution to the integral occurs along the wall of the capillary where most of the charge distribution is located. This region near the wall is also where the second-order effect of the hydraulic solution starts to dominate. Consequently, the frequency-dependent streaming potential exhibits high-frequency behavior that follows the form of the second-order effect of the hydraulic solution.

AC STREAMING POTENTIALS IN POROUS MEDIA

A model for ac streaming potentials in a porous medium derived from first principles was developed by Pride (12). Pride's version of the ac Helmholtz–Smoluchowski equation is given by

$$\begin{aligned} CP(\omega) &= \frac{\Delta V(\omega)}{\Delta P(\omega)} \\ &= \left[\frac{\varepsilon \zeta}{\eta \sigma} \right] \left[1 - i \frac{\omega}{\omega_t} \frac{m}{4} \left(1 - 2 \frac{d}{\Lambda} \right)^2 \left(1 - i^{\frac{3}{2}} d \sqrt{\frac{\omega \rho}{\eta}} \right)^2 \right]^{-\frac{1}{2}}, \end{aligned} \quad [29]$$

where $CP(\omega)$ represents the AC coupling coefficient for porous media using Pride's model, d is the Debye length, and Λ is a typical pore radius representing a weighted volume-to-surface ratio (19). The transition frequency

$$\omega_t \equiv \frac{\phi}{\alpha_\infty k_0} \frac{\eta}{\rho}, \quad [30]$$

as defined by Pride (12), separates the low-frequency viscous flow regime from the high-frequency inertial flow regime. In Fig. 2 the transition frequency occurs where the hydraulic imaginary curve intercepts the hydraulic real curve. Porosity is given by ϕ tortuosity by α_∞ , and the dc permeability by k_0 . The dimensionless number m is defined as

$$m \equiv \frac{\phi}{\alpha_\infty k_0} \Lambda^2, \quad [31]$$

which is also a function of the pore microgeometry, which reduces to 8 for a capillary. Closer examination of the ac porous media coupling coefficient, Eq. [29], reveals that to a first-order approximation the response is determined by the transition frequency ω_t and the dc coupling coefficient.

Figure 5 shows the normalized real and imaginary parts of the theoretical ac coupling for three capillaries of different radius using Eq. [16]. The obvious characteristic of the curves is that the transition frequency shifts with changing radius of the capillary. This relationship is easily evident when looking at the equation for the transition frequency, Eq. [30], and realizing that the dc permeability for a capillary is

$$k_0 = \frac{a^2}{8}. \quad [32]$$

The transition frequency for capillary then becomes

$$\omega_t(\text{cap}) = \frac{8}{a^2} \frac{\eta}{\rho}. \quad [33]$$

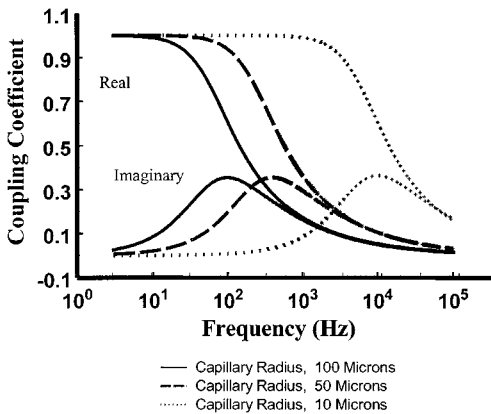


FIG. 5. Normalized coupling coefficient of the real and imaginary components for three capillaries of different radii.

COMPARISON OF PACKARD'S AND PRIDE'S MODELS

Pride (12) generated his model by creating low-frequency and high-frequency models separately and then combining them into a single combined model. Equations [34] and [35] show Pride's low-frequency and high-frequency models, respectively, when capillary geometry terms are used and second-order effects are neglected. Capillary geometry terms imply that $m = 8$, $\phi = 1$, $\alpha_\infty = 1$, $\Lambda = a$, and $k_0 = (a^2)/8$, where a is the capillary radius. Second-order effects occur when the Debye length is large compared to the capillary radius:

$$CP_{\text{LAC}} = \frac{\varepsilon \zeta}{\eta \sigma} \quad [34]$$

$$CP_{\text{HAC}} = -2 \left[\frac{\varepsilon \zeta}{\eta \sigma} \right] \frac{i^{\frac{1}{2}} \delta}{\Lambda}. \quad [35]$$

Equation [35] can be rewritten as

$$CP_{\text{HAC}} = \left[\frac{\varepsilon \zeta}{\eta \sigma} \right] \left[\frac{-2}{a} \sqrt{\frac{\eta}{\rho \omega}} \sqrt{i} \right], \quad [36]$$

where δ is the viscous skin depth and is given by $(2\eta/\rho\omega)^{1/2}$. Equation [36] can also be rewritten as

$$CP_{\text{HAC}} = \left[\frac{\varepsilon \zeta}{\eta \sigma} \right] \left[\frac{-2}{a} \sqrt{\frac{\eta}{\rho \omega}} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} i \right) \right], \quad [37]$$

which clearly shows that the real and imaginary parts of the high-frequency solution are identical to Eq. [25] in the Bessel function approximation. One possibility of combining the low-frequency and high-frequency solutions into a single model is

$$CP(\omega) = \left[\frac{\varepsilon \zeta}{\eta \sigma} \right] \left[1 - \frac{2}{a} \sqrt{\frac{\eta}{\rho \omega}} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} i \right) \right], \quad [38]$$

where capillary geometry is used and second order-effects are neglected.

The simplified version of Pride's model, Eq. [38], as opposed to the full equation, is compared to Packard's equation for two reasons. First, the second-order effects do not affect the model within the range of frequency or concentrations for which Pride's model is defined. Second, the simplified model of Pride, Eq. [38], is identical to the approximated form the Packard's model, Eq. [26]. This implies, when using capillary geometry terms and neglecting second-order effects, that Pride's model is identical to Packard's model when the series and asymptotic approximations are used.

A visual comparison of Packard's and Pride's models can be made by looking at Fig. 4, where Eqs. [16] and [38] are compared. A slight discrepancy between the two curves becomes apparent, which is addressed earlier in this paper. To completely approximate the Bessel function, the range where ka is greater than one but less than 10 must be addressed. Packard addresses

this range because the Bessel function form inherently covers the whole range. It appears from the comparison of Packard's Bessel function form to Pride's model that Pride's model does not completely address this region. When Fig. 4 is compared to actual measurements, however, it is difficult to resolve the difference between the two curves when experimental error is accounted for in the data. Consequently, Pride's model is adequate in this region.

EXPERIMENTAL APPROACH

The approach used to collect the streaming potential data was to hold the capillary or porous filter stationary and oscillate the fluid back and forth through the sample using a sinusoidal driving pressure at one end while having the other end open to the atmosphere. Silver silver-chloride electrodes were placed on either side of the sample in fluid ports to keep them out of the fluid flow path (22). The frequency response of the electrodes was measured using a four-electrode method and found to be flat in the region used in this experiment. The electrode measurements were also verified through the use of a capacitive coupled antenna placed around the outside of the sample. The pressure was monitored on the high-pressure and atmospheric-pressure sides of the sample by miniature hydrophones that have a flat frequency response from 1 to 20 kHz. The acrylic apparatus, which holds the sample, electrodes, and hydrophones, is shown schematically in Fig. 6. The enclosure around the sample and preamplifiers is constructed of Mu-metal. Mu-metal is used because of its superior electromagnetic shielding properties. The acrylic housing that holds the sample is additionally enclosed in an aluminum housing, and the electromechanical shaker that produces the driving pressure is enclosed in a separate steel box. This large quantity of shielding is required for two reasons. The first is that the laboratory is an electrically noisy environment. The second is that the pressure source is being generated by an electromechanical transducer, which is driven at the same frequency as the measured streaming potential. The driving transducer and associated leads emit enough electromagnetic signal (EMF) at the same frequency being measured to sometimes affect the measured results. Depending on the pore or capillary diameters, the resistance of the sample could be several hundred

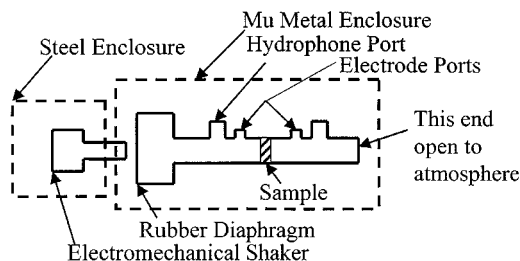


FIG. 6. A simplified schematic representation of the test cell and pressure source. The apparatus is constructed of Plexiglas and bolted to a heavy metal plate.

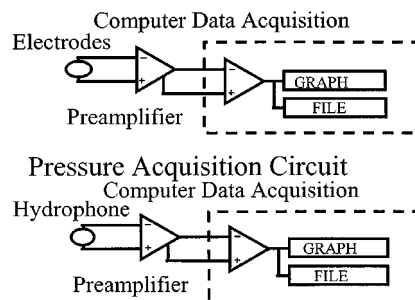


FIG. 7. A simplified representation of the streaming potential data acquisition circuit is shown. A Labview 12-bit AD board was used to acquire the data in the computer.

megaohms, which causes the sample to act as an antenna for the EMF fields produced by the electromechanical transducer. The acrylic housing has 50 kg of lead placed on it to reduce any sample vibration.

The waveforms were sampled using two instrument preamplifiers and a 12-bit analog-to-digital board with $24.8\text{-}\mu\text{V}$ resolution. Electrometers or FET preamplifiers were required as input buffers due to the high impedance of the samples ($.5\text{ M}\Omega$ to $1\text{ G}\Omega$). A schematic representation of the data acquisition system is shown in Fig. 7. The data were analyzed using spectral analysis with a Hanning window applied to the data prior to spectral analysis. In addition to spectral analysis, the time-domain voltage and pressure were monitored at each driving frequency. The raw data were saved for possible future processing. Cross-correlation analysis of the signals was also performed as a verification of the amplitude spectrum measurements.

DATA/RESULTS

To test Packard's and Pride's models experimentally, the pore diameters of the samples must be known. The pore diameter is used as an adjustable parameter in the models to fit the theory to the data. Actual pore diameters are then compared to the experimentally determined pore diameters. The pore diameters of the glass samples were provided by the manufacturer.

The glass samples consist of a capillary and two porous filters. Capillary 1 has an inner diameter that ranges from 0.8 to 1.1 mm, and a length of 60 cm. Porous Filter A has manufacturer-determined pore diameters ranging from 145 to $175\text{ }\mu\text{m}$, an overall diameter of 1 cm, and a thickness of 2 mm. Porous Filter B has manufacturer-determined pore diameters ranging from 70 to $100\text{ }\mu\text{m}$, an overall diameter of 1.9 cm, and a thickness of 1 cm.

The experiments were carried out using 10^{-3} and $10^{-3.7}$ M KCl with a pH of 5.5. Assuming a concentration of 10^{-4} M KCl, the Debye length was found to be much less than the radius of the pore space. Therefore, second-order effects can be neglected in Pride's model.

Figure 8 shows the measurements of the coupling coefficient versus frequency for Capillary 1 along with a theoretical curve

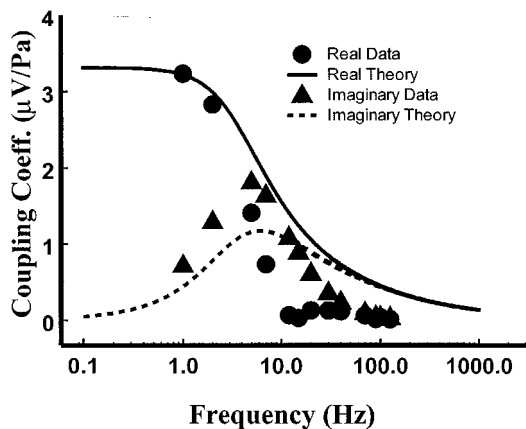


FIG. 8. Cross-coupling real and imaginary uncorrected data for Capillary 1 plotted along with the theory for a 0.8-mm capillary.

for Eq. [26]. It can be seen that the data do not have a satisfactory fit to the theory. In addition to the streaming potential data of Fig. 8, the frequency-dependent impedance of the sample and measuring circuit was also determined and is shown in Fig. 9. The impedance of the sample and measuring circuit was determined using a four-electrode method. The RC of the circuit was found to have a resistance of 1 G Ω and a capacitance of 25 pF. The resistance was determined to be on the order of the resistance of a 0.8-mm capillary filled with $10^{-3.7}$ M KCl. The capacitance was found to be on the order of the input capacitance of the amplifier and the capacitance of the wire leads. The data in Fig. 8 were then corrected for this impedance by normalizing the real and imaginary portions of the impedance at the particular frequencies of the streaming potential measurement. This normalized impedance was then divided into the data. The corrected data are shown in Fig. 10, where a good correlation between the theory and the data is evident. At the higher frequencies, signal-to-noise problems caused the fit to have a little more scatter. The transition frequency determined from the fit of the theoretical curve is 7.1 Hz, which gives a capillary diameter

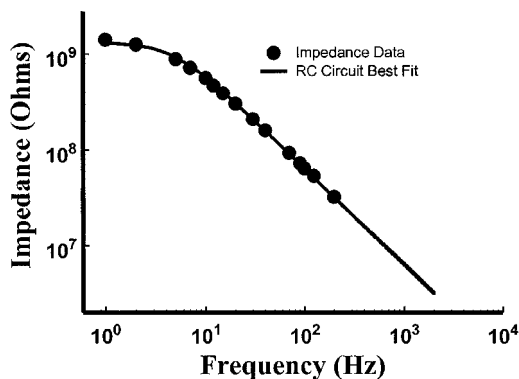


FIG. 9. Capillary 1 impedance data plotted with the best RC circuit response. The resistance and capacitance of the RC circuit are 1 G Ω and 25 pF, respectively.

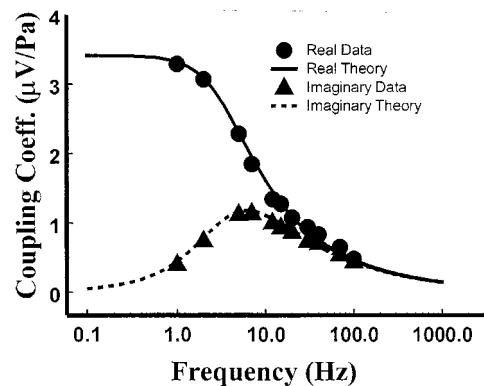


FIG. 10. Corrected cross-coupling real and imaginary data for 60-mm-long, 0.8-mm-diameter capillary plotted along with the theory for a 0.8-mm capillary. Error bars are not shown because they fall within the size of the data points.

of 0.8 mm. This is in agreement with the manufacturer-provided diameter of 0.8–1.1 mm for this type of capillary. Looking at Fig. 10, it can be seen that for the higher frequencies, inertial flow essentially dominates in Capillary 1.

The corrected data for Porous Filter A, shown in Fig. 11, are plotted against Eq. [26]. As can be seen, the theory fits the data well. Fitting the theoretical curve to the data for Porous Filter A gives a pore radius of 65 μm and a transition frequency of 269 Hz. The manufacture-provided pore radii for the Porous Filter A are 72.5–87 mm.

Figure 12 shows the corrected data for Porous Filter B along with the best-fit theoretical curve. Porous filter B has its best fit to Pride's theory, Eq. [26], using a 40- μm pore radius and a transition frequency of 710 Hz. Recall that the pore radii provided by the manufacturer for Porous Filter B are 35 to 50 μm . The data for Porous Filter B fits the theory very well in the low-frequency and the intermediate-frequency regions. At the higher frequencies, there is a little more scatter than at the low frequencies due to poor signal-to-noise ratio.

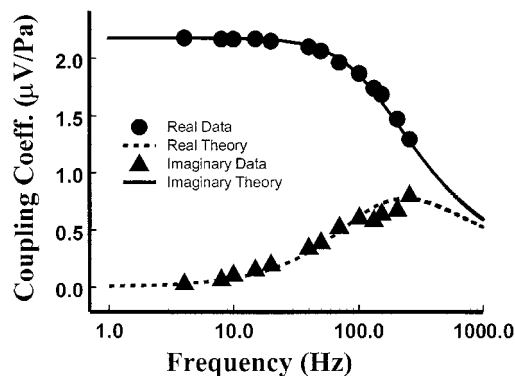


FIG. 11. Porous Filter A corrected cross-coupling real and imaginary data plotted with the theoretical response. The transition frequency was determined to be 269 Hz, which gives a pore radius of 65 μm . The manufacturer provided pore radius for porous filter B ranges from 72.5 to 87 μm . Error bars are not shown because they fall within the size of the data points.

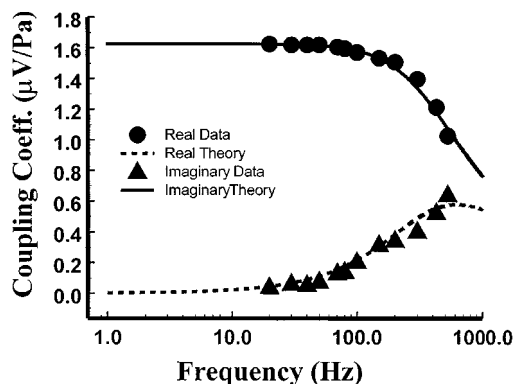


FIG. 12. Porous Filter B corrected cross-coupling frequency response data plotted along with the theoretical response. The transition frequency was determined to be 710 Hz, which gives a pore radius of 40 μm . The manufacturer provided pore radius for porous filter B ranges from 35 to 50 μm . Error bars are not shown because they fall within the size of the data points.

It appears from the data that the experimentally determined pore radii tend toward the low range of radii provided by the manufacturers. This may imply that the frequency-dependent streaming potential response is controlled by the smallest pore geometry in a fluid path. Further research is required to say for certain why the frequency response tends toward the smaller pore sizes. At the present time it appears to be related only to the hydrodynamic portion of the problem.

DISCUSSION AND APPLICATIONS

The emphasis of this study has been to examine, for the first time, complex streaming potentials and to extend the frequency range covered by this type of experiment. As a consequence of this study Packard's and Pride's theories have been verified over a range of pore sizes. Both Packard's and Pride's theories fit the data nearly identically. The transition frequency is the important parameter to be determined uniquely, for both capillaries and porous filters, from frequency-dependent streaming potential curves. From the transition frequency, the pore size can be determined. The transition frequency can be determined graphically from the data or by curve fitting the theory to the data and then determining the transition frequency from the theory. Packard's theory for a capillary is related to the capillary radius and the viscosity and density of the fluid. In Pride's model one additional parameter (m) is included if the second-order parameters are neglected and Λ is taken as the pore radius. The second-order effect parameters are important only where the pore sizes are very small and/or the electrolyte concentration is very low.

The data collected on glass filters with pore diameters much larger than the diffuse zone indicate that the curves generated using Pride's and Packard's theories cannot be distinguished from each other. This result confirms the analysis done earlier in the paper. What is different between the theories is that Pride's model includes second-order effects, which are eliminated in the approximation made during the solution of Eq. [16],

Packard's model. In this respect, Pride's model appears to be more complete and general, but the accuracy in these regions has not been confirmed. From the data thus far collected on porous filters, it can be concluded that the curves generated by Packard's and Pride's models can accurately fit the data. Therefore, both Packard's and Pride's models give identical results when used with realistic frequency ranges and solution chemistries as shown from the theoretical analysis and the experimental data.

SUMMARY/CONCLUSION

Real and imaginary parts of frequency-dependent streaming potentials were measured for the first time. A comparison of streaming potential data to proposed models has been presented for various pore sizes. Although there is a slight discrepancy between the capillary model and the generalized model, overall both Packard's and Pride's models fit the frequency-dependent streaming potential capillary data. For porous media, both Packard's capillary model and Pride's porous media model fit the data, which allows the pore size to be estimated. The difference between the two models comes from the model parameters used to fit the curve. When using Pride's model, if either m or Λ are known a priori, it allows determination of the other parameter. From the results thus far presented, the frequency streaming potential response may be governed by the smallest pore geometry in the fluid path.

Currently plans are underway to extend the frequency range of the experiment to 20–30 kHz. This will allow study of a wider range of samples, including rocks. Studying a broader sampling of rocks will help us to understand the relationship between frequency-dependent streaming potentials and permeability as alluded to by others (16, 17). A set of samples with well-defined geometries will better help to confirm how well the models can predict pore size and whether the response is governed by the pore-throat size. Last, performing frequency-dependent studies at higher temperatures and pressures will help us to ascertain what is happening to the solution chemistry and electrical double layer at *in situ* temperatures and pressures for rocks. This understanding has importance to the oil industry as well as to Earth scientists for the study of earthquake nucleation (23).

ACKNOWLEDGMENTS

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REFERENCES

1. Kurtz, R. J., Findl, E., Kurtz, A. B., and Stormo, L. C., *J. Colloid Interface Sci.* 57, 28–39 (1976).

2. Levine, S., Marriott, J. R., Neale, C., and Epstein, N., *J. Colloid Interface Sci.* **52**, 136–149 (1975).
3. Rice, C. L., and Whitehead, R., *J. Phys. Chem.* **69**(11) (1965).
4. Morgan, F. D., Williams, E. R., and Madden, T. R., *J. Geophys. Res.* **94**, 12,449–12,461 (1989).
5. Ishido, T., and Mizutani, H., *J. Geophys. Res.* **86**, 1763–1775 (1981).
6. Sears, A. R., and Groves, J. N., *J. Colloid Interface Sci.* **65**, 479 (1977).
7. Pengra, D. P., Shi, L., Li, S. X., and Wong, P., “MRS Symposium Proceedings, Boston, December 1995.”
8. Wong, P., U.S. Patent 5,417,104 (1995).
9. Li, S. X., Pengra, D. B., and Wong, P., *Phys. Rev. E* **51**, 5748–5751 (1995).
10. Packard, R. G., *J. Chem. Phys.* **21**, 303–307 (1953).
11. Cooke, C. E., *J. Chem. Phys.* **23**, 2299–2303 (1955).
12. Pride, S. R., *Phys. Rev. B* **50**, 15,678–15,696 (1994).
13. Thompson, A. H., and Gist, G. A., *The Leading Edge*, **December**, 1169 (1993).
14. Haartsen, M. W., and Pride, S. R., in “64th Ann. Internat. Mtg. Soc. Expl. Geophys., Expanded Abstracts,” pp. 1155–1158, 1994.
15. Zhu, Z., Cheng, C., and Toksöz, M. N., in “64th Ann. Internat. Mtg. Soc. Expl. Geophys., Expanded Abstracts,” pp. 26–29, 1994.
16. Mikhailov, O. V., Haartsen M. W., and Toksöz, M. N., *Geophysics* **62**, 1–9 (1997).
17. Reppert, P. M., and Morgan, F. D., in “Transactions of the 99th Annual Meeting of the American Geophysists Union,” T32H-04, 1998.
18. Overbeek, J. Th., in “Irreversible Systems” (H. R. Kruyt, Eds.), Vol. 1. Elsevier, New York, 1952.
19. Johnson, D. L., Koplik, J., and Schwartz, L. M., *Phys. Rev. Lett.* **57**, 2564–2567 (1986).
20. Crandall, I. B., “Theory of Vibrating Systems and Sound.” Van Norstrand, New York, 1926.
21. Abramowitz, M., and Stegun, I. A., “Handbook of Mathematical Functions With Formula, Graphs, and Mathematical Tables.” U.S. National Bureau of Standards, 1964.
22. Morgan, F. D., in “Lecture Notes in Earth Sciences” (S. Bhattacharji, G. M. Friedman, H. J. Neugebauer, and A. Seilacher, Eds.), Vol. 27, Springer-Verlag, Berlin/New York (1989).
23. Morgan, F. D., *Eos Trans. AGU*, F173 (1995).

Effects of Saturant Conductivity on Seismoelectric Conversion

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Abstract

When a seismic wave propagates in a fluid-saturated porous medium, a seismoelectric field can be induced in the medium due to an electric double layer at the interface between solid and fluid. The strength of the seismoelectric field depends on the characteristics of the double layer and the conductivity of the saturant fluid. In our experiments two kinds of seismoelectric fields, a radiating electromagnetic (EM) wave and a localized electric field, are induced with fractured borehole models. The amplitudes of the seismoelectric signals are recorded when the conductivity of the saturant varies from zero to 27 mS/cm. The results show that when the conductivity increases, the amplitude of the electric signals increases at a low conductivity area and decreases at a high conductivity area. In this paper we investigate the mechanisms of seismoelectric conversion. When a double layer is saturated by charges in fluid, the amplitude of seismoelectric signals is inversely proportional to the conductivity. Conversely, if it is not saturated, the amplitude is directly proportional to the conductivity.

1 Introduction

Seismoelectric conversion has been studied for many years. At the solid-fluid interface of a porous medium or a fluid-saturated fracture, adsorption of an electrical charge to the solid surface creates an excess of mobile ions of opposite charge to that of the fluid. Thus, a double layer is formed on the solid surface (Pride and Haartsen, 1996; Morgan et al., 1989). When a seismic wave propagates in a two-phase medium of solids and fluids, the particle vibration generates a movement of ions in the fluid that induces an electric field, a magnetic field, or an electromagnetic field (Haartsen, 1995). This phenomenon is referred to as seismokinetic or seismoelectric conversion. Inside a homogeneous, porous medium, the seismic wave induces localized seismoelectric and seismomagnetic fields which exist only in the area disturbed by the seismic wave (Zhu et al., 2000). At an interface of material with different properties, such as porosity, permeability, or lithology, the seismic wave induces a radiating seismoelectric or electromagnetic wave, which propagates with light speed and can be received anywhere (Zhu and Toksöz, 1999).

Seismoelectric conversion depends not only on the electric double layer, but also on the conductivity of fluid which saturates a porous medium. Some experiments (Zhu et al., 2000) show part of the relationship between the amplitude of seismoelectric signals and the resistivity of the saturant. When fluid resistivity increases, the seismoelectric amplitude increases. The electric amplitude, however, should be zero if the resistivity of the saturant is infinite and there is no double layer formed at a two-phase interface. Therefore, the proportional relationship does not govern this conversion in the whole range of the fluid resistivity.

Previous studies showed two double layer models (Morgan et al., 1989). Most minerals exhibit a negative surface charge and a zeta potential (Gouy-Chapman model). A seismic wave generates fluid flow, and excess ions in the fluid induce a convection current. In a steady state equilibrium, this convection current is balanced by a conduction current. The measurable potential is induced by these two currents.

In this study we make a borehole model with a fracture between two blocks. An acoustic transducer generates a compressional wave which propagates across the fracture and arrives at the borehole. An electrode records the radiating electromagnetic wave induced at the fracture and the localized electric field induced within the borehole. We record the electric signals as the saturant conductivity varies step by step. The

relationship between the amplitude and conductivity is explained qualitatively. These results are helpful for interpretation of an amplitude variation of a seismoelectric conversion.

2 Borehole Model With a Fracture

Figure 1 shows a borehole model with a fracture and a P-wave transducer on a Lucite block. There is a fracture between the Lucite block and rock block with a borehole. The plane P-wave transducer (1.5 inches in diameter) is excited by a square electric pulse with 1000 volts in amplitude and $5\ \mu s$ in width. The main acoustic energy focuses on the vertical direction (Figure 1). The thickness of the Lucite block and the rock block in the vertical direction are 5 cm and 10 cm, respectively. The borehole diameter is 1.27 cm. During the measurement the plane transducer is fixed on the location shown in Figure 1. The entire system is submerged in a water tank. An electrode made of cable or a hydrophone of 1.0 cm in diameter moves along the water-saturated borehole and records the electric signal or acoustic wave.

The acoustic wave generated by the transducer propagates through the Lucite block and excites a Stoneley wave in the fracture. The Stoneley wave induces a radiating electromagnetic wave at the fracture (Zhu and Toksöz, 2002). When the acoustic wave propagates to the borehole, it induces an electric field on the borehole wall. An electrode in the borehole can receive the electric component of the EM wave induced at the fracture and electric signals induced in the borehole. The EM wave is attenuated by the rock between the fracture and the borehole. The signals received by the electrode in the borehole are recorded by a digital oscilloscope after the signals pass through a preamplifier and filter ranging from 10 to 500 kHz. If the electrode is replaced with a hydrophone, we may record the acoustic wave arriving the borehole.

Figure 2 shows the electric signals recorded by the electrode moving in the borehole.

Comparing the arrival time of the electric signals to that of the acoustic waves, the signal at $22\ \mu s$ is an EM wave induced on the rock side of the fracture and arriving at the borehole through the rock. The signal at $34\ \mu s$ is induced by the acoustic waves within the borehole.

Using this borehole model and measurement system, we investigate the effect of the conductivity of the saturant fluid on seismoelectric signals.

3 Measurements With the Model

Previous investigations show that the electric signals induced by a seismic wave are inversely proportional to the saturant conductivity (Zhu et al., 2000). When the conductivity increases, the amplitude of the electric signals decreases. This relationship, in general cases, is correct; however, the amplitude should be close to zero if the conductivity decreases to zero. According to the basic principle of seismoelectric conversion, we know that conversion is based on a double layer due to adsorption of charges in the fluid onto the solid interface. If there is not a charge in the fluid (its resistivity is infinite), a double layer is not formed, and a seismoelectric field cannot be induced. This means the relationship of the inverse proportion is not true at a low conductivity area.

We saturate the fracture and borehole of the model shown in Figure 1 with pump oil. No electric signal was received. Because the oil is isolate fluid, there was no movable charge in the fluid. Thus, a seismic wave cannot induce an electric signal. Next, we built two rock borehole models, one of sandstone, the other of Westerly granite. We cleaned the rock models with pure water several times and saturated the models with pure water in a vacuumed system. We then put the models in the measurement system shown in Figure 1. About an hour later the conductivity of fluid in the tank remained stable. We recorded the electric signals with the electrode in the borehole. When we fixed the electrode at the position of trace 4 in Figure 2 and changed the conductivity by gradually adding tap water or sea water, we were able to record the electric signals. From the amplitude of the radiating EM wave and the localized electric field, we obtain the relationship between the electric amplitude and the fluid conductivity. Because the porosity and permeability of sandstone are larger than granite, the electric amplitude in sandstone is larger than granite.

Figures 3 and 4 respectively show the electric amplitudes in the sandstone and granite borehole models when the saturant conductivity increases gradually.

In our experiment the conductivity is increased to 27 mS/cm . Figures 3 and 4 show only the conductivity range of $0\text{--}2.0\text{ mS/cm}$. There is a peak of the amplitude in the sandstone borehole (solid line) at the conductivity C_m of 0.25 mS/cm in Figure 3 and a peak at the conductivity C_m of 0.06 mS/cm in Figure 4. In the following discussion, we call the area of the conductivity less or larger than C_m as section I and section II, respectively. The peak in the amplitudes of the EM waves (solid line in the figures) induced at the fracture is not clear. We focus on electrical signals induced in the sandstone and granite boreholes.

When the conductivity is zero (in the oil case), there is no electric signal. In section I, the electric amplitude increases when fluid conductivity increases. After amplitude reaches a maximum, electric amplitudes decrease when fluid conductivity increases. The maximum amplitudes are at a different conductivity in sandstone and granite boreholes. These amplitudes are approximately 0.25 mS/cm for sandstone and 0.06 mS/cm for granite. Below, we discuss the mechanism of seismoelectric conversion in the two sections.

4 Experimental Results

From the solid curves in Figures 3 and 4 we know there are two mechanisms that govern the relationship between seismoelectric conversion and fluid conductivity. Most minerals exhibit a negative surface charge and zeta potential (Gouy-Chapman model) when the surface is in contact with electrolytic pore fluid. Some excess positive ions shear off from the double layer. When an acoustic wave generates a relative motion between fluid and solid, the excess ions move in the direction where pressure decreases and form a convection current i_{conv} (Morgan et al., 1989; Reppert et al., 2001; Reppert and Morgan, 2002):

$$i_{conv} = -\frac{\epsilon\zeta}{\eta}G\Delta P$$

where ϵ is the permittivity of the fluid, G is the geometrical factor, ΔP is the pressure of the acoustic wave, η is the viscosity of the fluid, and ζ is the zeta potential.

If there are enough anions in the fluid, the electric force of the potential generated by the convection current creates a conductive current in the direction opposing the convection current by moving the anions. In steady state equilibrium, i_{conv} is balanced by a conduction current i_{cond} , given by

$$i_{cond} = \sigma G\Delta V$$

where σ is the fluid conductivity and ΔV is the voltage measured across the sample.

Equating these currents leads to the Helmholtz-Smoluchowski equation:

$$\Delta V = \frac{\epsilon\zeta}{\eta\sigma}\Delta P$$

From this equation we know that the voltage of seismoelectric signals is inversely proportional to the fluid conductivity. When conductivity increases, the amplitude of the electric signals decrease, as in section II in Figures 3 and 4. However, the Helmholtz-Smoluchowski equation cannot explain the relationship in section I in Figures 3 and 4.

The Gouy-Chapman model depends on two conditions: there are enough ions in the fluid to create a stable double layer on the interface, and there are both positive and negative ions in the diffuse layer with excess movable positive ions (Pride, 1994).

If not enough anions are adsorbed on the solid surface when the fluid conductivity is very low, there are no more movable anions, and only positive ions in the diffuse layer. In this case, a conductive current i_{cond} cannot be induced by the voltage, as there is only a convection current directly proportional to pressure ΔP . Therefore, the voltage of the electric signals is directly proportional to the fluid conductivity. When the conductivity increases, the positive ions in the diffuse layer and in the convection current increase in section I. If the fluid is an isolate (there is no electric double layer nor any positive or negative ion in the fluid), a convection current cannot be induced.

Different porous media need different amounts of anions in order to saturate the adsorbing layer. Because the porosity of sandstone is higher than granite, more anions are needed to saturate the solid surface of sandstone and thus, the conductivity C_m at the peak voltage in sandstone is larger than granite.

5 Conclusions

We perform experiments in a fractured borehole model to measure seismoelectric signals with an electrode in the borehole. There are two types of seismoelectric fields induced by an acoustic wave excited by a P-wave transducer. The first type is a radiating electromagnetic wave induced at the fracture, and the second type is a localized seismoelectric field induced within the borehole. The experiments show the effects of the saturant conductivity on the amplitudes of seismoelectric fields recorded in the borehole. We gradually change the conductivity from zero to 27 mS/cm with pump oil, pure water, tap water and sea water and record the amplitudes of the two seismoelectric fields. When the model is saturated by oil, no electric signal can be generated at the fracture or borehole. The relationship between the electric amplitude and the saturant (water) conductivity is measured at a different saturant conductivity. Results show that at low conductivity, the electric amplitude increases when the conductivity increases. At a high conductivity area, the electric amplitude decreases when the conductivity increases. On the relationship curve there is a maximum at conductivity C_m , which is related to the zeta potential of the rock. According to the model of a double layer, the amplitude of the electric field is directly proportional to the conductivity before the solid surface is saturated by enough charges, and the amplitude of the electric field is inversely proportional after saturation. Our results show the relationship between seismoelectric conversion and the saturant conductivity, and they provide experimental data to study the electric amplitude variation in seismoelectric conversion.

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References

- Haartsen, M. W. (1995). Coupled electromagnetic and acoustic wavefield modeling in poro-elastic media and its application in geophysical exploration. *Ph. D. Thesis, Massachusetts Institute of Technology*.
- Morgan, F. D., Williams, E. R., and Madden, T. R. (1989). Streaming potential properties of westerly granite with applications. *Journal of Geophysical Research*, 94:12449–12461.
- Pride, S. (1994). Governing equations for the coupled electromagnetics and acoustics of porous media. *Phys. Rev. B*, 50:15678–15696.
- Pride, S. R. and Haartsen, M. W. (1996). Electro seismic wave properties. *J. Acoust. Soc. Am.*, 100:1301–1315.
- Reppert, P. and Morgan, F. D. (2002). Frequency-dependent electroosmosis. *Journal of Colloid and interface Science*, 254:372–383.
- Reppert, P., Morgan, F. D., D.Lesmes, and Jouniaux, L. (2001). Frequency-dependent streaming potentials. *Journal of Colloid and interface Science*, 234:194–203.
- Zhu, Z., Haartsen, M. W., and Toksöz, M. N. (2000). Experimental studies of seismoelectric conversions in fluid-saturated porous media. *Journal of Geophysical Research*, 105:28,055–28,064.
- Zhu, Z. and Toksöz, M. N. (1999). Seismoelectric and seismomagnetic measurements in fractured borehole models. *SEG 69th Annual International Meeting Expanded Abstracts*, BH/RP 5.7:144–147.
- Zhu, Z. and Toksöz, M. N. (2002). Crosshole seismoelectric measurements in borehole models with fractures. *72th SEG Annual International Meeting Expanded Abstracts*, BH2.2:344–347.

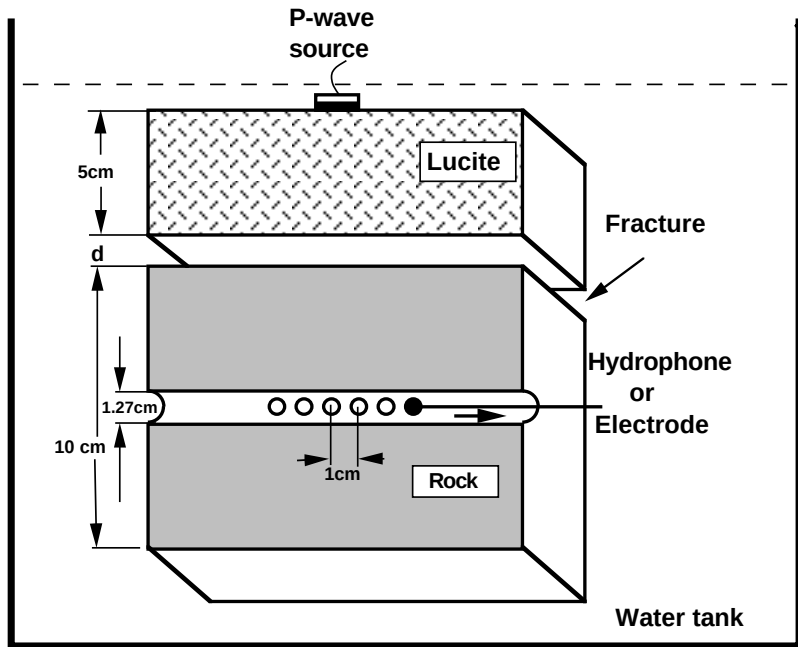


Figure 1: Borehole model with a fracture between rock and Lucite blocks. The model is saturated with water. A plane transducer is fixed on the Lucite block and generates an acoustic wave. A hydrophone or an electrode moves in the borehole and records the acoustic or electrical signals.

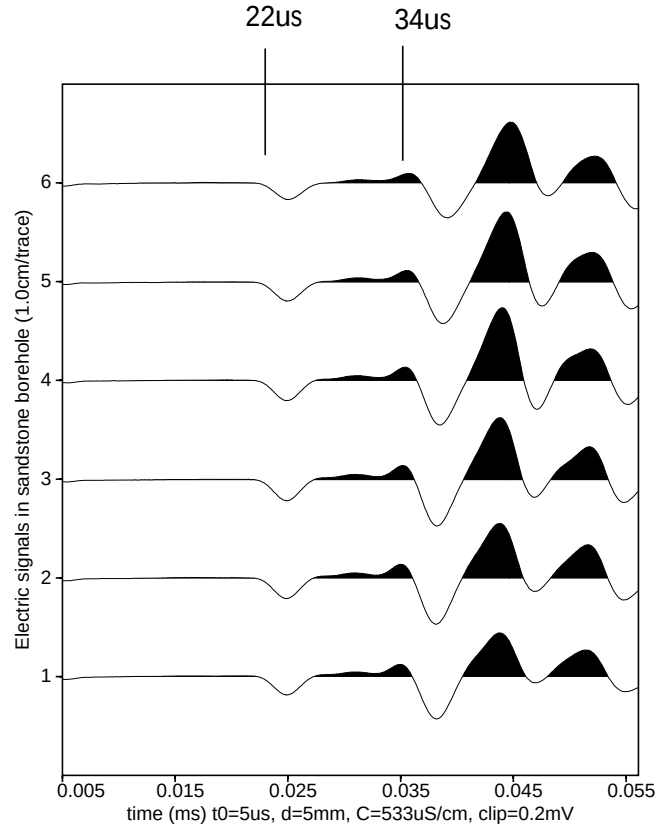


Figure 2: Seismoelectric signals recorded in the borehole in Figure 1. The amplitude in Figure 2a is normalized for each trace by $200\text{ }\mu\text{V}$. The fracture aperture is 5 mm. The conductivity of the saturant water is about 0.533 mS/cm .

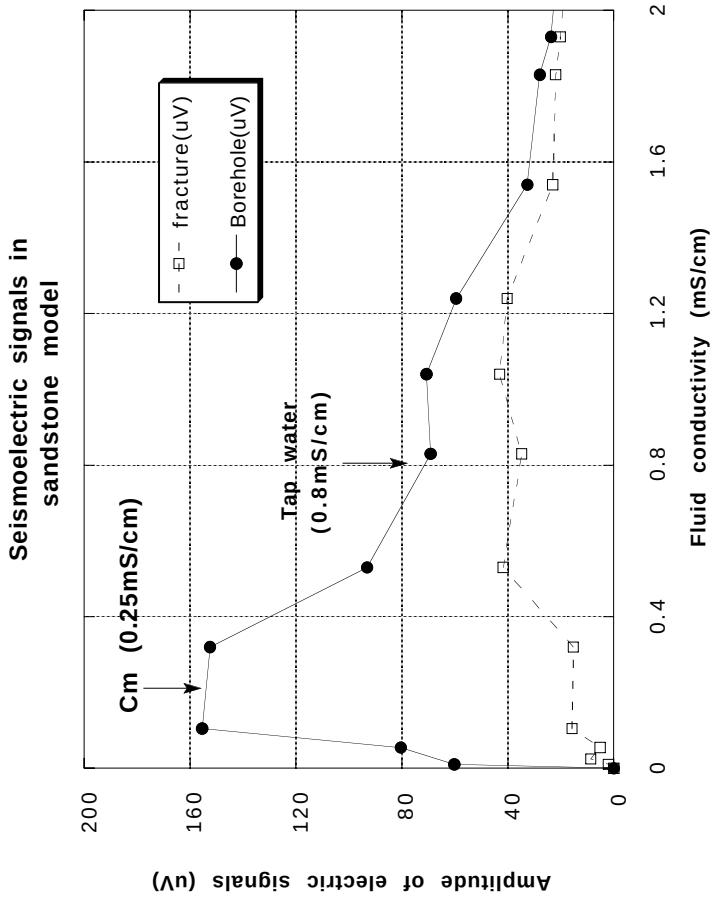


Figure 3: Relationship between water conductivity and the electric amplitude of the EM wave and seismo-electric field induced at the fracture and borehole of a sandstone model. Cm (0.25 mS/cm) is the conductivity where the amplitude is at maximum.

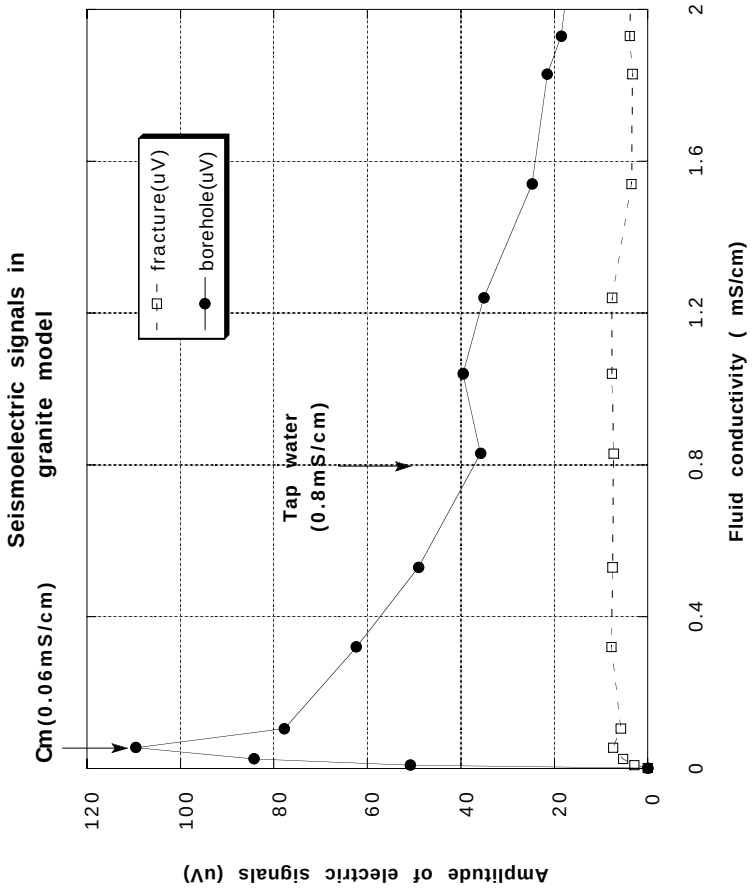


Figure 4: Relationship between water conductivity and electric amplitude of the EM wave and seismoelectric field induced at the fracture and borehole of the granite model. C_m (0.06 mS/cm) is the conductivity where the amplitude is at maximum.

The seismoelectric method

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Introduction

This paper follows the structure of the IFP Energies nouvelles e-book dedicated to electromagnetic methods in Geophysics. It is meant as a companion to the online course part devoted to the Seismoelectric method.

Back to the theory

The seismoelectric method is a geophysical imaging technique relying on electro-kinetic conversions between seismic and electromagnetic waves. When a seismic wave travels throughout a fluid-filled porous rock, it sets in motion the fluid in the pores with respect to the solid rock matrix: the relative motion of the ions in the free electrolyte with respect to those adsorbed along the pore walls triggers an electrical response. Therefore, by combining a seismic source with a string of dipole receivers, it becomes possible to measure the electric field associated with these seismoelectric conversions.

Other conversions between mechanical and electromagnetic energy exist, such as piezoelectricity and resistivity modulation. These phenomena were also applied to geophysical exploration: for instance, the piezoelectric effect was applied to quartz and pegmatite veins detection in the late 1960's (Volarovitch *et al.*, 1969). These other phenomena will not be further addressed in this chapter.

Last but not least, the seismoelectric method should not be mistaken with its electro-seismic counterpart, which relies on electro-osmotic phenomena, i.e. the fluid motion induced by an applied potential across a porous material (see for instance Thompson *et al.*, 2007).

The Streaming Potential Coefficient

Applying a pressure gradient ΔP [Pa] to a porous rock sample induces the fluid in the pore space to flow. A local convection current originates from the movement of some of the ions in the electrolyte; this current is counterbalanced by a conduction current, thus ensuring the electro-neutrality of the medium. This ohmic current creates a potential difference ΔV [V], called the streaming potential.

In the case of laboratory experiments, it is possible to measure both the applied ΔP and the measured ΔV (Figure 1).

The ratio between these quantities is called the streaming potential coefficient, or SPC, often noted C (Equation 1).

$$C = \frac{\Delta V}{\Delta P}$$

Equation 1: the streaming potential coefficient

Any phenomenon creating a relative motion between the liquid and solid phases should therefore create a streaming potential: gravity flow, fluid-filled fracture growth within a rock and the propagation of a seismic wave, which makes the seismoelectric method possible.

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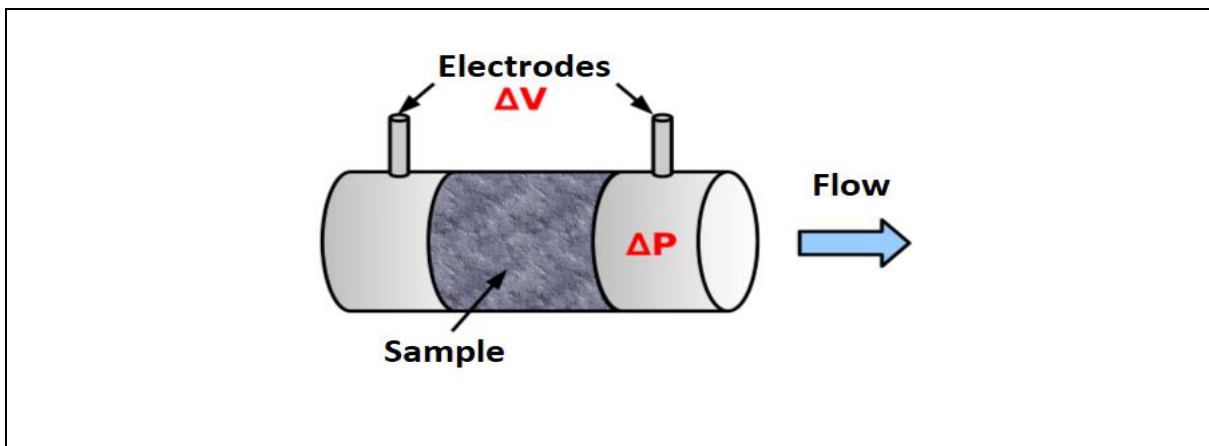


Figure 1: Experimental layout: the fluid flow induced by the pressure drop ΔP creates a potential difference ΔV measured by the electrode pair. From Warden 2012.

The Electrical Double Layer

Several models have been proposed to explain the mechanisms giving rise to the streaming coefficient. These models of Electrical Double Layer (EDL) or sometimes even Electrical Triple Layer (ETL) describe the way ions are organized in the pore fluid and along the mineral grains, at the microscopic scale.

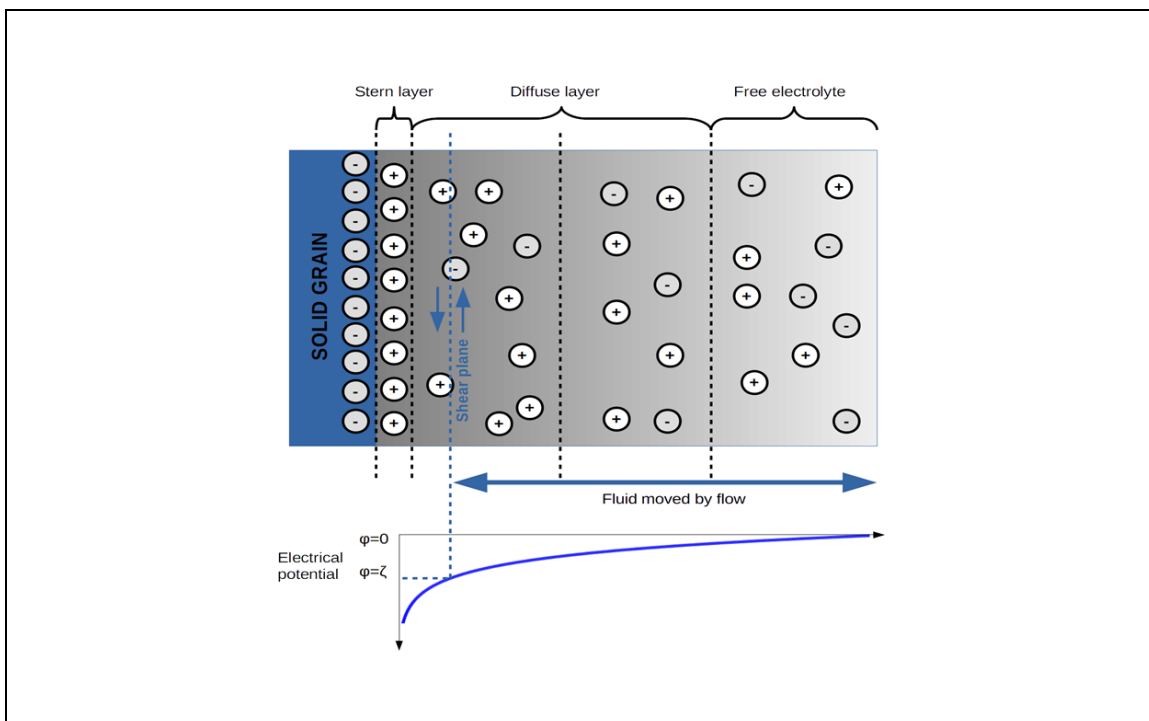


Figure 2: A simplified Electrical Double Layer (EDL) model.

Figure 2 shows such a model. The negatively charged mineral grain is depicted to the far left; to the right is the free electrolyte, where balance between positive and negative charges is achieved. Between these are a

series of intermediate layers.

The Stern layer contains positive ions (cations) adsorbed along the mineral grain. As the cations in the Stern layer do not compensate the negative surface charges, a diffuse layer exists, where cations outnumber the negative ions (anions). This excess of positive charges decreases exponentially until cations and anions are balanced in the free electrolyte. As a result, the electrical potential increases until it reaches 0 in the free electrolyte.

The different types of seismoelectric waves

Seismoelectric conversions mainly give rise to two types of waves:

- The **coseismic waves** (or type I waves).
- The **interface response** (or type II waves).

Let us consider a seismic source, such as a sledgehammer hitting a plate (see for instance Figure 3). When the subsurface contains fluid, the seismic wave gives rise to an accompanying electromagnetic wave that travels *within* the seismic disturbance. As it travels at the same velocity as the seismic wave that induces it, it is dubbed a **coseismic wave**.

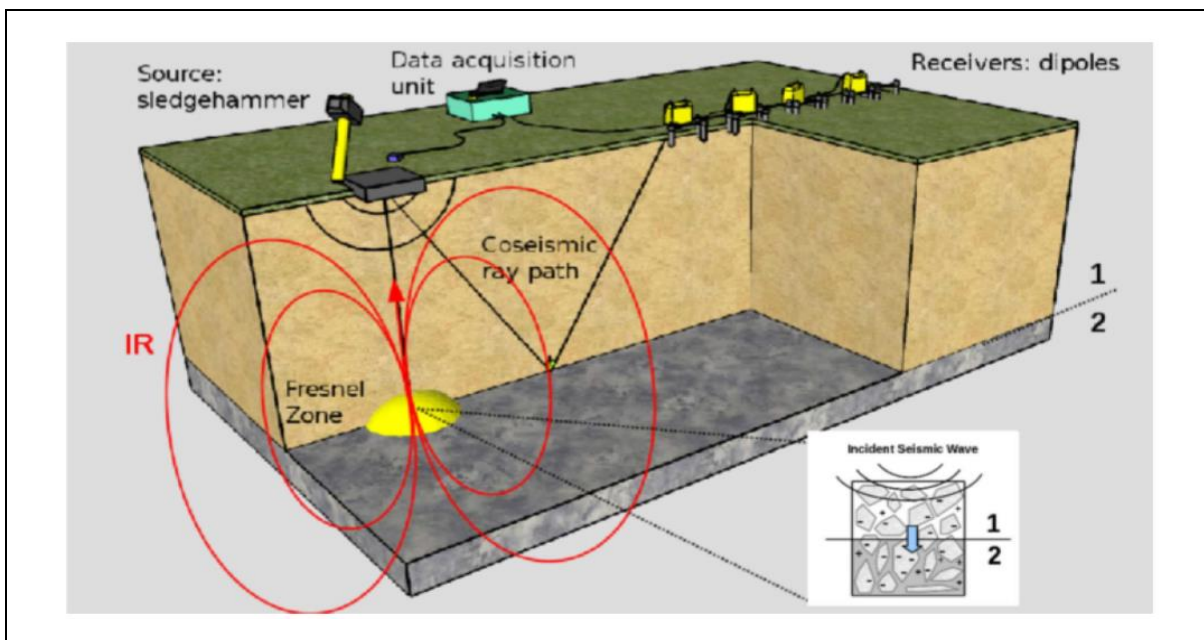


Figure 3: A simple seismoelectric data acquisition layout. The seismic source consists of a sledgehammer hitting a plate. The receivers are a string of dipoles connected to digitizers. A coseismic ray path is depicted in black in this Figure. When the incident seismic wave encounters the interface between layers 1 and 2, it gives rise to an interface response (IR) that can be modeled by a dipole oscillating at the first Fresnel zone: the IR is represented in red. From Warden et al., 2013.

Thus, taking a look at a seismic and a seismoelectric recording side by side (e.g. the synthetic recordings in Figure 4) reveals that some of the seismic arrivals show clear seismoelectric counterparts. In fact, seismoelectric dipole receivers were once foreseen as a possible alternative to conventional geophones.

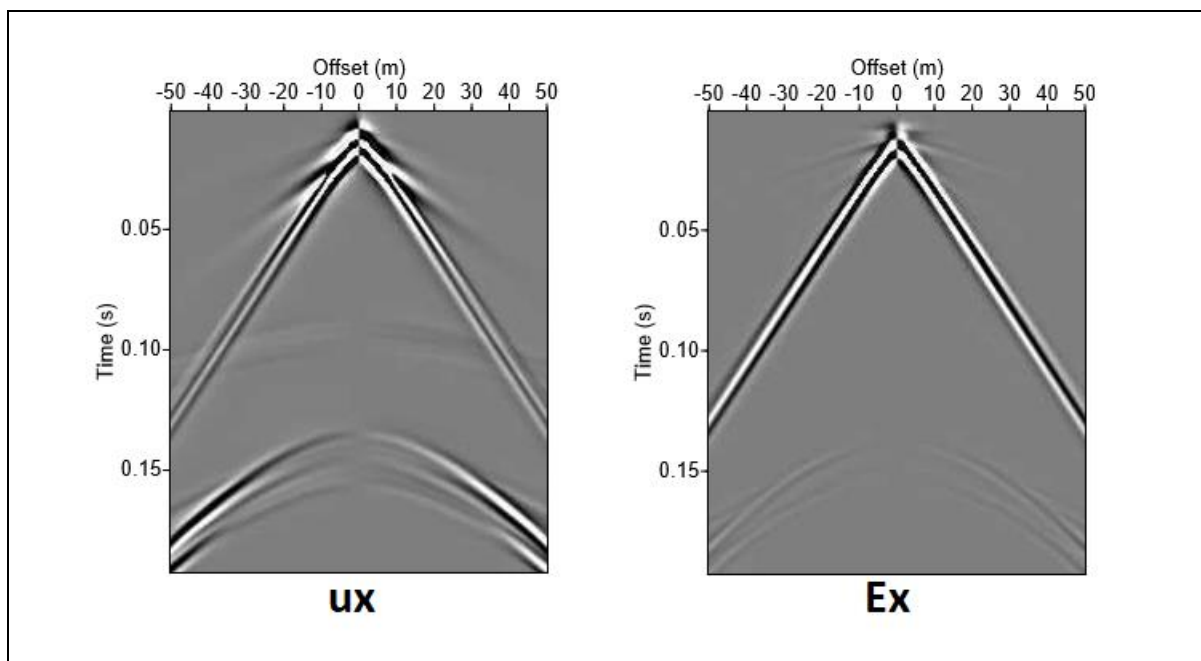


Figure 4: Synthetic seismic (to the left) and seismoelectric (to the right) recordings. These synthetic recordings were modeled for a simple two-layer tabular model consisting of a sand layer on top of a sandstone half-space. The source peak frequency is $f_{peak}=120\text{Hz}$. From Warden et al., 2012.

When the incident seismic wave encounters an interface between two media exhibiting different mechanical or electrical properties, it may give rise to an **interface response** (or IR), travelling with the velocity of an electromagnetic wave. As its velocity is much higher than seismic velocities, the IR looks like it seen almost instantaneously by all receivers; it therefore appears like a "horizontal" event in the seismoelectric recording.

An example of IR is given in Figure 5(c) for a synthetic recording. In this example, the amplitude of the IR has been artificially boosted; otherwise the IR would be concealed by the stronger coseismic arrivals seen in Figure 5(a) (the same recording as in Figure 4). In this example the IR can be approximated as a dipole oscillating at the first Fresnel zone at the interface: this means that its amplitude is not maximal right below the shot point, but instead reaches its maximum at a certain offset away from the source. The amplitude variations of the IR with respect to the offset are displayed in Figure 5(d). It can also be noted that the IR exhibits opposite polarities on either side of the shot point: this characteristic may be used to discriminate the IR from other arrivals.

Interface responses may provide valuable information about the subsurface at depth. However, their very weak amplitudes make them difficult to recover and analyze.

In the simple noise-free synthetic example detailed above (Figure 5(a)), the IR was concealed by the coseismic waves. One can imagine that it gets even trickier to recover such responses on noisy data acquired in the field. Section **Field Equipment and Data Processing** deals with the processing steps usually applied to extract IRs from seismoelectric recordings.

The seismoelectric theory of Pride (1994)

In this section, we briefly introduce Pride's theory (1994) for the coupled propagation of seismic and electromagnetic waves. One of Pride's key contribution was to consider the full set of Maxwell electromagnetic equations in order to derive the equations describing seismo-electromagnetic coupling. 50 years after the seminal yet incomplete work of Frenkel (1944), Pride's work rekindled the interest of the geophysical community for seismo-electric methods.

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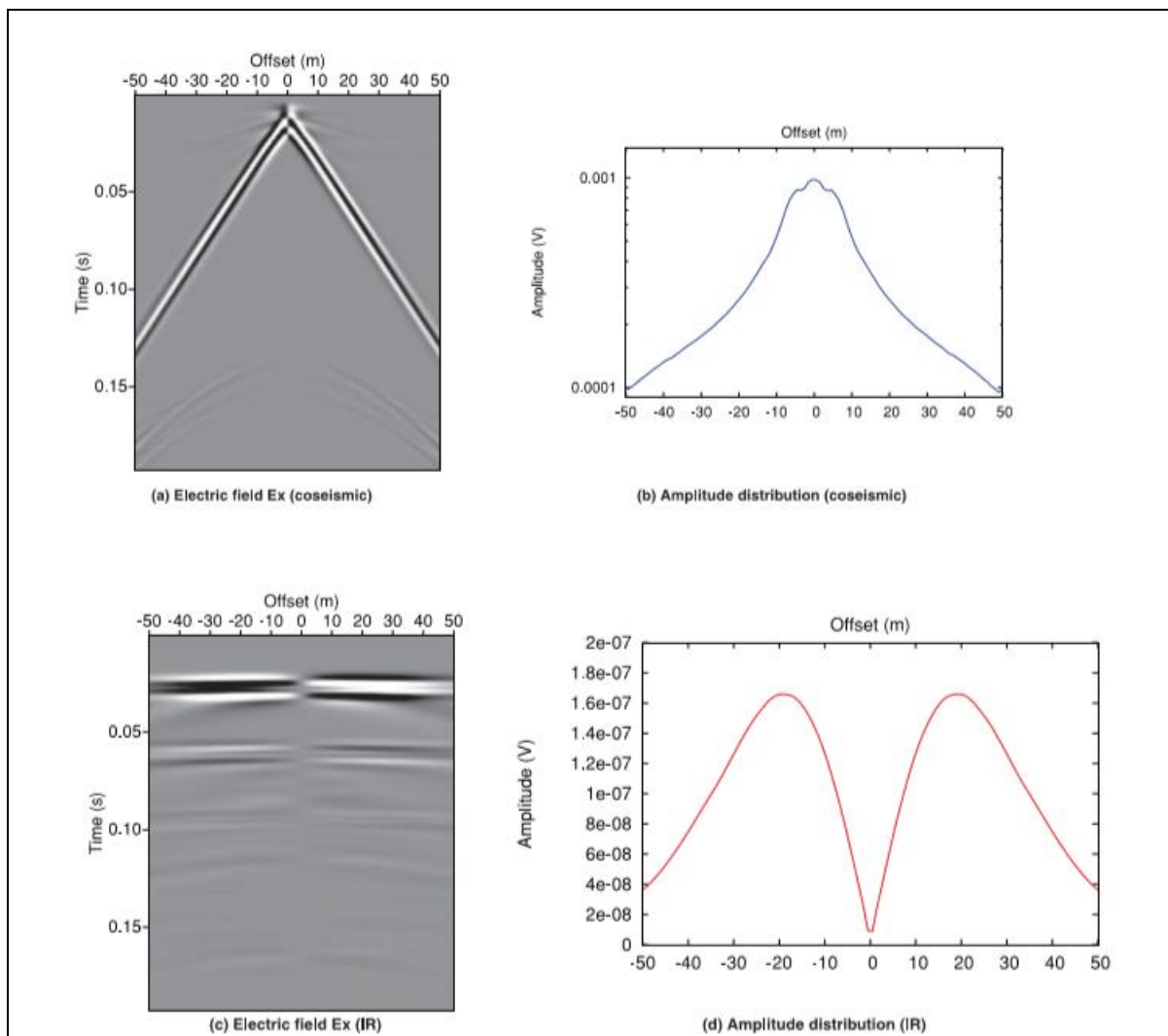


Figure 5: (a) Synthetic recording of the horizontal inline electric field reproduced from Figure 4. (b) Amplitude distribution of recording (a): the maximum absolute amplitude for each trace has been picked and plotted against offset. (c) Synthetic recording of the horizontal inline electric field for which the interface response was artificially highlighted. (d) Amplitude distribution of recording (c): the maximum absolute amplitude for each trace has been picked and plotted against offset.

Recently, several authors have proposed an alternate formulation of the seismoelectric theory, relying on the quasi-static limit of the electromagnetic fields. For instance, Revil and Jardani (2010) have applied this approach to model the seismoelectric response in heavy oil reservoirs. As highlighted by Grobbee and Slob (2016), this approach allows to simplify the system and might help develop the seismoelectric technique toward inversion. We will not further address this approach but direct the interested reader to the handbook of Revil *et al.* (2015).

Pride's theory relies on a number of assumptions, among which:

- The fluid is an ideal electrolyte, i.e. ions move independently from one another. This hypothesis is usually verified for low salinities (<1 mol/L).
- The medium is isotropic.
- The medium is fully saturated.
- Diffusion effects are negligible.
- The wavelength is much larger than the grain size.

In order to describe the conversion mechanisms between seismic and electromagnetic waves, Pride combined Maxwell's equations with Biot's equations for poroelasticity describing the propagation of seismic waves throughout a saturated porous medium. These two subsets of equations are linked through the two following coupling equations:

$$\mathbf{J} = \sigma(\omega)\mathbf{E} + L_{12}(\omega) (-\nabla p + \omega^2 \rho_f \mathbf{u}_s) \quad \text{Equation 2}$$

$$-i\omega \mathbf{w} = L_{21}(\omega)\mathbf{E} + \frac{k(\omega)}{\eta} (-\nabla p + \omega^2 \rho_f \mathbf{u}_s) \quad \text{Equation 3}$$

$\mathbf{J}$ is the electrical current density [A.m^{-2}], expressed as the sum between conduction and electro-filtration currents. The first term on the right-hand side of Equation 2 corresponds to ohmic currents: σ is the conductivity [S/m] and $\mathbf{E}$ the electric field [V/m]. The second term on the right-hand side of Equation 2 is related to the electro-filtration currents. These may be induced by fluid pressure p [Pa] or by the acceleration of the solid matrix $\omega^2 \rho_f \mathbf{u}_s$ [m.s^{-2}], where ω is the angular frequency [rad.s^{-1}], ρ_f is the fluid volumic mass [kg.m^{-3}] and $\mathbf{u}_s$ is the displacement of the solid phase. L_{12} is the seismoelectric coupling coefficient; note that when $L_{12}=0$, Equation 2 falls back to Ohm's law.

$\mathbf{w}$ [m] is the relative displacement between fluid and solid phases; its derivative $-i\omega \mathbf{w}$ [m/s] corresponds to Darcy's velocity. η [Pa.s] is the viscosity and k is the frequency-dependent dynamic permeability [m^2]. L_{21} is the electro-seismic coupling coefficient.

Pride derived that $L_{12}=L_{21}$, i.e. the seismoelectric coupling equals the electro-seismic coupling. This is called the Onsager reciprocity and was already verified for electrokinetic and electro-osmotic phenomena. One can therefore write: $L_{12}=L_{21}=L$.

This coupling coefficient $L(\omega)$ can be written as the product of a static term L_0 with a frequency-dependent term.

Field equipment and processing

In this section, we present a dataset acquired in a sedimentary environment by Garambois & Dietrich (2001), as well as the key processing steps applied to this data, as described in Warden *et al.* (2012).

Source and receivers key features

The minimal field equipment for a seismoelectric experiment consists of:

- A seismic source: a plate hit by a sledgehammer, a weight drop or even a vibrating source, as detailed in the next section.
- Sensors: pairs of electrodes forming a dipole over which a voltage can be measured.
- Cables conveying the signal back to the recording system.
- The recording system itself.

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- Preamplifiers are sometimes used to buffer the signal before it is recorded.

In order for the digitizer not to affect the current travelling within the subsurface between the electrode pair, its resistance must be much higher than that of the ground. However, the ground contact resistance may reach several tens of k Ω which may exceed the device's input impedance, which must therefore be increased through the use of pre-amplifiers. Pre-amplifiers also allow to multiply the weak (several mV to several μ V) seismoelectric signals before those undergo a stacking procedure. On the other hand, they ensure a high input impedance.

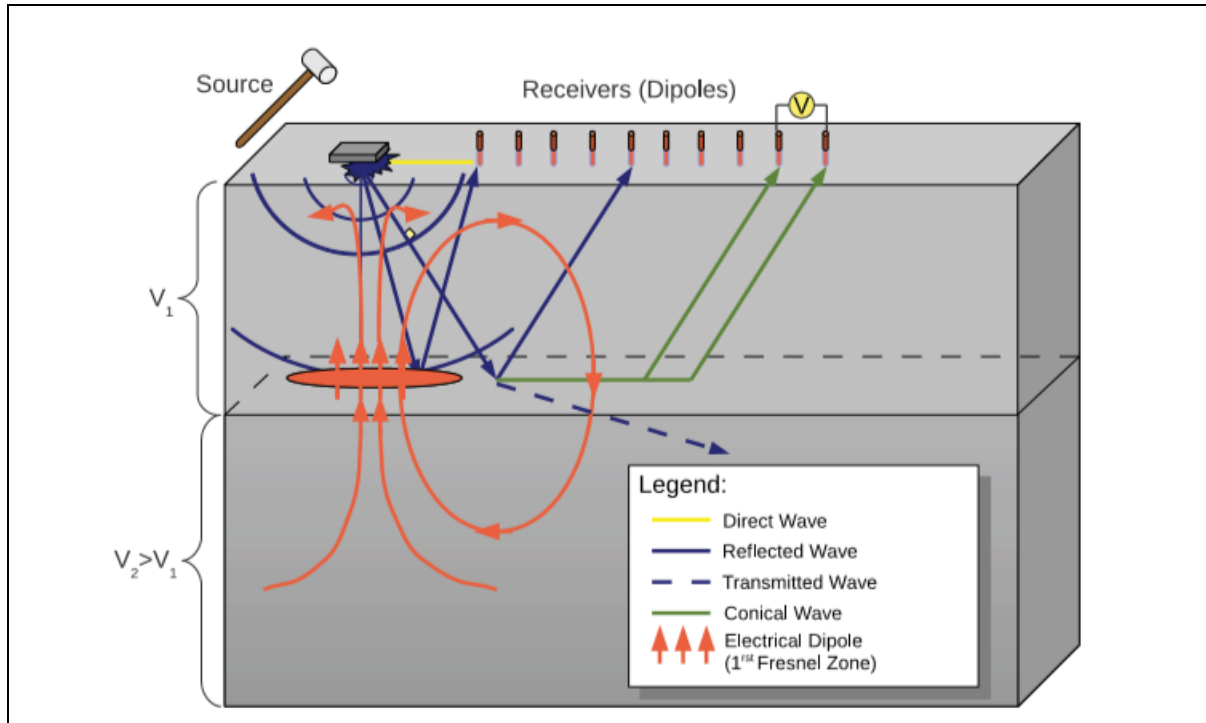


Figure 6: Typical seismoelectric layout. From Warden et al., 2012.

Acquisition

In Figure 7 we present seismoelectric data acquired in a sedimentary environment by Garambois & Dietrich (2001).

These data were collected in a river catchment area by the Fier river near the Annecy area. The stratigraphy of the site is marked by alluvial deposits with alternating layers of gravel, sand, and clay from the surface down to 150m depth. At the time of the experiment, the water table was located at a depth of about 1.5m. The source used here was a 200g dynamite charge buried at a depth of 1m. The data were recorded by 12 unevenly spaced dipoles, deployed on either side of the shot point, between offsets $x=\pm 5$ m and $x=\pm 30$ m.

Data processing: Anthropic noise reduction

Seismoelectric data are often contaminated with anthropic noise, mainly 50 or 60Hz power-line noise, depending on the survey area.

Several techniques have been proposed to reduce this power-line noise.

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The **block subtraction technique** (Butler & Russell, 2003) implies recording for a few seconds while the seismic source is not active, typically just before activating the source. This "passive" recording, assumed to consist of pure harmonic noise, is used to estimate the ambient noise: it can then be duplicated over the entire duration of the actual seismoelectric recording before being subtracted from the data.

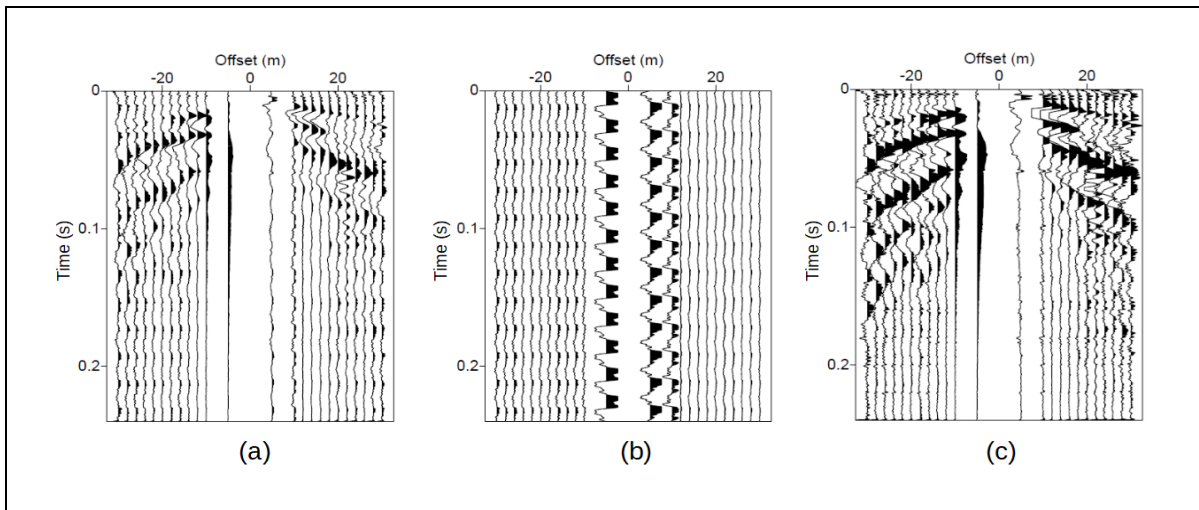


Figure 7: Seismoelectric data acquired by the Fier river. From Garambois & Dietrich, 2001. Results of the sinusoidal subtraction technique: (a) raw data. This recording is dominated by coseismic Rayleigh surface waves.; (b) power line noise estimate; (c) processed data. Power line noise was estimated up to the order $n=15$.

The drawback with this simple approach is that any non-harmonic noise component present over the initial recording interval will be subtracted from the data and will thus contaminate the processed recording, where it will appear with opposite polarity.

Also, extra care must be taken when duplicating the initial noise interval over the entire duration of the seismoelectric recording. Ideally, the noise block should be shifted by a multiple of $1/f_0$, where f_0 is the fundamental powerline noise frequency: this step may require resampling the data.

For the example given in Figure 7, we did not have access to data acquired before the source was activated: we therefore assumed the last 40ms of the recording consisted of pure harmonic noise. The noise estimate is represented in Figure 7(b). The processed data appear in Figure 7(c).

Minimizing the power-line noise magnifies some of the seismoelectric arrivals at late time and large offsets.

The **sinusoidal subtraction technique** consists in finding the amplitude and the phase of the n -th noise harmonic contaminating the seismoelectric recording through a least-square approach. The harmonic noise $p(t)$ can be written as a sum of cosine and sine functions such as:

$$p(t) = \sum_{k=1}^{\infty} a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)$$

Equation 4

Where t is the time, ω_0 is the fundamental angular frequency and k is the order of the harmonic.

For each order k , the amplitude and phase of the k -th harmonic can be deduced from the coefficients a_k and b_k . These coefficients are unknown, but can be estimated by minimizing the difference between the data and the sinusoid of frequency nf_0 in a least-square sense.

Data processing: Interface response extraction

The previous section describes how to reduce the power-line noise often contaminating seismoelectric data. The processed seismoelectric recording contains signals of both types: type I (coseismic waves) and type II (interface response), as described in the section **Back to the Theory**.

The coseismic signals provide information about the medium in the vicinity of the receivers, that is, in the very near surface. The interface response, on the other hand, provides information about the subsurface at depth.

Unfortunately, the interface response may be hard to recover, because its amplitude is generally much weaker than that of the coseismic response. Furthermore, in the case of surface geometries, the interface response usually shows up at about half the two-way travel time needed by the associated coseismic wave to reach the receivers: the total travel time of the interface response corresponds to the time needed for the direct seismic wave to reach the interface, plus the time needed by the converted electromagnetic wave to travel back to the surface, usually much smaller. Therefore, the weak interface response is often "masked" by the stronger coseismic signals, as seen in Figure 8 in the case of synthetic data.

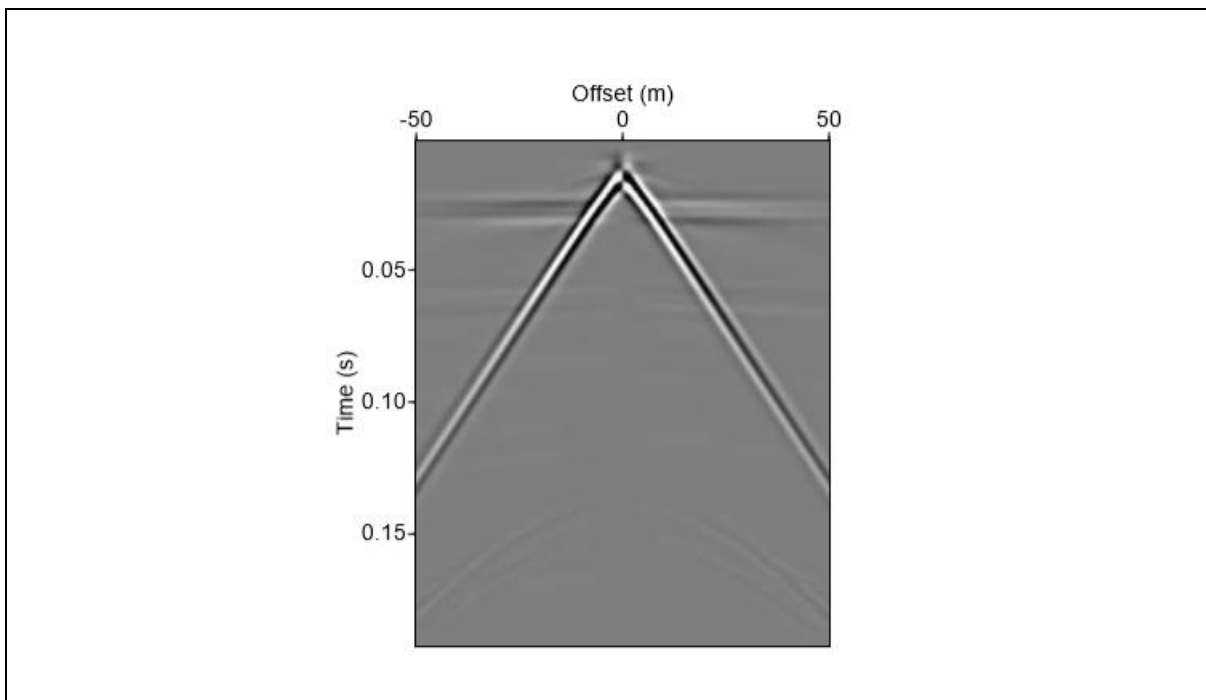


Figure 8: Synthetic seismoelectric recording modeled for a two-layers tabular medium consisting of a sand layer on top of a sandstone half-space. The horizontal event seen at about $t=0.02s$ corresponds to the interface response. From Warden et al., 2012.

The interface response can be extracted through digital filtering techniques. Most of these filtering strategies are based on the interface response's very high velocity, which translates into the fact that the interface response appears as a horizontal event on the seismoelectric recording.

Frequency-wavenumber ($f - k$) domain filtering is one of these dip-based techniques: this method commonly applied to seismic data to separate events based on their slope. For instance, it is used to filter out the ground roll from seismic data. Several authors (Haines et al. 2007; Strahser 2007; Warden et al., 2012) have applied this strategy to IR extraction.

Let us consider the same seismoelectric dataset that was processed for power-line noise in the previous section: this recording is represented in Figure 9 a.

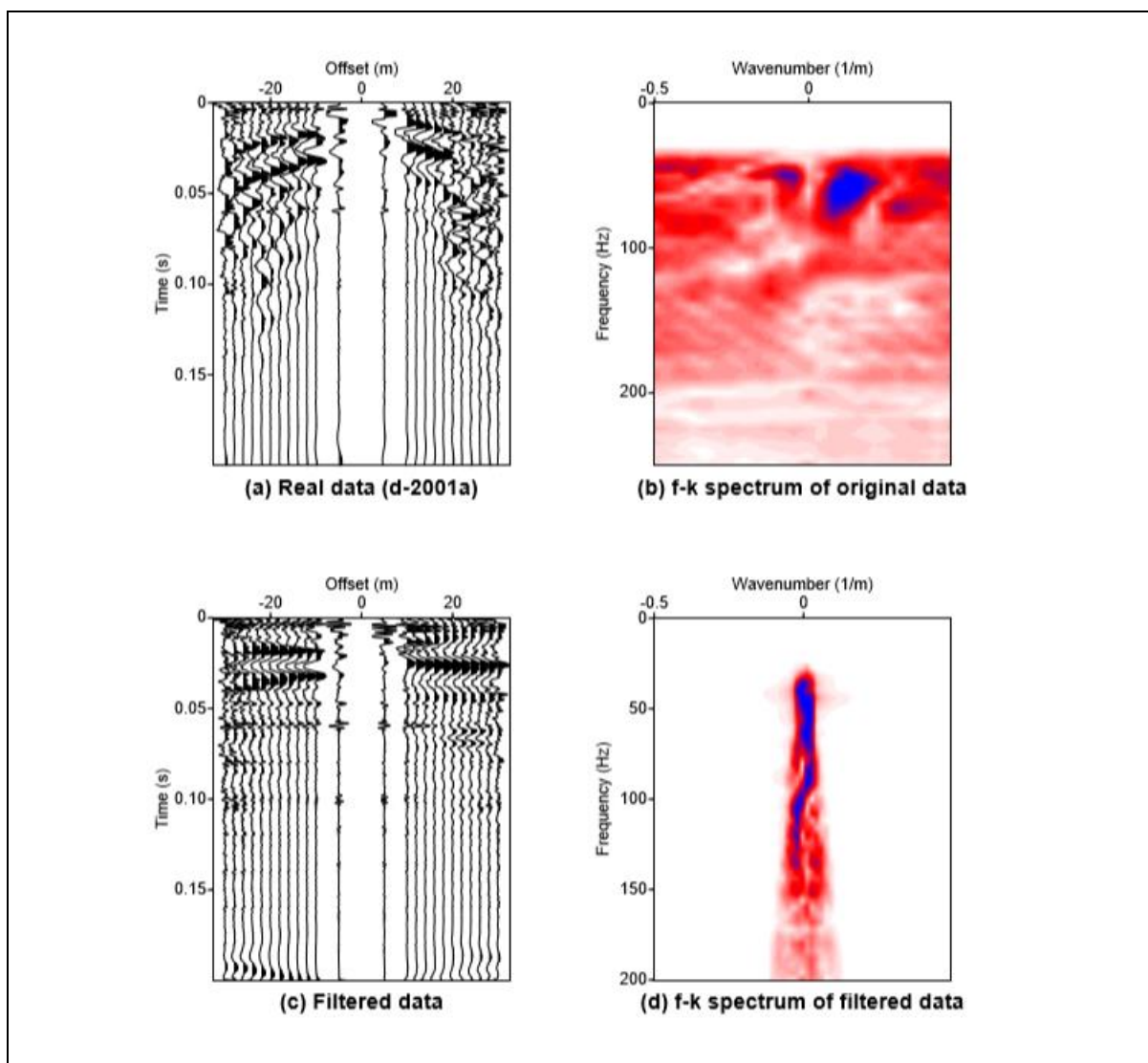


Figure 9: (a) Original seismoelectric dataset after power-line noise removal, represented in the time-distance ($t-x$) domain. (b) Dataset from a) represented in the frequency wavenumber domain. (c) Data in $t-x$ domain after IR extraction. (d) Data in $f-k$ domain after IR extraction using a "pie-slice" filter selecting high-velocity events.

$f-k$ filtering consists in applying a spatial and temporal Fourier transform to the original data: the data in $f-k$ domain are represented in Figure 9 b. In this representation, events are plotted along lines whose slopes are

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given by their apparent velocities $v=df/dk$. Thus, zero-slowness arrivals such as type II converted waves align along the vertical wavenumber axis $k=0$ and can be recovered using a "pie-slice" filter; see for instance Figure 9 d. The data are then Fourier transformed back to the conventional time-space ($t-x$) domain, as shown in Figure 9 c. Interface responses can be clearly seen on selected time windows after f-k filtering (figure 10).

f-k filtering is easy to apply but suffers a number of limitations: it requires a good spatial resolution, i.e. a large number of traces, and it does not filter out the flat portions of the type I coseismic signal, such as reflection hyperbolas near the shot-point. Last but not least, it may alter the amplitudes of the extracted IR.

Other filtering techniques (Warden *et al.*, 2012) have been devised to circumvent these problems: some of them are dip-based, such as Radon domain filtering, while others rely upon filtering in the curvelet domain. Curvelets are a type of wavelets that allow to optimally describe seismic or seismoelectric wave-fronts. After decomposing both the seismic and seismoelectric recordings using this multi-scale and multi-direction transform, it is then possible to apply a mask in the curvelet domain taking advantage of the correlation between seismic acceleration and coseismic data.

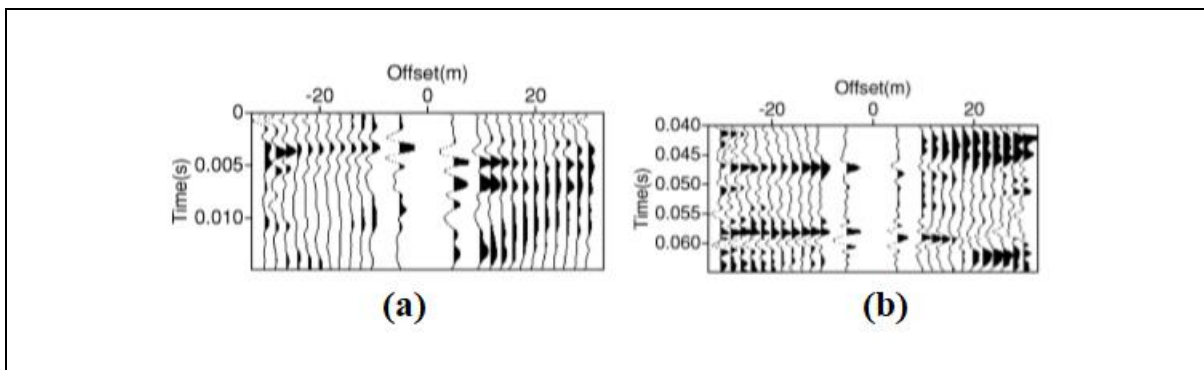


Figure 10: Close-up view on Figure 9 c. These two figures show some of the interface responses extracted through f-k filtering. Note the opposite polarities on either side of the shot point at offset $x=0$.

Last but not least, some authors (e.g. Haines, 2004) have suggested using fan-shaped acquisition geometries to achieve "natural" separation between coseismic and interfacial signals. Setting source and receivers on either side of the interface of interest allows the interface response to arrive before the associated coseismic waves.

Survey example

We present here an experiment designed to overcome the low signal-to-noise ratio (SNR) associated with seismoelectric conversions. The use of limited energy sources such as a weight drop or a sledgehammer hitting a plate usually produces weak signals and requires a large amount of stacking to achieve an acceptable SNR, even at shallow depths of several meters. In this experiment, resorting to a hydraulic vibrator helped overcome this difficulty.

This work was a joint effort between WesternGeco, Schlumberger Water Services and the Curtin University; it was presented at the SEG 2012 annual meeting.

Experiment layout

This "brute-strength" experiment was conducted at an aquifer storage and recovery (ASR) site in the Abu Dhabi Emirate. ASR consists in injecting water into an aquifer in order to produce it later on: a wide variety of seismic and non-seismic methods can be used to help delineate the aquifer. In April 2011, a seismic survey

was conducted at this ASR site: a seismic receiver line consisting of both one-component (1C) and three-components (3C) geophones was deployed over 1,2km. The source line was 3,6km long, extending 700m West and 1,700m East of the receiver line.

The key idea behind this experiment was to record seismoelectric data simultaneously with seismic data to take advantage of the powerful seismic source used for seismic imaging. Three separate seismoelectric spreads were deployed, each array consisting of 49 electrodes separated by 4m. The first two spreads were deployed at the eastern end of the seismic receiver line, while the third spread was centered on the receiver line.

The source used here was an 80,000lb peak force tracked DX-80 Desert Explorer Vibrator.

Data processing

50Hz power-line noise was the main noise source here. It was attenuated using a harmonic subtraction algorithm such as the one described in Section "Data processing: Anthropogenic noise reduction". Vibroseis correlation was also performed on the data.

Results

Figure 11-a shows the seismoelectric and seismic super-gathers created from 8 records acquired with a 1m source spacing. Both recordings look fairly similar as several seismic events have seismoelectric counterparts, exhibiting similar slope and intercept time values: these are coseismic or type-I waves. But horizontal arrivals can also be seen on the seismoelectric recordings, as highlighted by the white arrows at 32ms and 47ms: these arrivals correspond to waves travelling at velocities several orders of magnitude greater than seismic velocities. Also, they exhibit reversed polarities on either side of the shot point at $x=0$. Last but not least, they appear at early times, that is, before the reflected coseismic waves reaches the receivers at the surface. These characteristics indicate that these arrivals correspond to the type-II interface response.

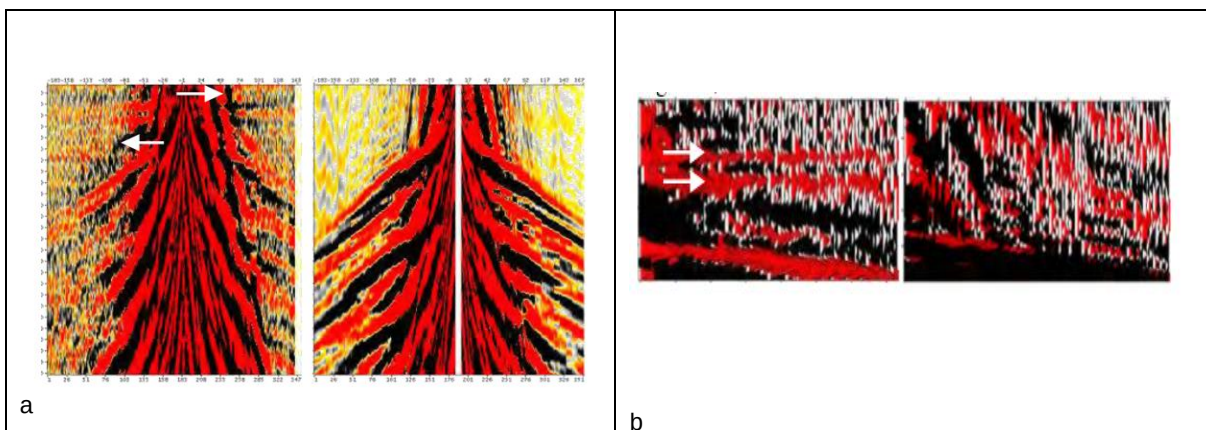


Figure 11: Field example. (a) Seismoelectric (left) and seismic (right) data super-gathers. These gathers were obtained by combining 8 records. The trace spacing is 1m, with a maximum offset at 160m. The recording duration is 0.3s. (b) Close-up view on the recordings from Figure 11 a: seismoelectric traces are displayed on the left-hand side, while seismic traces are seen on the right-hand side. The first 100ms are displayed here, for offsets comprised between $x=60$ and $x=100$ m. The traces are sorted by offset. The white arrows indicate two interface responses.

Figure 11.b shows a close-up view of the recordings seen in Figure 11.a. The first 100ms for offsets comprised between $x=60$ and $x=100$ m are displayed here for the seismoelectric recording on the left-hand side. The corresponding seismic recording is displayed to the right. The white arrows indicate two interface responses.

Note that these events of zero slowness have no equivalent in the seismic recording. The recording shown here is a super-gather, which means that it is a combination of several recordings: random electrical noise is cancelled through this stacking process which confirms the flat events highlighted here are indeed interface responses.

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The author would like to thank Stéphane Garambois and Tim Dean for allowing to reproduce figures from their work.

Suggested readings

A comprehensive review of the seismoelectric method was recently written by Jouniaux & Zyserman (2016). "A review on electro-kinetically induced seismo-electrics, electro-seismics, and seismo-magnetics for Earth sciences". This work compiles several field and lab examples and is a good starting point for the interested reader.

For those who wish to go further, we also suggest reading the handbook "The Seismoelectric Method: Theory and Application" by Revil *et al.* (2015).

References

Dean T., C. Dupuis, R. Herrmann and J. Valuri, 2012, A brute strength approach to improving the quality of seismoelectric data. SEG extended abstract. SEG Las Vegas.

Garambois S. and M. Dietrich, 2001, Seismoelectric wave conversions in porous media: field measurements and transfer function analysis, *Geophysics*, 66(5), 1417–1430.

Grobbe N. and E. Slob, 2016, Seismo-electromagnetic thin-bed responses: Natural signal enhancements?, *Journal of Geophysical Research – Solid Earth*, 121, 2460–2479. doi: 10.1002/2015JB012381.

Haines S., S. Pride, S. Klemperer. & B. Biondi, 2007, Seismoelectric imaging of shallow targets, *Geophysics*, 72(2), G9–G20, 2007. doi:10.1190/1.2428267.

Jouniaux L. and F. Zyserman, 2016, A review on electrokinetically induced seismo-electrics, electro-seismics, and seismo-magnetics for Earth sciences. *Solid Earth*, 7, 249–284.

Revil A. and A. Jardani, 2010, Seismoelectric response of heavy oil reservoirs: theory and numerical modeling, *Geophysical Journal International*, 180, 781–797. doi:10.1111/j.1365-246X.2009.04439.x.

Revil A., A. Jardani, P. Sava and A. Haas, 2015, *The seismoelectric method: theory and application*. Wiley-Blackwell. ISBN: 978-1-118-66026-3.

Strahser M., 2007, Polarization and slowness of seismoelectric signals: a case study, *Near Surface Geophysics*, 5, 97–114. doi:10.3997/18730604.2006022.

Thompson AH., J. R. Sumner, and S. C. Hornbostel, 2007, Electromagnetic-to-seismic conversion. *The Leading Edge*, 26(4) :428–435. doi: 10.1190/1.2723205.

Volarovich M. and G. Sobolev, 1969, Direct piezoelectric prospecting of quartz and pegmatitic veins. *Geoexploration*, 7(4) :241–246. ISSN 0016-7142. doi: 10.1016/00167142(69)90029-5.

Warden S., S. Garambois, P. Sailhac, L. Jouniaux, and M. Bano, 2012, Curvelet-based seismoelectric data processing. *Geophysical Journal International*, 1533–1550. ISSN 0956540X.

Warden S., S. Garambois, L. Jouniaux, D. Brito, P. Sailhac and C. Bordes, 2013, Seismoelectric wave

IFP Energies Nouvelles ebook: JL Mari et al., *Electromagnetic methods in Geophysics*, 2016,

DOI: 10.2516/ifpen/2017001

<http://books.ifpennergiesnouvelles.fr/ebooks/ifpen-electro/>

propagation numerical modeling in partially saturated materials. Geophysical Journal International, 1498-1513.
ISSN 0956-540X.

IFP Energies Nouvelles ebook: JL Mari et al., Electromagnetic methods in Geophysics, 2016,
DOI: 10.2516/ifpen/2017001
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Modelling pH-Dependent and Microstructure-Dependent Streaming Potential Coefficient and Zeta Potential of Porous Sandstones

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Abstract In this paper, we have significantly modified an existing model for calculating the zeta potential and streaming potential coefficient of porous media and tested it with a large, recently published, high-quality experimental dataset. The newly modified model does not require the imposition of a zeta potential offset but derives its high salinity zeta potential behaviour from Stern plane saturation considerations. The newly modified model has been implemented as a function of temperature, salinity, pH, and rock microstructure both for facies-specific aggregations of the new data and for individual samples. Since the experimental data include measurements on samples of both detrital and authigenic overgrowth sandstones, it was possible to model and test the effect of widely varying microstructural properties while keeping lithology constant. The results show that the theoretical model represents the experimental data very well when applied to model data for a particular lithofacies over the whole salinity, from 10^{-5} to 6.3 mol/dm^3 , and extremely well when modelling individual samples and taking individual sample microstructure into account. The new model reproduces and explains the extreme sensitivity of zeta and streaming potential coefficient to pore fluid pH. The low salinity control of streaming potential coefficient by rock microstructure is described well by the modified model. The model also behaves at high salinities, showing that the constant zeta potential observed at high salinities arises from the development of a maximum charge density in the diffuse layer as it is compressed to the thickness of one hydrated metal ion.

Keywords Microstructure · Porosity · Streaming potential · Zeta potential · Electrical rock properties

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List of Symbols

C_f	Pore fluid salinity (mol/dm ³)
C_{sp}	Streaming potential (coupling) coefficient with respect to pressure (mV/MPa)
F	Formation factor, $F = \phi^{-m}$ (–)
K_1	Equilibrium constant for solution of CO ₂ in water. $pK_1 = -\log_{10} K_1$ (–)
K_2	Equilibrium constant for solution of CO ₂ in water. $pK_2 = -\log_{10} K_2$ (–)
$K_{(-)}$	Disassociation constant for dehydrogenization of silanol surface sites. $pK_{(-)} = -\log_{10} K_{(-)}$ (–)
K_{me}	Binding constant for cation adsorption. $pK_{me} = -\log_{10} K_{me}$ (–)
K_w	Disassociation constant of water. $pK_w = -\log_{10} K_w$ (–)
N	Avogadro's number, $\sim 6.022 \times 10^{+23}$ /mol (mol ^{–1})
pH_{expt}	pH at which the experiments were run (arithmetic mean value) (–)
$pH_{\text{st.dev}}$	Standard deviation in pH_{expt} (–)
T	Temperature (°C or K)
a	1. Fitting parameter in Eq. (1) for zeta potential (mV). 2. Constant in the Theta Transformation, $= 8/3$ for clastic rocks (–)
b	Fitting parameter in Eq. (1) for zeta potential (mV)
d_{gr}	Modal grain diameter (m)
e	Elementary charge, $\sim 1.602 \times 10^{-19}$ C (C)
k	Boltzmann's constant, $\sim 1.3806 \times 10^{-23}$ J/K (J/K)
m	Cementation exponent (–)
β_{Na+}	Na ⁺ Ionic fluid mobility (m ² /s/V)
β_{H+}	H ⁺ Ionic fluid mobility (m ² /s/V)
β_{Cl-}	Cl [–] Ionic fluid mobility (m ² /s/V)
β_{OH-}	OH [–] Ionic fluid mobility (m ² /s/V)
β_{Stern}	Stern layer mobility
ϵ_r	Relative dielectric permittivity of the pore fluid (–)
ϵ_o	Absolute dielectric permittivity of a vacuum, $\sim 8.854 \times 10^{-12}$ F/m (F/m)
ζ	Zeta potential (mV)
η_f	Pore fluid viscosity (Pa s)
σ_s	Measured surface conductivity (–)
ϕ	Total porosity (–)
$\chi\zeta$	Shear plane distance (m)
Γ_s^o	Density of mineral surface sites for complexation (sites/nm ²)
Λ	Characteristic length scale of the pores (m)
Σ_s^{Prot}	Protonic surface conductance/specific conductivity (S)
Σ_s	Total surface conductance/specific conductivity (S)

1 Introduction

The only existing theoretical model for the streaming potential coefficient (C_{sp}) and zeta potential (ζ) in porous media was proposed by Glover et al. (2012) and introduced a parameter called the zeta potential offset. However, the data on which it was tested were not of sufficient quality to allow validate it fully, especially as a function of pH and rock microstructure. There also remained the difficulty of interpreting the physical meaning of the zeta potential offset parameter. This empirical parameter was included in the Glover et al. (2015) model because

its inclusion allowed the model to work well. However, at the time of its inclusion it had no satisfactory formal interpretation.

Consequently, an experimental campaign was initiated to produce a large database of 1253 measurements of C_{sp} and ζ of the highest possible quality, using pore fluids which were fully equilibrated with the rock samples, where pore fluid salinity, conductivity and pH were all accurately measured, which were supported by a full suite of associated petrophysical measurements made on individual samples, and where the dataset contains a large number (324) of measurements with a salinity $C_f > 1 \text{ mol/dm}^3$ (Walker and Glover 2017). The availability of this dataset offers the opportunity to carry out a detailed modelling campaign.

This paper reports that modelling campaign, and is, we believe, the most comprehensive modelling of C_{sp} and ζ carried out so far. The first aim of the work described in this paper was to modify the existing C_{sp} and ζ theoretical model (Glover and Déry 2010; Glover et al. 2012, 2015) in order to fully represent pH variation in electrolytes. In previous models, the pH was allowed to vary as a function of temperature and with the addition of acids and bases but not as a function of salinity. The second aim was to replace the unjustified zeta potential offset with a modelling approach that takes into account the development of a maximum charge density in the diffuse layer as it is compressed to the thickness of one hydrated metal ion in thickness at high salinities. The third aim was to attempt to model the data of Walker and Glover (2017) both by facies and for individual rock samples and, in so doing, confirm (1) the operation of the low salinity control of microstructure on C_{sp} , (2) examine the role of pH variations on C_{sp} and ζ , and (3) investigate the possible physical causes of the zeta potential offset hypothesized by Jaafar et al. (2009) and Vinogradov et al. (2010).

There have been several attempts to model the ζ and C_{sp} of rocks. Pride and Morgan (1991) produced an empirical fit to 35 existing ζ data as a function of salinity in the low salinity range, $C_f < 0.1 \text{ mol/dm}^3$. They obtained

$$\zeta = a + b \log_{10}(C_f), \quad (1)$$

where ζ is in volts and C_f is in mol/dm^3 , $a = +8 \text{ mV}$ and $b = +26 \text{ mV}$. Although this relationship describes the main aspects of the data to which it was fitted well, it provides positive ζ at salinities $> 0.5 \text{ mol/dm}^3$, whereas the physical model of the mineral/fluid interface would predict a negative value trending towards zero, and all measurements that have been made so far at high salinities provide extremely small but negative values (Jaafar et al. 2009; Vinogradov et al. 2010; Walker et al. 2014; Walker and Glover 2017). Similar empirical fits have been carried out by a number of authors with similar results: Walker and Glover (2017) provided $a = +3.505 \pm 28.823 \text{ mV}$ and $b = +11.33 \pm 4.06 \text{ mV}$ and for their entire dataset as well as carrying out fits to individual facies, Vinogradov et al. (2010) obtained $a = +9.67 \text{ mV}$ and $b = +19.02 \text{ mV}$, Jaafar et al. (2009), which share some data with Vinogradov et al. (2010), suggested $a = +6.43 \text{ mV}$ and $b = +20.85 \text{ mV}$. Bolève et al. (2007) suggested $a = +14.6 \text{ mV}$ and $b = +29.1 \text{ mV}$, while Revil et al. (1999a) have estimated the values at $a \approx +10 \text{ mV}$ and $b \approx +20 \text{ mV}$. However, these fits are simply empirical relationships. Their similarity (as shown in Fig. 1) underlines a general trend. However, all of these fits produce positive ζ at high salinities, whereas none of the 400 or so measurements for $C_f > 1 \text{ mol/dm}^3$ so far are positive (Jaafar et al. 2009; Vinogradov et al. 2010; Walker and Glover 2017); they all have small constant negative values instead. Clearly the empirical approach is limited, and a physically based theoretical model is required.

In the late 1990s, a series of papers allowed a theoretical model for ζ to be constructed (Revil and Glover 1997, 1998; Revil et al. 1999a). Equation (1) is obtained theoretically in the last of these papers. Revil et al. (1999a) applied a number of simplifying assumptions to

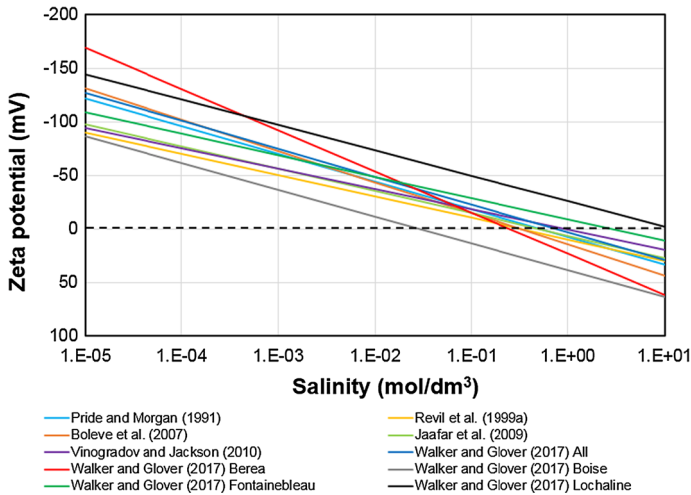


Fig. 1 Representation of 10 existing empirical fits of ζ as a function of pore fluid salinity showing approximately the same behaviour but with unexplained high salinity behaviour and variability with pH

their more general equations, which were (1) that $\text{pH}=7$, (2) that the fluid is an Na^+ or K^+ symmetric electrolyte, (3) that the influence of H^+ and OH^- ions on the ionic strength of the solution saturating the pores can be neglected (Revil and Glover 1997), and (4) that direct adsorption of K^+ and Na^+ ions upon the silica surface can also be neglected.

Under these conditions

$$a = \frac{2kT}{3e} \ln \left\{ \frac{\sqrt{8000 \varepsilon_r \varepsilon_0 k T N}}{2e \Gamma_s^\circ K_{(-)}} 10^{-\text{pH}} \right\}, \quad (2)$$

and

$$b = \frac{kT}{3e} \log_e 10, \quad (3)$$

where k is Boltzmann's constant ($\sim 1.3806 \times 10^{-23}$ J/K), T is the temperature (in K), e is the elementary charge ($\sim 1.602 \times 10^{-19}$ C), ε_r is the relative dielectric permittivity of the pore fluid, ε_0 is the absolute dielectric permittivity of a vacuum ($\sim 8.854 \times 10^{-12}$ F/m), N is Avogadro's number ($\sim 6.022 \times 10^{23}$ /mol), Γ_s° is the density of surface sites, and $K_{(-)}$ is the disassociation constant for dehydrogenization of silanol surface sites. The gradient of ζ versus $\log_{10}(C_f)$ then depends only on temperature with $b=+19.38$ mV when $T=20^\circ\text{C}$, while the offset at the same temperature is $a = +16.89$ mV for reasonable choices of its input parameters ($\text{pH} \equiv 7$, $\Gamma_s^\circ = 10$ sites/ nm^2 , $K_{(-)} = 10^{-8}$, and assuming $\varepsilon_r = 80$), which are discussed later in this paper. This model gives values of ζ which remain negative for the entire salinity range, tending towards zero at high salinities ($C_f \approx 7.4$ mol/ dm^3 , which is just higher than the saturation limit for NaCl in water at $C_{\text{sat}} \approx 6.16$ mol/ dm^3) as required by the physical model.

Although the ζ could be calculated from the Revil et al. (1999a) model without the restrictions used to produce Eqs. (2) and (3), or from a more complex calculation that accounts for a variable pH, calculation of C_{sp} remained more difficult. One approach would be to use the classical Helmholtz–Smoluchowski (H–S) equation (e.g., Glover et al. 2015; Walker and Glover 2017), but as the H–S equation has only been validated for capillary tubes, the

result would not take into account the properties of the rock such as porosity, grain size and cementation exponent. In 2010, Glover and Déry (2010) provided the equations required to calculate the C_{sp} of individual rocks, taking into account their individual microstructural properties. That paper, which considered packs of glass beads, implemented a full theoretical description of ζ (Revil et al. 1999a), and C_{sp} was then calculated for the grain size, porosity and cementation exponent of the glass bead packs. The modelling was capable of representing the sigmoidally shaped variation of streaming potential with grain size (Glover and Déry 2010), but was restricted to the ideal matrix geometry imposed by the bead packs.

The theory was developed further and published in a form which could be applied to porous granular media (Glover et al. 2012) and was validated against a database of 290 C_{sp} and 269 ζ measurements made by a large number of researchers. Unfortunately this database did not contain sufficient information about the salinity, electrical conductivity and pH of the fluids in contact with the grains of the rock at the time that the streaming potential measurement was made. In many cases, the experimental temperature was not measured or reported and information about the microstructural properties of the samples that were being measured was not measured or given. Although this dataset represented all the data that were available, it was not considered a stringent test of the theoretical model. Consequently, new experimental approaches (Walker et al. 2014) were developed to make a large number of well-constrained measurements for testing the model, which are reported in full in Walker and Glover (2017) and form the experimental base with which the modelling in this paper is compared.

2 Reference Data

A full description of the reference data can be found in Walker and Glover (2017). This modelling uses the entire dataset of 1253 C_{sp} measurements and their derived ζ , representing 14 samples of 4 sandstone facies (Berea, Boise, Lochaline and Fontainebleau sandstones). Several separate examples of Fontainebleau and Lochaline sandstones composed of sub-rounded detrital grains, and consisting of detrital grains with euhedral quartz overgrowths, were measured (distinguished by the additional letter 'D' and 'Q' in the sample codes, respectively). A brief summary of sample properties measured during initial characterisation is shown in Table 1 of Walker and Glover (2017). The quality of the experimental data is generally good, having been measured at pore fluid equilibrium and with well-defined salinity and pH. However, the early data from one of the Berea sandstone (BR1) and one of the Boise sandstone samples (B1II) are less good and this is due to immature experimental protocols, as described in Walker and Glover (2017).

3 Electro-kinetic Modelling

The theoretical model used in this work is based on the electro-chemical approach to modelling the electrical double layer (Pride 1994; Revil and Glover 1997; 1998; Revil et al. 1999a; Glover et al. 2012). It has been modified for this paper by (1) the inclusion of a variable pH that is a function of temperature, added acid, added base, and salinity, (2) improvements to the calculation and temperature and salinity dependencies of the fluid properties, and (3) the implementation of implicit modelling of the high salinity ζ obviating any need for the zeta potential offset as in previous models (e.g., Glover et al. 2012).

Table 1 Model parameters

Parameter type	Parameters	Status and value	Sources
<i>Fixed parameters</i>			
Experimental conditions	Temperature T (°C and K)	Fixed: From experimental data	Walker and Glover (2017)
Fundamental physical constants	Permittivity in vacuo	Fixed: 8.850×10^{-12} F/m	Lide (2017)
	Boltzmann's constant	Fixed: 1.381×10^{-23} J/K	Lide (2017)
	Avagadro's constant	Fixed: 6.022×10^{23} /mol	Lide (2017)
	Electronic charge	Fixed: 1.602×10^{-19} C	Lide (2017)
	Na ⁺ Ionic fluid mobility	Fixed: 5.20×10^{-8} m ² /s/V	Crow (1988)
Pore fluid parameters	H ⁺ Ionic fluid mobility	Fixed: 3.63×10^{-7} m ² /s/V	Crow (1988)
	Cl ⁻ Ionic fluid mobility	Fixed: 7.90×10^{-8} m ² /s/V	Crow (1988)
	OH ⁻ Ionic fluid mobility	Fixed: 2.05×10^{-7} m ² /s/V	Crow (1988)
	Disassociation constant K_w	Fixed: 7.801×10^{-15} mol/dm ³	Calculated in the model using Lide (2017)
	Equilibrium constant pK_1 (–)	Fixed: 2.95121×10^{-8}	Wu et al. (1991) and Revil et al. (1999a, b)
	Equilibrium constant pK_2 (–)	Fixed: 5.01187×10^{-11}	Wu et al. (1991) and Revil et al. (1999a, b)
	Stern layer ionic mobility β_{Stern}	Fixed: 5×10^{-9} m ² /s/V	Revil and Glover (1997)
Fixed mineral/pore fluid interface parameters	Surface site density Γ_o	Fixed: 10 sites/nm ²	Revil and Glover (1998)
	Cation adsorption binding constant pK_{Me}	Fixed: 7.5	Kosmulski (1996)
	Shear plane distance $\chi\zeta$	Fixed: 2.4×10^{-9} m	Revil et al. (1999a, b)
	Protonic surface conduction Σ_{prot} (S)	Fixed: Calculated in the model	Calculated in the model
Microstructural parameters	Modal grain size, d_{gr} (m)	Fixed: From experimental data	Walker and Glover (2017)
	Porosity, ϕ (–)	Fixed: From experimental data	Walker and Glover (2017)
	Cementation exponent, m (–)	Fixed: From experimental data	Walker and Glover (2017)
	Formation factor, F (–)	Fixed: Calculated in the model	Calculated in the model
	Characteristic pore size, Λ (m)	Fixed: Calculated in the model	Calculated in the model
<i>Variable parameters</i>			
Model variables	Pore fluid salinity (mol/dm ³)	Varied between 10^{-5} and 10 mol/dm ³	Experimental range in Walker and Glover (2017)
	Pore fluid pH (–)	For fixed values: pH 6, 6.5, 7, 7.5, 8	
Variable parameters	Silanol dehydrogenization constant pK (–)	Variable: between 7.3 and 8.1	Dove and Rimstidt (1994), Kosmulski (1996), Hiemstra and van Riemsdijk (1990), Rustad et al. (1998)

It should be noted that modelling has been carried out at the temperature at which the individual data were measured, but instead of implementing one modelling curve for the experimental pH, a range of different models with different pHs have been implemented and shown as 5 modelling curves which divide the zeta potential–pore fluid salinity space and streaming potential–pore fluid salinity space. This approach has the advantage of informing the reader how the model behaves with both changes in pore fluid salinity and pH in one plot. The experimental data are then added to the plot following a trend which lies on or is parallel to a curve for one of the modelled pHs. Conformity of the model to the experimental data can then be judged by comparing the experimentally derived pH (provided in the data tables) with that indicated by how the experimental data fall on the network of modelled curves for different pHs. Despite there being many variables in the model, only one, $K_{(-)}$, was varied to improve the conformity of the model curves to the experimental data, and then only over a very restricted range supported by independent measurements for this parameter (please see Sect. 3.7).

3.1 Modelling Procedure

The modelling has the following steps:

1. Define the temperature, fluid salinity range and fluid pH over which the modelling is to take place.
2. Calculate the density of pure water in kg/m^3 at the modelling temperature T in ($^{\circ}\text{C}$) using

$$\rho_w = 1000 \frac{1 - (T + 288.9414)}{508929.2 \times (T + 68.12963)} (T - 3.9863)^2. \quad (4)$$

3. Calculate the density of the NaCl solution (in g/cm^3) at the given salinity (in mol/dm^3) and temperature using

$$\rho_f = 58.44 C_f + \frac{\rho_w}{1000} \left(1000 - \frac{58.44 C_f}{2.16} \right). \quad (5)$$

4. Calculate the molality of the NaCl solution (in mol/kg) at the given salinities (in mol/dm^3) and temperature using

$$c_f = \frac{C_f}{(\rho_f - (58.44 C_f / 1000))}. \quad (6)$$

5. Calculate the pore fluid electric permittivity using Olhoeft's empirical equation (Revil et al., 1999b)

$$\varepsilon_f = \varepsilon_o \varepsilon_r = \varepsilon_o (a_0 + a_1 T + a_2 T^2 + a_3 T^3 + c_1 C_f + c_2 C_f^2 + c_3 C_f^3), \quad (7)$$

where $a_0 = 295.68$, $a_1 = -1.2283/\text{K}$, $a_2 = -2.094 \times 10^{-3}/\text{K}^2$, $a_3 = -1.41 \times 10^{-6}/\text{K}^3$, $c_1 = -13.00 \text{ dm}^3/\text{mol}$, $c_2 = -1.065 (\text{dm}^3/\text{mol})^2$, $c_3 = -0.03006 (\text{dm}^3/\text{mol})^3$, the temperature is in Kelvin and is valid in the range 273–373 K, the salinity is in mol/dm^3 and the permittivity *in vacuo* $\varepsilon_o = 8.854 \times 10^{-12} \text{ F/m}$ (Lide 2017).

6. Calculate the pore fluid electrical conductivity σ_f using Sen and Goode's method (Sen and Goode 1992a, b).
7. Calculate the pore fluid viscosity η_f using Phillips et al.'s method (Phillips et al. 1978; Glover et al. 2012). This method requires the use of the pore fluid molality (Eq. (6)).

8. Calculate the ionic concentration of the pore fluid including contributions from acids and bases that are added to perturb the pH to give the pH to be modelled. This step uses the imposed salinity with the addition of the concentrations of hydrogen and/or base ions to arrive at the desired pH value. It is not sufficient to calculate these from the desired pH and pOH because the pore fluid is also in equilibrium with atmospheric carbon dioxide. Consequently, the pH depends on the concentrations of hydrogen and base ions according to a cubic law. We solve the cubic law automatically in our model using Cardano's method (Glover et al. 2012). This approach also requires knowledge of the disassociation constant for water K_w , for which an empirical equation is available in Lide (2017) and described in full in Glover et al. (2012).
9. Calculate the ζ of the sample (Glover et al. 2012).
10. Calculate the C_{sp} of the sample (Glover et al. 2012).

The model outputs for this work are C_{sp} and ζ . The calculation of C_{sp} requires three additional parameters, which are all microstructural parameters that are specific to the individual rock sample. The ζ is calculated for a given mineralogy, fluid type (electrolyte type, salinity and pH) and temperature and is usually considered to be independent of the microstructure of the individual rock, while it is clear from experiments that C_{sp} depends upon the microstructure of the individual rock, at least at low salinities (Glover et al. 2015; Walker and Glover 2017).

All parameters are discussed below, separating them into groups. The parameters used in this paper for the modelling by rock type and for individual samples can be found in Tables 2 and 3, respectively, together with a note of their sources. It should be noted that while many parameters contribute to the model, almost all are fixed either by independent experimental measurements made in this paper or by other researchers or perform as modelling variables.

3.2 Modelling Variables

There are three parameters which are modelling variables. These are temperature, salinity and pH. In this work, the salinity and pH have been varied between 10^{-5} and 10 mol/dm^3 , and between pH 5.5 and pH 8.5, respectively. The salinity is used to calculate the concentration of ions in the solution together with the concentration of H_3O^+ and OH^- , which are calculated from the fluid pH. The model is not strongly sensitive to temperature. Nevertheless, the temperature for each modelling run has been fixed to the mean experimental temperature given in Walker and Glover (2017).

3.3 Pore Fluid Parameters

There are six parameters which describe the pore fluid. Four describe the ionic mobilities. Since it was assumed that the electrolyte was a simple symmetric monovalent NaCl solution in this work, the four ionic mobilities are those for Na^+ , Cl^- , H^+ and OH^- (Crow 1988). The disassociation constant K_w is required in the model, but it is not an input parameter because we calculate it from the temperature using the empirical relationship of Lide (2017). The disassociation constant of water varies little in the range of temperatures encountered in the experimental data modelled by this work, which were from 20.1 to 27.2 °C, and takes values from $10^{-14.17}$ to $10^{-13.97}$, respectively. The two remaining input parameters are equilibrium constants which describe the reaction of the electrolyte with atmospheric carbon dioxide (Glover et al. 2012); fixed at $pK_1 = 7.53$ and $pK_2 = 10.3$ after Wu et al. (1991) and Revil et al. (1999a).

Table 2 The mean values of parameters used in the modelling of the ζ and C_{sp} for each rock type together with their source

Parameter	Units	Source	Berea sandstone	Boise sandstone	Detrital Fontainebleau sandstone	Overgrown Fontainebleau sandstone	Detrital Lochaline sandstone	Overgrown Lochaline sandstone
Samples			BR1, BR2 and BR3 (a)	B1III, B2II and B3I (b)	F1D and F2D (c)	F3Q and F4Q (d)	L1D and L2D (e)	L3Q and L4Q (f)
Part of Figs. 2 and 3								
<i>Fluid parameters</i>								
Fluid type			NaCl	NaCl	NaCl	NaCl	NaCl	NaCl
β_{Na+}	$\times 10^{-8} \text{ m}^2/\text{s/V}$	Crow (1988)	5.20	5.20	5.20	5.20	5.20	5.20
β_{H+}	$\times 10^{-8} \text{ m}^2/\text{s/V}$	Crow (1988)	36.3	36.3	36.3	36.3	36.3	36.3
β_{Cl-}	$\times 10^{-8} \text{ m}^2/\text{s/V}$	Crow (1988)	7.90	7.90	7.90	7.90	7.90	7.90
β_{OH-}	$\times 10^{-8} \text{ m}^2/\text{s/V}$	Crow (1988)	20.5	20.5	20.5	20.5	20.5	20.5
pK_1	(-)	Revil and Glover (1998)	7.53	7.53	7.53	7.53	7.53	7.53
pK_2	(-)	Revil and Glover (1998)	10.3	10.3	10.3	10.3	10.3	10.3
<i>Interface parameters</i>								
β_{Stem}	$\times 10^{-9} \text{ m}^2/\text{s/V}$	Revil and Glover (1997)	5	5	5	5	5	5
Γ_s^o	Sites/nm ²	Revil and Glover (1998)	10	10	10	10	10	10
pK_{me}	(-)	Kosmulski (1996)	7.5	7.5	7.5	7.5	7.5	7.5
$pK(-)$	(-)	See text for references	7.5	8.0	7.5	7.65	7.55	7.5
χ_ζ	$\times 10^{-9} \text{ m}$	Revil et al. (1999a)	2.40	2.40	2.40	2.40	2.40	2.40

Table 2 continued

Parameter	Units	Source	Berea sandstone	Boise sandstone	Detrital Fontainebleau sandstone F1D and F2D	Overgrown Fontainebleau sandstone F3Q and F4Q	Detrital Lochaline sandstone L1D and L2D	Overgrown Lochaline sandstone L3Q and L4Q
Samples			BR1, BR2 and BR3	B1II, B2II and B3I	F1D and F2D	F3Q and F4Q	L1D and L2D	L3Q and L4Q
Part of Figs. 2 and 3			(a)	(b)	(c)	(d)	(e)	(f)
Σ_{prot}	$\times 10^{-8}$ S	Calculated	15.1	50	0.24	0.24	0.23	0.68
<i>Microstructural rock parameters (only used for C_{sp} modelling)</i>								
d_{gr}	$\times 10^{-4}$ m	Measured	1.657	12.9	2.33	2.53	2.655	2.775
m	(-)	Measured	1.72	1.78	1.695	1.615	1.525	1.325
F	(-)	Measured	10.44	6.42	11.13	532.74	12.74	91.83
ϕ	(-)	Measured	0.256	0.3516	0.2415	0.0205	0.1885	0.033
<i>Model variables</i>								
T	$^{\circ}\text{C}$	Measured	22.47	26	24.6	25	26.65	25.85
pH_{expt}	(-)	Measured	7.5	6.58	6.405	6.495	7.265	7.135
$pH_{\text{st.dev.}}$	(-)	Measured	± 0.72	± 0.84	± 0.35	± 0.20	± 0.175	± 0.50

'Measured' indicates an independent laboratory measurement or a parameter directly calculated from such measurements. 'Calculated' indicates that the parameter is calculated in the model (Figs. 2 and 3)

3.4 Mineral/Fluid Interface Parameters

There are six parameters which describe the mineral/fluid interface. Of these, the ionic mobility in the Stern layer β_{Stern} , the surface site density Γ_o , the binding constant for cation adsorption pK_{Me} , and the shear plane distance χ_ζ have all been held constant throughout this work (Table 1). Each, however could be adjusted within the limits imposed by the independent experimental measurements that are noted in Walker et al. (2014) and discussed fully in Glover et al. (2012). The Stern layer mobility has been taken as $\beta_{\text{Stern}} = 5 \times 10^{-9} \text{ m}^2/\text{s/V}$ from Revil and Glover (1997), the surface site density has been taken as $\Gamma_o = 10 \text{ sites/nm}^2$ from Revil and Glover (1998) and the binding constant for cation adsorption has been taken as $pK_{\text{Me}} = 7.5$ from Kosmulski (1996). The shear plane distance (χ_ζ). The model is sensitive to this parameter because it defines where in the diffuse layer the ζ is measured. In this work, it was held constant at $\chi_\zeta = 2.4 \times 10^{-9} \text{ m}$ for all the samples.

The fifth interface parameter is the specific surface conduction. In this paper we use it to provide a measurement of the protonic surface conduction (Σ_{prot}) for each of the rock samples. The specific surface conduction was calculated from the measured surface conductivity σ_s using the modal grain diameter d_{gr} , and the cementation exponent m and formation factor F for each sample. The surface conductivity was taken to be the conductivity that was measured when the rock was saturated with the lowest salinity equilibrium fluid. These surface conductivities were converted to surface conductances Σ_s using the relationship

$$\sigma_s = 2\Sigma_s / \Lambda, \quad (8)$$

where Λ is a characteristic length scale for the pores of the rock that was introduced by Johnson et al. (1986) and which can be calculated with the equation (Revil and Cathles 1999)

$$\Lambda = d_{\text{gr}} / 2mF. \quad (9)$$

Consequently, Eqs. (8) and (9) provide individual values of surface conductance Σ_s for each sample. There are three contributions to surface conduction, arising from (1) the movement of protons on the surface, (2) ionic mobility in the Stern plane, and (3) ionic mobility in the diffuse layer. We have assumed that the value of surface conductance calculated using Eqs. (8) and (9) above is the same as the protonic surface conduction. This assumption has been justified by using the model to calculate and compare the three contributions to surface conduction as a function of salinity and pH. In all cases the calculated contributions to surface conduction from the diffuse layer and Stern layer were more than two orders of magnitude less than the calculated total surface conductance at all values of pH and salinity. Hence, it was possible to say that the remaining contribution, that of the protonic surface conduction, makes up the great majority of the surface conduction, and that $\Sigma_{\text{prot}} \approx \Sigma_s$. Consequently, for the purposes of this paper, Σ_{prot} is completely defined by independent measurements of σ_s , d_{gr} , m and F . The surface conductivity can also be obtained empirically from the formation factor measured at ultra-low salinities and the pore fluid conductivity, as implemented by Lorne et al. 1999a, b) and others, but this approach is more suited to experimental research rather than modelling.

The last interface parameter is the disassociation constant for dehydrogenization of silanol $pK_{(-)}$. It has been taken to vary in the restricted range $8.1 > pK_{(-)} > 7.3$, which fall in the middle of the range of values given by other researchers: $pK_{(-)} = 6.8$ (Dove and Rimstidt 1994), $pK_{(-)} = 6.5$ (Kosmulski 1996), $pK_{(-)} = 7.5$ (Hiemstra and van Riemsdijk 1990), and $pK_{(-)} = 8.5$ (Rustad et al. 1998). In this paper we have varied it only within this restrictive range in order to obtain an optimum fit of the model curves to the experimental data. It is the

only parameter which has been varied in this way since the modified model does not now contain a zeta potential offset parameter as described below.

3.5 The Zeta Potential Offset

In earlier modelling (Glover and Déry 2010; Glover et al. 2012; Glover et al. 2015) there was an additional mineral/fluid interface parameter called the zeta potential offset (ζ_o). In this work, the zeta potential offset has been replaced by a procedure which holds ζ constant at high salinities, which models the attainment of a maximum charge density in the diffuse layer as it is compressed to the thickness of one hydrated metal ion in thickness according to the hypothesis of Jaafar et al. (2009). This modelling procedure recognises the onset of such behaviour and keeps the ζ constant at salinities higher than this onset value. The values of constant ζ obtained using this approach vary with pH, as noted in the experimental measurements of Walker and Glover (2017) and produce values which are in the same range (−8 to −20 mV) as those obtained in earlier modelling (Glover and Déry 2010; Glover et al. 2012, 2015).

The previous method was essentially the addition of an ‘ad hoc’ parameter that allowed the model to work at high salinities while retaining its precision at low and medium salinities. The addition was not carried out with regard to any underlying physics and could be viewed as a parameter that could be varied to ensure a better fit of the model to the data. However, values of zeta potential offset found in this way had a very restricted range, indicating that they were the result of some unknown physical process (Glover and Déry 2010; Glover et al. 2012). The new method accepts the hypothesis of Jaafar et al. (2009) and has the advantages of (1) removing the only arbitrary parameter from the model, (2) reducing the number of variables in the model (to one in this work), and (3) having the modified model based more soundly on our understanding of the physical process occurring in the porous rocks.

3.6 Rock Microstructural Parameters

Finally, there are five parameters which describe the microstructure of the rock sample. Only three of them, however, are independent. The parameters are the modal grain size d_{gr} , porosity ϕ , cementation exponent m (Glover and Déry 2010), formation factor F , and characteristic pore size Λ . They are inter-related by Eq. (9) and Archie’s first law (e.g., Glover et al. 2015)

$$F = \phi^{-m}. \quad (10)$$

It is worth noting that it is also possible to express the theoretical model in terms of pore size d_p (Glover and Walker 2009) and pore throat size d_{pt} (Glover and Déry 2010) using the relationships

$$d_p = d_{gr} \sqrt{\frac{8\phi^{2m}}{am^2}}, \quad (11)$$

$$d_{pt} \approx d_p / 1.665, \quad (12)$$

respectively. In this work, we have opted to use grain size, porosity and cementation exponents as the three independent parameters to describe the microstructure of each rock sample.

We have obtained all of the microstructural parameters independently of the electrokinetic measurements by measuring (1) the electrical conductivity of the sample at each of the experimental salinities, (2) the electrical conductivity of each of the experimental fluids after equilibration, (3) the porosity of the sample with a range of different methods, and (iv)

the modal grain size by laser diffractometry. The microstructural parameters that were used in the modelling are shown in tables later in the paper.

3.7 Degrees of Freedom

Consequently, the model has a number of different types of parameter as summarised in Table 1. There are 3 model variables, which are parameters to which C_{sp} and ζ are known to be sensitive and which have been varied to explore these relationships, 4 fundamental physical quantities, 12 pore fluid and interface parameters which are defined by external measurements and held constant during the modelling, 5 microstructural parameters that are defined by petrophysical measurements that have been carried out on the samples and consequently also held constant during the modelling. There is only one parameter, the disassociation constant for the dehydrogenisation of silanol $pK(-)$, which has been varied to improve the fit of the model to the experimental data, and then only within the very restricted range ($8.1 > pK(-) > 7.3$) which has been fixed by independent measurements from other authors. Consequently, the degree to which the model curves presented in this work fit the experimental data is not a consequence of varying a large number of parameters until some sort of fit is attained, but the result of varying one parameter over a small range and having all the other parameters fixed by independent measurements on the particular samples or other parameters measured by independent researchers.

4 Theoretical Modelling Results

Since we know the microstructural (porosity, cementation exponent and modal grain size) and the experimental fluid parameters (equilibrated salinity, electrical conductivity and pH) for each sample, we have been able to carry out modelling of individual samples as well as for aggregated lithofacies.

In this section the figures and discussion of the ζ modelling are followed by those related to the C_{sp} , which is the order in which they are calculated in the theoretical model. Despite this order, it should be remembered that for the experimental results, it is the C_{sp} that is derived first, directly from streaming potential and pressure difference measurements, while the ζ is derived subsequently from the C_{sp} using additional parameters.

4.1 Modelling by Lithofacies

Figures 2 and 3 show, respectively, all of the ζ and C_{sp} modelling curves we have calculated for each of the six lithofacies (Berea, Boise, and detrital and overgrowth forms of both Fontainebleau and Lochaline sandstones) from this paper compared with all of the data for these lithofacies from Walker and Glover (2017). The modelled curves extend for the complete relevant salinity range (10^{-5} – 10 mol/dm³) and from pH 6 to pH 8 in increments of half a pH unit. In each case the curves are calculated for the temperature at which the experimental measurements were taken. The lithofacies modelling was carried out with mean values of the microstructural parameters for each lithofacies and which are shown in Table 2.

Figure 2 shows the calculated ζ model curves as a function of salinity and pH together with the data from Walker and Glover (2017) and arranged by lithofacies. The model curves use mean microstructural parameters for the samples composing each lithofacies.

A number of conclusions may be drawn from comparison of the modelled curves with the experimental data. First, that the quality of the experimental data is better than was previously

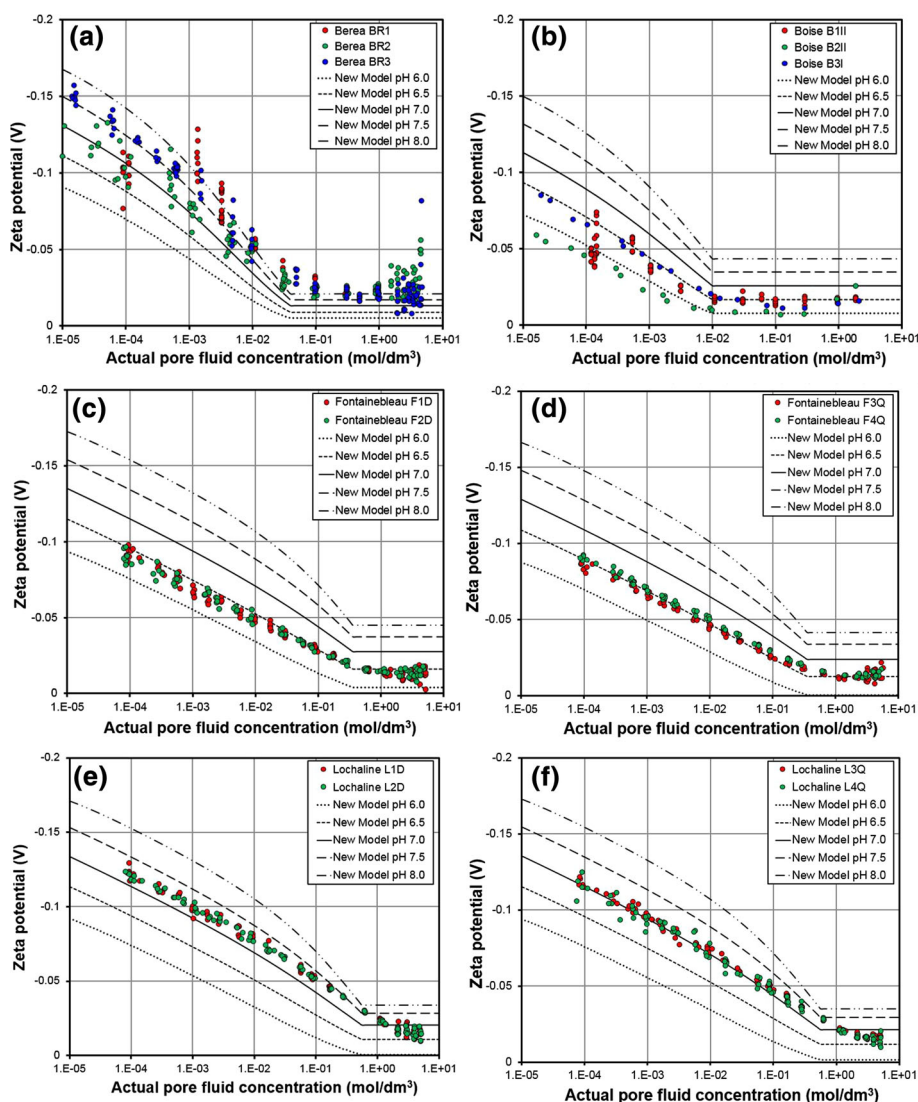


Fig. 2 Modelled ζ (curves) for all facies types as a function of pore fluid salinity and pH , compared with the experimental measurements of ζ (symbols) from this paper. **a** Berea sandstone, **b** Boise sandstone, **c** detrital Fontainebleau sandstone, **d** overgrown Fontainebleau sandstone, **e** detrital Lochaline sandstone, and **f** overgrown Lochaline sandstone. All model parameters are shown in Table 2. The ζ part of the model has 16 independent parameters, of which 3 are model variables (temperature, fluid salinity and pH), and 13 are predefined by the electro-chemistry of the fluid and fluid–mineral interface. ζ does not depend on the microstructure of the rock in this implementation of the theoretical model. The model retains only 1 variable parameter ($pK_{(-)}$), which was allowed to vary in the experimentally restricted range $8.0 > pK_{(-)} > 7.5$

available [collated in Glover et al. (2012)], both in its smaller uncertainty (scatter) and because the temperature, salinity and pH of the equilibrated pore fluids have been measured reliably and reported together with the C_{sp} and ζ data. This enables the modelling values for a particular temperature, pore fluid salinity and pH to be compared directly with their measured values.

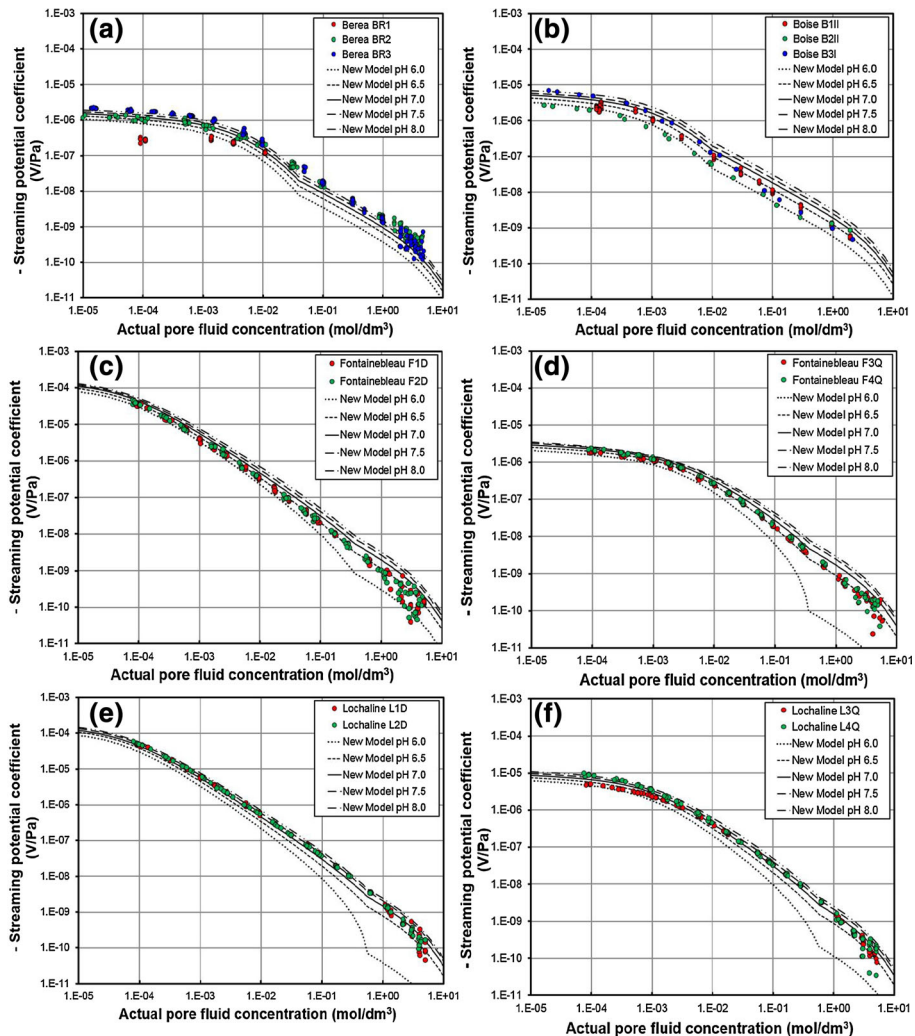


Fig. 3 Modelled C_{sp} (curves) for all facies types as a function of pore fluid salinity and pH , compared with the experimental measurements of C_{sp} (symbols) from this paper. **a** Berea sandstone, **b** Boise sandstone, **c** detrital Fontainebleau sandstone, **d** overgrown Fontainebleau sandstone, **e** detrital Lochaline sandstone, and **f** overgrown Lochaline sandstone. All model parameters are shown in Table 2. In this model, there are 19 independent parameters, of which 3 are model variables (temperature, fluid salinity and pH), 13 are predefined by the electro-chemistry of the fluid and fluid-mineral interface, and 3 are predefined by the rock microstructure (d_{gr} , m , ϕ). The model retains only 1 variable parameter ($pK(-)$), which was allowed to vary in the experimentally restricted range $8.0 > pK(-) > 7.5$

Second, the modelled ζ curves reproduce all of the main features of the ζ data, with magnitude of the ζ increasing as the logarithm of the salinity as pore fluid salinity is reduced. The ζ model curves are also extremely sensitive to pore fluid pH , as was noted when Walker and Glover (2017) discussed the experimental dataset. This is shown by the spread of the model curves as a function of pH and the degree to which the experimentally measured ζ data concord with the modelled ζ curve for the experimentally determined value of pH , for each sample. There is a greater degree of scattering of the experimental data at high salinities, where

measurements are difficult to make (Vinogradov et al. 2010; Walker et al. 2014; Walker and Glover 2017). However, the model ζ curves agree well with the mean of this cloud of points.

Figure 3 shows the calculated C_{sp} model curves as a function of pore fluid salinity and pH together with the data from Walker and Glover (2017) and arranged by lithofacies.

Once again, it is clear that the C_{sp} model curves reproduce all of the main features of the experimentally measured data, with the value of C_{sp} increasing in an approximately power law fashion as pore fluid salinity is reduced. The C_{sp} model curves are also sensitive to pore fluid pH , but not to the same degree as the ζ curves, and this is also noted in the experimental data.

It was hypothesised by Glover et al. (2012) that the flattening of C_{sp} at low salinities, which is clear in experimental data (e.g., Walker and Glover 2017) is controlled by the development of surface conduction and the microstructure of the rocks on the basis of a comparison of their model with the experimental data then available. Comparison of the modelling carried out in this paper with the new data in Walker and Glover (2017) has confirmed the effect. Our C_{sp} modelling for each of the two microstructural forms of Lochaline and Fontainebleau sandstone (i.e., between parts (c) and (d) and parts (e) and (f) in Fig. 3) shows that variations in their independently measured microstructural parameters (modal grain size, porosity, cementation exponent and/or formation factor) and surface conduction (in the absence of any mineralogical differences) are sufficient to explain the degree of flattening occurring in the experimental C_{sp} data. In each case the quartz overgrown version shows a greater degree of low salinity flattening, and this is well-modelled by smaller porosities and cementation exponents which occur for these samples (the measured grain sizes and values of surface conduction are not significantly different).

It should be noted that there is no microstructural control on zeta potential because it is a property of the electrical double layer (EDL) which occurs between the mineral grain and the pore fluid and can be regarded as a property acting on a grain surface scale, independent of how those grains are arranged in a microstructure. This is discussed in more detail at the end of Sect. 4.2.

4.2 Sample-by-Sample Modelling

The model has been implemented for each sample measured by Walker and Glover (2017) for a range of salinities and pH at the experimental temperature and with the microstructural and electrical parameters (modal grain size, porosity, cementation exponent, and formation factor) that were measured for that sample.

As in the case of the modelling by lithofacies, the electro-chemical parameters that describe the interface between the rock matrix and the pore fluid could be adjusted to ensure a best fit. However, once again, only one was varied ($pK_{(-)}$). The full sample parameters and the modelling parameters are shown in Table 3. Figures 4 and 5 show the sample-specific ζ and C_{sp} curves from the theoretical model as well as the relevant experimental data from that sample, for C_f ranging from 10^{-5} to 10 mol/dm^3 and for five values of pH in the range $6 < pH < 8$. The pH of the pore fluid was controlled and measured during the experiments of Walker and Glover (2017) to within ± 0.2 , which is given as pH_{expt} in the data tables in this paper. This value should be used to interpret how the values of the experimental data in Figs. 4 and 5 compare with the set of modelled ζ and C_{sp} curves for the five values of pH for which modelling has been implemented.

Overall, both theoretical ζ and C_{sp} model curves agree extremely well with the experimental data for individual samples; better than when calculated for the generic lithofacies. This implies that modelling ζ and C_{sp} benefits from knowledge of individual sample properties. Accord between the experimental data and the modelling was particularly good for

Table 3 The parameters used in the modelling of the C_{sp} and ζ for individual samples (Figs. 4 and 5)

Parameter	Units	Source	Berea sandstone			Boise sandstone			Fontainebleau sandstone				Lochaline sandstone			
			BR1	BR2	BR3	B1III	B2II	B3I	F1D	F2D	F3Q	F4Q	L1D	L2D	L3Q	L4Q
Part of Figs. 4 & 5			(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
	Fluid parameters															
	Fluid type		NaCl	NaCl	NaCl	NaCl	NaCl	NaCl	NaCl	NaCl	NaCl	NaCl	NaCl	NaCl	NaCl	NaCl
	β_{Na+}	$\times 10^{-8} \text{ m}^2/\text{s/V}$	Crow (1988)	5.20	5.20	5.20	5.20	5.20	5.20	5.20	5.20	5.20	5.20	5.20	5.20	5.20
	β_{H+}	$\times 10^{-8} \text{ m}^2/\text{s/V}$	Crow (1988)	36.3	36.3	36.3	36.3	36.3	36.3	36.3	36.3	36.3	36.3	36.3	36.3	36.3
β_{Cl-}	$\times 10^{-8} \text{ m}^2/\text{s/V}$	Crow (1988)	7.90	7.90	7.90	7.90	7.90	7.90	7.90	7.90	7.90	7.90	7.90	7.90	7.90	
β_{OH-}	$\times 10^{-8} \text{ m}^2/\text{s/V}$	Crow (1988)	20.5	20.5	20.5	20.5	20.5	20.5	20.5	20.5	20.5	20.5	20.5	20.5	20.5	
pK_1	(-)	Revil and Glover (1998)	7.53	7.53	7.53	7.53	7.53	7.53	7.53	7.53	7.53	7.53	7.53	7.53	7.53	
pK_2	(-)	Revil and Glover (1998)	10.3	10.3	10.3	10.3	10.3	10.3	10.3	10.3	10.3	10.3	10.3	10.3	10.3	
Interface parameters																
β_{Stem}	$\times 10^{-9} \text{ m}^2/\text{s/V}$	Revil and Glover (1997)	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	
Γ_s^o	Sites/nm ²	Revil and Glover (1998)	10	10	10	10	10	10	10	10	10	10	10	10	10	
pK_{me}	(-)	Kosmulski (1996)	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	
$pK_{(-)}$	(-)	See text for references	7.5	7.5	7.3	8.0	8.1	8.1	7.5	7.5	7.7	7.6	7.6	7.6	7.6	
χ_ζ	$\times 10^{-9} \text{ m}$	Revil et al. (1999a)	2.40	2.40	2.40	2.40	2.40	2.40	2.40	2.40	2.40	2.40	2.40	2.40	2.40	

Table 3 continued

Parameter	Units	Source	Berea sandstone			Boise sandstone			Fontainebleau sandstone				Lochaline sandstone			
			BR1	BR2	BR3	B1II	B2II	B3I	F1D	F2D	F3Q	F4Q	L1D	L2D	L3Q	L4Q
Part of Figs. 4 & 5			(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Σ_{prot}	$\times 10^{-8}$ S	Measured	50.1	15.1	15.1	49.6	50.1	40.1	0.29	0.3	0.25	0.25	0.262	0.252	0.601	0.68
<i>Microstructural rock parameters (only used for C_{sp} modelling)</i>																
d	$\times 10^{-4}$ m	Measured	1.8	1.55	1.62	7.39	9.91	21.4	2.35	2.31	2.51	2.55	2.63	2.68	2.77	2.78
m	(-)	Measured	1.79	1.72	1.66	1.569	1.677	1.811	1.71	1.68	1.60	1.63	1.50	1.55	1.33	1.32
F	(-)	Measured	10.22	10.63	10.47	6.53	5.65	7.48	11.64	10.62	567.54	503.33	13.09	12.4	142.67	65.66
ϕ	(-)	Measured	0.272	0.253	0.243	0.3024	0.3561	0.3292	0.238	0.245	0.019	0.022	0.180	0.197	0.024	0.042
<i>Model variables</i>																
T	$^{\circ}\text{C}$	Measured	25.8	20.1	21.5	26	26	26	24.5	24.7	25.1	24.9	27.2	26.1	25.9	25.8
pH_{expt}	(-)	Measured	8.03	7.05	7.41	7.2	5.94	6.61	6.40	6.41	6.48	6.51	7.27	7.26	7.15	7.12
$pH_{\text{st.dev.}}$	(-)	Measured	± 0.41	± 0.21	± 0.11	± 0.54	± 0.24	± 0.086	± 0.15	± 0.20	± 0.10	± 0.10	± 0.20	± 0.15	± 0.20	± 0.30

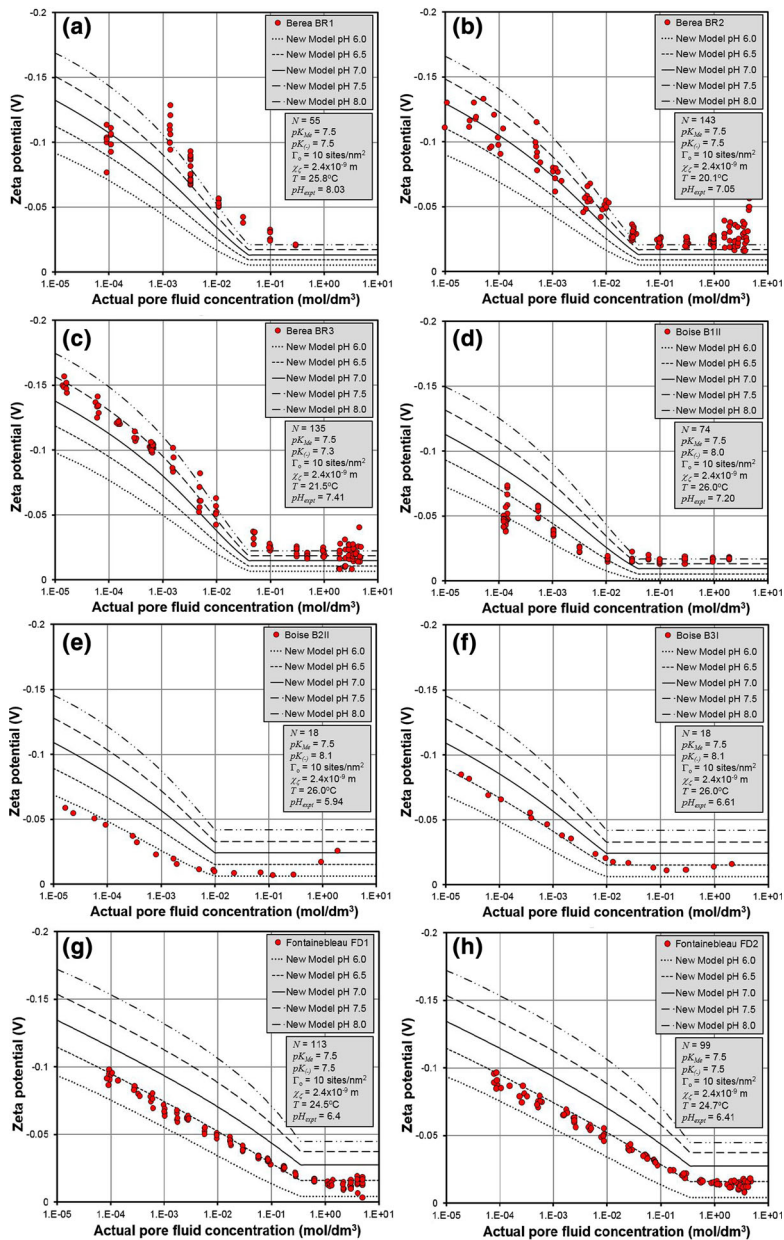


Fig. 4 Modelled ζ (curves) for individual samples as a function of pore fluid salinity and pH , compared with the experimental measurements of ζ (symbols) from this paper. **a** Berea (BR1), **b** Berea (BR2), **c** Berea (BR3), **d** Boise (B1II), **e** Boise (B2II), **f** Boise (B3I) **g** Fontainebleau (F1D), **h** Fontainebleau (F2D), **i** Fontainebleau (F3Q), **j** Fontainebleau (F4Q), **k** Lochaline (L1D), **l** Lochaline (L2D), **m** Lochaline (L3Q), and **n** Lochaline (L4Q). All model parameters are shown in Table 3. In this model, there are 16 independent parameters, of which 3 are model variables (temperature, fluid salinity and pH), and 13 are predefined by the electro-chemistry of the fluid and fluid-mineral interface. ζ does not depend on the microstructure of the rock in this implementation of the theoretical model. The model retains only 1 variable parameter ($pK_{(-)}$), which was allowed to vary in the experimentally restricted range $8.1 > pK_{(-)} > 7.3$. Please note that there are multiple experiments at the same salinity carried out for sample BR1 as this sample was used to develop the experimental protocol

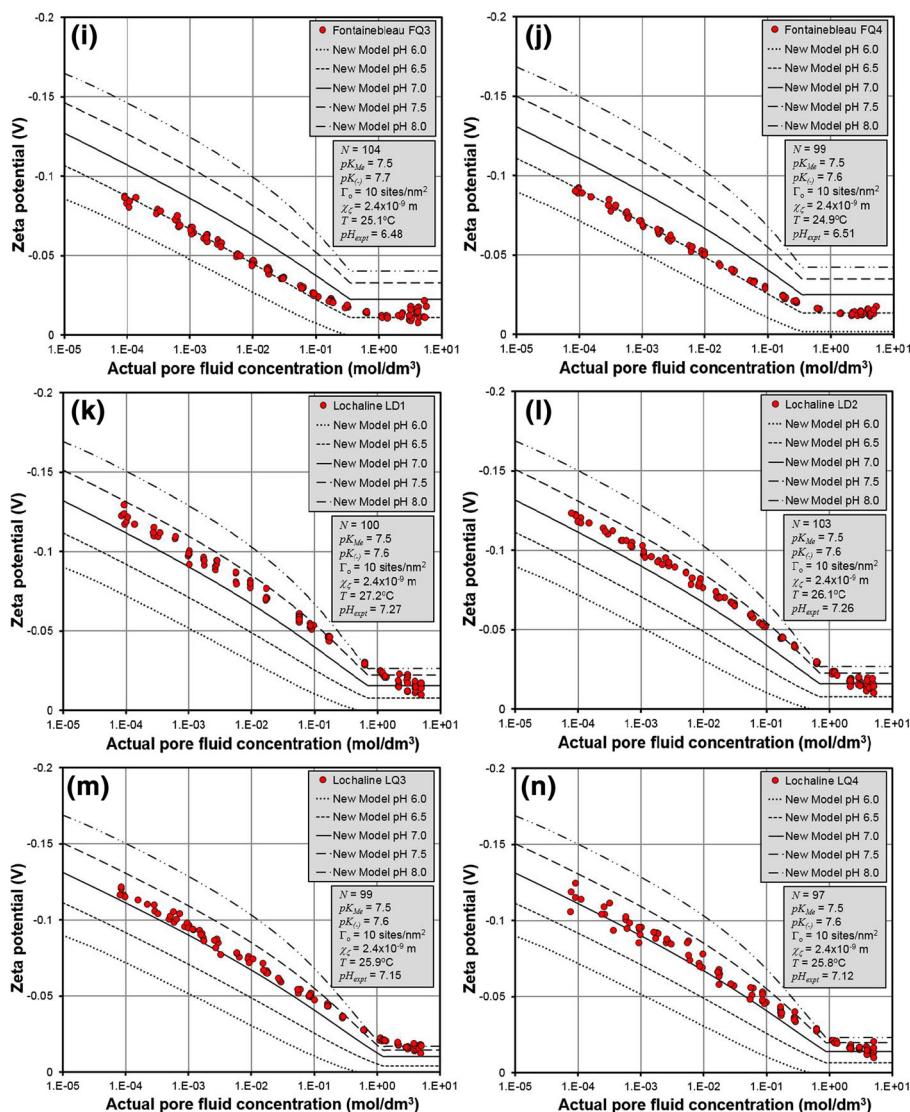


Fig. 4 continued

the Fontainebleau and Lochaline sandstone samples. For example, the data for the sample of detrital Fontainebleau sandstones F1D and F2D, shown in Fig. 4g, h, fall clearly on the curve for a $pH=6.5$ in each diagram. The experimentally measured pH for these measurements was $pH_{exp} = 6.40$ and 6.41 . The data for the Berea and Boise sandstones have a greater scatter, however, even here the mean behaviour agrees very well with the theoretical curves. For example, the Berea sandstone sample BR3 has ζ and C_{sp} data that follow a trend the mean behaviour of which would lie on a theoretical curve for pH 7.5, while the measured pH during the experiment was $pH_{exp} = 7.41$.

Considering the ζ modelling, Fig. 4 shows the calculated curves modelled for individual samples. It is clear that the ζ model curves reproduce the overall low to medium salinity logarithmic behaviour of the experimental data well. The high salinity behaviour of the

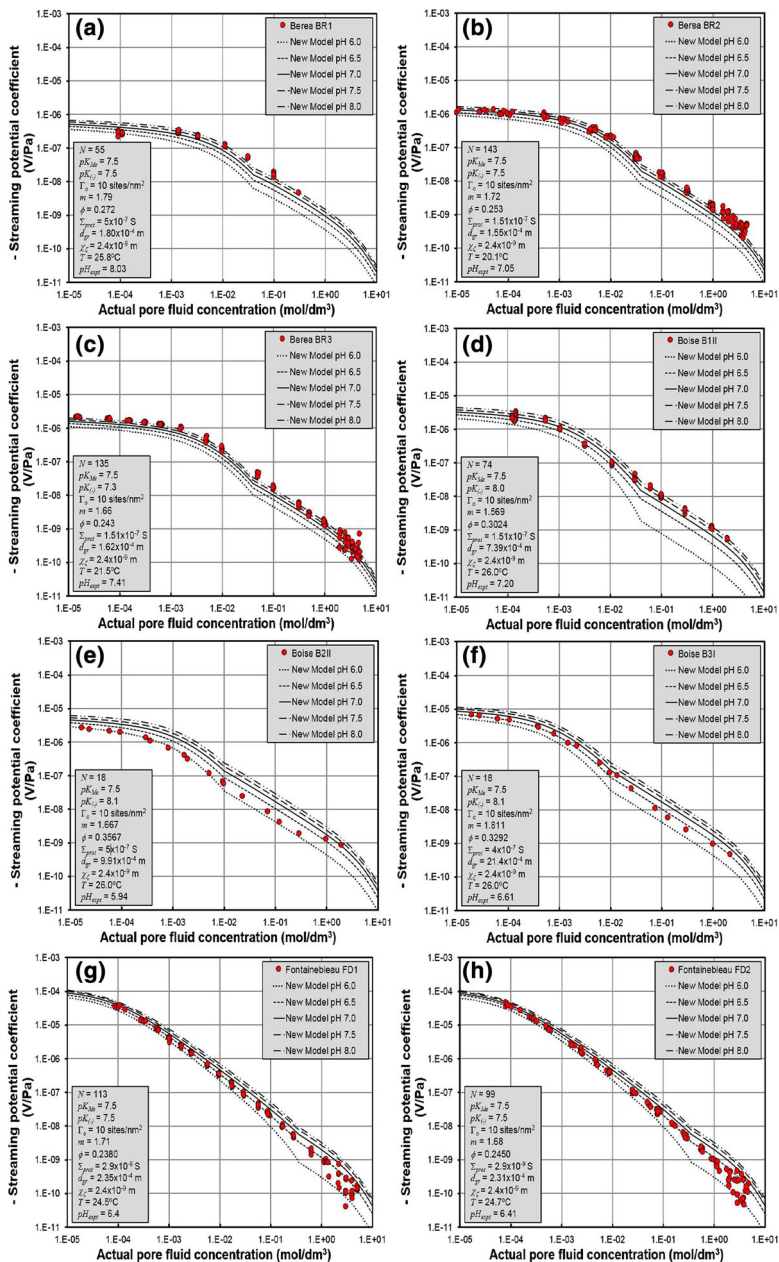


Fig. 5 Modelled C_{sp} (curves) as a function of pore fluid salinity and pH, compared with the experimental measurements of C_{sp} (symbols). **a** Berea (BR1), **b** Berea (BR2), **c** Berea (BR3), **d** Boise (B1II), **e** Boise (B2II), **f** Boise (B3I) **g** Fontainebleau (F1D), **h** Fontainebleau (F2D), **i** Fontainebleau (F3Q), **j** Fontainebleau (F4Q), **k** Lochaline (L1D), **l** Lochaline (L2D), **m** Lochaline (L3Q), and **n** Lochaline (L4Q). All model parameters are shown in Table 3. In this model, there are 19 independent parameters, of which 3 are model variables (temperature, fluid salinity and pH), 13 are predefined by the electro-chemistry of the fluid and fluid-mineral interface, and 3 are predefined by the rock microstructure (d_{gr} , m , ϕ). The model retains only 1 variable parameter ($pK(-)$), which was allowed to vary in the experimentally restricted range $8.1 > pK(-) > 7.3$. Apparently incomplete curves represent negative values that cannot be plotted on a logarithmic axis

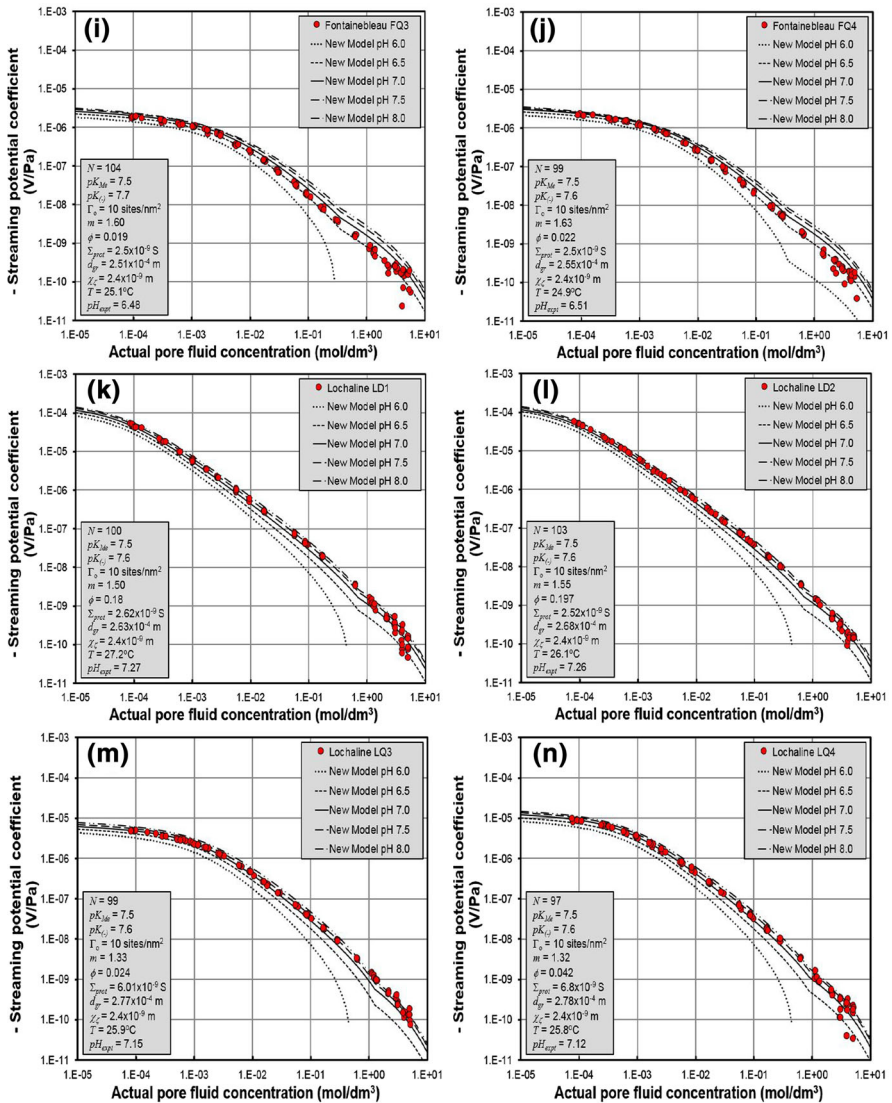


Fig. 5 continued

experimental data is also modelled well by replacing the ad hoc ζ offset that was used in previous models with our new approach, which keeps the ζ at a constant level for salinities above which the thickness of the diffuse layer is approximately the same or less than the diameter of a hydrated metal ion.

It is clear from the experimental data that the ζ is highly sensitive to pore fluid pH . The results of modelling reproduce this pH -dependence well with the mean of the experimental data falling close to the theoretical curves for the appropriate experimental pH in all cases. For example, the mean pH of the Fontainebleau and Lochaline samples is 6.45 and 7.2, respectively. For the four samples of Fontainebleau sandstone, Fig. 4g–j shows clearly that the experimentally determined ζ fall just below the modelled curve for pH 6.5 theoretical

line, while for Lochaline sandstones, Fig. 4k–n shows the experimentally determined ζ fall between the modelled curves for pH 7 and pH 7.5.

We have calculated coefficients of determination for the comparison of the ζ data for each sample and the theoretical curve calculated for the pH at which the measurements were made (pH_{expt}) and find that the coefficient of determination (R^2) values are in all cases > 0.89 . Hence the experimental observation that pH is one of the major controls on the ζ , and that even small changes in pH (e.g., pH 7 ± 1) can lead to a large change in ζ (Walker and Glover, 2017) is now supported by modelling.

Considering the C_{sp} modelling, Fig. 5 shows the modelled C_{sp} curves as a function of salinity and pH for individual samples, taking into account their individual petrophysical properties. The C_{sp} model curves reproduce all of the main features of the C_{sp} data extremely well. The overall approximate power law behaviour is overlain with a flattening at low salinities and curvature at moderate salinities for some values of pH , both of which conform to the experimental data. In general, the use of sample-specific parameters improves the agreement between the theoretical model curves and the experimental data compared to the modelling in Fig. 3, and the agreement between the theory and the experimental data is in many cases excellent.

Disagreement occurs for specific salinities where there may be systematic errors in the experimental data (e.g., for 10^{-4} mol/dm³ for Berea sandstone BR1, Fig. 5a), and there is an overestimation of the theoretical model at low salinities for the Boise sandstone samples (i.e., for B1II, B2II and B3I, Fig. 5d–f), which probably arises from using inappropriate modelling parameters. The agreement could be improved if we had taken the approach of fitting the theoretical model to the experimental data, but the goal of this work was not to fit the theoretical model to the data but to see how robust the theoretical model is when compared to high-quality data. The agreements between theory and experiment in the sample-by-sample study represented by Figs. 3 and 5 are better than any published so far.

The C_{sp} model curves are sensitive to pore fluid pH just as in Fig. 3, but now it is possible to see clearly that the experimental data fall close to the theoretical curve for the experimental value of pH (pH_{expt}), which is given for reference on each plot and in Table 3, together with the uncertainty in its measurement. A typical example is for Berea sample BR3 (Fig. 5c), where the experimental data fall almost exactly on the curve for pH 7.5 and for which the mean experimental pH was 7.41 ± 0.11 (Table 3). We have calculated coefficients of determination for the comparison of the C_{sp} data for each sample and the theoretical curve calculated for the pH at which the measurements were made (pH_{expt}), and find that the R^2 values are in all cases > 0.91 .

The microstructural control of the flattening of the C_{sp} model curves at low salinities is completely clear in the sample-by-sample modelling. Comparison between the detrital Fontainebleau sandstones (F1D and F2D, Fig. 5g, h) with the overgrown Fontainebleau sandstones (F3Q and F4Q, Fig. 5i, j), and similarly for the detrital and overgrown forms of Lochaline sandstone (L1D and L2D, Fig. 5k, l with L3Q and L4Q, Fig. 5m, n) shows that there is a much stronger degree of flattening at low salinities for the authigenic overgrown forms of the sandstone than the detrital sandstones. The implementation of these microstructural parameters in each theoretical curve allows the curves to match the experimental data and take into account the effect of the microstructural differences on the measured C_{sp} data. The microstructural differences between the detrital and authigenic overgrown sandstones is marked, with the overgrown samples of either rock type having (1) a much smaller porosity, (2) a much larger formation factor, (3) a slightly larger grain size, and (4) a similar or slightly smaller cementation exponent. These changes indicate that, from an electrical point of view, the development of overgrowths fills most of the pores without reducing the degree to which

the pores are connected, leaving the cementation exponent relatively unchanged. After the overgrowths are developed, electrically patent pathways still occur along thin passageways between the euhedral grains, but their sensitivity to closure, which is measured by the large formation factor, is high. The increase in formation factor is primarily due to the significant reduction in porosity than a change in cementation exponent. There is little change in grain size because the overgrowths are filling-in pore space rather than increasing the size of the grain in all directions. This is akin to comparing the diameter of a sphere with a co-centred cube of the same linear dimensions—the cube has a volume that is almost twice as big ($6/\pi = 1.9099$), yet seems to occupy a similar amount of space.

Once again, it should be noted that there is no microstructural control on zeta potential. The zeta potential is a property of the electrical double layer (EDL) which occurs between the mineral grain and the pore fluid. It is calculated from the Stern potential which itself is obtained by assuming that the EDL is thin compared to its extent (Fixman 1980, 1983). Consequently, it can be regarded as a property acting on a grain surface scale and should be independent of how those grains are arranged in a microstructure. That will certainly be the case when the EDL is extremely small at high pore fluid salinities. However, there is a possibility that the zeta potential might vary in a rock at a local scale when the pore fluid salinity is so low that the EDL is of the same scale as the characteristic scale of the pores. In this case, the shear plane might vary from location to location within the microstructure as fluid flow conforms to that microstructure. This will give rise to local, transient potential differences which can be regarded as microstreaming potentials. However, the electro-neutrality requirement will ensure that most of these microstreaming potentials cancel each other out so that we are left with the macroscopic overall streaming potential. A similar effect can also be imagined to arise from a rock composed of a mixture of minerals with different Stern potentials and consequently different zeta potentials. Consequently, the effect will have an influence on how the zeta potential from two different minerals in a two-mineral mixture mix to provide an effective zeta potential.

5 Conclusions

A modified form of the theoretical model of Glover et al. (2012) has been applied to all of the data from Walker and Glover (2017) for each lithofacies and for each individual sample. The major modifications to the model comprised (1) implementing a calculation for pore fluid pH that is a function of all the ionic contributions to the electrolyte, (2) removing the zeta potential offset as a parameter forcing a constant ζ at high salinities, (3) implementing a variable threshold ζ at high salinities where the thickness of the diffuse layer is comparable to the size of the hydrated metal ions and representing, therefore, a maximum charge density for the diffuse layer. The modelling was carried out as a function of salinity and pH allowing only one parameter (the disassociation constant for dehydrogenation of silanol) to vary over a small range constrained by independent measurements made by other researchers.

The modelled ζ curves were found to agree well with the experimental data for each lithofacies and extremely well for each individual sample. The modelled ζ is highly sensitive to pH and salinity, and it was found that the experimental data fell on the curves expected from the experimental pH, in most cases to within ± 0.2 pH points, contrasting with the widespread scatter of previously existing ζ measurements which suffered a lack of pH control.

The modelled C_{sp} curves were also found to agree well with the experimental data for each lithofacies and extremely well for each individual sample. It has been confirmed that

the low salinity behaviour of the C_{sp} measurements is caused by the pore structure and best exemplified by the difference results obtained when measuring either the detrital or overgrown forms of the Fontainebleau and Lochaline data, each of which is chemically indistinguishable.

The replacement of the zeta potential offset with an implementation of the maximum charge density hypothesis of Vinogradov et al. (2010) for generating a constant value of ζ at high salinities (to match the experimental observations) has led to a modelling which concords extremely well with all of the data in the Walker and Glover (2017) dataset.

It should be noted that this paper concerns itself only with sandstones. The model it describes does not take into the account the complex mineralogy of rocks that comprise clays, which affect the mineral–electrolyte interfacial chemistry. In order to make the model applicable to a wider range of rocks, a proper surface complexation model (e.g., Datta et al., 2009; Li et al., 2016) should be incorporated in order to yield the equilibrium zeta potential of each type of mineral before calculating the effective zeta potential value for each rock sample and then calculating the streaming potential from the individual rock sample microstructural parameters. The surface complexation models, when validated against experimental data, could provide a route for evaluating the zeta potential under varying concentration, composition and temperature.

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References

- Bolève, A., Crespy, A., Revil, A., Janod, F., Mattiuzzo, J.L.: Streaming potentials of granular media: influence of the Dukhin and Reynolds numbers. *J. Geophys. Res.* **112**, B08204 (2007). <https://doi.org/10.1029/2006JB004673>
- Crow, D.R.: Principles and Applications of Electrochemistry. Chapman and Hall, London (1988)
- Datta, S., Conlisk, A.T., Li, H.F., Yoda, M.: Effect of divalent ions on electroosmotic flow in microchannels. *Mech. Res. Commun.* **36**(1), 65–74 (2009)
- Dove, P.M., Rimstidt, J.D.: Silica–water interactions. *Rev. Mineral. Geochem.* **29**, 259–308 (1994)
- Fixman, M.: Charged macromolecules in external fields. I. The sphere. *J. Chem. Phys.* **72**(9), 5177–5186 (1980)
- Fixman, M.: Thin double layer approximation for electrophoresis and dielectric response. *J. Chem. Phys.* **78**(3), 1483–1491 (1983)
- Glover, P.W.J.: Electrical Properties, 2nd ed. Treatise Geophys, vol. 189. Elsevier, London (2015)
- Glover, P.W.J., Déry, N.: Dependence of streaming potential on grain diameter and pore radius for quartz glass beads. *Geophysics* **75**(6), F225–F241 (2010). <https://doi.org/10.1190/1.3509465>
- Glover, P.W.J., Walker, E.: A grain size to effective pore size transformation derived from an electro-kinetic theory. *Geophysics* **74**(1), E17–E29 (2009)
- Glover, P.W.J., Walker, E., Jackson, M.D.: Streaming-potential coefficient of reservoir rock, a theoretical model. *Geophysics* **77**(2), D17–D43 (2012). <https://doi.org/10.1190/GEO2011-0364.1>
- Hiemstra, T., Van Riemsdijk, W.H.: Multiple activated complex dissolution of metal (hydr)oxide: a thermodynamic approach applied to quartz. *J. Colloid Interface Sci.* **136**, 132–150 (1990). [https://doi.org/10.1016/0021-9797\(90\)90084-2](https://doi.org/10.1016/0021-9797(90)90084-2)
- Jaafar, M.Z., Vinogradov, J., Jackson, M.D.: Measurement of streaming potential coupling coefficient in sandstones saturated with high salinity NaCl brine. *Geophys. Res. Lett.* **36**, L21306 (2009). <https://doi.org/10.1029/2009GL040549>

- Johnson, D.L., Koplik, J., Schwartz, L.M.: New pore-size parameter characterizing transport in porous media. *Phys. Rev. Lett.* **57**(20), 2564–2567 (1986). <https://doi.org/10.1103/PhysRevLett.57.2564>
- Kosmulski, M.: Adsorption of cadmium on alumina and silica: analysis of the values of stability constants of surface complexes calculated for different parameters of triple layer model. *Colloids Surf. A* **117**, 201–214 (1996). [https://doi.org/10.1016/0927-7757\(96\)03706-5](https://doi.org/10.1016/0927-7757(96)03706-5)
- Li, S., Leroy, P., Heberling, F., Devau, N., Jougnot, D., Chiaberge, C.: Influence of surface conductivity on the apparent zeta potential of calcite. *J. Colloid Interface Sci.* **468**, 262–275 (2016)
- Lide, D.R.: *Handbook of Chemistry and Physics*, 98th edn. Taylor Francis, London (2017). ISBN 1439880492
- Phillips, S.L., Ozbek, H., Otto, R.J.: Basic energy properties of electrolytic solutions database: sixth International CODATA Conference Santa Flavia (Palermo), Sicily, Italy, May 22–25, accessed 10 June 2010. <https://www.osti.gov/bridge/purl.cover.jsp?jsessionid=3954E775156A8BC0FA35DB5CE5B402D4?purl=6269880-iPJPhB/> (1978)
- Lorne, B., Perrier, F., Avouac, J.: Streaming potential measurements: 1. Properties of the electrical double layer from crushed rock samples. *J. Geophys. Res.: Solid Earth* **104**(B8), 17857–17877 (1999a)
- Lorne, B., Perrier, F., Avouac, J.: Streaming potential measurements: 2. Relationship between electrical and hydraulic flow patterns from rock samples during deformation. *J. Geophys. Res.: Solid Earth* **104**(B8), 17879–17896 (1999b)
- Pride, S.: Governing equations for the coupled electromagnetics and acoustics of porous media. *Phys. Rev. B* **50**(21), 15678–15696 (1994)
- Pride, S.R., Morgan, F.D.: Electrokinetic dissipation induced by seismic waves. *Geophysics* **56**, 914–925 (1991). <https://doi.org/10.1190/1.1443125>
- Revil, A., Cathles, L.M.: Permeability of shaly sands. *Water Resour. Res.* **35**, 651–662 (1999)
- Revil, A., Glover, P.W.J.: Theory of ionic surface electrical conduction in porous media. *Phys. Rev. B* **55**, 1757–1773 (1997). <https://doi.org/10.1103/PhysRevB.55.1757>
- Revil, A., Glover, P.W.J.: Nature of surface electrical conductivity in natural sands, sandstones, and clays. *Geophys. Res. Lett.* **25**, 691–694 (1998). <https://doi.org/10.1029/98GL00296>
- Revil, A., Pezard, P.A., Glover, P.W.J.: Streaming potential in porous media. I. Theory of the zeta-potential. *J. Geophys. Res.* **104**(B9), 20021–20031 (1999a). <https://doi.org/10.1029/1999JB900089>
- Revil, A., Schwaeger, H., Cathless, L.M., Manhardt, P.D.: Streaming potential in porous media 2. Theory and application to geothermal systems. *J. Geophys. Res.* **104**, 20033–20048 (1999b). <https://doi.org/10.1029/1999JB900090>
- Rustad, J.R., Wasserman, E., Felmy, A.R., Wilke, C.: Molecular dynamics study of proton binding to silica surfaces. *J. Colloid Interface Sci.* **198**, 119–129 (1998). <https://doi.org/10.1006/jcis.1997.5195>
- Sen, P., Goode, P.: Influence of temperature on electrical conductivity on shaly sands. *Geophysics* **57**, 89–96 (1992a)
- Sen, P., Goode, P.: Errata, to: influence of temperature on electrical conductivity of shaly sands. *Geophysics* **57**, 1658 (1992b)
- Vinogradov, J., Jaafar, M.Z., Jackson, M.D.: Measurement of streaming potential coupling coefficient in sandstones saturated with natural and artificial brines at high salinity. *J. Geophys. Res.* **115**, B12204 (2010). <https://doi.org/10.1029/2010JB007593>
- Walker, E., Glover, P.W.J.: A transient method for measuring the DC streaming potential coefficient of porous and fractured rocks. *Transp. Porous Media* (2017). <https://doi.org/10.1002/2013JB010579>
- Walker, E., Glover, P.W.J., Ruel, J.: A transient method for measuring the DC streaming potential coefficient of porous and fractured rocks. *J. Geophys. Res.* (2014). <https://doi.org/10.1002/2013JB010579>
- Wu, L., Forsling, W., Schlindler, P.W.: Surface complexation of calcium minerals in aqueous solution. *J. Colloid Interface Sci.* **147**, 178–185 (1991). [https://doi.org/10.1016/0021-9797\(91\)90145-X](https://doi.org/10.1016/0021-9797(91)90145-X)

Measurements of the Relationship Between Microstructure, pH, and the Streaming and Zeta Potentials of Sandstones

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Abstract A large number (1253) of high-quality streaming potential coefficient (C_{sp}) measurements have been carried out on Berea, Boise, Fontainebleau, and Lochaline sandstones (the latter two including both detrital and authigenic overgrowth forms), as a function of pore fluid salinity (C_f) and rock microstructure. All samples were saturated with fully equilibrated aqueous solutions of NaCl (10^{-5} and 4.5 mol/dm^3) upon which accurate measurements of their electrical conductivity and pH were taken. These C_{sp} measurements represent about a fivefold increase in streaming potential data available in the literature, are consistent with the pre-existing 266 measurements, and have lower experimental uncertainties. The C_{sp} measurements follow a pH-sensitive power law behaviour with respect to C_f at medium salinities ($C_{sp} = -1.44 \times 10^{-9} C_f^{-1.127}$, units: V/Pa and mol/dm^3) and show the effect of rock microstructure on the low salinity C_{sp} clearly, producing a smaller decrease in C_{sp} per decade reduction in C_f for samples with (i) lower porosity, (ii) larger cementation exponents, (iii) smaller grain sizes (and hence pore and pore throat sizes), and (iv) larger surface conduction. The C_{sp} measurements include 313 made at $C_f > 1 \text{ mol/dm}^3$, which confirm the limiting high salinity C_{sp} behaviour noted by Vinogradov et al., which has been ascribed to the attainment of maximum charge density in the electrical double layer occurring when the Debye length approximates to the size of the hydrated metal ion. The zeta potential (ζ) was calculated from each C_{sp} measurement. It was found that ζ is highly sensitive to pH but not sensitive to rock microstructure. It exhibits a pH-dependent logarithmic behaviour with respect to C_f at low to medium salinities ($\zeta = 0.01133 \log_{10}(C_f) + 0.003505$, units: V and mol/dm^3) and a limiting zeta potential (zeta potential offset) at high salinities of $\zeta_o = -17.36 \pm 5.11 \text{ mV}$ in the pH range 6–8, which is also pH dependent. The sensitivity of both C_{sp} and ζ to pH and of C_{sp} to rock microstructure indicates that C_{sp} and ζ measurements

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can only be interpreted together with accurate and equilibrated measurements of pore fluid conductivity and pH and supporting microstructural and surface conduction measurements for each sample.

Keywords Microstructure · Porosity · Streaming potential · Zeta potential · pH · Electrokinetic rock properties

1 Introduction

The last 15 years have seen the publication of a number of laboratory studies of the streaming potential coefficient (C_{sp}) and zeta potential (ζ) of rocks, predominantly silica-rich mineralogies (e.g. [Jouniaux and Pozzi 1995a,b, 1997](#); [Alkafeef and Alajmi 2007](#); [Vinogradov et al. 2010](#)). [Glover et al. \(2006\)](#) collated a database of C_{sp} and ζ determinations as a function of salinity, which is reproduced in [Fig. 1](#) with supplementary data from AC measurements on Ottawa sand ([Tardif et al. 2011](#); [Glover et al. 2012a](#)), and omitting carbonate samples. Four aspects of the data are worth mentioning explicitly.

First, at low salinities there is a spread of C_{sp} values encompassing up to three orders of magnitude. Some of this spread is almost certainly due to experimental uncertainties, perhaps with the measurement itself, but most of the scatter is likely to be associated with the lack of pore fluid equilibration or knowledge of the real electrical conductivity, salinity, and pH of the fluid in the sample at the time of the measurement ([Walker et al. 2014](#)).

Second, the spread of C_{sp} at low salinities is also due to the effect of the rock microstructure ([Jouniaux and Pozzi 1995b](#); [Glover and Déry 2010](#); [Glover et al. 2012b](#)).

Third, the behaviour of both C_{sp} and ζ changes depending on the salinity. Attempts have been made previously to characterize the medium and high salinity behaviours of both C_{sp} and ζ , the former as a power law with respect to salinity (e.g. [Vinogradov et al. 2010](#)), and the latter as a logarithmic relationship with respect to salinity in the medium salinity range (e.g. [Pride and Morgan 1991](#); [Bolève et al. 2007](#); [Revil et al. 1999](#); [Jaafar et al. 2009](#); [Vinogradov et al. 2010](#)) while taking a constant value in the high salinity range (e.g. [Johnson et al. 1999a,b](#); [Dukhin et al. 2005](#); [Jaafar et al. 2009](#); [Vinogradov et al. 2010](#)). Higher-quality measurements are needed to confirm and improve such relationships, as well as allowing their dependence on other factors such as pH to be studied.

Fourth, it is extremely difficult to make measurements of the C_{sp} with high salinity fluids. Highly saline fluids are difficult to handle in a stable fashion, precipitating with small changes in temperature ([Vinogradov et al. 2010](#)). Furthermore, the values of C_{sp} are so low ([Fig. 1a](#)) that it is difficult to make accurate measurements of the resulting small streaming potentials.

There are a number of questions which arise from [Fig. 1](#) and the points made above. These are: (i) to what extent in the scatter observed in the C_{sp} and ζ data caused by uncertainties in measurement and lack of control of parameters upon which C_{sp} and ζ depend? (ii) What are the main controlling factors on C_{sp} and ζ , and can relationships between C_{sp} , ζ , and salinity be defined accurately? (iii) How does pH affect C_{sp} and ζ and their relationship with salinity? (iv) How does the microstructure of the samples affect C_{sp} and ζ ? (v) Can the high salinity behaviour of C_{sp} and ζ observed by [Jaafar et al. \(2009\)](#) and [Vinogradov et al. \(2010\)](#) be confirmed and quantified?

Consequently, the primary aim of the work described in this paper was to make the best possible measurements of C_{sp} and determinations of ζ as a function of salinity on sandstones over the largest range of salinities and especially including salinities greater than 1 mol/dm³.

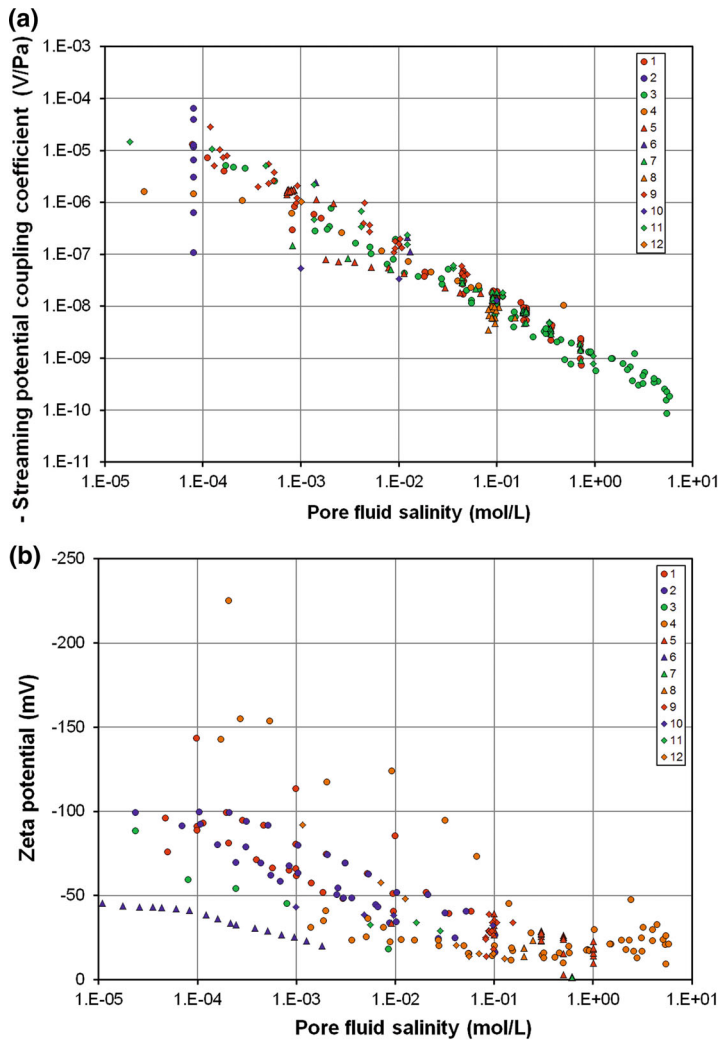


Fig. 1 **a** A compilation of 266 experimentally measured values of the streaming potential coefficient as a function of pore fluid salinity. **b** A compilation of 269 experimentally measured values of the zeta potential coefficient as a function of pore fluid salinity. All values are negative in both parts of the figure. Temperature 20–25 °C. pH5–9 for all measurements. **a** [1] sandstone with NaCl (Sprunt et al. 1994; Jouniaux and Pozzi 1995a, 1997; Li et al. 1995; Jiang et al. 1998; Pengra et al. 1999); [2] sandstone with NaCl as a function of permeability/microstructure (pH5) (Jouniaux and Pozzi 1995b); [3] St. Bees, Stainton, and Fontainebleau sandstones with NaCl (Jaafar et al. 2009; Vinogradov et al. 2010); [4] sandstone with KCl (Alkafeef and Alajmi 2007); [5] sand with NaCl (Guichet et al. 2003; Block and Harris 2006); [6] granite with NaCl (Morgan et al. (1989)); [7] glass with NaCl (Pengra et al. 1999; Block and Harris 2006); [8] zeolitized tuffs with NaCl (Revil et al. 2002); [9] basalt with NaCl (Revil et al. 2003); [10] granite with KCl (Tosha et al. 2003); [11] silica nanochannel with KCl (Heyden et al. 2006); [12] Ottawa sand (Tardif et al. 2011; Glover et al. 2012a). **b** [1] Quartz with NaCl (Pride and Morgan (1991)); [2] silica with NaCl (Gaudin and Fuerstenau 1955; Li and Bruyn 1966; Kirby and Hasselbrink 2004); [3] glass beads with NaCl (Bolève et al. 2007); [4] St. Bees, Stainton, and Fontainebleau sandstones with NaCl (Jaafar et al. 2009; Vinogradov et al. 2010); [5] clay minerals with NaCl (Kosmulski and Dahlsten 2006; Avena and Pauli 1998); [6] sandstone with KCl (Lorne et al. 1999); [7] quartz with NaCl (Kosmulski et al. 2002); [8] kaolin-coated sandstone with NaCl (Pengra et al. 1999); [9] tuff samples containing clays and zeolites (Revil et al. 2002); [10] kaolinite with NaCl (Poirier and Cases 1985); [11] mica with NaCl (Will and Nover 1986); [12] sandstone with NaCl (Alkafeef and Alajmi 2007)

These high-quality measurements would then allow us answer the questions posed above as well as to provide high-quality experimental data against which the theoretical model of Glover and Déry (2010) and Glover et al. (2012b) could be rigorously tested. This paper addresses the questions posed above and reserves the modelling to a forthcoming paper.

2 Sample Material

Fourteen samples of four sandstones were measured, three samples of each of Berea and Boise sandstone, and four samples of each of Lochaline; Fontainebleau sandstones. Both Fontainebleau and Lochaline sandstones have two grades, one composed of subrounded detrital grains, the other consisting of detrital grains with additional euhedral quartz overgrowths. Our samples consisted of two of each type and can be distinguished by the additional letter 'D' for the detrital type and 'Q' for quartz overgrown type in the sample codes.

Each sample was submitted to a battery of basic measurements. The grain size distribution of crushed offcuts of each sample was measured by laser diffractometry using a Malvern Mastersizer 2000. The theta transformation (Glover and Walker 2009) was then used to calculate the pore size distribution of the rock and hence calculate the pore throat diameter with the method of Glover and Déry (2010). This approach is more accurate than using X-ray computed microtomography data. The pore throat diameter distribution was measured independently by mercury injection. Porosity measurements were taken using at least three techniques for most samples. These core porosity measurements are helium pycnometry, mercury injection porosimetry, and fluid saturation/Archimedes' porosity. Some samples were also subjected to image analysis, X-ray computed tomography (CT), X-ray computed microtomography (μ CT), and NMR-based porosity measurements. A simple klinkenberg nitrogen permeability measurement was also taken on each sample. Fluid permeametry and impedance spectroscopy were also carried out during the main experimental runs. Data for all the samples are summarized in Table 1, while SEM photomicrographs of each rock are shown in Fig. 2.

We have carried out the procedure of making resin casts of the pore networks of both types of Fontainebleau and Lochaline sandstones. Initially, the samples were injected with low-viscosity epoxy resin by submerging them in resin in a vacuum oven heated to 50 °C. The viscosity of the resin is sufficiently low at 50 °C to allow vacuum impregnation into pores as small as 1 μ m. In this work we also added a small confining pressure of 500 psi (3.45 MPa) using an old heated Hassler cell in order to fill the smallest spaces possible. Even then, we found that it was difficult to impregnate the thinnest intergranular pore sheets between the euhedral grains of the overgrown sandstones, the evidence of which can be found in the subcircular windows in the centre of some of the resin sheets (see Fig. 2j). These windows represent pore space too thin for the resin to access. The samples were then cut into small 10-mm cubes, placed in plastic containers, and treated with dilute hydrochloric acid to remove any carbonates. The latter step is required because insoluble fluorides may be formed if carbonates are present when the samples are placed in hydrofluoric acid. Finally, the samples were thoroughly cleaned and then submerged in warm (50 °C) hydrofluoric acid for several weeks in order to remove all traces of the silicate minerals. In practice, we found that there always remained a small core of undigested silicates at the centre of the sample, but this did not impede the imaging of the surface structures. Finally, the samples were thoroughly cleaned and then prepared for the SEM in the usual way.

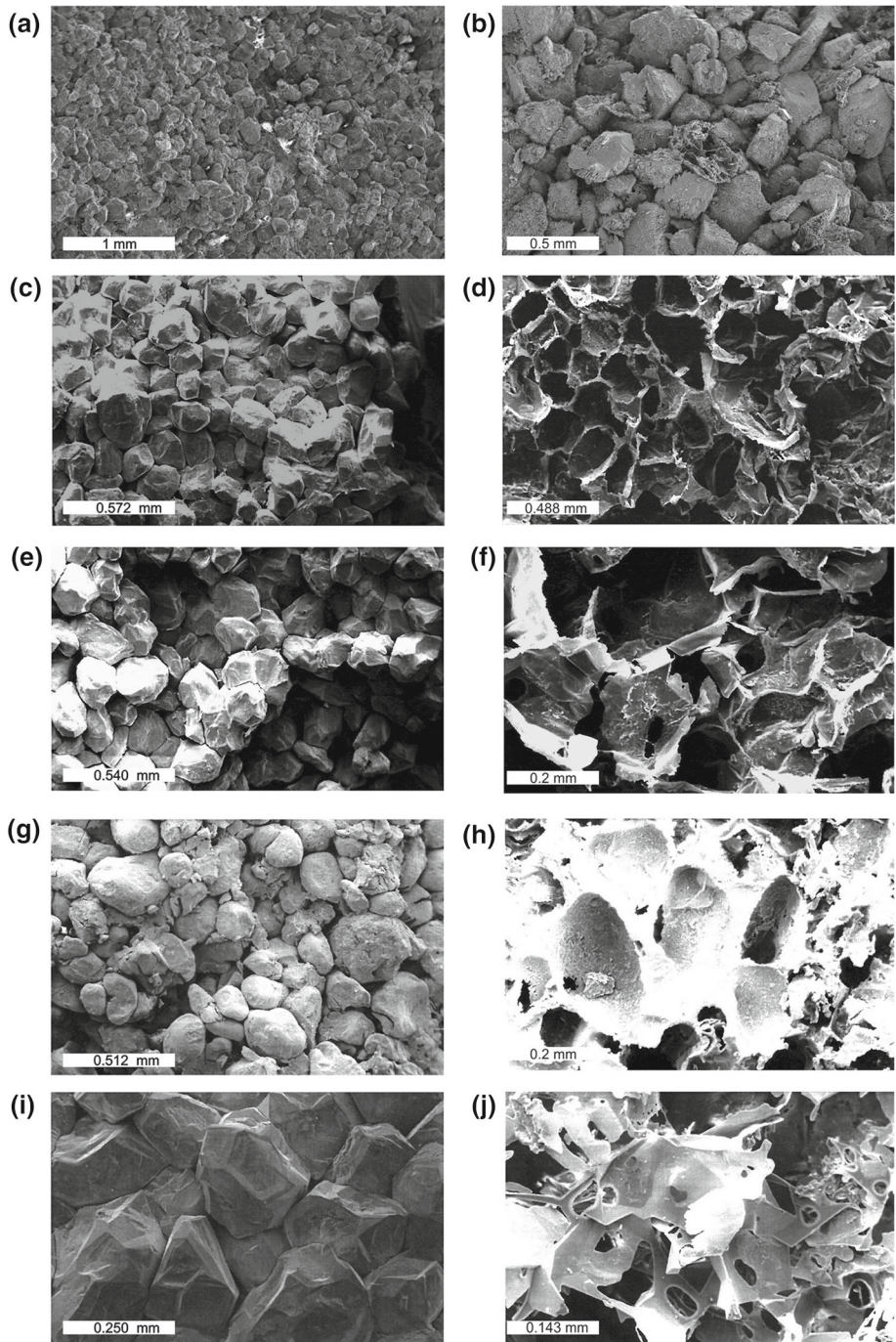


Fig. 2 Photo-micrograph SEM images of **a** Berea sandstone (BR1), **b** Boise sandstone (B3I), **c** Fontainebleau sandstone (detrital, F1D) together with **d** its resin cast (F1D), **e** Fontainebleau sandstone (quartz overgrown, F4Q) together with **f** its resin cast (F4Q), **g** Lochaline sandstone (detrital, L2D) together with **h** its resin cast (L2D), **i** Lochaline sandstone (quartz overgrown, L3Q) together with **j** its resin cast (L3Q)

The resulting ‘negative’ of the rock shows how the distribution of the pore space differs between the uncemented and cemented forms of the rock. Figure 2d, g, h, j shows SEM images of the resulting pore networks. In the detrital forms of both the Fontainebleau and Lochaline sandstones, the grain casts are subrounded and there is a significant volume of epoxy between the grain casts, indicating that the pore space is well-developed and interconnected, especially for the Lochaline sandstone (Fig. 2h). The saturation of this relatively large interdrital porosity with conductive fluids results in high permeabilities and electrical conductivities, as well as cementation exponents that approach the values expected for spherical grains ($m = 1.5$), as shown in Table 1, and explaining the low mechanical strength.

By contrast the overgrown samples of both Fontainebleau and Lochaline sandstones show interlocking euhedral grains, which manifest themselves as thin sheets of porosity where two crystal faces grow together. Figure 2g, j shows these fine planar pore spaces, some of which are small enough not to allow the ingress of epoxy. These thin but highly connected pore spaces result in porosities ten times lower than in the detrital rocks (Table 1) with concomitantly smaller permeabilities. The slightly lower cementation exponents in these rocks are associated with the increased connectivity of the pore space, while interlocking grains ensure that the rock has a relatively high mechanical strength compared to the detrital form.

3 Apparatus and Methodology

The experimental cell and measurement set-up is described in Walker et al. (2014). The pore fluids were made up to approximately the correct salinity using distilled, deionized water that had been previously degassed. The electrical conductivity and pH of the initial solution and that measured after equilibration throughout the measurements were measured using bench-top fluid conductivity and pH meters (VWR SB70C and SB70P, respectively), and the salinity of the solution was derived using the method of Sen and Goode (1992a, b) as a function of temperature. Once the samples were loaded and full saturation had been attained, the process fluid was circulated through the rock in a closed system until it had attained complete equilibrium with the sample. This was carried out while measuring the electrical conductivity and pH of fluids leaving the sample using both an in-line technique and by testing aliquots of the fluid. Initially we allowed 50 pore volumes (about 2000 cm^3) and the entire recycling of the input reservoir twice (8000 cm^3), whichever was greater, for this to occur. At the highest pump flow rate this would take about 16 h to complete. However, the measurements on the fluids leaving the cell were a useful monitor of when equilibration was attained, and we were often able to reduce the equilibration time to about 5 h, especially when we were using highly permeable Boise sandstone and a pressurized reservoir to flow through the sample at a very high rate.

Figure 3 shows the equilibration of a sample of Berea sandstone. The time at which equilibration is achieved is difficult to define. In our case we considered equilibration to have taken place when the difference between two consecutive running means over 10 measurements was less than the standard deviation in those same, but detrended, measurements. This is equivalent to noting that stability is judged when the systematic change is significantly less than the random errors in the measurements. This approach can be automated and avoids setting a threshold gradient below which equilibrium is achieved, which needs to be considered carefully and altered according to the magnitude of the measurements. In practice, we applied the method conservatively, often waiting for many more pore volumes to pass

Table 1 Sample properties

Parameter Sample codes	Symbol	Units	Berea sandstone			Boise sandstone			Fontainebleau sandstone			Lochaline sandstone				
			BR1 ET5	BR2	BR3	B III	B II	B3I	F1D	F2D	F3Q	F4Q	L1D O6	L2D O2	L3Q H3	L4Q H1
Modal grain diameter (laser diffractometry)	d_{grain}	μm	180	155	162	739	991	2140	235	231	251	255	263	268	277	278
Modal pore throat diameter (MICP)	d_{pore}	μm	8.35	6.27	14.24	34.8	34.8	74	13.2	14.8	0.46	0.68	15.5	18.2	1.19	2.58
Modal pore diameter [calculated, Glover and Walker (2009)]	d_{pore}	μm	17.04	14.68	16.14	125	181	274	20.4	22.4	0.48	0.54	23.2	24.1	0.99	1.99
Modal pore throat diameter [calculated, Glover and D��ry (2010)]	d_{pt}	μm	10.23	8.82	9.69	75.3	109	165	12.3	13.5	0.29	0.32	13.95	14.53	0.60	1.20
Cementation exponent	m	(-)	1.79	1.72	1.66	1.569	1.677	1.811	1.71	1.68	1.60	1.63	1.5	1.55	1.54	1.48
Formation factor using m and ϕ_{sat}	F	(-)	10.22	10.63	10.47	6.53	5.65	7.48	11.6	10.6	567	503	13.1	12.4	312	163
Helium porosity	ϕ_{He}	(-)	0.274	0.258	0.252	0.370	0.342	0.333	0.241	0.249	0.020	0.026	0.187	0.208	0.027	0.035
Saturation porosity	ϕ_{sat}	(-)	0.272	0.253	0.243	0.302	0.356	0.329	0.238	0.245	0.019	0.022	0.180	0.197	0.024	0.032
Mercury porosity	ϕ_{Hg}	(-)	0.254	0.232	0.228	0.300	0.259	0.257	0.235	0.235	0.015	0.018	0.210	0.205	0.022	0.030
Surface conductivity	Σ_s	$\times 10^{-4} \text{ S/m}$	285	66.5	50.2	13.6	32	13.4	2.44	2.31	62.2	55.4	1.96	1.81	21.85	10.1
pH	pH	(-)	8.03	7.05	7.41	7.2	5.94	6.61	6.40	6.41	6.48	6.51	7.27	7.26	7.15	7.12
Number of measurements	N	(-)	55	143	135	74	18	18	113	99	104	95	100	103	99	97

Note: Further information concerning the general properties of Boise, Berea, Fontainebleau and Lochaline sandstones may be found in [Churcher et al. \(1991\)](#); [Gomez et al. \(2010\)](#), [Pasqualini et al. \(2007\)](#), and [Worden and Morad \(2009\)](#)

than was strictly necessary. The black line in Fig. 3a shows the conductivity of the pore fluid exiting the core, monitored every minute by an in-line device. Initially it is high as previously stagnant fluid is washed out of the sample, dropping towards the conductivity of the fluid in the large reservoir (dashed line), then increasing towards stability as the reservoir fluid comes into equilibrium with the rock by dissolution. The bump at 769 min was caused by a change in the thermal stability of the laboratory as winter heating was activated. The red line is the input conductivity, again monitored every minute by an in-line device. It starts at the reservoir (bulk) fluid conductivity and increases only gradually due to the reservoir's large volume, and is relatively unaffected by the initial washout. The black diamonds are checks on this value made by taking aliquots of the reservoir fluid for measurement with a dipmeter. The difference between the conductivities of the input and output fluids is shown in Fig. 3b as a function of the conductivity of the input fluid and shows, as would be expected, that there is more dissolution when the input fluid is less saline, but also shows that the relationship between input fluid conductivity and the enhancement in the conductivity of that fluid by passing it through the rock is approximately linear. We conclude that equilibrium will be reached much faster if the input pore fluid has a conductivity and hence salinity which is close to that when equilibrium is achieved. This figure just confirms what we would expect by intuition. Figure 3 also shows the variation of pH during equilibration, which has a more complex and currently unexplained form.

The streaming potential and pressure difference measurements were taken during transient increases of the pressure difference and then during transient falls. Each of these measurements took about 100 s and logged over 10,000 individual measurements. Streaming potential measurements were taken with leak-free nonpolarizing reference Ag/AgCl electrodes (Warner Instruments 69-0053) and differentially amplified before being logged using a National Instruments DAQBoard NI USB6229 USB logger and PC running LabVIEW Signal Express. The streaming potential measurement system was calibrated using a Time Electronics 1017 Voltage/Current/Resistance Calibrator with a resolution of 10 nV, allowing us to make measurements to within an error of $\pm 1.2\%$ when the streaming potential was over 1 mV. The measurement error dropped to $\pm 4.4\%$ when the streaming potential was at the limit of its measurement at about 30 nV, which was caused by the signal dropping below the ambient noise and occurred for a salinity of about 4.5 mol/dm^3 . The pressure measurements were taken using Omega PX302 transducers calibrated to an Endress and Hauser Deltabar S PMD75 secondary calibration standard and were also amplified by individual custom-designed amplifiers before being logged. The error in the differential measurement ranged from $\pm 3.4\%$ at the lowest pressure differences measured to $\pm 1.2\%$ at the highest. The electrical conductivity of the rock samples was made using the existing Ag/AgCl electrodes and a Solartron 1260A impedance spectrometer with the frequency fixed at 1 kHz.

Since the individual transient experiments were so easy and quick to carry out, we often carried out three or four in each direction (increasing and decreasing inlet pressure) in order to obtain very high-quality data. The fitting procedure also generates an individual fitting error, which we have combined with the measurement errors to obtain an overall experimental error for each measurement. The streaming potential data were plotted against the pressure difference measurements, all zeroes having been adjusted for. The streaming potential coefficient was then calculated from the gradient of the plot. This procedure was described in Walker et al. (2014), where such a plot can be found as their Fig. 2. The permeability was also measured when a steady pressure difference was attained at the end of the upward leg of the transient measurement. The electrical conductivity of the sample was measured immediately after at the end of the downward leg of the transient measurement after the vestiges of pressure difference had been removed using an equalizing valve.

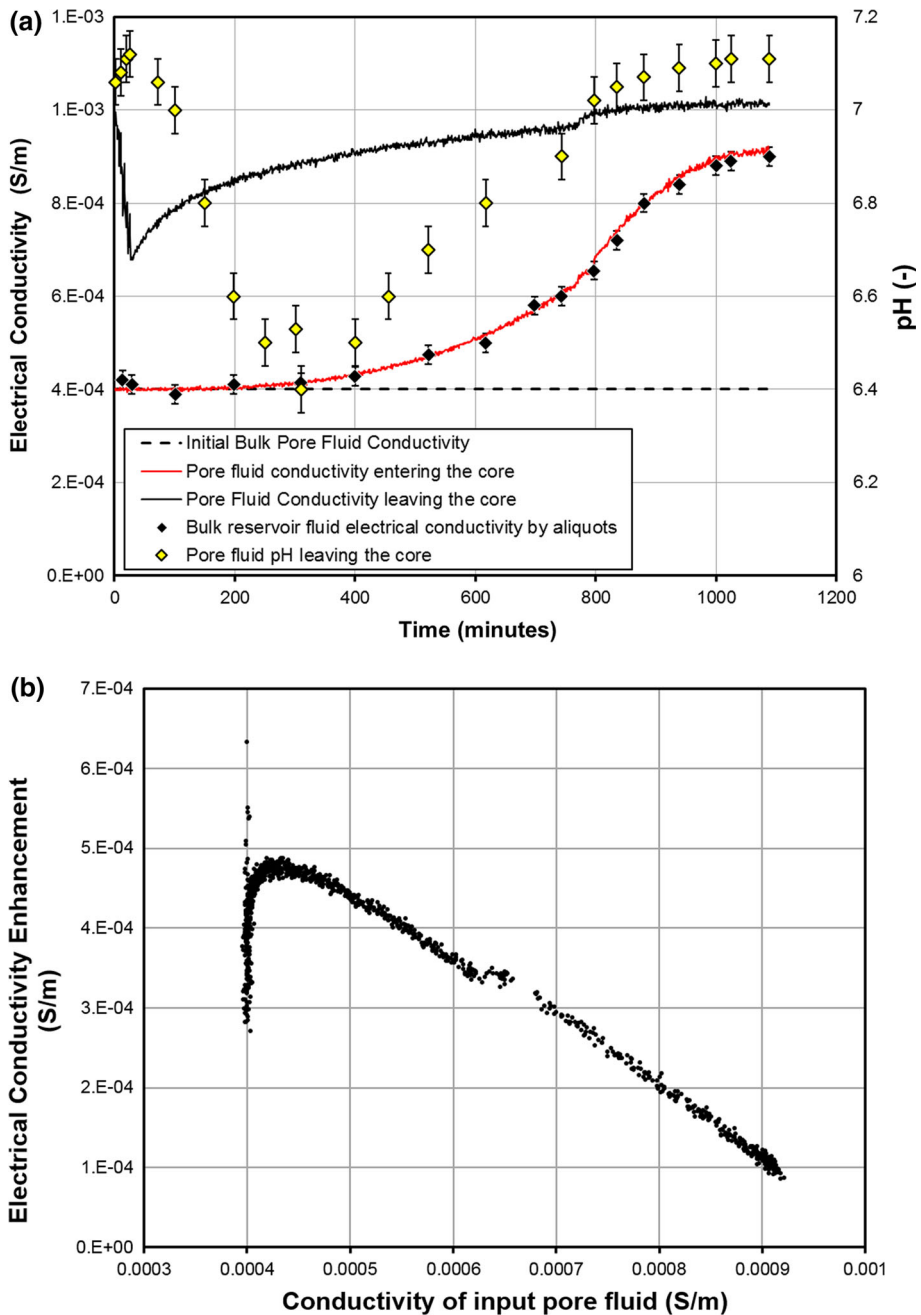


Fig. 3 Example equilibration of electrical conductivity and pH for one of the measurement campaigns of sample BR2

4 C_{sp} Results and Discussion

4.1 Data Quality and Uncertainties

Figure 4 shows the measured streaming potential coupling coefficients for all the samples as a function of the salinity of the fully equilibrated pore fluid that was in the sample at the time of the measurement. This figure contains 1253 individual measurements of streaming potential coefficient at a given fluid salinity, temperature and fluid pH, each of which is the result of about 10,000 determinations of streaming potential and fluid pressure difference. This data set represents a significant increase in the number of measurements available in the literature. The only previously available data of comparable quality to that generated in this work are for Fontainebleau sandstone by [Vinogradov et al. \(2010\)](#). These data have been included in Fig. 4c for comparison with the new determinations, showing that the two data sets are mutually consistent. Figure 5 shows a comparison of the new measurements with the database of 266 streaming potential coefficient measurements that were collated by [Glover et al. \(2012b\)](#) and which represent the majority of previously available measurements.

In general, it is clear that most of the new data are of relatively high quality with measurements for individual samples describing a smooth curve with little scatter. Berea sandstone BR1 and Boise sandstone BIII show lower-quality data than the other new data because they were made at the start of our measurement campaign and before our experimental protocols were fully developed. Nevertheless, the quality of these measurements also surpasses that of the previously existing data.

The uncertainties in all the measurements have been assessed and propagated where necessary. These uncertainties are not shown explicitly in Figs. 4 and 5 to avoid complicating already complex plots. However, all experimental errors are generally much smaller than most previous studies. At low and medium salinities the error in C_{sp} is approximately $\pm 5\%$ of the measured value, while at high salinities, the error in C_{sp} is significantly higher, at about $\pm 45\%$. Nevertheless, the strong power law behaviour of C_{sp} ensures that its variation with salinity is clear. This observation is consistent with the scatter of each of the streaming potential trends.

The behaviour of the data can be split into three regimes: low salinity ($C_f < 0.01 \text{ mol/dm}^3$), medium salinity ($0.01 \text{ mol/dm}^3 < C_f < 1 \text{ mol/dm}^3$), and high salinity ($C_f > 1 \text{ mol/dm}^3$).

4.2 Low Salinity Behaviour

At low salinities there is a flattening of the curves that has been ascribed to the microstructural properties of the rocks ([Glover et al. 2012b](#)). The flattening is more pronounced for the samples where there has been authigenic quartz overgrowth for both the Fontainebleau and Lochaline sandstones.

Comparison of the data with the supporting measurements shown in Table 1 shows that the flattening is more pronounced for samples with (i) lower porosity, (ii) larger cementation exponents, (iii) smaller grain sizes (and hence pore and pore throat sizes), and (iv) larger surface conduction.

This is consistent with the theoretical model of ([Glover and Déry 2010](#); [Glover et al. 2012b](#)) and also implies that low salinity flattening of C_{sp} is more pronounced for rocks with larger formation factors F and electrical tortuosities τ_e and hence for smaller connectedness G and connectivities χ ([Glover 2010, 2017](#)). The best example of this behaviour is the comparison of the results for the authigenic overgrowth forms of the Fontainebleau and Lochaline sandstones

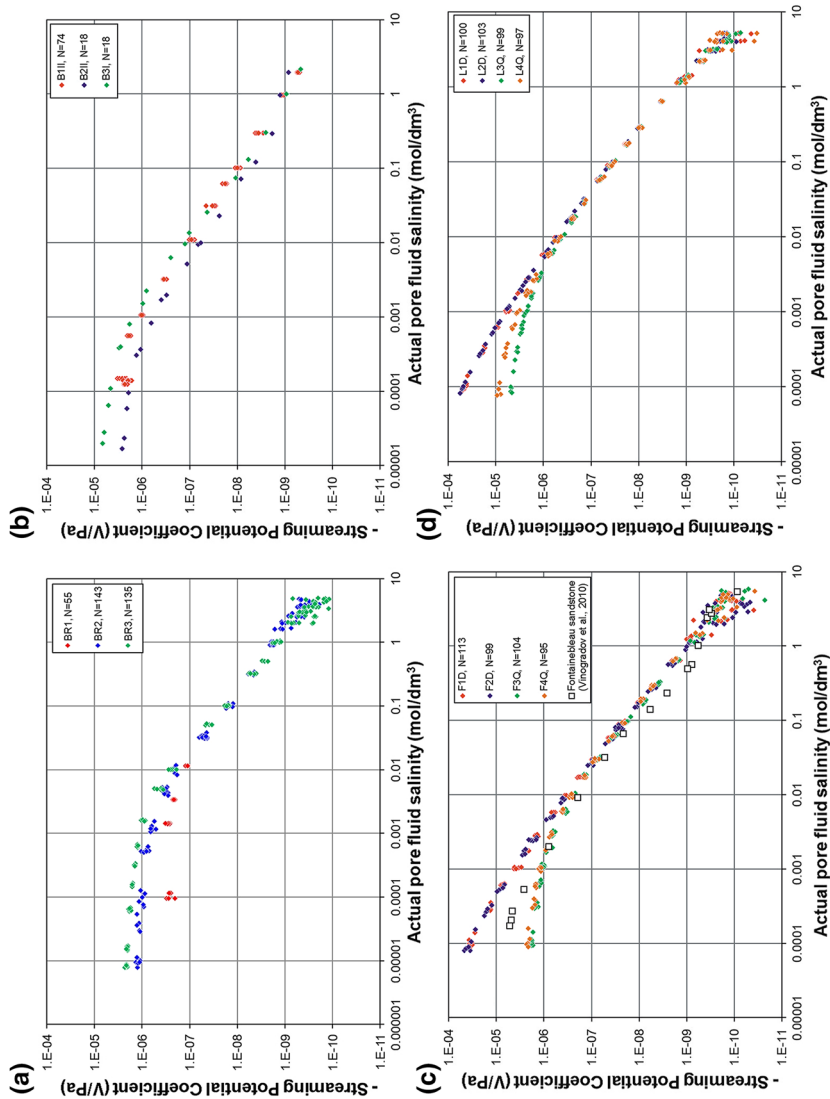


Fig. 4 Measured streaming potential coefficients for samples of Berea, Boise, Fontainebleau, and Lochaline sandstones as a function of the salinity of the pore fluid in the rock at the time of the streaming potential measurement. **a** Berea sandstone (BR1, BR2, and BR3), **b** Boise sandstone (B1I, B2I, and B3I), **c** Fontainebleau sandstone (F1D, F2D, F3Q, and F4Q) together with the best previous data (Vinogradov et al. 2010) as open squares and **d** Lochaline sandstone (L1D, L2D, L3Q, and L4Q). In the cases of Fontainebleau and Lochaline sandstones red and blue symbols represent detrital form of the rock, while orange and green symbols represent the quartz overgrown type

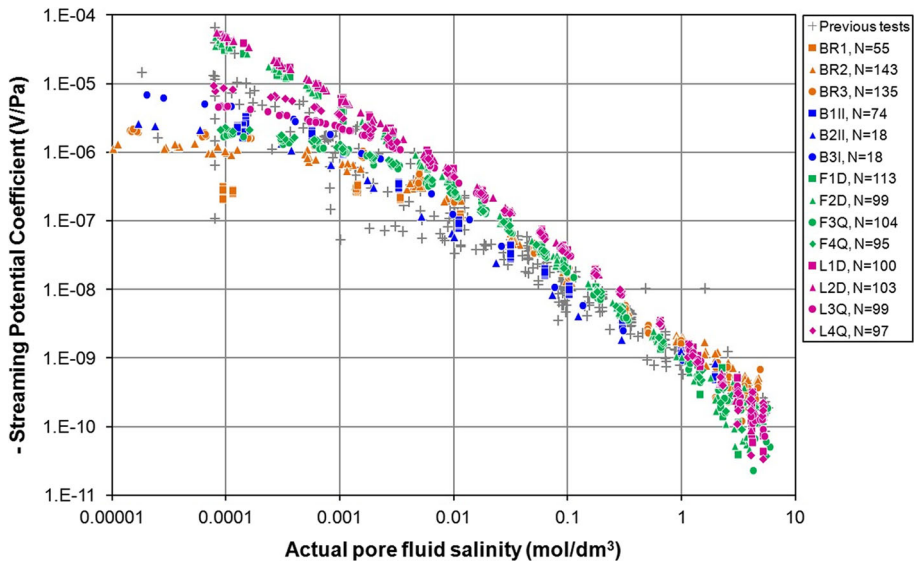


Fig. 5 Aggregation of the new streaming potential coefficient measurements with those from the existing database. The existing data are shown undifferentiated by grey crosses, and the new data are shown as coloured symbols. Berea (red), Boise (blue), Fontainebleau (green), and Lochaline (purple)

with respect to their chemically identical detrital forms. The four samples of overgrown sandstones are subject to a clear and pronounced flattening at high salinities compared to their analogous detrital samples. Reference to Table 1 shows that the authigenic quartz samples for both the Fontainebleau and Lochaline sandstones have (i) significantly lower porosities (a factor of 0.11 times lumping all porosity measurement types for both Fontainebleau and Lochaline samples), (ii) much higher formation factors (by a factor of 32.39), (iii) similar cementation exponents, (iv) similar grain sizes but much smaller pore throat sizes (by a factor of 0.062), and (v) much larger surface conduction (increased by a factor of 17.55). All of these differences are consistent with expectations from the theoretical model of Glover et al. (2012b).

Such functional dependencies would suggest that the sample measured by Vinogradov et al. (2010) and shown in Fig. 4c was predominantly detrital with incomplete development of authigenic quartz and a porosity and formation factor approximately midway between 2.4 and 22% and between 12 and 386, respectively. The values of these parameters reported by Vinogradov et al. (2010) were 7.2% and 157, respectively, which is consistent with our data and the expectations of the theoretical model.

It is worth noting that much of the scatter of the previously existing measurements at low salinities (shown by the grey crosses in Fig. 5) can be explained by invoking differing degrees of flattening that we now associate with the different porosity, cementation exponent, grain size, and specific surface conductivity of each sample. Since most of these parameters are not known because they either were not measured or reported, it is difficult to assess the degree to which the unknown microstructure of samples has led to the apparent large scatter of the previous streaming potential determinations. However, a sensitivity analysis of the Glover et al. (2012b) model should show the extent to which varying each parameter might lead to flattening in an attempt to be more confident in the assertion that this is the cause.

4.3 Medium Salinity Behaviour

Our streaming potential coefficient data taken at medium salinities follow a smooth curve which approximates to a power law and manifests itself as a straight line with a negative gradient on bilogarithmic axes as shown in Figs. 4 and 5. There are slight indications that the streaming potential trends in this medium salinity range have additional structure, which has been noted in some samples before (Glover et al. 2012b).

We have carried out a simple power law fit to all of the new data in the salinity range $0.01 \text{ mol/dm}^3 < C_f < 1 \text{ mol/dm}^3$ and present the results in Table 2. The behaviour of these fits can be represented by the relationship

$$C_{sp} = aC_f^b, \quad (1)$$

where the values of the coefficients a and b are given in Table 2 and the mean values for all our measurements are $\langle a \rangle = (-1.440 \pm 0.517) \times 10^{-9}$ and $\langle b \rangle = -1.127 \pm 0.125$. The large number of measurements in the data set, combined with their quality, ensures that the coefficients of determination for all individual power law fits are high, always exceeding 0.97. A fit to the same relationship carried out by Vinogradov et al. (2010) provided $a = -1.36 \times 10^{-9}$ and $b = -0.9123$ with $R^2 = 0.982$ with no uncertainties given and once their mV/MPa unit system has been converted into that used in this work (V/Pa). These values are consistent with our results. The extent to which such mean values are useful is, however, limited. One of the main controlling influences on streaming potential is zeta potential and that, as we will see in the next section, depends strongly on pH. Since the pore fluids in each of the measured samples has a distinct pH, it is perhaps unreasonable to read much into the mean values in Table 2 without knowing the pore fluid pH and its effect. Consequently, we have plotted the coefficients in Eq. (1) for individual samples as a function of the measured pore fluid pH in the first part of Fig. 6. This figure includes propagated experimental and fitting uncertainties for the coefficients and experimental uncertainties for the pH. It is clear that the factor coefficient a is sensitive to pH, varying between about 0.75 and about 2.2 over the pH range from 6 to 8, while the exponent coefficient b seems not to be sensitive to changes in pH in this range, taking a mean value of -1.127 ± 0.125 . The consequence of these two dependencies is that increasing the pH tends to displace the streaming potential curves in Figs. 4 and 5 upwards without appreciably altering their slope (when plotted on bilogarithmic axes) as shown schematically in Fig. 6b.

5 High Salinity Behaviour

The measurements for salinities greater than 1 mol/dm^3 for all samples are in good agreement with the measurements of Jaafar et al. (2009) and Vinogradov et al. (2010) up to 5.55 mol/dm^3 (the maximum salinity at which measurements were taken in this work). The behaviour is characterized by a continuation of the power law relationships that exists at medium salinities, which is manifested in bilogarithmic plots as a continuing reducing linear trend and in linear space as a tendency towards zero as the salinity approaches full saturation. This behaviour ensures that the streaming potential does not change polarities at high salinities for the range of pHs studied in this work.

It should be noted that the measurements of streaming potential in the high salinity regime (Figs. 4, 5) have a much higher uncertainty than at lower salinities. This is due to the extreme

Table 2 Medium salinity behaviour of streaming potential and zeta potential

Sample code	Rock type	<i>pH</i>	<i>a</i> ($\times 10^{-9}$) Equation (1)		<i>b</i>	<i>R</i> ²	<i>c</i> Equation (3)		<i>d</i>	<i>R</i> ²	Zeta offset (mV)
BR1	Berea sandstone	8.030	1.890		− 0.923	0.9891	23.66		52.88	0.8127	− 26.00
BR2		7.050	1.620		− 1.008	0.9916	12.04		8.64	0.9239	− 25.52
BR3		7.410	1.406		− 1.110	0.9946	14.45		8.89	0.9723	− 21.05
Mean		7.497	1.639		− 1.014	—	16.72		23.47	—	− 24.19
SD	Boise sandstone	0.496	0.243		0.093	—	6.133		25.47	—	2.73
B1II		7.200	1.120		− 0.9757	0.9945	14.1		57.3	0.9085	− 25.40
B2II		5.940	0.818		− 0.9059	0.9745	8.05		30.59	0.9795	− 12.03
B3I		6.610	0.743		− 1.101	0.9899	10.37		28.76	0.9958	− 13.79
Mean		6.583	0.894		− 0.994	—	10.84		38.88	—	− 17.07
SD		0.630	0.200		0.099	—	3.052		15.98	—	7.26
F1D	Fontainebleau sandstone	6.400	1.047		− 1.259	0.9966	8.88		− 8.76	0.9882	− 12.64
F2D		6.410	1.045		− 1.275	0.9997	8.54		− 9.97	0.9881	− 13.64
F3Q		6.480	1.014		− 1.207	0.9981	8.53		− 6.87	0.9928	− 12.64
F4Q		6.510	1.131		− 1.212	0.9984	8.76		− 9.36	0.9962	− 13.10
Mean		6.450	1.059		− 1.238	—	8.678		− 8.74	—	− 13.01
SD		0.054	0.050		0.034	—	0.172		1.341	—	0.48

Table 2 continued

Sample code	Rock type	pH	Equation (1)		Equation (3)		Zeta offset (mV)	
			$a (\times 10^{-9})$	b	c	d		R^2
L1D	Lochaline sandstone	7.270	2.171	-1.200	10.46	-27.51	0.9887	-17.13
L2D		7.260	2.122	-1.215	10.44	-27.32	0.9935	-16.89
L3Q		7.150	2.053	-1.186	10.19	-24.48	0.9943	-17.39
L4Q		7.120	1.984	-1.202	10.09	-23.72	0.9779	-15.84
Mean		7.200	2.083	-1.201	10.29	-25.76	-	-16.81
SD	Overall behaviour	0.076	0.082	0.012	0.184	1.940	-	0.68
Mean		6.917	1.440	-1.127	11.33	3.505	-	-17.36
SD		0.544	0.517	0.125	4.060	28.823	-	5.11
SD standard deviation								

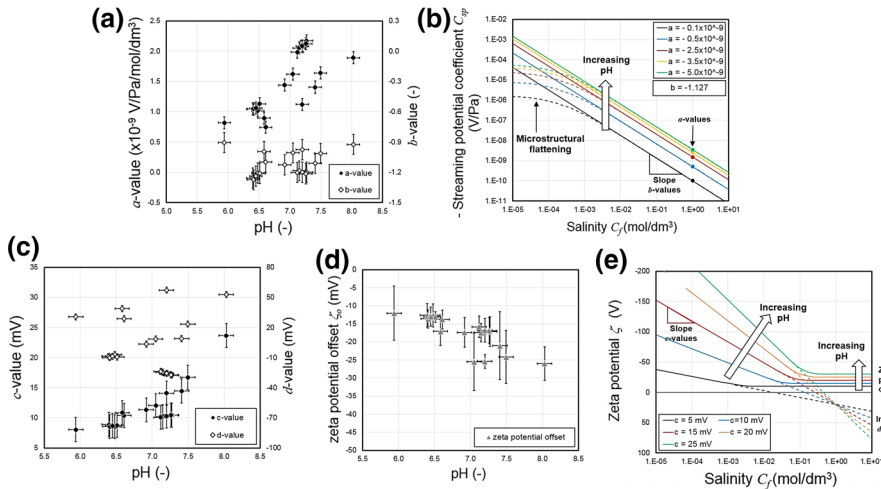


Fig. 6 **a** The factor and exponent coefficients of the power law fit characterizing the medium salinity behaviour of streaming potential as a function of the pH of the pore fluid that was in equilibrium with the rock sample. **b** Schematic diagram of the variation of C_{sp} with increasing pH. The value of a becomes larger, displacing the curves upwards, while the d remains approximately constant leading to no overall change of slope. **c** The factor and offset coefficients of the logarithmic law fit characterizing the low to medium salinity behaviour of zeta potential as a function of the pH of the pore fluid that was in equilibrium with the rock sample. **d** The mean zeta potential offset calculated for all measurements with salinities greater than a visually defined threshold (Berea sandstone: 0.1 mol/dm³, Boise sandstone: 0.01 mol/dm³, Fontainebleau sandstone: 0.1 mol/dm³, Lochaline sandstone: 1 mol/dm³). Uncertainties represent the standard deviation of these data. **e** Schematic diagram of the variation of zeta potential with increasing pH. The value of c becomes larger, increasing the slope, the intercept d has a complex behaviour that can be positive or negative, and the zeta potential offset ζ_0 becomes more negative

difficulty in making such measurements, particularly in keeping the pore fluid stable, as noted by Jaafar et al. (2009) and Vinogradov et al. (2010).

6 Zeta Potentials

6.1 Calculation Methodologies

Zeta potentials have been calculated from each streaming potential coefficient measurement using the Helmholtz–Smoluchowski law (Glover 2015a) with Overbeek’s correction (Jouniaux and Pozzi 1995; Walker et al. 2014), together with experimentally measured supporting data and empirical models. This calculation requires the knowledge of the electrical conductivity (σ_f), permittivity (ϵ_f), and viscosity (η_f) of the equilibrated pore fluid as well as the formation factor of the rock both at high salinities (F_o) and at each of the salinities used in the measurement (F_i). Since the effective conductivity of the saturated sample was also measured independently both at the experimental salinity (σ_i) and at high salinity (σ_{eff}), the formation factor at each experimental salinity ($F_i = \sigma_f/\sigma_i$) and that at high salinity ($F_o = \sigma_f/\sigma_{eff}$) could be calculated. The ratio of the formation factor at high salinity (i.e. when the contribution of surface conduction is negligible) to that at the given salinity of the experiments (i.e. F_o/F_i) is termed the Overbeek’s correction. The full equation for calculating the zeta potential including the Overbeek correction is given by

$$\zeta = \frac{C_{sp}\eta_f\sigma_f}{\varepsilon_f} \frac{F_o}{F_i}. \quad (2)$$

Other parameters used in the calculation were electrical permittivity, which was calculated for an aqueous NaCl solution for each salinity at 25 °C using the unpublished method of Gary Olhoeft (e.g. [Glover et al. 2012b](#)) and pore fluid viscosity, which was calculated for a NaCl solution for each salinity at 25 °C using the method of [Phillips et al. \(1978\)](#).

7 Data Quality and Uncertainties

Figure 7 shows the resulting zeta potential as a function of the salinity of the fully equilibrated pore fluid for all the samples. These measurements are compared with the database of 269 derived zeta potential values that were collated from the literature by [Glover et al. \(2012b\)](#) in Fig. 8.

Figure 7 shows that the new zeta potential data for each of the rock types form trends, where trends were often difficult to make out in the previously published data due to inaccurate measurements or the lack of control of key parameters. The knowledge and control of pH was a particular problem in these previously published data, as exemplified in Fig. 1b. We ascribe the relatively small scatter in our data to the control we have imposed on the pH of the solutions we used. For example, the main reason for the contrast between the values of zeta potential for the Fontainebleau sandstone (green symbols) and Lochaline sandstone (purple symbols) in Fig. 8 arises from a difference in the pH of the solutions with which each was saturated. These solutions had a pH of 6.45 ± 0.05 and 7.2 ± 0.08 (mean $\pm$ range), respectively (Tables 1, 2). In fact, modelling shows that it is the pH that is the main control on the zeta potential (e.g. [Hunter 1981](#); [Glover et al. 2012b](#)). We hypothesize that the scatter in all of the existing measurements (shown by the grey crosses in Fig. 8), can be explained by suggesting that these measurements were taken at a variable but unknown pore fluid pH somewhere between 5 and 9. The papers from which these previous data were gathered often mention that the fluid pH was within this range but rarely specify the measured value, remaining sparse, ambiguous and approximate with their pH information (Fig. 1b).

Unfortunately, the sparsity of pH data in the previous publications ensures that we cannot test the hypothesis that it is the pH that causes the scatter in the previous data because the fluid pH is either ill-controlled or not disclosed. We can, however, test the hypothesis by examining our new data, where the pH is well-controlled, and comparing it with theoretical models ([Glover and Déry 2010](#); [Glover et al. 2012b](#)). We consider the variation of zeta potential with pH later in this paper, but unfortunately there is insufficient space to include full modelling of the new data in this paper. However, such modelling will be reported in a forthcoming paper. A simple example of the control that pH has on the zeta potential is given by comparing the curves for our Boise B2II and Boise B3I samples (Fig. 5b). The mean and standard deviations for the pH of the equilibrated fluids for these two samples were 5.94 ± 0.24 and 6.61 ± 0.086 , respectively, leading to a clear separation in the zeta potential curves.

It is worth noting that, unlike the C_{sp} , the zeta potential does not depend on rock microstructure at low salinities. This implies that the cause of the low salinity flattening in the streaming potential coefficient for some rock samples is to be found in the way the rock microstructure affects surface conductance rather than being associated with changes in the surface conductivity itself.

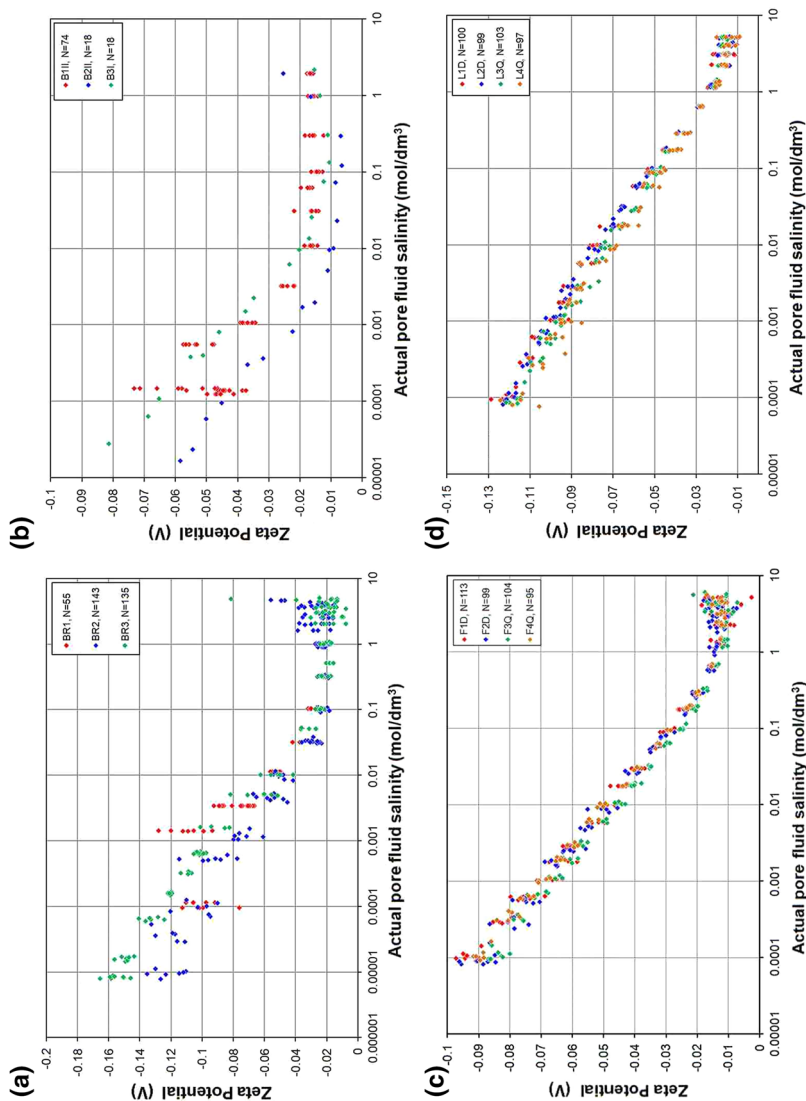


Fig. 7 Derived zeta potentials for samples of Berea, Boise, Fontainebleau, and Lochaline sandstones as a function of the salinity of the pore fluid in the rock at the time of the streaming potential measurement. **a** Berea sandstone (BR1, BR2, and BR3), **b** Boise sandstone (B1II, B2II, and B3I), **c** Fontainebleau sandstone (F1D, F2D, F3Q, and F4Q), and **d** Lochaline sandstone (L1D, L2D, L3Q, and L4Q). For Fontainebleau and Lochaline sandstones solid symbols represent the detrital form of the rock, while open symbols represent the quartz overgrown type

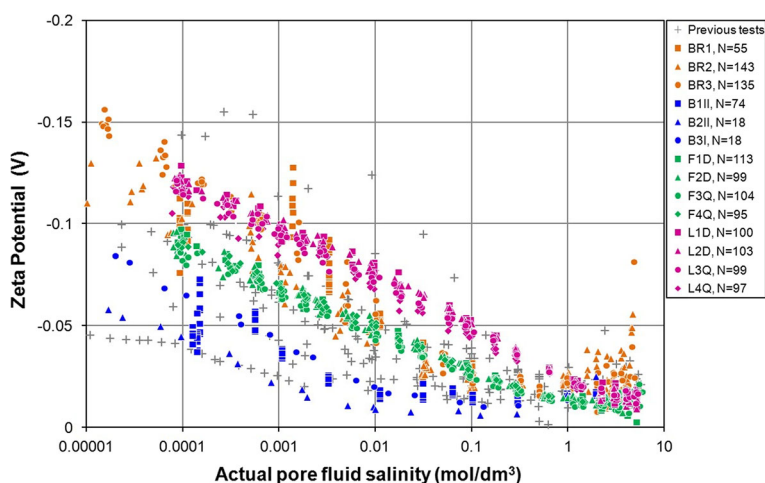


Fig. 8 Aggregation of the new derived zeta potential measurements with those from the existing database. The existing data are shown undifferentiated by grey crosses; the new data are shown as coloured symbols. Berea (red), Boise (blue), Fontainebleau (green), and Lochaline (purple)

The consequence of the lack of a different zeta potential behaviour at low salinities is that the zeta potential can be split into two regimes, one which describes the behaviour in the low to medium salinity range, and the other which is valid for high salinities.

7.1 Low to Medium Salinity Behaviour

For the low to medium salinities the zeta potential decreases linearly with the logarithm of salinity. We have carried out a simple logarithmic fit to all of the new data in the salinity range for which Fig. 7 shows this behaviour to occur. This was $C_f < 0.1 \text{ mol/dm}^3$ for Berea sandstone, $C_f < 0.01 \text{ mol/dm}^3$ for Boise sandstone, and $C_f < 1 \text{ mol/dm}^3$ for the Fontainebleau and Lochaline sandstones. The mean behaviour of these fits can be represented by the relationship

$$\zeta = c \log_{10}(C_f) + d, \quad (3)$$

where the values of the coefficients c and d are given in Table 2 and the mean values for all our measurements are $\langle c \rangle = 11.33 \pm 4.06 \text{ mV}$ and $\langle d \rangle = 3.505 \pm 28.823 \text{ mV}$. Once again the large number of measurements in the data set combined with their quality ensures that the coefficients of determination for all individual power law fits are high, usually exceeding 0.97, but dropping below that value for two samples where the data were not as high quality.

A number of authors have fitted Eq. (3) or its formal equivalent to their data. Vinogradov et al. (2010) obtained $c = 19.02 \text{ mV}$, $d = -9.67 \text{ mV}$, Pride and Morgan (1991) obtained $c = 26 \text{ mV}$ and $d = -8 \text{ mV}$, Bolève et al. (2007) suggested $c = 29.1$ and $d = -14.6 \text{ mV}$, while Revil et al. (1999) have estimated the values at $c \approx 20 \text{ mV}$ and $d \approx -10 \text{ mV}$. Jaafar et al. (2009), which shares some of its data with Vinogradov et al. (2010) suggested $c = 20.85 \text{ mV}$ and $d = -6.43 \text{ mV}$. In all of these works, the uncertainty on the values was not calculated and often the values obtained are for an aggregation of different samples and rock types at uncontrolled pH. Nevertheless, all values of c occur in the 10–30 mV range and all values of d occur in the -5 to -15 mV range and are not obviously consistent with the values we obtain, which we ascribe to a pH dependence.

While it is clear that zeta potential depends strongly on pH from Figs. 7 and 8 and Table 2 alone, we wished to examine how the zeta potential depended upon pH in the pH 6–8 range. Consequently, we have plotted the coefficients in Eq. (3) against the measured pH in the third part of Fig. 6. This figure includes propagated experimental and fitting uncertainties in the same way as the analogous diagram for streaming potential that was described previously. It is clear from Fig. 6c that the coefficient c varies in a well-defined manner that all samples follow and which is extremely sensitive to pH, varying nonlinearly between about 8.05 mV and about 23.66 mV over the pH range from 6 to 8 and covering the range of values obtained by other authors. By contrast the offset coefficient d shows a more scattered behaviour, with different rock types producing different trends, and both positive and negative values. There is some indication of a weak dependence on pH for the Berea and the Boise sandstones, with positive values, while the Fontainebleau and Lochaline samples provide mean values of d that are consistent with the values from other authors.

The consequence of these two dependencies is that increasing the pH tends to increase the slope of the low to medium salinity zeta potential curves in Figs. 7 and 8, while producing both positive and negative values for the intercept as shown in the schematic diagram in Fig. 6e.

7.2 High Salinity Behaviour

Continuation of the trend described in the last section to high salinities would result in the zeta potential changing polarity (from negative to positive). However, this has not been observed in any of the experimental determinations made in this work or in that of Jaafar et al. (2009) and Vinogradov et al. (2010). Instead, at high salinities the zeta potential appears to level off at a small constant negative value or even increase slightly (i.e. become more negative). Previously such behaviour had not been noted by Pride and Morgan (1991), Bolève et al. (2007) or Revil et al. (1999) because they had not considered high salinities and had consequently not observed the levelling-off behaviour.

The cause of the asymptotic behaviour of the zeta potential has been discussed by Jaafar et al. (2009) and Vinogradov et al. (2010). They noted that the start of the constant zeta potential behaviour occurs at about 0.4 mol/dm^3 , at which salinity the Debye length is about 0.47 nm and approximates to the size of a hydrated sodium ion. This observation leads them to suggest that the constant zeta potential at high salinities reflects the maximum charge density in the diffuse layer which is reached when the diffuse layer thickness approaches the diameter of the counterions. In this model at high salinities the double layer resembles the Helmholtz model, where there is a single layer of hydrated, immobile ions in the Stern layer, and a single layer of hydrated, mobile ions composing the diffuse layer (i.e. a parallel plate capacitor model). They noted that this hypothesis is supported by their data, with the limiting value of the zeta potential occurring at about -20 mV (Jaafar et al. 2009) and -17 mV (Vinogradov et al. 2010). Previous measurements made by Dukhin et al. (2005) on α -alumina provided a value of about -10 mV , while those of Johnson et al. (1999a,b), also for α -alumina, gave values between -20 and -30 mV at 1 mol/dm^3 . In this work we have obtained values of the high salinity zeta potential which are $-24.19 \pm 2.73 \text{ mV}$ for Berea sandstone, $-17.07 \pm 7.26 \text{ mV}$ for Boise sandstone, $-13.01 \pm 0.48 \text{ mV}$ for Fontainebleau sandstone, and $-16.81 \pm 0.68 \text{ mV}$ for Lochaline sandstone. The mean value for all of our high salinity measurements taken together is $-17.36 \pm 5.11 \text{ mV}$, all of which agree well with measurements made by other researchers.

Supporting information for the existence of this constant value of zeta potential at high salinities is provided from modelling studies. It has been observed that the electrochemical

model developed by Glover and Déry (2010), Glover et al. (2012b), and Glover (2015b) cannot provide good fits to zeta potential and streaming potential data unless a so-called zeta potential offset is introduced, which is formally the same as the zeta potential at high salinities. The parameter is necessary because the equations for zeta potential (Revil and Glover 1997, 1998; Revil et al. 1999) cannot model the Helmholtz behaviour. When included in modelling Glover et al. 2012b found that zeta potential offsets between -10 and -35 mV allowed model curves to fit the experimental data, with values depending on the rock type, while Glover (2015b) found a value of -15 mV fitted the data of Luong and Sprik (2014) well.

Nevertheless, the data in all of our tests and previous tests were made on rocks that were predominantly silica and with an aqueous NaCl solution. It would be expected that they would provide similar zeta potential offsets. Consequently, we hypothesized that pH could also have a role in determining the zeta potential offset (i.e. the value of the zeta potential at high salinity). We have plotted the zeta potential offset against the pH that was measured in this work on fully equilibrated pore fluid. The mean value of zeta potential offset for each sample is shown in Fig. 6d with uncertainties based on the standard deviation of the measurements made at high salinities. This figure suggests a slight but clear increase in the magnitude of the zeta potential offset with increasing pH, from about -5 mV to about -35 mV. Consequently, it is possible to explain the entire range of high salinity zeta potential offset behaviour observed by all the authors who have observed it by invoking changes in the pore fluid pH, and the pH-dependent behaviour of the zeta potential offset is consistent with the pH dependence found in the medium to low salinity behaviour discussed previously. The zeta potential offset behaviour has been incorporated in the schematic diagram showing the effect of increasing pH on the zeta potential curves (Fig. 6e).

8 Conclusions

High-quality streaming potential measurements have been taken on fourteen samples of four types of sandstone (Berea, Boise, Fontainebleau, and Lochaline) as a function of pore fluid salinity from 10^{-5} to 4.5 mol/dm³, amounting to over 1253 individual measurements in total. This database easily surpasses the total number of previous measurements on silica-based rocks and provides an effective set of data for the testing of theoretical models, which will be the subject of a forthcoming paper. Measurements were taken on sandstones with a number of different compositions and pore structures, represented by detrital sandstones and sandstones with authigenic quartz overgrowths in both Fontainebleau and Lochaline sandstones.

The 313 measurements of streaming potential coefficient that were taken at salinities greater than 1 mol/dm³ (an 8.83-fold increase in available data) confirm the behaviour observed by Jaafar et al. (2009) and Vinogradov et al. (2010), and are consistent with their interpretation of the development of a maximum charge density at high salinities. The low salinity behaviour of streaming potential coefficient shows a sensitivity to the microstructural properties of the rock (porosity, grain size, formation factor), which is particularly clear in the samples with authigenic quartz overgrowths. The streaming potential coefficient is sensitive to the pore fluid pH.

Zeta potential was calculated from each of the streaming potential coefficient measurements and showed a variation that was highly sensitive to both salinity and pH. The database of 269 existing measurements had been widely spread and difficult to interpret, and though it was suspected that pore fluid pH was a factor, the lack of pH control, measurement, and discussion in previous experimental papers meant that it was impossible to quantify the effect

of pH. This data set shows the pH dependence clearly and suggests that the constant high salinity zeta potential (zeta potential offset) also depends on pH.

While the data presented in this work represent, we believe, a significant step forward in the availability of data which can be used to test the developing models, much remains to be done experimentally. There is the need for experiments using divalent brines and at raised temperatures. In addition, clay minerals are also thought to be important in controlling many salinity-dependent processes in sandstones, such as formation damage and controlled salinity water injection. Consequently, more experimental data examining the effect of clays on the streaming potential coefficients and zeta potentials of clastic rocks. Finally, few measurements are available in carbonates, and these experiments are complicated by rock–fluid interaction needs to be carried out. In summary, there is scope for much more experimental work, all of which will have important implications for water, hydrocarbon, and waste disposal industries.

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References

- Alkafef, S.F., Alajmi, A.F.: The electrical conductivity and surface conduction of consolidated rock cores. *J. Colloid Interface Sci.* **309**(2), 253–261 (2007). <https://doi.org/10.1016/j.jcis.2007.02.004>
- Avena, M.J., de Pauli, C.P.: Proton adsorption and electrokinetics of an Argentinean montmorillonite. *J. Colloid Interface Sci.* **202**, 195–204 (1998). <https://doi.org/10.1006/jcis.1998.5402>
- Block, G.I., Harris, J.G.: Conductivity dependence of seismoelectric wave phenomena in fluid-saturated sediments. *J. Geophys. Res.* **111**, B01304 (2006). <https://doi.org/10.1029/2005JB003798>
- Bolève, A., Crespy, A., Revil, A., Janod, F., Mattiuzzo, J.L.: Streaming potentials of granular media: influence of the Dukhin and Reynolds numbers. *J. Geophys. Res.* **112**, B08204 (2007). <https://doi.org/10.1029/2006JB004673>
- Churcher, P.L., French, P.R., Shaw, J.C., Schramm, L.L.: Rock properties of Berea sandstone, Baker dolomite, and Indiana limestone. *Soc. Pet. Eng.* (1991). <https://doi.org/10.2118/21044-MS>
- Dukhin, A., Dukhin, S., Goetz, P.: Electrokinetics at high ionic strength and hypothesis of the double layer with zero surface charge. *Langmuir* **21**, 9990–9997 (2005)
- Gaudin, A.M., Fuerstenau, D.W.: Streaming potential studies: quartz flotation with anionic collectors. *Trans. Am. Inst. Min. Metall. Eng.* **202**, 66–72 (1955)
- Glover, P.W.J.: A generalized Archie's law for n phases. *Geophysics* **75**(6), E247–E265 (2010)
- Glover, P.W.J.: *Electrical Properties. Treatise on Geophysics*, 2nd edn. Elsevier, London (2015a)
- Glover, P.W.J.: Comment on “examination of a theoretical model of streaming potential coupling coefficient”. *Int. J. Geophys.* (2015b). <https://doi.org/10.1155/2015/941246>
- Glover, P.W.J.: A new theoretical interpretation of Archie's saturation exponent. *Solid Earth Discuss.* (2017). <https://doi.org/10.5194/se-2017-5>
- Glover, P.W.J., Déry, N.: Dependence of streaming potential on grain diameter and pore radius for quartz glass beads. *Geophysics* **75**(6), F225–241 (2010). <https://doi.org/10.1190/1.3509465>
- Glover, P.W.J., Walker, E.: A grain size to effective pore size transformation derived from an electro-kinetic theory. *Geophysics* **74**(1), E17–E29 (2009)
- Glover, P.W.J., Zadjali, I.I., Frew, K.A.: Permeability prediction from MICP and NMR data using an electrokinetic approach. *Geophysics* **71**(4), F49–F60 (2006)
- Glover, P.W.J., Walker, E., Ruel, J., Tardif, E.: Frequency-dependent streaming potential of porous media—part 2: experimental measurement of unconsolidated materials. *Int. J. Geophys.* (2012a). <https://doi.org/10.1155/2012/728495>

- Glover, P.W.J., Walker, E., Jackson, M.D.: Streaming-potential coefficient of reservoir rock: a theoretical model. *Geophysics* **77**(2), D17–D43 (2012b). <https://doi.org/10.1190/GEO2011-0364.1>
- Gomez, C., Dvorkin, J., Vanorio, T.: Laboratory measurements of porosity, permeability, resistivity and velocity on Fontainebleau sandstones. *Geophysics* **75**(6), E191–E204 (2010). <https://doi.org/10.1190/1.3493633>
- Guichet, X., Jouniaux, L., Pozzi, J.P.: Streaming potential of a sand column in partial saturation conditions. *J. Geophys. Res.* **108**(B3), 2141 (2003). <https://doi.org/10.1029/2001JB001517>
- Hunter, R.J.: *Zeta Potential in Colloid Science*. Academic Press, New York (1981)
- Jaafar, M.Z., Vinogradov, J., Jackson, M.D.: Measurement of streaming potential coupling coefficient in sandstones saturated with high salinity NaCl brine. *Geophys. Res. Lett.* **36**, L21306 (2009). <https://doi.org/10.1029/2009GL040549>
- Jiang, Y.G., Shan, F.K., Jin, H.M., Zhou, L.W.: A method for measuring electro-kinetic coefficients of porous media and its potential application in hydrocarbon exploration. *Geophys. Res. Lett.* **25**(10), 1581–1584 (1998). <https://doi.org/10.1029/98GL01102>
- Johnson, S.B., Scales, P.J., Healy, T.W.: The binding of monovalent electrolyte ions on α -alumina: I. Electroacoustic studies at high electrolyte concentrations. *Langmuir* **15**, 2836–2843 (1999a)
- Johnson, S.B., Franks, G.V., Scales, P.J., Healy, T.W.: The binding of monovalent electrolyte ions on α -alumina: II. The shear yield stress of concentrated suspensions. *Langmuir* **15**, 2844–2853 (1999b)
- Jouniaux, L., Pozzi, J.P.: Streaming potential and permeability of saturated sandstones under triaxial stress: consequences for electrotelluric anomalies prior to earthquakes. *J. Geophys. Res.* **100**, 10197–10209 (1995a). <https://doi.org/10.1029/95JB00069>
- Jouniaux, L., Pozzi, J.P.: Permeability dependence of streaming potential in rocks for various fluid conductivities. *Geophys. Res. Lett.* **22**, 485–488 (1995b)
- Jouniaux, L., Pozzi, J.P.: Laboratory measurements anomalous 0.1–0.5 Hz streaming potential under geochemical changes: implications for electrotelluric precursors to earthquakes. *J. Geophys. Res.* **102**, 15,335–15,343 (1997). <https://doi.org/10.1029/97JB00955>
- Kirby, B.J., Hasselbrink, E.F.: Zeta potential of microfluidic substrates. 1. Theory, experimental techniques, and effects on separations. *Electrophoresis* **25**, 187–202 (2004). [https://doi.org/10.1002/\(ISSN\)1522-2683](https://doi.org/10.1002/(ISSN)1522-2683)
- Kosmulski, M., Dahlsten, P.: High ionic strength electrokinetics of clay minerals. *Colloids Surf. A Physicochem. Eng. Asp.* **291**, 212–218 (2006). <https://doi.org/10.1016/j.colsurfa.2006.06.037>
- Kosmulski, M., Mączka, E., Janusz, W., Rosenholm, J.B.: Multi-instrument study of the electrophoretic mobility of quartz. *J. Colloid Interface Sci.* **250**, 99–103 (2002). <https://doi.org/10.1006/jcis.2002.8330>
- Li, H.C., de Bruyn, P.L.: Electrokinetic and adsorption studies on quartz. *Surf. Sci.* **5**, 203–220 (1966). [https://doi.org/10.1016/0039-6028\(66\)90082-3](https://doi.org/10.1016/0039-6028(66)90082-3)
- Li, S.X., Pengra, D.B., Wong, P.Z.: Onsager's reciprocal relation and the hydraulic permeability of porous media. *Phys. Rev. E* **51**, 5748–5751 (1995). <https://doi.org/10.1103/PhysRevE.51.5748>
- Lorne, B., Perrier, F., Avouac, J.-P.: Streaming potential measurements: 1. Properties of the electrical double layer from crushed rock samples. *J. Geophys. Res.* **104**, 17857–17877 (1999). <https://doi.org/10.1029/1999JB900156>
- Luong, D.T., Sprik, R.: Examination of a theoretical model of streaming potential coupling coefficient. *Int. J. Geophys.* (2014). doi:<https://doi.org/10.1155/2014/471819>
- Morgan, F.D., Williams, E.R., Madden, T.R.: Streaming potential properties of Westerly Granite with applications. *J. Geophys. Res.* **94**, 12449–12461 (1989). <https://doi.org/10.1029/JB094iB09p12449>
- Pasqualini, D., Heitmann, K., TenCate, J.A., Habib, S., Higdon, D., Johnson, P.A.: Nonequilibrium and nonlinear dynamics in Berea and Fontainebleau sandstones: low-strain regime. *J. Geophys. Res.* **112**, B01204 (2007). <https://doi.org/10.1029/2006JB004264>
- Pengra, D.B., Li, S.X., Wong, P.Z.: Determination of rock properties by low-frequency AC electrokinetics. *J. Geophys. Res.* **104**, 29485–29508 (1999). <https://doi.org/10.1029/1999JB900277>
- Phillips, S.L., Ozbek, H., Otto, R.J.: Basic energy properties of electrolytic solutions database. In: Sixth International CODATA Conference Santa Flavia (Palermo), Sicily, Italy, May 22–25. <http://www.osti.gov/bridge/purl.cover.jsp?sessionid=3954E775156A8BC0FA35DB5CE5B402D4?&purl=6269880-iPJPhB/> (1978). Accessed 10 June 2010
- Poirier, J.E., Cases, J.M.: Sur l'origine et la nature de l'interaction adsorbat-adsorbant dans les systèmes à interactions faibles. *Solid-liquid interactions in porous media*; Technip Editions 447–462 (1985)
- Pride, S.R., Morgan, F.D.: Electrokinetic dissipation induced by seismic waves. *Geophysics* **56**, 914–925 (1991). <https://doi.org/10.1190/1.1443125>
- Revil, A., Glover, P.W.J.: Theory of ionic surface electrical conduction in porous media. *Phys. Rev. B* **55**, 1757–1773 (1997). <https://doi.org/10.1103/PhysRevB.55.1757>
- Revil, A., Glover, P.W.J.: Nature of surface electrical conductivity in natural sands, sandstones, and clays. *Geophys. Res. Lett.* **25**, 691–694 (1998). <https://doi.org/10.1029/98GL00296>

- Revil, A., Pezard, P.A., Glover, P.W.J.: Streaming potential in porous media. I. Theory of the zeta-potential. *J. Geophys. Res.* **104**(B9), 20021–20031 (1999). <https://doi.org/10.1029/1999JB900089>
- Revil, A., Hermitte, D., Spangenberg, E., Cochemé, J.J.: Electrical properties of zeolitized volcanoclastic materials. *J. Geophys. Res.* **107**(B8), 2168 (2002). <https://doi.org/10.1029/2001JB000599>
- Revil, A., Saracco, G., Labazuy, P.: The volcano-electric effect. *J. Geophys. Res.* **108**(B5), 2251 (2003). <https://doi.org/10.1029/2002JB001835>
- Sen, P., Goode, P.: Influence of temperature on electrical conductivity on shaly sands. *Geophysics* **57**, 89–96 (1992a)
- Sen, P., Goode, P.: Erratum to: Influence of temperature on electrical conductivity of shaly sands. *Geophysics* **57**, 1658 (1992b)
- Sprunt, E.S., Mercer, T.B., Djabbarah, N.F.: Streaming potential from multiphase flow. *Geophysics* **59**, 707–711 (1994)
- Tardif, E., Glover, P.W.J., Ruel, J.: Frequency-dependent streaming potential of Ottawa sand. *J. Geophys. Res.* **B 116**, B04206 (2011)
- Tosha, T., Matsushima, N., Ishido, T.: Zeta potential measured for an intact granite sample at temperatures to 200°C. *Geophys. Res. Lett.* **30**(6), 1295 (2003). <https://doi.org/10.1029/2002GL016608>
- van der Heyden, F.H.J., Stein, D., Besteman, K., Lemay, S.G., Dekker, C.: Charge inversion at high ionic strength studied by streaming currents. *Phys. Rev. Lett.* **96**, 224502 (2006)
- Vinogradov, J., Jaafar, M.Z., Jackson, M.D.: Measurement of streaming potential coupling coefficient in sandstones saturated with natural and artificial brines at high salinity. *J. Geophys. Res.* **115**, B12204 (2010). <https://doi.org/10.1029/2010JB007593>
- Walker, E., Glover, P.W.J., Ruel, J.: A transient method for measuring the DC streaming potential coefficient of porous and fractured rocks. *J. Geophys. Res.* (2014). <https://doi.org/10.1002/2013JB010579>
- Will, G., Nover, G.: Measurement of the frequency dependence of the electrical conductivity and some other petro-physical parameters of core samples from the Konzen (West Germany) drill hole: *Ann. Geophys.* **4**(B2), 173–182 (1986)
- Worden, R., Morad, S.: *Quartz Cementation in Sandstones*, Special Publication 29 of the IAS, p. 352. Wiley, New York (2009)



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Seismo-electric conversion in shale: Experiment and analytical modelling

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Seismo-electric conversion in shale: Experiment and analytical modelling

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Abstract

The development of seismo-electric exploration techniques relies critically upon the strength of the seismo-electric conversion. However, there have been very few seismo-electric measurements or modelling on shales, despite shales accounting for the majority of unconventional reservoirs. We have carried out seismo-electric measurements on Sichuan Basin shales (permeability 0.00147 – 0.107 mD), together with some comparative measurements on sandstones (permeability 0.2–60 mD). Experimental results show that the amplitudes of the seismoelectric coupling coefficient in shales are comparable to that exhibited by sandstones, and are approximately independent of frequency in the seismic frequency range (<1 kHz). Numerical modelling has also been used to examine the effects of varying (i) dimensionless number, (ii) porosity, (iii) permeability, (iv) tortuosity, and (v) zeta potential on seismo-electric conversion in porous media. It was found that while changes in dimensionless number and permeability seem to have little effect, seismoelectric coupling coefficient is highly sensitive to changes in porosity, tortuosity and zeta potential. Numerical modelling suggests that the cause of the seismo-electric conversion in shales is enhanced zeta potentials caused by clay minerals, which are highly frequency dependent. This is supported by a comparison of our numerical modelling with our experimental data, together with an analysis of seismo-electric conversion as a function of clay mineral composition from XRD measurements. The sensitivity of seismo-electric coupling to the clay minerals suggests that seismo-electric exploration may have potential for the characterization of clay minerals in shale gas and shale oil reservoirs.

Keywords: Seismo-electric conversion, shales, frequency-dependence, porosity, permeability, tortuosity

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55 **1 Introduction**

56 Seismo-electric conversion occurs when the passage of an elastic wave through an aqueous fluid
57 saturated porous medium causes that fluid to move, and where the resulting fluid movement
58 separates charge such that an electrical potential is developed across the medium (e.g., [Haines et al.](#)
59 [2007](#); [Glover, 2012a](#); [2012b](#)). Seismo-electric effects occur in all aqueous fluid-filled porous media,
60 including sandstones and shales. Laboratory experiments (e.g., [Glover & Déry 2010](#); [Peng et al.](#)
61 [2016](#); [Schakel et al. 2011a](#); [2011b](#); [Zhu et al. 2008](#); [2013](#); [2015](#); [Wang et al. 2015a, b](#); [2016](#)) of
62 electro-kinetic and seismo-electric coupling conducted in sandstones indicate that a relative strong
63 seismo-electric conversion can be generated in sandstones, especially when the pore fluid salinity is
64 low ([Walker & Glover 2018](#)). Currently, there is a good and growing body of experimental
65 measurements on sandstones, limestones, sands and glass beads. By contrast, there has been less
66 seismo-electric research on shales. The strength of seismo-electric conversion in shales, and the
67 parameters which control it, remain uncertain.

68 [Thompson & Gist \(1993\)](#) first presented the results of seismo-electric profiling to detect the signals
69 generated at interfaces between high permeability water sands and low permeability shales at depths
70 of up to 300 m. In their study, the media on each side of the interface produced a strong electric
71 contrast as well as different rock properties, which together enhanced the seismo-electric conversion.
72 However, the specific influences of shale on seismo-electric conversion were either not studied or
73 were not reported. Numerical modelling of seismo-electric processes for a similar interface was
74 carried out by [Revil et al. \(2015\)](#). In their study, the seismo-electric wavefield induced at the
75 shale-sand interface during the passage of a seismic P-wave was also characterized by the difference
76 in electrical conductivity between the shale and the sandstone layers. Meanwhile, modelling of the
77 seismo-electric properties of a sand channel surrounded by clay ([Haines & Pride 2006](#)) and clay
78 lenses embedded in sandstone ([Kröger et al. 2009](#)) have also indicated that the seismo-electric
79 method can detect clay or shale interfaces as well as interbedded layers and lenses of clays and
80 sandstones. In addition, [Jougnot et al. \(2010\)](#) studied the streaming potential coupling coefficient of
81 clay rocks as a function of saturation experimentally and showed that the coupling coefficient was a
82 power law function of the relative water saturation.

83 Field measurements have also suggested that electric signals induced by seismic waves through
84 seismo-electric coupling may be used to image the subsurface boundaries of sand and clay layers
85 ([Butler et al. 1996](#); [2002](#); [Thompson et al. 2007](#); [Dupuis et al. 2009](#); [Glover & Jackson 2010](#)). The
86 sparse experimental and field data suggest that shale exhibits a sizeable electric coupling coefficient.
87 However, since no authoritative value is currently available, numerical modelling studies assume a
88 significant electric coupling coefficient for shales.

Clay minerals are a major component of shales. They are known to have a strong impact on the hydraulic and electrical properties of rocks (e.g., [Arulanandan 1969](#); [Alali 2007](#); [Santiwong et al. 2008](#)). Consequently, it is expected that the fraction of clay minerals will also exert a partial control on the seismo-electric properties of shales. The processes leading to seismo-electric conversion occur in the electric double layer (EDL) ([Shaw 1992](#); [Pride 1994](#); [Glover 2015](#)) which forms at the solid-fluid interface in porous media. Theoretical developments ([Revil & Glover 1997](#); [1998](#); [Revil et al. 1999](#)) have shown that the size of the Stern potential developed at the mineral-fluid interface depends upon the composition of the mineral grains and fluid ([Leroy & Revil 2004](#); [Tournassat et al. 2009](#); [2013](#); [Leroy et al. 2015](#)). Since the surface charge controls the Stern zeta potential, which, together with the rock microstructure, controls the streaming potential coefficient ([Glover & Déry 2010](#); [Glover 2012](#); [Li et al. 2016](#)), there is a functional link between clay minerals and seismoelectric coupling coefficient ([Revil et al. 2005](#); [Peng et al. 2018b](#)). It is known that the phase due to induced polarization and Maxwell Wagner polarization (MWP) is very strong in shales (several hundreds of milliradians), which plays an important role in the seismoelectric coupling coefficient ([Leroy and Revil, 2009](#); [2015](#)). This effect is not accounted for in the [Pride \(1994\)](#) model, which was formulated for silica-based rocks. Furthermore the Pride model does not take account of the Stern Layer which can contain the majority of counterions. This creates a strong frequency dependent complex conductivity because of induced polarization. Consequently, the Pride model should be considered to represent the bottom limit of possible frequency-dependence in shales.

Another difficulty is that many of the models developed for describing both steady-state ([Glover et al. 2012b](#); [Walker & Glover 2014](#); [2018](#)) or frequency-dependent ([Peng et al. 2019](#); [Glover 2018](#)) electro-kinetic coupling measurements make the assumption that the EDL is thin compared to the pore size. This has been a restriction since the first models ([Glover et al. 1994](#); [Revil & Glover 1995](#)) and provides difficulties for porous media with either small pores (such as shales), or medium to large pores when the pore fluid has very low salinities. In practice, the latter effect is not observed because mineral-fluid equilibrium ensures that sufficiently low salinities cannot be attained in rocks ([Walker & Glover 2014](#)). Instead, it is the pore microstructure that dominates the low salinity behaviour of streaming potential coupling ([Glover et al. 2012b](#); [2018](#)). The effect of the assumption is that rocks with very small pores will have their pore space dominated by the EDL, and any coupling will subsequently be limited, leading to measured coupling coefficients that are less than would have occurred if the rock had been composed of larger pores. In this work we implement a modification which allows thick double layers to be modelled ([Kozak & Davis 1987](#)).

In this paper, we present experimental measurements of seismo-electric conversion for 14 shale samples and 5 sandstone samples, the latter of which are primarily included for comparison. In addition, we have compared the differences in seismo-electric coupling between sandstones and shales by carrying out limited numerical modelling of the seismo-electric coupling as a function of

frequency using [Pride's \(1994\)](#) equation with no MWP and Stern Layer component. Importantly, both experimental results and modelling show that the intensity of the seismo-electric conversion in the low-permeability shale is comparable to that produced in high-permeability sandstones. However, we acknowledge that new modelling will be necessary to extend the [Pride \(1994\)](#) model to include Stern Layer/Maxwell-Wagner effects before the experimental measurements can be fully accounted for.

2 Theory

Coupling between the acoustic field and electromagnetic fields of porous media depends on the existence of the electric-double layer (EDL), which is formed at the solid-fluid boundary in the pore space of porous media. The EDL includes a diffuse layer, which is electrically polarised, having unequal concentrations of positive and negative ions. Only the part of the diffuse layer closest to the mineral surface will be immobile, while the remainder will remain capable of flow. The propagation of seismic waves in the fluid-filled porous media causes movement of mobile fluids relative to the solid matrix ([Glover et al. 2012b](#)). Consequently, ions in the electrically neutral bulk fluid are moved together with electrically unbalanced ions in the mobile part of the diffuse layer. The resulting separation of charge leads to the development of a streaming potential, and a streaming counter-current which flows to balance the induced charge separation. The transient current flow is capable of generating electro-magnetic waves at discontinuous boundaries. Also the seismoelectric coupling coefficient $L(\omega)$ is developed to describe the coupling between the seismic and electromagnetic fields ([Pride 1994](#); [Haartsen & Pride 1997](#); [Haines et al. 2007](#); [Jouniaux & Zyserman 2016](#)).

[Pride \(1994\)](#) derived the governing equations of the seismo-electric conversion for saturated porous media expressing the frequency-dependent seismo-electric coupling coefficient as

$$L(\omega) = L_0 \left[1 - i \frac{\omega m^*}{\omega_c^4} \left(1 - 2 \frac{\tilde{d}}{\Lambda} \right)^2 \left(1 - i^{3/2} \frac{\tilde{d}}{\delta} \right)^2 \right]^{-1/2}, \quad (1)$$

where, L_0 is the low frequency seismo-electric coupling coefficient, δ is the skin depth, $\delta = \sqrt{\frac{\eta}{\omega \rho_f}}$, η is the pore fluid viscosity, ρ_f is the density of the pore fluid, $\tilde{d}$ is the Debye length, Λ is the characteristic length scale of the pores ([Johnston et al. 1987](#)). It should be noted that Pride model does not account for surface conductivity in the Stern layer band hence does not include induced polarization (IP) effects.

In Equation (1) the transition frequency, ω_c , separating low-frequency viscous flow and high-frequency inertial flow, is given by

$$\omega_c = \frac{\phi\eta}{\alpha_{\infty}\rho_f k_0}, \quad (2)$$

where ϕ is the porosity, k_0 is the steady-state permeability of the porous medium, and α_{∞} is the tortuosity of the pore space.

The tortuosity is a term used to describe the sinuosity and interconnectedness of the pore space as it affects transport processes through porous media. Tortuosity has no simple or universal definition (Clennell 1997). We may understand the tortuosity in a simplified but not comprehensive way that the tortuosity of a pathway is defined as the ratio of the length of the pathway to the distance between the ends of it (Epstein 1989). There are many models for tortuosity estimation including hydraulic models ($\alpha_{\infty h}$), electric models ($\alpha_{\infty e}$) and diffusive models ($\alpha_{\infty d}$). Values for hydraulic electrical and diffusional tortuosity are in general different from one another. Electrical tortuosity is defined in terms of conductivity whereas hydraulic tortuosity is usually defined geometrically, and diffusional tortuosity is typically computed from temporal changes in concentration (Clennell 1997). Usually, it is considered that $\alpha_{\infty d} \approx \alpha_{\infty e} < \alpha_{\infty h}$ holds (Saomoto & Katagiri 2015; Thanh et al. 2019). In this paper, we use the α_{∞} to represent tortuosity and consider $\alpha_{\infty} = \alpha_{\infty d} = \alpha_{\infty e}$. Another definition was made by Glover (2009) in his physical interpretation of the cementation exponent, m as the rate of change of connectedness (G) of a porous medium with porosity (ϕ) and connectivity (χ), as $m = \frac{d^2 G}{d\chi d\phi}$, where the tortuosity is the inverse of the connectivity, $\alpha_{\infty} = 1/\chi$.

Each of the models for tortuosity are a function of porosity despite of the difference between their background concepts (Saomoto & Katagiri 2015). The electrical tortuosity can be expressed as $\alpha_{\infty e} = \phi^{1-m}$ (David 1993; Glover 2012), where m is the cementation exponent. We do not use this equation to obtain the tortuosity in this work because it introduces a new unknown parameter m into the calculations. Similar expressions of tortuosity exist which are based on the experimental data. Bruggeman (1935) defined $\alpha_{\infty e} = \phi^{-\alpha}$, with α usually chosen as 0.5, valid for packs of grains of porosity $\phi > 0.2$. this particular definition infers that $m=1.5$ for packs of glass beads, a result which has been validated. In this work we have used the modified Weissberg model (Boudreau 1996) to calculate the tortuosity

$$\alpha_{\infty} = \sqrt{1 - 2.02 \ln(\phi)}. \quad (3)$$

This relationship is an empirical equation which shows the best least-squares fits to experimental data.

The [Pride \(1994\)](#) model contains a dimensionless number (m), which we relabel as m^* in order not to confuse it with the cementation exponent. The dimensionless number m^* is given by ([Pride 1994](#); [Walker & Glover 2010](#))

$$m^* = \frac{\phi}{\alpha_\infty k_0} \Lambda^2, \quad (4)$$

The value of m^* is assumed to be 8 for sandstones ([Pride 1994](#)), and 12 if shale is modelled as a layered porous medium ([Johnson et al. 1987](#)). It should be noted that for most geological media the pore fluid is sufficiently saline for the Debye length to be much smaller than the characteristic length scale (i.e., $\tilde{d} \ll \Lambda$). However, the pore size of shales may be comparable to the thickness of the double layer, hence, the thick double layer approximation have been developed. The assumption of thin double layer has been extended to account for thicker double layers, which still allows significant simplifications to be made ([Kozak & Davis 1987](#)).

The low-frequency seismoelectric coupling coefficient L_0 can be written as

$$L_0 = -\frac{\phi \varepsilon_f \zeta}{\alpha_\infty \eta} \left[1 - 2 \frac{\tilde{d}}{\Lambda} \right], \quad (5)$$

where ζ is the zeta potential, ε_f represents the fluid permittivity. If $\tilde{d} \ll \Lambda$, L_0 can be simplified to

$$L_0 = -\frac{\phi \varepsilon_f \zeta}{\alpha_\infty \eta}, \quad (6)$$

which is a statement of the classical Helmholtz-Smoluchowski equation (e.g., [Glover 2012](#)).

3 Experimental Measurement

3.1 Experimental Methodology

The apparatus we have used to measure seismo-electric conversion in shale is shown in [Figure 1](#). Both the seismo-electric and acoustic signals are collected by this apparatus, which operates in a water tank ([Peng et al. 2017; 2018a](#)). A compressional wave transducer (Panametrics-NDT V101, central frequency 0.5 MHz) produces acoustic waves, which propagate across a sample (4 cm×4 cm×2 cm). These waves induce transient pressure differences across the sample, which give rise to the seismoelectric coupling in the sample.

The transient pressures at each side of the sample (P1 and P2) are measured by transient pressure transducers (also Panametrics-NDT V101), which are placed on each side of the rock sample. The

transient pressure difference ΔP ($\Delta P = P_1 - P_2$) is amplified by a pre-amplifier (Olympus Ultrasonic Preamplifier 5660c) and monitored using an oscilloscope (Agilent Technology DSO6012A).

The transient seismoelectric voltage signals at each side of the sample are measured by two non-polarising Ag/AgCl electrodes (V1 and V2). These electrodes cover the entirety of two opposite faces of the sample (i.e., each are 4 cm×4 cm in area) and are 0.1 cm thick. The electrodes are placed between the pressure transducers and the sample, and are positioned 0.1 cm away from the sample surface to avoid the surface conductivity in our experiments. The instantaneous seismo-electric potential ΔV is the instantaneous difference between these two potential measurements ($\Delta V = V_1 - V_2$).

The seismoelectric coupling coefficient $L(\omega)$ calculated by the amplitude of the seismo-electric signals (V1) divided by the acoustic pressure difference ($V_1/\Delta P$) (Zhu et al. 2013, 2015). The instantaneous ratio of $\Delta V/\Delta P$ is the experimental value of the streaming potential coefficient of the rock sample (Zhu et al. 2008). In the experimental procedure we improve the signal to noise ratio of the measured seismoelectric signals (V1 and V2) by stacking 1024 measured values .

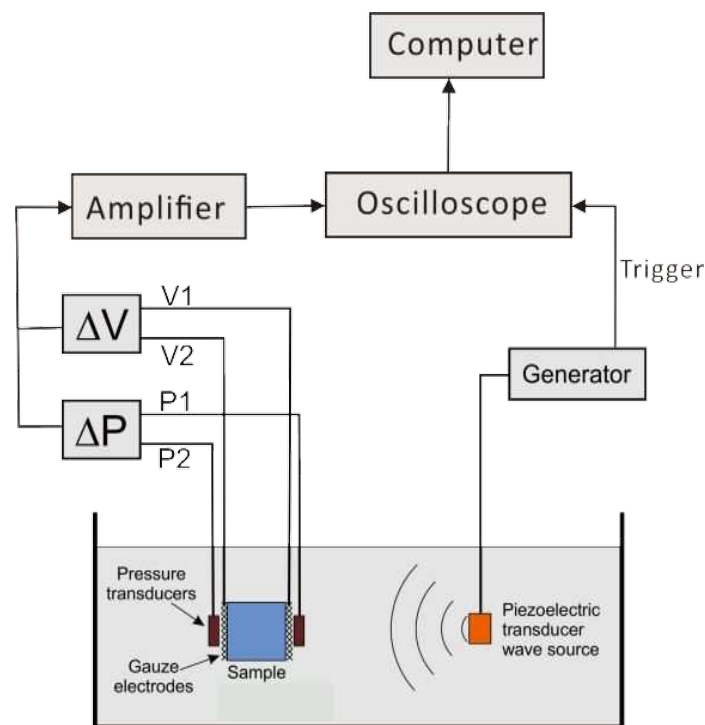


Figure 1. Schematic diagram of the seismo-electric apparatus. The wave source, sample and all transducers and electrodes are placed in a water tank.

In this work we have measured 14 samples of shale from the Sichuan Basin in China. Measurements were made both parallel and perpendicular to the observed bedding plane to provide $L(\omega)$ in the two

directions, as shown diagrammatically in [Figure 2](#). We used X-ray diffraction techniques to obtain the composition of the shale samples. The samples have a characteristic porosity less than 5% ([Deng et al. 2015](#)) and permeability of approximately less than 1 mD ([Fu 2017](#)). The shales have variable composition with 17%wt. to 29%wt. of various clay minerals together with variable amounts of quartz, K-feldspar and plagioclase feldspar. Full data on this shale samples are available in [Table 1](#).

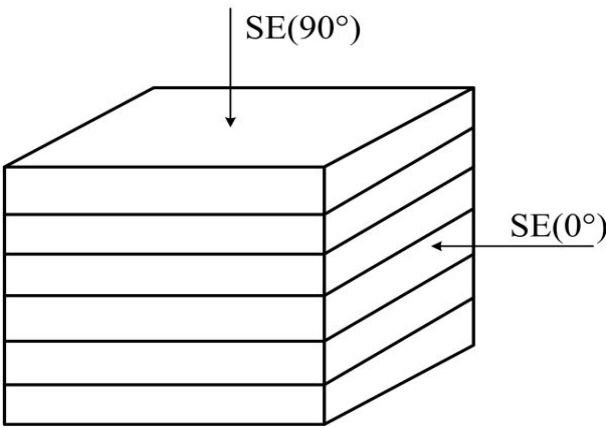


Figure 2. Schematic diagram of shale rock samples. The symbols SE(0°) and SE(90°) represent the seismo-electric measurements parallel and perpendicular to shale bedding, respectively.

We have also measured 5 samples of sandstone from a tight gas reservoir in northwest China. These sandstones have a characteristic porosities and permeabilities in the ranges 1% to 13.7% and 0.235 mD to 58.7 mD, respectively. The sandstones have variable composition with 59.3%wt. to 71.4%wt. of quartz together with variable amounts of feldspar, calcite and muscovite, and less than 5%wt. of clay minerals (wt represents the mass percentage). Full data on the sandstone samples are provided in [Table 2](#).

Table 1. Parameters of the shale samples

Sample	Porosity	0°	90°	Mean	Quartz (%)	K-Feldspar (%)	Plagioclase (%)	Dolomite (%)	Clay (%)	Calcite (%)	TOC (%)
		Permeability (mD)	Permeability (mD)	Permeability (mD)							
Shale 1	0.0442	0.00074	0.00686	0.00380	61.6	12.4	3.5	-	22.5	-	3.61
Shale 2	0.0247	0.000442	0.00946	0.00495	72.1	7.4	2.6	-	17.9	-	4.01
Shale 3	0.0429	0.000894	0.00205	0.00147	58.9	11.7	4.6	-	24.8	-	3.60
Shale 4	0.0365	0.00300	0.211	0.107	73.7	6.0	1.9	-	18.4	-	4.63
Shale 5	0.0494	0.00120	0.0148	0.00800	57.3	12.3	1.6	-	28.8	-	3.96
Shale 6	0.0145	0.000257	0.00195	0.00110	41.4	7.9	1.9	30.0	18.8	-	3.81
Shale 7	0.0124	0.00130	0.00170	0.00150	36.2	6.4	2.0	36.3	19.1	-	3.04

Shale 8	0.0360	0.000760	0.00264	0.00170	73.3	3.5	3.0	-	20.2	-	3.40
Shale 9	0.0400	0.000491	0.00571	0.00310	67.7	7.4	2.6	-	24.3	-	4.05
Shale 10	0.0410	0.00091	0.00449	0.00270	68.6	6.2	2.4	-	22.8	-	4.00
Shale 11	0.0130	0.00172	0.00256	0.00214	67.7	6.9	2.5	-	22.9	-	3.30
Shale 12	0.0320	0.00151	0.00233	0.00192	48.0	14.8	5.1	4.9	23.2	4	0.88
Shale 13	0.0170	0.00278	0.00506	0.00392	76.1	3.9	2	-	18.0	-	5.02
Shale 14	0.01800	0.00067	0.00253	0.00160	30.1	6.8	1.3	45.0	16.8	-	0.81

Note: Permeability is Klinkenberg-corrected permeability, average permeability of shale means the average permeability tested parallel (0°) and perpendicular (90°) to shale bedding. The porosity is effective porosity acquired by helium porosimetry measurements with the gas volume method. TOC = Total Organic Carbon.

Table 2. Parameters of the sandstone samples

Sandstone samples	Porosity	Permeability (mD)	Quartz (%)	Feldspar (%)	Calcite (%)	Muscovite (%)	Clay (%)
Sandstone 1	0.0621	0.914	64.9	27.4	2.8	2.5	2.4
Sandstone 2	0.124	11.4	71.4	23.6	1.1	1.8	2.1
Sandstone 3	0.137	58.7	64.2	30.2	1.3	1.7	2.6
Sandstone 4	0.0390	10.1	59.3	28.5	4.5	3.5	4.2
Sandstone 5	0.0100	0.235	68.8	26.1	1.8	1.4	1.9

Note: Permeability is Klinkenberg-corrected permeability. The porosity is effective porosity acquired by helium porosimetry measurements with the gas volume method.

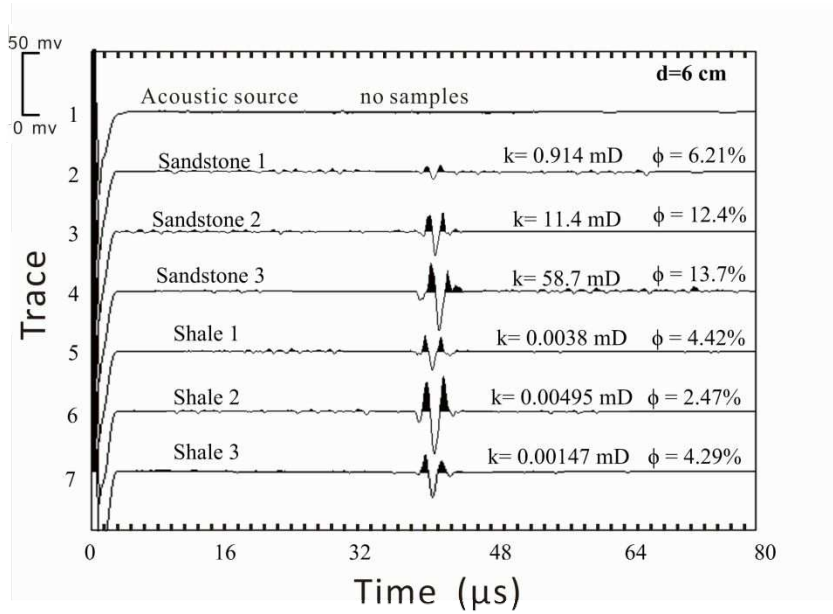
All the samples are saturated with NaCl concentration by vacuuming for 8 hours, so that the solution and the fluid in pore space of samples are fully mixed, and then the samples stand for one day to keep the stability of the whole sample and fluid system before measurements.

3.2 Experimental Results

[Figure 3](#) shows a direct comparison of the measured seismoelectric signals for three shale samples and three sandstone samples of varying porosity and permeability together with a reference trace with no seismic source present.

In the absence of a sample (Trace 1) no seismo-electric signal is present but with an interference signal in Trace 1 at the starting moment, this is in accordance with Wang et al. (2020) observed in their experiments. The seismo-electric signals appearing in other 6 traces are those where a sample has been applied, demonstrating that the observed transient electric signals are due to seismo-electric conversion in the rock samples under the excitation of acoustic pressure. This observation is confirmed by calculating the expected arrival time of the seismo-electric signal. In these experiments the source-sample distance was 6 ± 0.1 cm and the pressure wave velocity in the water is about 1500

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4 272 m/s. Consequently, a seismoelectric signal would be expected to occur $40 \pm 0.67 \mu\text{s}$ after the source
5 273 pulse. In these experiments the measured electric potential arrived at $39.04 \pm 0.5 \mu\text{s}$, where the
6 274 uncertainty reflects the spread of measurements and the experimental uncertainty, and this is the one
8 275 way travel-time of the pressure wave from the source to the sample.
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11 276 In Figure 3 the size of the seismo-electric signal increases with the increasing porosity and
12 277 permeability for sandstones. For example, the seismo-electric coupling is relatively weak for
13 278 Sandstone 1 ($k=0.914 \text{ mD}$), but larger for the sandstone samples which have higher permeabilities.
15 279 On this basis, one might expect the amplitude of the seismo-electric signals of the much lower
16 280 permeability shales (0.00147 to 0.00495 mD) to be extremely small and perhaps unmeasurable.
18 281 However, that is not the case. Figure 3 shows seismo-electric signals for shales of 0.00495 mD
19 282 permeability (Shale 2; Trace 6) with amplitudes as large as those for samples with permeabilities 4
21 283 orders of magnitude larger (58.7 mD ; Sandstone 3; Trace 4). In summary, although the magnitude of
22 284 the seismoelectric signals is related to porosity and permeability for both sandstones and shales, the
24 285 seismoelectric signals of shales are comparable to those of the sandstones even though the shales
25 286 have much lower permeabilities.



288 **Figure 3.** The measured seismoelectric signals (V1) of sandstones and shales at $T=25^\circ\text{C}$, in water and with a
289 source-sample distance of 6 cm. The floating y-axis scale refers to all traces and is relative to the baseline of
290 each trace. The permeability, k , and porosity, ϕ , of each sample is also given.

293 We hypothesize the existence of some other characteristic that enhances seismo-electric conversion
294 in shales so they can provide comparable seismo-electric signals' amplitudes despite the shales
295 having permeabilities approximately 5 orders of magnitude smaller than the sandstones.

The experimental measurements show that the amplitude of the induced seismo-electric signal is also dependent on anisotropy. **Figure 4** shows the seismo-electric signals measured in each of two orthogonal directions on 3 shale samples (**Figure 4a**) and on 3 sandstones (**Figure 4b**). In this experiment the distance between the acoustic source and the rock sample was 8 ± 0.1 cm. Taking the acoustic velocity in water to be about 1500 m/s, as before, predicts the seismo-electric conversion signal to arrive at about 53.33 ± 0.67 μ s. Experimentally measured value, which can be read from **Figure 4** was 52.5 ± 0.5 μ s.

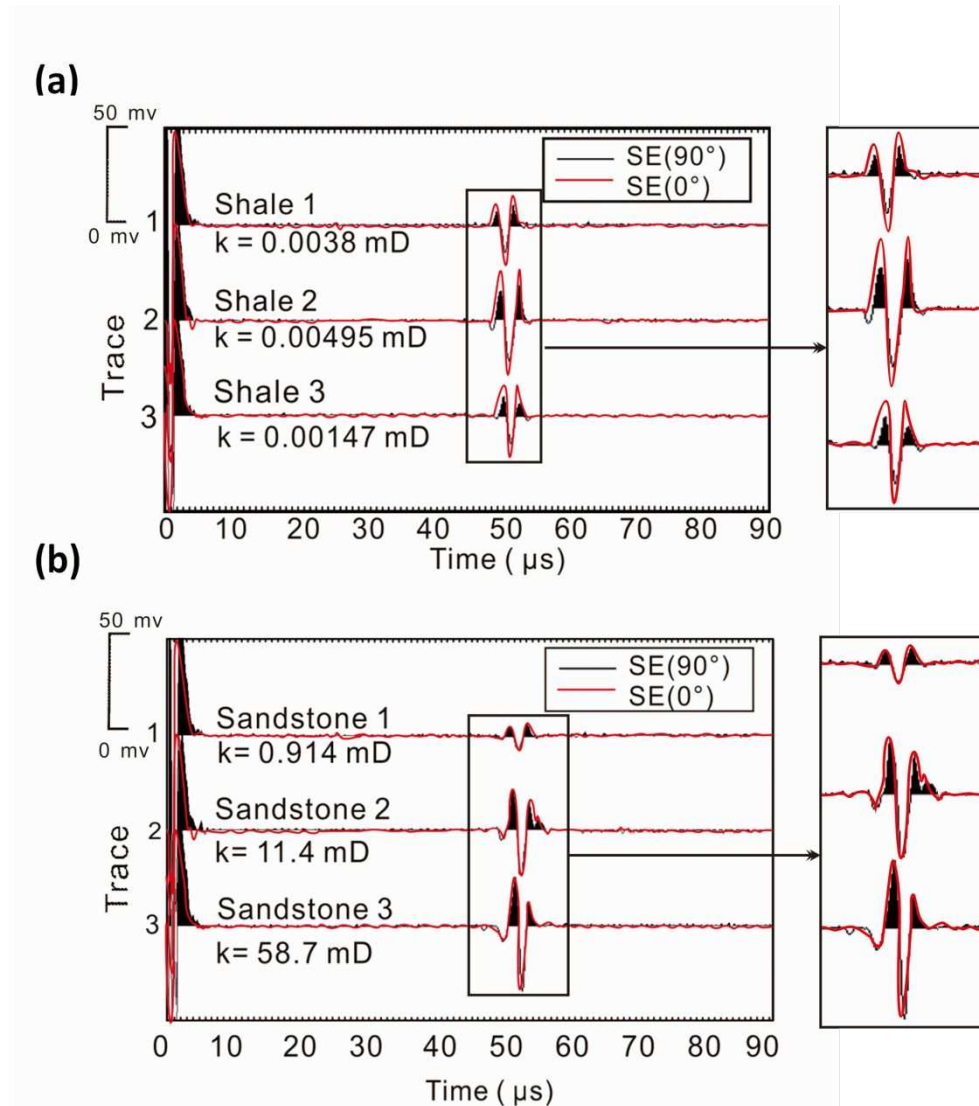


Figure 4. The seismoelectric signals of (a) 3 shale samples, and (b) 3 sandstone samples. The signals measured parallel to the shale bedding (red line) are much larger than those measured perpendicular to the shale bedding (in-filled black line) in the case of the shales, but there is no difference for the sandstones.

In **Figure 4a** the measurements for shale samples made parallel to bedding ($SE(0^\circ)$, red line) and vertical to the shale bedding ($SE(90^\circ)$, in-filled black line) are different, with the measurements parallel to the shale bedding (red line) being significantly larger than those measured perpendicular

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to the shale bedding (in-filled black line). Clearly, the anisotropy of the shale has a measureable effect on the strength of seismo-electric conversion. This contrasts with the situation for sandstones. For sandstones measured in this work (Figure 4b), no bedding was observable. Consequently, the two orthogonal measurements given in the figure, though made perpendicular to each other, are not related to a bedding plane. These two measurements behave in the same way and have the same intensity. Consequently, it might be said that measurements of the amplitude of the seismo-electric signals in the sandstones exhibit isotropic seismo-electric conversion, as would be expected.

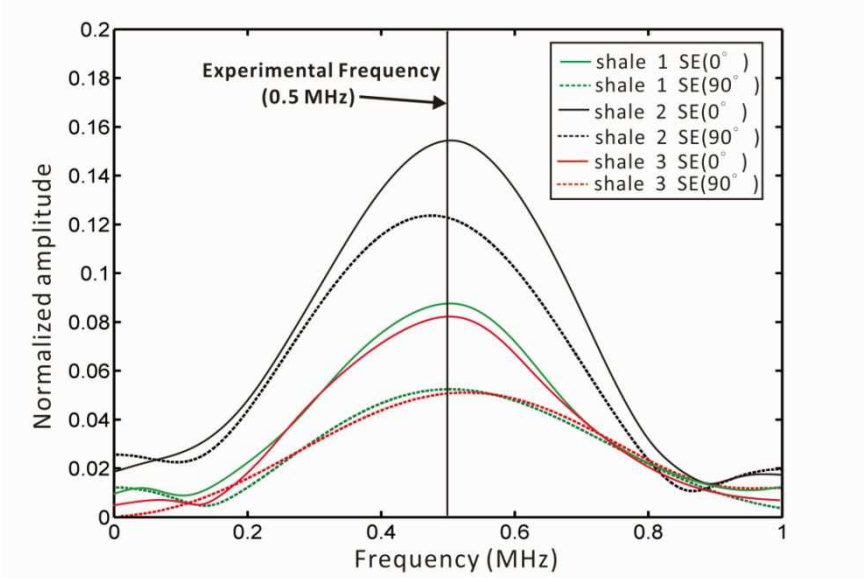


Figure 5. The frequency spectra of seismoelectric signals for the 3 shale samples whose data appear in Figure 4.

Analysis of the frequency spectra of the induced seismoelectric signal has been carried out for all shale samples and for both directions of measurement. Figure 5 shows the seismoelectric signal frequency spectra for those sample traces shown in Figure 4. In all cases the modal frequencies of the spectra correspond approximately to the frequency of transmitting transducer (0.5 MHz). For measurements parallel to the bedding the modal peak falls in the range 0.5 ± 0.008 MHz. By contrast, the frequency spectra of seismoelectric signals made perpendicular to bedding exhibit smaller amplitudes, as also seen in Figure 4, and are broader, with a modal peak value which falls in the broader range 0.5 ± 0.05 MHz. The process which broadens the spectra for measurements made perpendicular to the bedding in shales is not currently known.

Most of the experimental results conform to what might been expected. The most interesting unexplained observation is that seismoelectric coupling, while it clearly depends on porosity and permeability for both shales and sandstones, is equally strong for shales than for sandstones even though shales have a much lower permeability. We have carried out numerical modelling of the Pride (1994) theory of frequency-dependent seismo-electric conversion in porous media in order to

try to understand the effect. Numerical modelling has also been carried out to ascertain whether it is possible to model our experimental measurements on both shales and their corresponding sandstones. An associated objective was to understand the sensitivities of the amplitude and phase of the generated seismoelectric signals to a number of important rock parameters, including porosity, permeability, tortuosity, zeta potential and dimensionless number occurring in Equation (1).

4 Numerical modelling

4.1 Modelling methodology

This section is dedicated to examining how the frequency dependence of the seismo-electric coupling coefficient $L(\omega)$ depends on major rock parameters. In order to do this the Helmholtz-Smoluchowski (Equation (6)) and the Pride (Equation (1)) relationships have been implemented in the frequency domain using numerical modelling with Matlab.

The Helmholtz-Smoluchowski relationship for C_{sp0} (Equation (6)) and the Pride relationship for $L(\omega)$ (Equation (1)) imply that there are many factors affecting the seismo-electric conversion. These include the transition frequency ω_c , the Debye length $\tilde{d}$, the skin depth δ , the characteristic length scale of the pores Λ , the dimensionless number m^* , porosity ϕ , permeability k_0 , tortuosity α_∞ and zeta potential ζ .

In this work we assume the Debye length $\tilde{d}$ of shale and sandstone in NaCl solution are the same, equal to 3.0709×10^{-6} m, and can be written as (Debye & Hückel 1923)

$$\tilde{d} = \sqrt{\frac{\epsilon_f k_B T}{2000 N e^2 I_f}}, \quad (7)$$

where ϵ_f is the absolute permittivity of the fluid, k_B is Boltzmann's constant, T is absolute temperature, N is the Avogadro's number defined as $N = 6.022 \times 10^{23}$ in molarity, e is the electric charge, and I_f is the ionic strength ($I_f = \frac{1}{2} \sum_{i=1}^n c_i z_i^2$), where Z_i is the ionic valence ($Z = 1$ for an NaCl solution), and C_i is the molar concentration of the ions in the fluid. For a NaCl solution, $I_f = C$, where C is the fluid concentration. This assumption remains valid providing the temperature and salinity of the pore fluid remains unchanged. We also assume that the skin depth δ ($\delta = \sqrt{\frac{\eta}{\omega \rho_f}}$) for shale and sandstone are also identical (i.e., the same pore fluid with the same density and viscosity is present in both rocks). Hence, we present numerical modelling results for the variation of frequency-dependent seismo-electric coupling coefficient $L(\omega)$ as a function of frequency and m^* , porosity, permeability, tortuosity and zeta potential.

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4 365 In addition to the variables examined by numerical modelling, there are also some variables which
5 366 are either fundamentally constant or that we have held constant for the purposes of the modelling.
6 367 These values are shown in [Table 3](#).

8
9 368 **Table 3.** Parameters used in numerical modelling of frequency-dependent seismo-electric coupling
10 369 coefficients.

Parameters	Symbol	Values used in modelling	Unit
Fluid viscosity	η^1	0.001	N.s/m ²
Fluid concentration	C	1×10^{-5}	mol/L
Water density	ρ_f^1	1000	kg/m ³
Absolute temperature	T^1	298	K
Ion concentration	N^1	6.022×10^{23}	ions/m ³
Charge of an electron	e^1	1.602×10^{-19}	C
Boltzmann's constant	k_B^1	1.381×10^{-23}	J/K
Fluid permittivity	ϵ_f^1	$80 \times 8.854 \times 10^{-12}$	F/m
Sample permittivity	ϵ_r^1	$4 \times 8.854 \times 10^{-12}$	F/m
Tortuosity	α_∞	[2, 4, 6, 8, 10]	-
Porosity	ϕ	[0.0001, 0.001, 0.05, 0.1, 0.5]	-
Permeability	k_0	[0.001, 0.01, 1, 50, 1000]	mD
The dimensionless number	m^*	[6, 8, 12, 14, 18]	-
the characteristic length scale of the pores	$\mathcal{A}$	[0.01, 0.05, 0.1, 0.3, 1]	μm
Zeta potential	ζ	[0.01, 0.1, 1, 10, 100]	mV

37 370 ¹[Zhu et al. \(2015\)](#)

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40 371 4.2 The effect of dimensionless number

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42 372 The dimensionless number m^* describes the geometric characteristics of pore space and consists
43 373 only of the pore-space geometry terms ([Equation \(4\)](#)). [Figure 6](#) shows the effect of the dimensionless
44 374 number m^* on seismo-electric coupling coefficient $L(\omega)$. Values of $m^*=8$ and $m^*=12$ are the
45 375 characteristic for sandstone and shale ([Johnson et al. 1987](#)). We chose to model $m^*=6, 8, 12, 14, 18$
46 376 to see the influence of dimensionless number m^* on $L(\omega)$ and including the accepted values of m^*
47 377 for sandstone and shale. In modelling $L(\omega)$ as a function of frequency and m^* , porosity,
48 378 permeability, tortuosity and zeta potential were all kept constant, with $\phi=10\%$ (0.1), $k_0=10$ mD,
49 379 $\alpha_\infty=3$, and $\zeta=-40$ mV.
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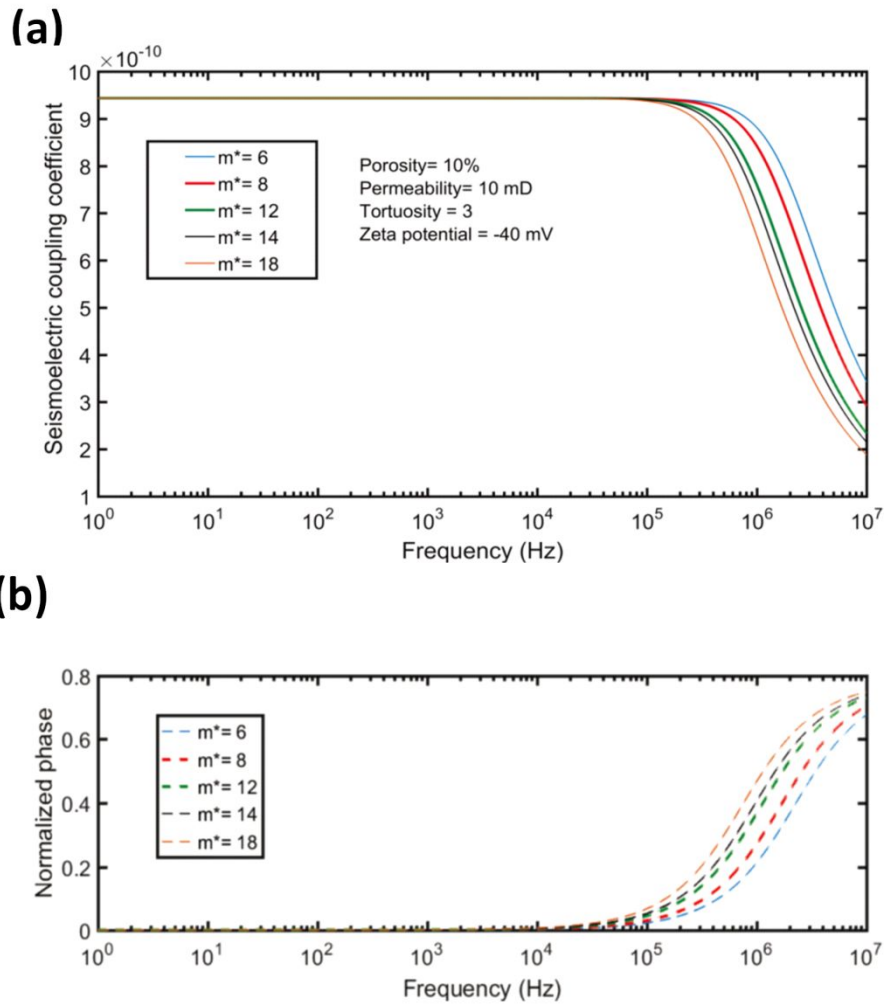
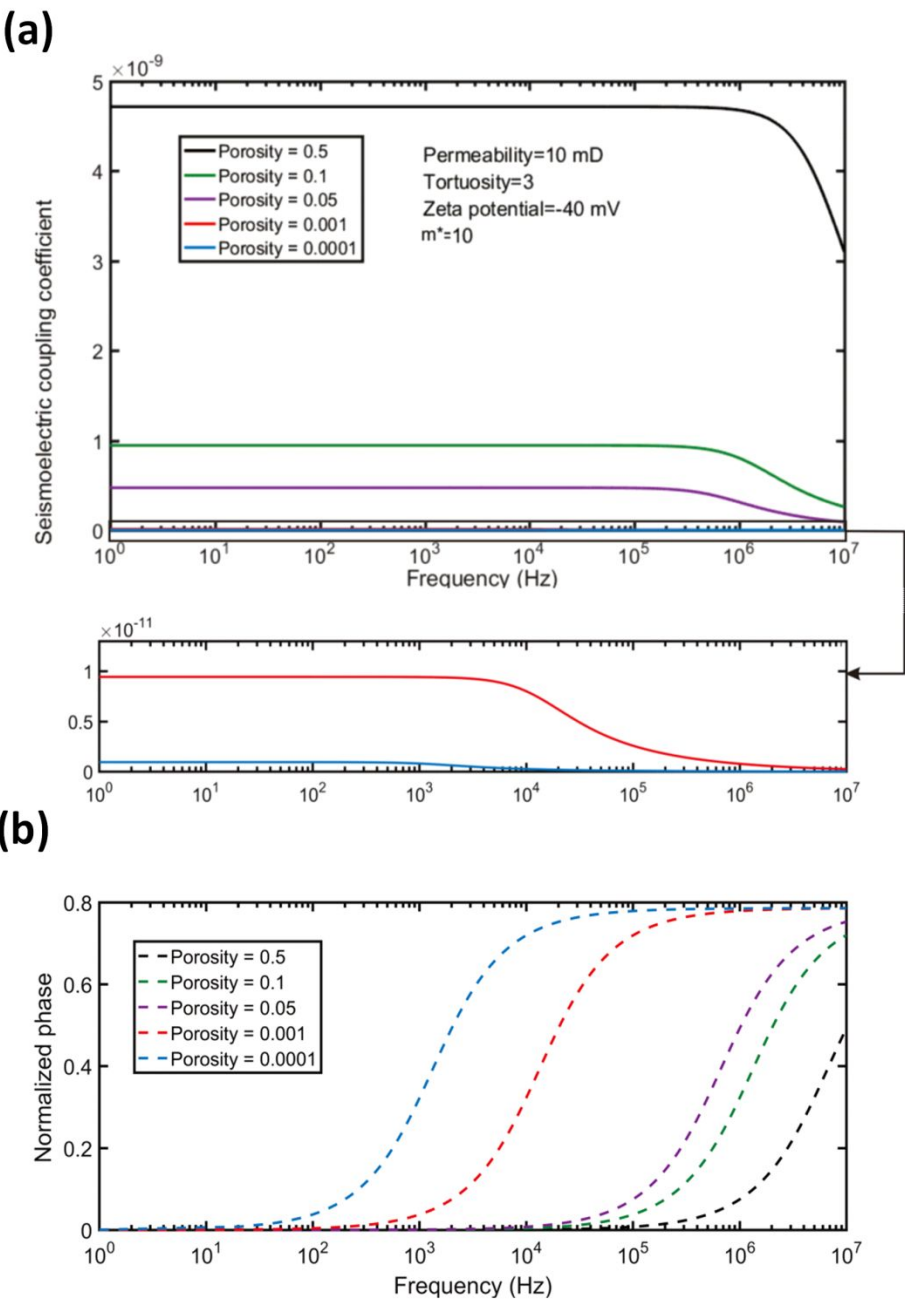


Figure 6. The effect of varying the dimensionless number m^* on modelled frequency-dependent $L(\omega)$, with $\phi=10\%$ (0.1), $k_0=10$ mD, $\alpha_\infty=3$, and $\zeta=-40$ mV all held constant. The two bold curves $m^*=8$ and $m^*=12$ are the reported values for sandstone and shale, respectively (Johnston et al. 1987), the solid line (a) and dashed line (b) are the amplitudes and the phases of the $L(\omega)$, respectively.

Figure 6 clearly shows that the amplitude and phase of the seismo-electric coupling coefficient are independent of m^* when the frequency is lower than 100 kHz. Above this frequency an increase in m^* shifts the dispersion in the $L(\omega)$ to lower frequencies, decreasing the critical frequency. However, the effect is small. For example, at 1000 kHz the difference in amplitude of the seismo-electric coupling coefficient resulting from the change of dimensionless number from $m^*=8$ to $m^*=12$ is only about 6×10^{-11} . Hence, the dimensionless number has a little influence on the seismo-electric coupling over a large frequency range (0-1000 kHz). One interpretation of this phenomenon is that the seismo-electric coupling coefficient is not affected by large scale microstructure and disorder in a

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4 395 porous medium (Charlaix et al. 1988). However, it is possible that other mechanisms might give rise
5 396 to frequency-dependence at low frequencies such as Stern plane polarization.
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51 398 **Figure 7.** The effect of changing porosity on (a) the amplitude, and (b) the phase of the frequency-dependent
52 $L(\omega)$ as a function of frequency. Other parameters, $m^*=10$, $k_0=10$ mD, $\alpha_\infty=3$, and $\zeta=-40$ mV held constant.
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4.3 The effect of porosity

When modelling the effects of porosity we have to cover a wide range of values. The porosity of conventional sandstones is generally in the range 5% to 30% (0.05 to 0.30). By comparison shales may have an even wider range of porosities, from Sichuan shales which have values less than 5% (Deng et al. 2015) to some gas shales with porosities approaching 30% (0.30). Consequently, we have carried out modelling from 0.01% to 50% (0.0001 to 0.5) in order to capture all possible values. During the modelling $L(\omega)$ as a function of frequency and porosity, the dimensionless number, permeability, tortuosity and zeta potential were all kept constant, with $m^*=10$, $k_0=10$ mD, $\alpha_\infty=3$, and $\zeta=-40$ mV.

Figure 7 shows how varying the porosity of the porous medium affects the amplitude and phase of the frequency-dependent seismo-electric coupling coefficient. The range of modelled values is so great that the results for the smaller porosities are shown on an expanded scale in the inset of the figure. It is clear that values of the amplitude of the $L(\omega)$ are very small ($<5 \times 10^{-11}$) for porous media whose porosity is less than 5% (purple line). As porosity increases there is an increase in the amplitude of the $L(\omega)$ by about 3.5 orders of magnitude (from about 1×10^{-12} to just under 5×10^{-9}) as well as an increase in the critical frequency from about 1 kHz for $\phi=0.0001$ (0.01%) to about 10 MHz for $\phi=0.5$ (50%). Hence, the numerical modelling results indicate that porosity has a strong effect on the amplitude of low frequency seismo-electric coupling as well as the frequency at which dispersion occurs.

4.4 The effect of the characteristic length scale of the pores (Λ)

Figure 8 is the image analysis of scanning electron microscopy from the shale samples in the southern Sichuan Basin (Wang et al. 2014). The shale in Longmaxi formation in Sichuan Basin is mainly composed of intergranular pores, inter-crystalline pores, organic matter pores and fossil interpores. The first four pore types belong to inorganic type, while the latter two pore types are organic type.

The pore size in the shale samples from Sichuan Basin is mainly from nanometer to micrometer scale. The matrix pore size is generally less than 2 μm , and its main pore length is in the range of 0.1-1 μm . The pore size of organic matter is mostly from 0.01 μm to 0.3 μm . The pore morphology is mainly round, elliptical and irregular and the pore characteristics of Sichuan shale are similar to the Barnett shale in North America (Wang et al. 2014; Zhang et al. 2014; Liu et al. 2019).

Consequently, we have carried out modelling for characteristic length scale of the pores (Λ) covering the entire range of observed length scales, viz. 0.01 μm , 0.05 μm , 0.1 μm , 0.3 μm , 1 μm , in order to examine the effect of altering the length scale of pores on the seismo-electric coupling. Figure 9 shows the effect of the characteristic length scale of the pores in numerical results. The red curve

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corresponds to the smallest pore size ($0.01\mu\text{m}$), and shows the smallest seismo-electric coupling coefficients. The low frequency seismo-electric coupling coefficient increases as pore length scale increases, seeming to reach a limiting behaviour. Increasing the value of pore length reduces the value of the critical frequency, producing dispersive behavior at lower frequencies as the pore scale increases. This corresponds with structures at larger pore scales causing lower frequency (longer period) dispersion, from which we infer a dispersive process with a fixed mobility or transport rate.

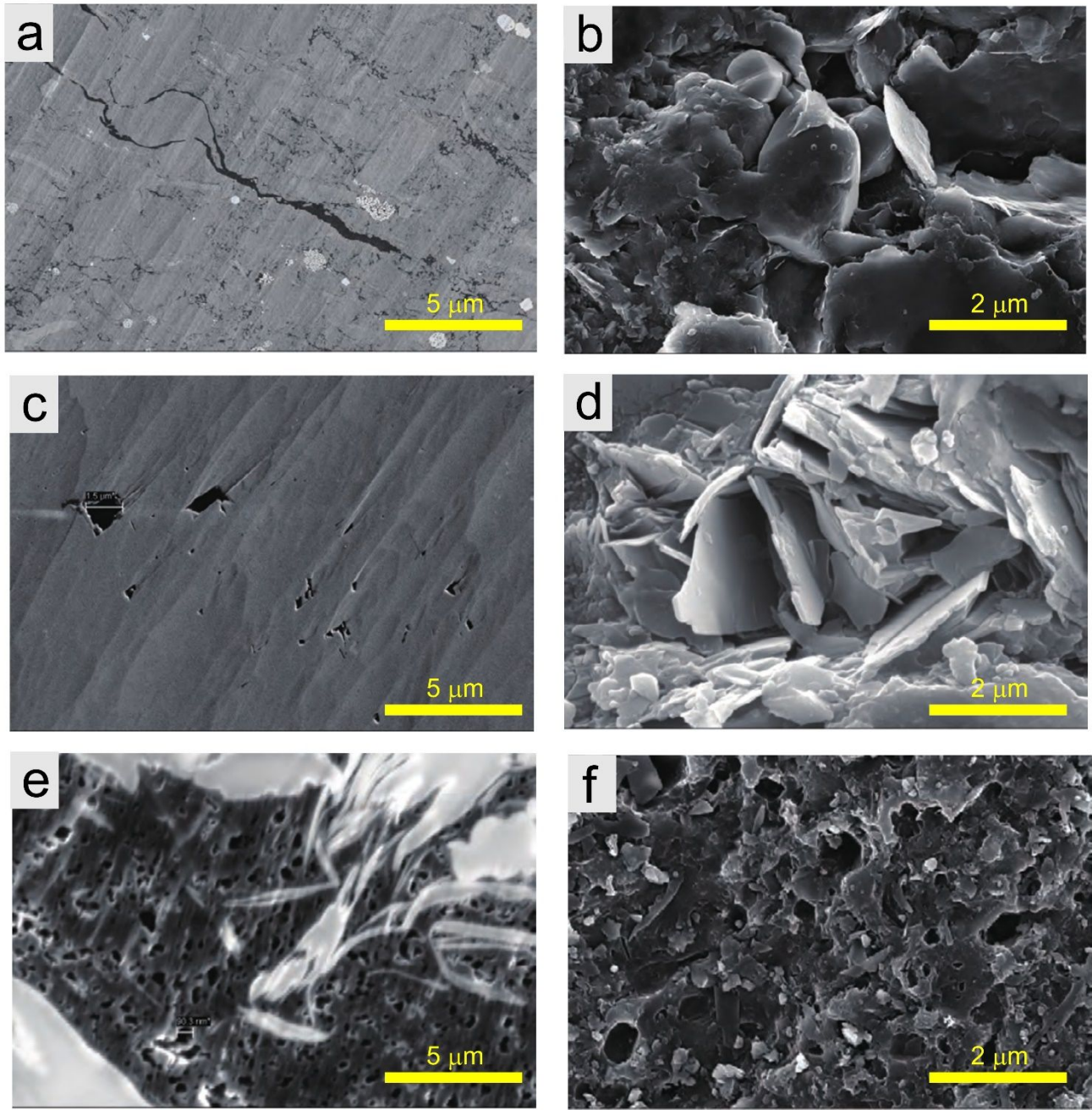
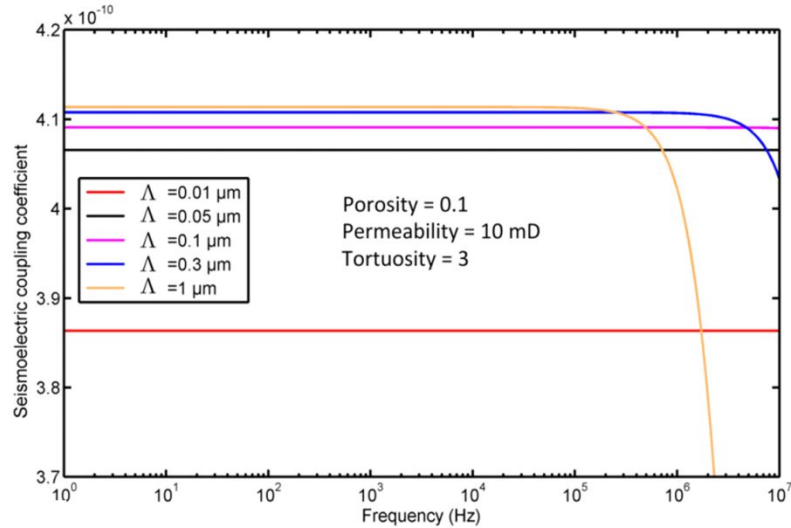


Figure 8. Image analysis of scanning electron microscopy from the shale samples in the southern Sichuan Basin. (a) Microfractures filled with organic matter, (b) inter-granular pores of quartz, (c) inter-granular pore

after argon ion polishing, (d) inter-crystalline pores, (e) pores of organic matter, (f) internal pores of graptolite (Wang et al., 2014).

(a)



(b)

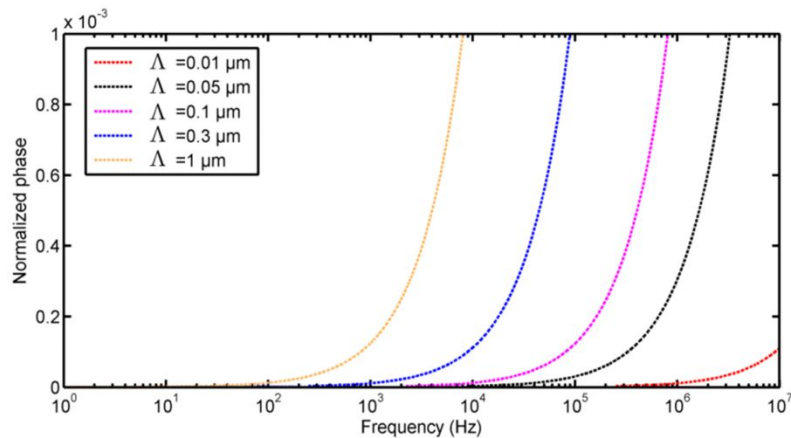
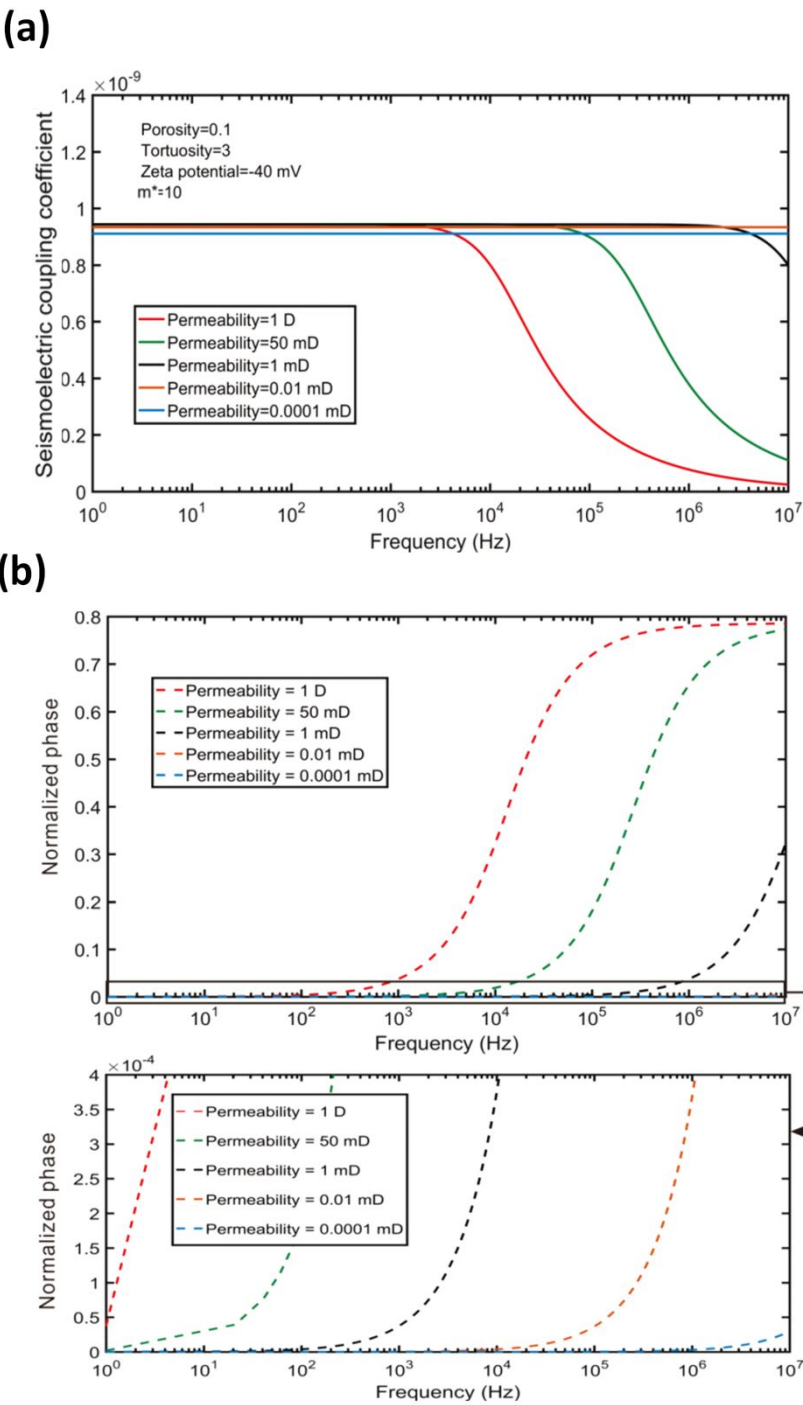


Figure 9. The effect of changing pore length on (a) the amplitude, and (b) the phase of the frequency-dependent $L(\omega)$. Other parameters, $\phi=10\%$ (0.1), $\alpha_\infty=3$, and $\kappa=10$ mD held constant.

4.5 The effect of permeability

Numerical modelling has also shown that the frequency-dependent seismo-electric coupling coefficient depends upon permeability in a complex way. Modelling has been carried out for five values of permeability as a function of frequency. The modelled numerical values of permeability are 100 nD, 10 μ D, 1 mD, 50 mD and 1 D, while keeping the dimensionless number, porosity, tortuosity and zeta potential constant at $m^*=10$, $\phi=10\%$ (0.1), $\alpha_\infty=3$, and $\zeta=-40$ mV. This range of

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4 457 permeabilities covers most sandstones and shales, with most conventional reservoir sandstones
5 458 falling in the range $50 \text{ mD} < k_0 < 1 \text{ D}$, tight sandstones and carbonates falling in the range 50
6 459 $\mu\text{D} < k_0 < 50 \text{ mD}$, and most shales having $k_0 < 10 \mu\text{D}$ and usually in the hundreds of nanodarcies. The
8 460 permeability of Sichuan shale is less than 1 mD (Fu 2017).
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57 462 **Figure 10.** The effect of changing permeability on (a) the amplitude, and (b) the phase of the
58 463 frequency-dependent $L(\omega)$. Other parameters, $m^*=10$, $\phi=10\%$ (0.1), $\alpha_\infty=3$, and $\zeta=-40 \text{ mV}$ held constant.
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The numerical modelling results displayed as [Figure 10](#) show that, for the sandstone permeabilities ($k_0=1$ D, 50 mD; red line, green line, respectively), the amplitude of the seismo-electric coupling coefficient is independent of frequency at frequencies below about 1 kHz, which is also indicated by the phase of the seismo-electric coupling coefficient being close to zero for these frequencies. The amplitude of this low frequency behaviour depends very little on permeability, sharing the same value. Dispersion occurs at higher frequencies with the critical frequency occurring at about 10 kHz for $k_0=1$ D and increasing to about 200 kHz for $k_0=50$ mD. The same pattern continues into the tight clastic rock range of permeabilities ($k_0=1$ mD, 10 μ D; black line, brown line, respectively) and then for the shale permeability range ($k_0=100$ nD; blue line), with the range of frequency independent behavior increasing, the value of the critical frequency increasing and surpassing 10 MHz, and seeing a small decrease in the amplitude of the seismo-electric coupling coefficient for the lowest permeabilities.

This behavior is important as it implies that the strength of seismo-electric coupling in shales is not affected by their low permeabilities, sharing approximately the same value of seismo-electric coupling coefficient as rocks with permeabilities 7 orders of magnitude larger (providing they share the same values of dimensionless number, porosity, tortuosity and zeta potential). Such behavior accords well with our experimental results, and confirms that the similar measured seismo-electric coupling coefficient for shales and sandstones is relatively independent of their very different permeabilities, but rather controlled by their porosities which are different, but not to the same degree that the permeabilities vary.

4.6 The effect of tortuosity

Tortuosity also affects both the amplitude and the phase of the seismo-electric coupling coefficient, as shown in [Figure 11](#). We have carried out numerical modelling of the amplitude and phase of the seismo-electric coupling coefficient for 5 values of tortuosity ($\alpha_\infty=2, 4, 6, 8, 10$) while keeping the other parameters constant ($m^*=10$, $\phi=10\%$ (0.1), $k_0=10$ mD, and $\zeta=-40$ mV). This range of tortuosities is equivalent to a range of cementation exponents of $m=1.2$ to 2 for a porous medium of 10% porosity, according to the relationship $\alpha_\infty=\phi^{1-m}$ ([Glover 2012](#)).

The amplitude of the seismo-electric coupling coefficient is independent of frequency at low frequencies and undergoes dispersion at higher frequencies. The value of the low frequency (steady-state) seismo-electric coupling coefficient is very sensitive to changes in tortuosity, decreasing from about 1.4×10^{-9} to 0.3×10^{-9} as the tortuosity increases from 2 to 10. The onset of dispersive behaviour is also sensitive to tortuosity, but less so than the steady-state seismo-electric coupling coefficient. If we take critical frequency as a measure of the frequency at which dispersion

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occurs, the critical frequency decreases from about 2 MHz to about 300 kHz as tortuosity increases from 2 to 10.

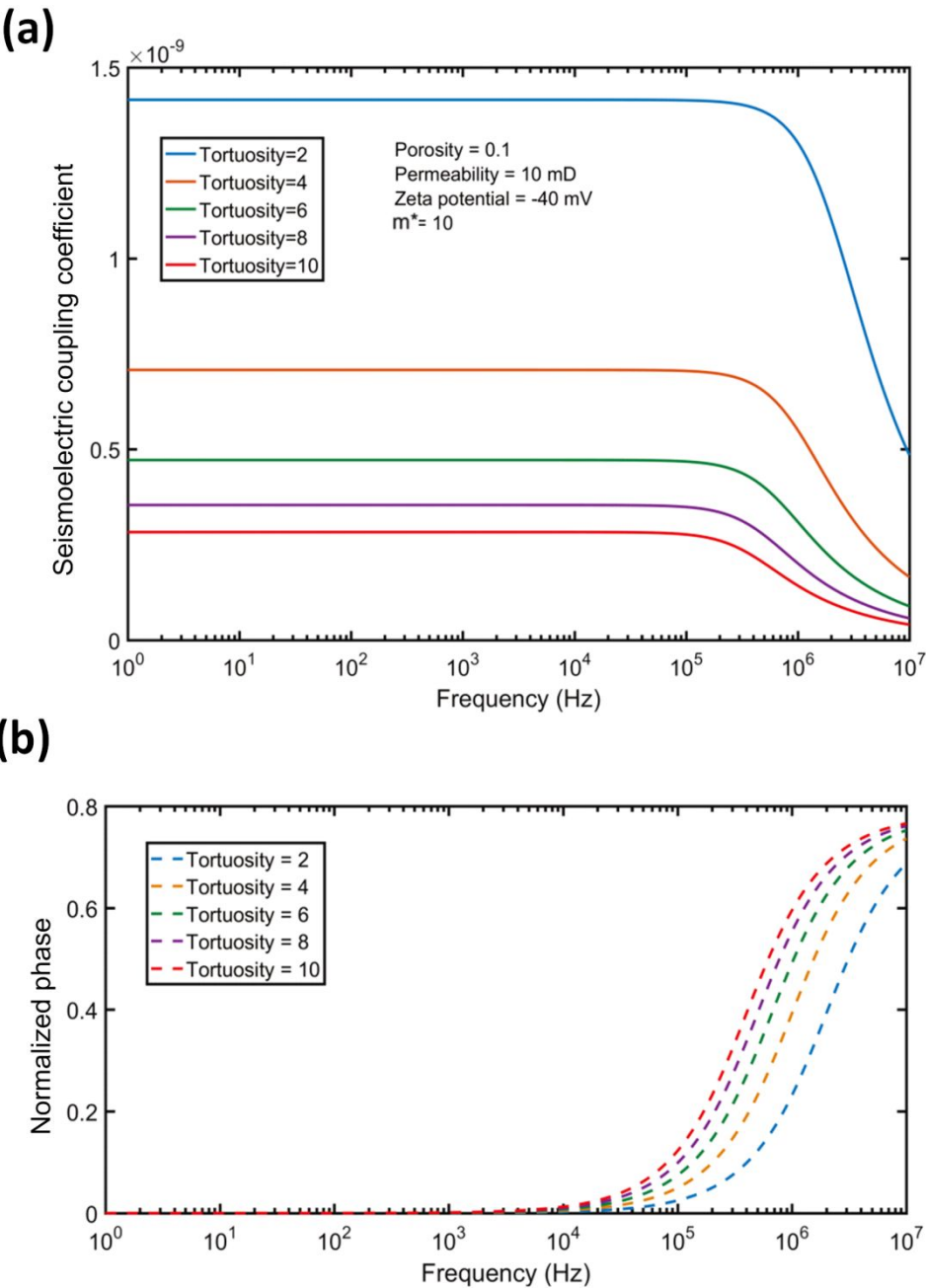


Figure 11. The effect of changing tortuosity on (a) the amplitude, and (b) the phase of the frequency-dependent $L(\omega)$. Other parameters, $m^*=10$, $\phi=10\%$ (0.1), $\kappa=10$ mD, and $\zeta=-40$ mV held constant.

The variation of $L(\omega)$ with tortuosity has implications for measurements made on shales. Shales generally have higher values of tortuosity, which would be expected to lead to a lower strength of seismo-electric coupling, and for a smaller frequency range. However, given that the critical frequency for a tortuosity of 10 is as high as 300 kHz, this restriction is very unlikely to be important as it is much higher than seismic frequencies.

4.7 The effect of zeta potential

Zeta potential strongly affects the amplitude of the seismo-electric coupling coefficient but has no effect on its phase, as shown in [Figure 12](#). Numerical modelling of the amplitude and phase of the seismo-electric coupling coefficient has been carried out for 5 values of zeta potential from -0.01 mV to -100 mV in order of magnitude steps, while keeping all other parameters constant ($m^*=10$, $\phi=10\%$ (0.1), $\kappa=10$ mD, and $\alpha_\infty=3$). This range of zeta potentials covers almost the entire range of values measured experimentally ([Walker & Glover 2018](#)), where the maximum zeta potential value is about -150 mV and the minimum is about -2 mV.

Once again the amplitude of the seismo-electric coupling coefficient is independent of frequency at low frequencies and undergoes dispersion at higher frequencies. The value of the low frequency (steady-state) seismo-electric coupling coefficient is approximately linearly sensitive to changes in tortuosity, increasing over 4 orders of magnitude from about 2×10^{-13} to almost 2.5×10^{-9} as the zeta potential also increases by four orders of magnitude, from -0.01 mV to -100 mV.

Unlike the other variables examined in our modelling, varying the zeta potential has no effect on the critical frequency. This is expected since the zeta potential does not occur in the frequency-dependent terms in [Equation \(1\)](#).

The numerical modelling confirms the expectation that the zeta potential is critical to the control of the steady-state seismo-electric coupling coefficient, defining its low frequency behaviour, but has no role in controlling the frequencies at which dispersion occurs as characterised by the critical frequency.

4.8 Low frequency behaviour

The seismoelectric coupling coefficient is a function of frequency but only at high frequencies. The value of the constant, low frequency behavior depends strongly on porosity, tortuosity and zeta potential, but very little on permeability, and is independent of the dimensionless number. This behaviour is consistent with [Equation \(6\)](#), which describes the low frequency behavior of the $L(\omega)$. However, [Figure 8](#) shows that there is a slight dependence of $L(\omega)$ on permeability for very low permeabilities. This may be due to the assumption $\tilde{a} \ll \Lambda$ breaking down as Λ becomes very small in low permeability shales. Consequently, we may expect that the low permeability may also reduce the strength of seismo-electric coupling in shales.

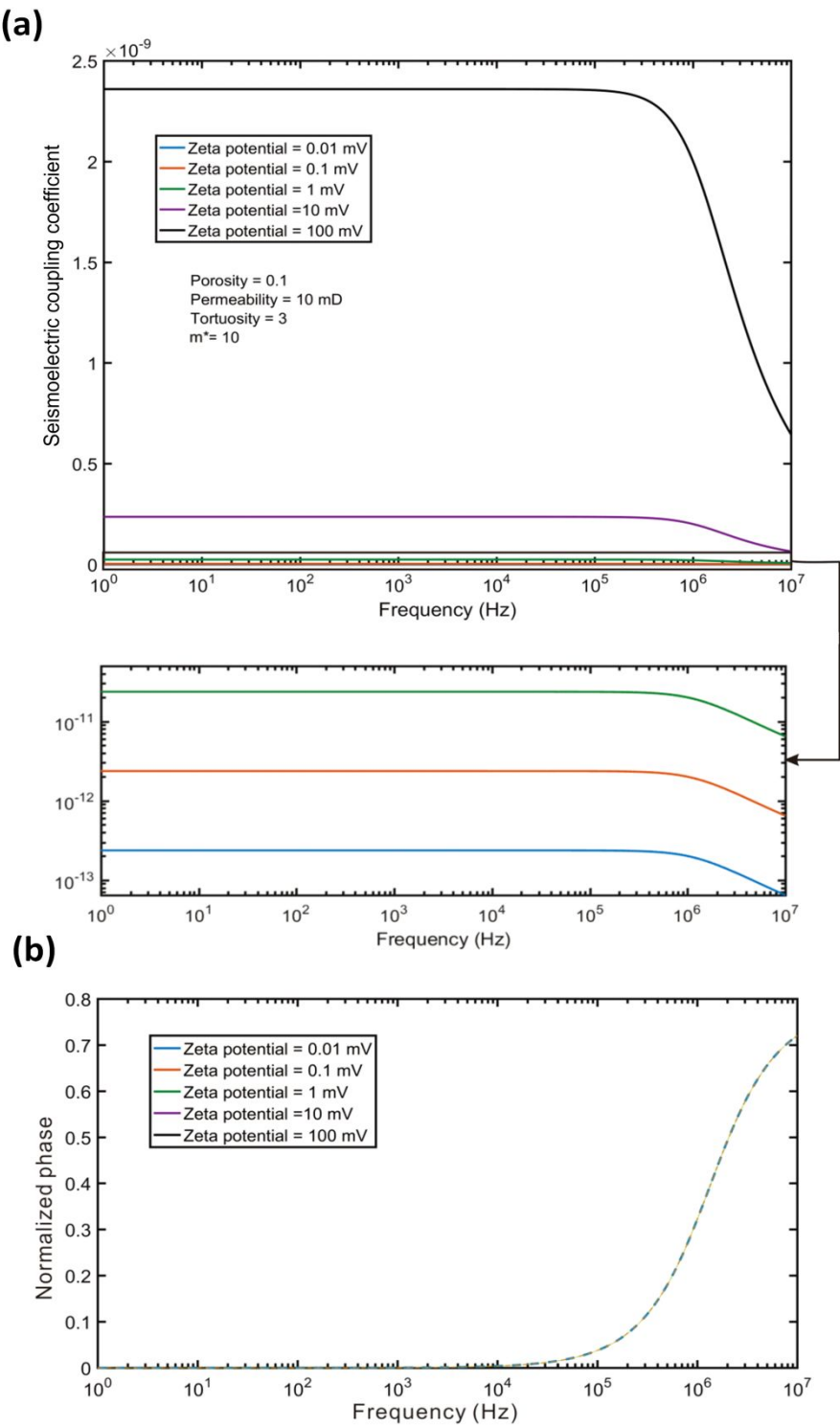


Figure 12. The effect of changing zeta potential on (a) the amplitude, and (b) the phase of the frequency-dependent $L(\omega)$. Other parameters, $m^*=10$, $\phi=10\%$ (0.1), $\kappa=10$ mD, and $\alpha_\infty=3$ held constant.

Examination of the variables controlling the seismo-electric coupling coefficient $L(\omega)$ in this section, leads us to recognize that for most relevant frequencies, porosity, tortuosity and zeta potential are most important. However, the porosity of the Sichuan shales is low in this work, and their tortuosity is high (Chen et al. 2015). Both of these observations conspire to reduce the strength of seismo-electric coupling. That seismo-electric coupling is at least as great as that found in sandstones is observed experimentally implies, therefore, that the shales must exhibit a higher zeta potential. Furthermore, such an enhanced zeta potential should be related to the mineralogical composition of the rock. This hypothesis is explored in the last section of this paper.

Typical shale values (say, $m^*=12$, $\phi=5\%$ (0.05), $k_0=0.005$ mD, $\alpha_\infty=2.66$ ($\alpha_\infty=(1-2.02\ln 0.05)^{1/2}=2.66$, and $\zeta=-100$ mV) lead to a low-frequency seismo-electric coupling coefficient of approximately 1.344×10^{-9} at 100 Hz. This value represents a strong seismo-electric effect which can explain the experimental observations (Figure 3) described earlier in this paper. We have carried out further numerical modelling matched to the experimental conditions in order that the modelling results can be compared directly with the experimental measurements. This work is described in the next section.

4.9 High frequency behaviour

At higher frequencies dispersion occurs, reducing the amplitude of the $L(\omega)$ and increasing its phase ultimately to $\pi/2$ radians. The frequency at which this behavior occurs is characterized by a critical frequency, which is defined as the frequency where the phase angle is $\pi/4$ radians. Figures 6 to 10 show clearly that all of the variables tested in this work except the zeta potential affect the critical frequency, sometimes by several orders of magnitude. This is consistent with Equation (1) which contains all of the variables except the zeta potential in the frequency-dependent part of the equation. However, and remembering that we were careful to ensure that our input variables covered the whole range expected for conventional and unconventional reservoir rocks, the critical frequencies are so high that they occur several orders of magnitude above those frequencies encountered in practical exploration seismology. The frequency range of seismo-electric exploration is determined by the seismic and acoustic source. The source frequency of seismic exploration is mainly less than 1000 Hz, and the emission frequency of acoustic source used in well logging is mainly in the range of 2~20 kHz. The seismic source and well logging acoustic source are both used as the source in seismo-electric exploration, hence, the frequency range for the seismo-electric exploration will not exceed 20 kHz. Consequently, it is unnecessary to use frequency-dependent $L(\omega)$ when dealing with seismo-electric applications except when the characteristic frequency exceeds at least 20 kHz.

5. Comparing numerical results with experimental data

We have compared the frequency-dependent amplitude of the seismoelectric coupling coefficient $L(\omega)$ of sandstone and shale samples obtained experimentally earlier in this paper with results from

numerical modelling of the [Pride \(1994\)](#) model but constrained by independently obtained sample characteristics such as porosity, permeability, tortuosity etc. The porosities and permeabilities were measured in the laboratory. The calculation parameters are shown in [Table 4](#).

The zeta potential of solid porous materials is difficult to measure experimentally. It requires the measurement of the seismoelectric coupling coefficient, and then calculation of the zeta potential assuming that the Helmholtz-Smoluchowski equation is valid ([Walker et al. 2014](#)). Consequently, in many applications empirical relationships between the zeta potential and the pore fluid salinity are used.

[Pride and Morgan \(1991\)](#) first produced an empirical fit of zeta potential ζ to pore fluid salinity C using 35 data points. They obtained the relationship of the form

$$\zeta = a + b \log_{10}(C), \quad (8)$$

with $a = -8$ mV and $b=26$ mV, which was later found to arise from a theoretical model for ζ ([Revil & Glover 1997; 1998; Revil et al. 1999a](#)) if a number of simplifying assumptions applied. These were (i) that the pH of pore fluid was approximately 7, (ii) that the pore fluid is a Na⁺ or K⁺ symmetric electrolyte, (iii) that the influence of H⁺ and OH⁻ ions on the ionic strength of the solution saturating the pores could be neglected ([Revil & Glover 1997](#)), and (iv) that direct adsorption of K⁺ and Na⁺ ions upon the silica surface could also be neglected. Consequently, it was implied that the rock would be a sandstone. Recently a fit has been implemented on a very large dataset of sandstones (1253 measurements) ([Walker & Glover 2018](#)), which confirms [Equation \(8\)](#) and provides $a = -3.505 \pm 28.823$ mV and $b=11.33 \pm 4.06$ mV. The uncertainty of a and b is because the values obtained are for an aggregation of different samples and rock types at uncontrolled pH.

However, good relationships of the form of [Equation \(8\)](#) for silica-based rocks, and despite the fact that [Equation \(8\)](#) can be derived from EDL equations for silica-based rocks, it cannot be assumed that an equation of this form will be valid for shales, where the presence of stern-plane polarization in clays is an important additional factor. [Figure 13](#) shows 13 measurements of the zeta potential in shales saturated with NaCl and KCl solutions from four sources together with a best fit of [Equation \(8\)](#). The combined datasets shows a considerable scatter, and it is not possible to say that [Equation \(8\)](#) is indicated by the data. If [Equation \(8\)](#) is fitted to the shale data, we obtain $a=19.14 \pm 24.32$ mV, $b=21.07 \pm 8.17$ mV, with the coefficient of determination $R^2=0.874$, and is shown by the red line in [Figure 13](#). Many more measurements on shale are required before we can even say that [Equation \(8\)](#) is a valid empirical approximation let alone define its parameters.

[Figure 14](#) shows the numerical and experimental amplitude of the seismoelectric coupling coefficient $L(\omega)$ of sandstones and shale samples as a function of frequency. The experimental data includes an assessment of uncertainty for each measurement based on the propagation of experimental

uncertainties associated with the measurement of the transient seismoelectric potential and transient pressure differences. In each case the experimental measurements agree fairly well with the numerical modelling, indicating that, while not ideal, a model based on the Pride (1994) approach captures the features of the experimental data well.

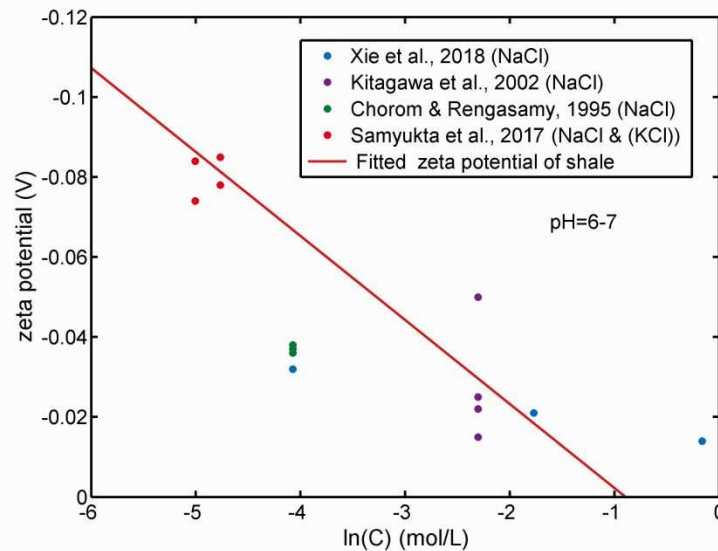


Figure 13. Zeta potential of shales (orange line). The symbols are experimental values of zeta potential measured on shale and clay minerals extracted from previous research for which pH= 6-7.

The fits shown in Figure 14 show that the amplitude of the seismoelectric coupling coefficient for shales is of comparable size and sometimes surpasses that of the sandstones despite the low porosities and permeabilities exhibited by the shales. This phenomenon depends upon the shales having larger zeta potentials. Indeed, the modelling shown in Figure 14 uses an imposed zeta potential function for shales that results in greater zeta potentials than the zeta potential function imposed for the sandstones (see Table 4) for all values of fluid concentration less than 0.02482 mol/dm³. At the fluid concentration used for the experiments and the modelling (10⁻⁵ mol/dm³) the zeta potential for the shale modelling is 1.622 times larger than that used for the sandstone modelling.

6. Seismo-electric coupling and clay mineralogy

Since the experimental and modelling results given in this paper strongly suggest that shales exhibit a larger zeta potential, we have examined their mineralogical composition. Analysis of the mineralogical composition of the 14 samples of Sichuan Basin shale used in this work was carried out by X-ray diffraction (XRD).

Table 4. The input parameters used in numerical modelling of frequency-dependent seismoelectric coupling coefficients of shales and sandstones

Parameters	Symbol	Values of shale used in modelling	Values of sandstone used in modelling	Unit
Fluid concentration	C	1×10^{-5}		mol/dm ³
Tortuosity	α_{∞}	$\alpha_{\infty} = \sqrt{1 - 2.02 \ln(\phi)}$		-
Sample Porosity	ϕ	0.0442, 0.0247, 0.0429, 0.0494, 0.0365	0.137, 0.124, 0.0621, 0.0390, 0.0100	-
Sample Permeability	k_0	0.00380, 0.00495, 0.00147, 0.008, 0.107	58.7, 11.4, 0.914, 10.1, 0.235	mD
Dimensionless number	m^*	12	8	-
Zeta potential	ζ	$\zeta = 19.14 + 21.07 \log_{10}(C)$	$\zeta = 3.505 + 11.33 \log_{10}(C)$	mV

Note. The value for C , ϕ , and k_0 were measured in the laboratory. The value of k_0 was Klinkenberg-corrected and obtained using an ultra-low-permeability measuring instrument (PDP-200). The values for other parameters are from [Zhu et al. \(2015\)](#) and shown in [Table 3](#).

The main minerals composing the shales measured in this report were quartz, plagioclase, K-feldspar and a range of clay minerals, including kaolinite, illite and smectite. [Figure 15](#) shows the relationship between the contents of non-clay and clay minerals, total organic carbon (TOC) and the seismo-electric signals obtained parallel and perpendicular to bedding. The non-clay components of the Sichuan shales are predominantly quartz with smaller amounts of K-feldspar and plagioclase feldspar. [Figure 15a](#) indicates that there is no clear relationship between the seismo-electric conversion and the weight fraction of any of the minerals indicating that their presence has no influence over the zeta potential of the rock or the resulting seismo-electric coefficient.

The seismo-electric coefficient is, however, sensitive to the clay composition. [Figure 15b](#) shows that the amplitude of the seismo-electric signal increases approximately 60% (from 16 ± 4 mV to 27 ± 4 mV) when the clay mineral fraction increases from 16.8%vol. to 28.8%vol. This indicates that seismo-electric coupling is strongly influenced by the clay content, increasing with the increases in the clay mineral fraction, and implying that the zeta potential also increases significantly as clay content increases.

Zeta potential is directly related to the amount and distribution of the charges at the solid-liquid interface, and characterizes the electric double layer. Shale has the mechanism of generating surface charge ([Müller 1996](#); [Fosu-Duah et al. 2016](#)). The surface potential of clay minerals is large compared to that of other reservoir rock-forming minerals such as quartz, calcite and dolomite. So a higher clay mineral fraction will lead to higher surface potentials, thus producing a higher zeta potential ([Leroy & Revil 2009](#); [Fosu-Duah et al. 2017](#)). Consequently, the close relationship between increasing clay content and increasing seismo-electric conversion indicates that clay mineral-fluid interactions are the main cause of high zeta potentials and leads to strong seismo-electric conversions in clay-rich shales.

[Figure 16](#) shows the effects of the main components of the clay minerals on seismo-electric (SE) conversion. In our samples the content of kaolinite in the clay is low (0-4%) and there is no evidence for the seismo-electric coupling coefficient depending on the amount of kaolinite in the sample ([Figure 16a](#)). There is a substantially greater contribution of mixed illite/smectite, up to about 60%, but here too there is no convincing evidence that this clay structure affects the seismo-electric coupling ([Figure 16c](#)). Illite is present in the samples in amounts ranging from 25% to 69%, and as the illite content increases within this range seismo-electric conversion increases by approximately 75% from 16 mV to 28 mV ([Figure 16b](#)). This strong functional dependence seems to show that the presence of illite is the main control on the zeta potential in shales.

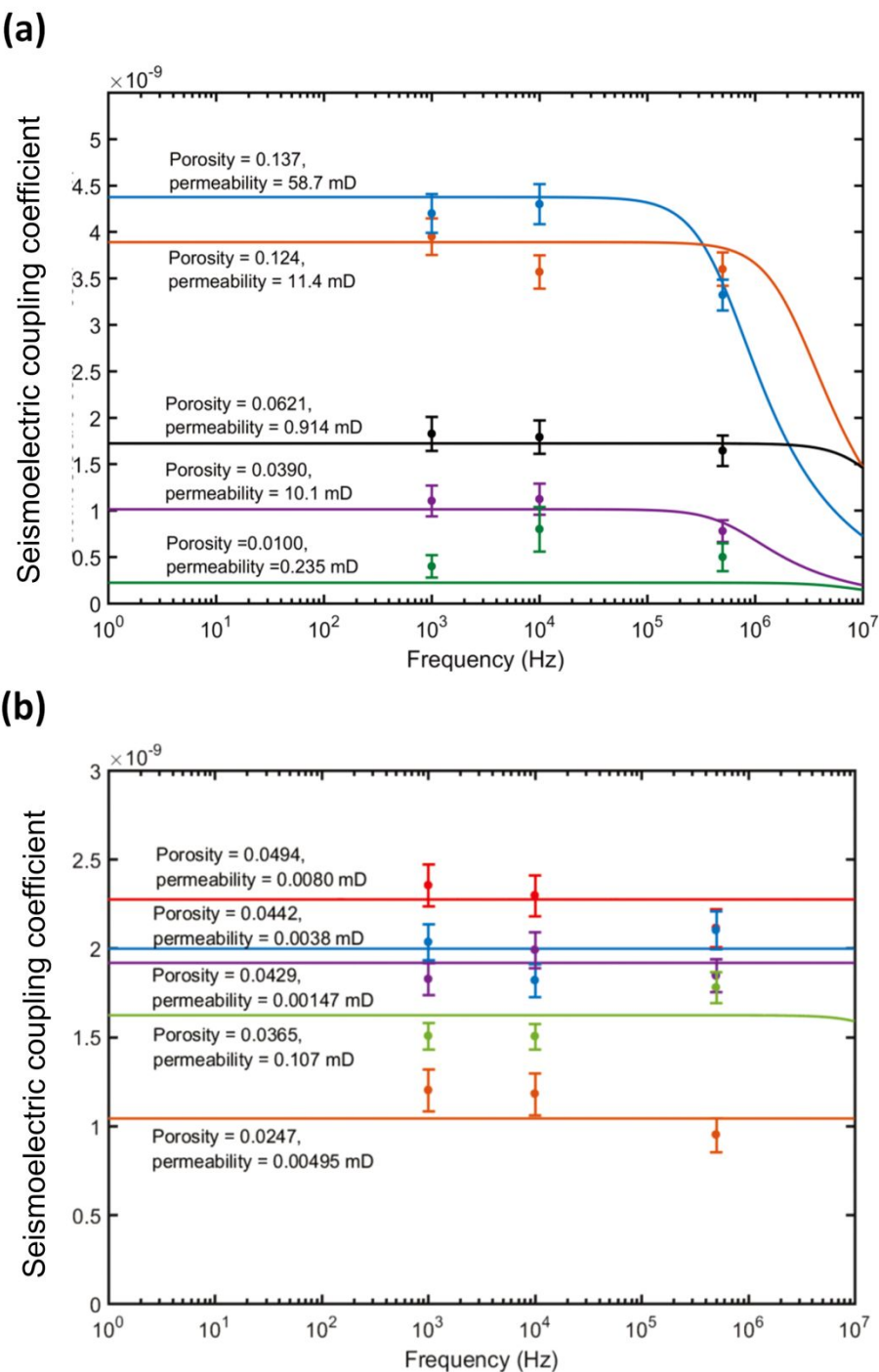


Figure 14. Seismo-electric coupling coefficient as a function of frequency for (a) sandstones, and (b) shale samples. The symbols are the experimental values of $L(\omega)$ which are obtained by taking the arithmetic mean of $L(\omega)$ values obtained parallel and vertical to the shale bedding measured at 1 kHz, 10 kHz and 500 kHz. The error bars represent an uncertainty of about $\pm 5\%$. The curves represent modelled seismo-electric coupling coefficient using the Pride theory represented by Equation (1).

We hypothesise that the difference in the efficacy of different clay minerals with regard to seismo-electric coupling is due to differences in their specific surface areas. Kaolinite has a low specific surface area ($4 - 16 \text{ m}^2/\text{g}$ (Bhattacharyya & Gupta 2009)) and hence provides low zeta potentials that cause only small seismo-electric coupling, while Illite has a much higher specific surface area (about $83 \text{ m}^2/\text{g}$ according to Macht et al. (2011)). There was no Montmorillonite in our samples, but the specific surface area of this clay ($346 \text{ m}^2/\text{g}$ according to Macht et al. (2011)) that it is likely that it would have an extremely large zeta potential and consequent seismo-electric coupling coefficient.

7. Seismo-electric coupling and TOC

The organic carbon component of rocks is expressed by the total organic carbon (TOC) measurement. The value of TOC is expected to be negligible or low in sandstones, but raised in shales because the organic component of some shales provide a source of hydrocarbons and some are exploitable as shale oil and shale gas resources. The lower limit of TOC for gas shales in the USA is considered to be between 2% and 3% (Rezaee & Rothwell 2015; Lu et al. 2012), while some US plays have TOC values extending to 10% (Rezaee & Rothwell 2015). Since the presence of organic carbon is likely to interfere with water-wet flow pathways, we hypothesise that it is possible, even likely, that the effectiveness of seismo-electric coupling will decrease as TOC increases. Assuming that the main applications of seismo-electric methods will be in conventional, non-conventional and shale oil and gas plays, the dependence of the seismo-electric coupling on TOC is important.

We have carried out TOC measurements on our shale samples. All but two of our samples have a TOC in the range 3% to 5%, which makes them potential gas shales. The exceptional two samples had a low TOC, less than 1%. Figure 15c shows the seismo-electric coupling coefficient as a function of TOC. Considering samples with $\text{TOC} > 1\%$ (i.e., those points in the box), the seismo-electric coupling coefficient is inversely proportional with respect to TOC, as hypothesised. At present the cause of the behaviour is not clear. It may be that the presence of kerogen either decreases the porosity of the sample or increases its tortuosity. Either of these effects will lead to a reduction in seismo-electric coupling, as indicated by Equation (6). Indeed, it is most likely that both effects occur because processes that tend to reduce the porosity of rocks also lead to their tortuosity increasing. It is also possible that the zeta potential is decreased for organically-rich shales. It has been observed that the zeta potential is diminished in organoclays due to the formation of inner-sphere complexes (Zacur et al. 2011, 2015).

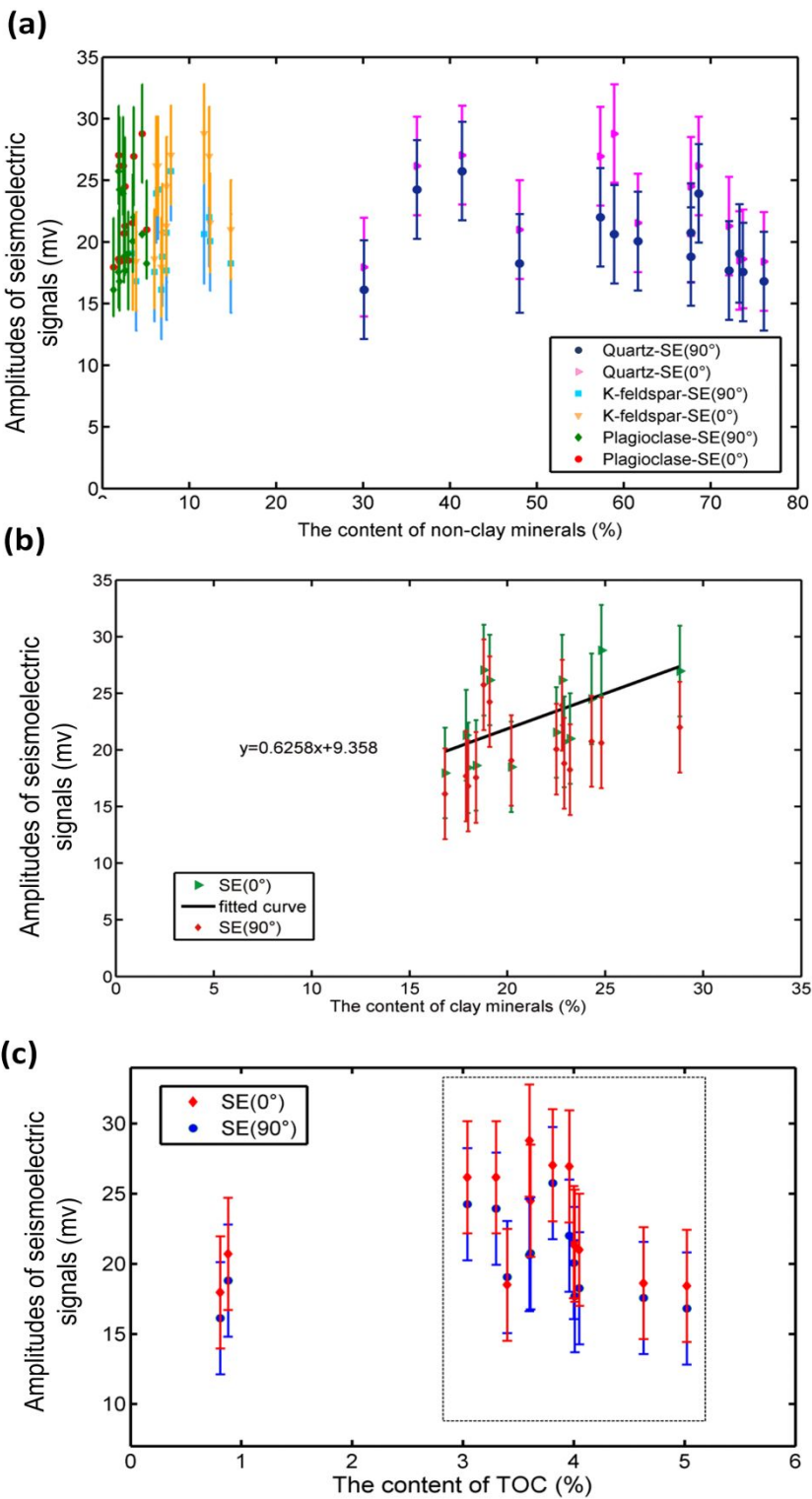


Figure 15. The amplitude of seismo-electric signals versus (a) the fraction of non-clay minerals (quartz, plagioclase, K-feldspar), (b) the fraction of clay minerals and (c) the fraction of TOC (total organic matter). The y-axis is the amplitudes of the seismo-electric signals measured by electrode V1. The measured data have error bars which are all about ± 4 mV.

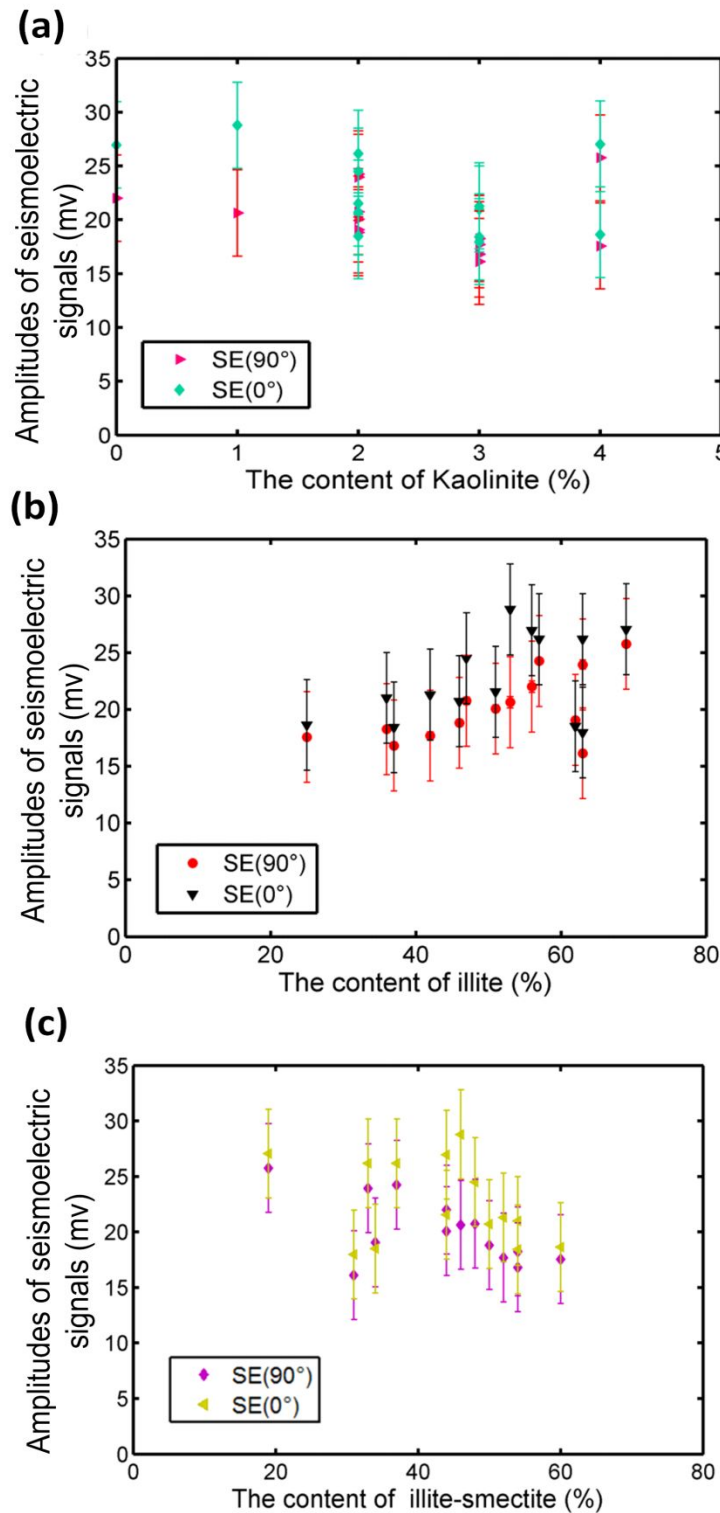


Figure 16. The influence of kaolinite, illite and illite/smectite mixed-layer on seismo-electric coupling. The x -axis is the percentage of minerals in the clay mineral. The y -axis is the amplitudes of the seismo-electric signals measured by electrode V1. The measured data have error bars which are all about ± 4 mV. (a) Kaolinite, (b) Illite, and (c) Mixed Illite/Smectite.

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717 **8. Discussion**

718 In this paper, we focused on what caused the strong seismo-electric conversion in low permeability
719 shale and the influencing factors of seismo-electric conversion in shale.

720 We obtained the relationship between the seismo-electric conversion and clay minerals in shale, but
721 the relationship in sandstones containing clay (shaly sand etc) need to be further studied. The
722 increase of clay will increase the seismoelectric coupling coefficient. We think that no matter
723 whether the clay is in shale or dispersed in sandstone, it will enhance seismo-electric conversion.
724 Hence, the clay contents increase in sandstone will also increase the value of seismoelectric coupling.
725 However, the relationships in shale and sandstones may be not the same. Because the structure of
726 shale and sandstones are totally different, with shale having a much more anisotropic structure. In
727 addition, the mineralogical components of sandstone and shale are different and will give rise to
728 differences in the zeta potential. The zeta potential data of shale in the literature (Figure 13) is
729 relatively small, so the empirical formula fitting out is not comprehensive. Proper zeta potential data
730 of shale need to exclude the influence of surface conductivity, and the types of solutions should be
731 the same, so there is not much data. Further work should carried out to both increase the size of the
732 shale dataset and improve the seismo-electric model to include those aspects not included in the
733 present model.

734 Further experimental measurements of seismo-electric conversion are needed. In this paper, we only
735 examined NaCl solution as the experimental pore fluid. Further experiments changing the type of
736 pore fluid (e.g., oil/water mixtures, pore fluids with different concentrations, saturations and
737 compositions) may improve our knowledge of the fluid effects on seismo-electric conversion. In
738 particular, experiments with multiphase pore fluids are needed to examine if it is possible to obtain
739 seismo-electric signals induced at oil/water and gas/water interfaces; an ability which would help
740 identify the oil/gas target and monitor its production using seismo-electric techniques. This paper
741 reports seismo-electric anisotropy. A fuller study of this may provide parameters which are useful in
742 characterizing the reservoir quality of the rock. This paper has also revealed a link between
743 seismo-electric conversion, clay mineralogy and TOC, all of which need fuller study in order to fully
744 characterize the processes that are occurring with the goal of using them to predict reservoir rock
745 properties from remote seismo-electric measurements in the field.

746 We recognize that the experiments presented here can be improved upon. There would be a
747 particular benefit in characterizing the experimental samples more fully, with additional
748 measurements to determine the CEC and specific surface areas of the samples. Fuller analysis would
749 be possible if the frequency-dependent complex conductivity of the samples were known such that
750 phase lags, surface conductivities and the formation factors of the samples could be calculated. Also,
751 the estimate of the tortuosity of the medium can also be obtained through the formation factor

(Clennell 1997; Jougnot et al. 2009), which will greatly reduce the uncertainty of influencing factors in this study. However, the samples have been used to measure other parameters after the seismoelectric measurements, such as TOC, test of shale elastic parameters and mineral component content tests, which had destroyed the samples, so we are not able to retrieve the conductivity data.

The relationship between the seismoelectric coupling and shale anisotropy need to gain a deep insight in future. In the comparison of numerical simulation and experimental results, the values of the seismoelectric coupling coefficients are the average value of parallel bedding and vertical bedding of shale, which is to simplify the result analysis. In fact, it is easy to find out from the results of this paper (Figure 5, Figure 15, Figure 16) that the amplitude of the seismoelectric signals of parallel bedding and vertical bedding are different. The signals' amplitudes of parallel bedding are larger than those of vertical bedding, which may be mainly related to the anisotropy of shale. Some may say it may be caused by the permeability anisotropy of shale, but we can see from the results of this paper that permeability is not the main factor that caused the strong seismo-electric conversion in shale. The permeability of shale in Sichuan Basin is less than 1 mD (Fu 2017). Comparing the curves below 1 mD (blue, orange and black curves) in Figure 10a, we can see that the three curves almost share the same seismo-electric value in a large range of frequency ($<10^6$ Hz). The seismo-electric conversion in the low permeability shale can be considered as a quasi-static behavior, there is little frequency dispersion, the seismo-electric conversion remains unchanged with the increasing frequency. Hence, the effect of permeability on seismo-electric conversion is not that great in shale, and the observed differences in seismoelectric signals measured parallel and perpendicular to bedding are not caused by the permeability anisotropy. That is to say, the difference of seismo-electric conversion in different directions of shale reflects the anisotropy of shale. We can call this phenomenon the seismo-electric anisotropy of shale. We all know that shale anisotropy can be reflected by elastic waves, while it is first proposed that shale anisotropy can be reflected by seismo-electric conversion. This needs to be confirmed and studied in detail.

Finally, we recognize that although much has been gained by comparing our experimental measurements with a model based on the Pride (1994) equation, that equation does not fully describe shales and that future work must expand the numerical model to include processes occurring in the Stern layer and which lead to Maxwell-Wagner and Interfacial polarization.

9. Conclusions

This paper investigates the characteristics of seismo-electric conversion in shales using experimental measurements and analytical modelling.

Experimental measurements have shown that the passage of acoustic waves through samples of sandstones and shales results in the generation of a seismoelectric voltage signal whose amplitude

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4 786 and phase is a function of frequency and which can be characterized by a seismo-electric coupling
5 787 coefficient. Experimental results show that the amplitude of the seismoelectric coupling coefficient
6 788 in shales is comparable to that exhibited by sandstones. Furthermore, anisotropy which arises from
7 789 bedding in the shales results in anisotropy in the amplitude of the seismoelectric signals.
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10 790 Numerical modelling has been used to examine the effects of varying five parameters in the [Pride](#)
11 (1994) model of seismo-electric conversion in porous media. The parameters investigated were (i)
12 791 dimensionless number, (ii) porosity, (iii) permeability, (iv) tortuosity, and (v) zeta potential. Each of
13 792 these parameters has different effects on the magnitude of the low frequency behaviour of the
14 793 seismoelectric coupling coefficient and/or the frequency at which dispersive behaviour occurs as
15 794 characterised by the critical frequency. While changes in dimensionless number and permeability
16 795 seem to have little effect, seismoelectric coupling coefficient is highly sensitive to changes in
17 796 porosity, tortuosity and zeta potential. Numerical modelling suggests that the cause of the
18 797 seismo-electric conversion in shales, which otherwise have low porosities and permeabilities, is
19 798 enhanced zeta potentials caused by clay minerals. Comparison of numerical modelling with the
20 799 experimental data has supported that conclusion together with an analysis of seismo-electric
21 800 conversion as a function of clay mineral composition.
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29 802 Numerical modelling has also shown that the frequency-dependent behaviour can be safely ignored
30 for most natural porous media because it occurs only at frequencies much higher than those
31 803 encountered in conventional seismology (usually >10 kHz).
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35 805 The sensitivity of seismo-electric coupling to the content of clay minerals and the TOC in shale
36 806 suggests that seismo-electric exploration may have future potential for the characterization of clay
37 807 minerals in conventional reservoirs as well as in characterising shale gas and shale oil reservoirs.
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51 814 **References**

52
53 815 Alali, F., 2007. Dependence of NMR and SIP parameters on clay content, master's thesis, Berlin University of
54 816 Technology.
55
56 817 Arulanandan, K., 1969. Hydraulic and electrical flows in clays, *Clays & Clay Minerals*, **17**(2), 63–76.
57 818 Bhattacharyya, K.G. & Gupta, S.S., 2009, Adsorptive accumulation of Cd(II), Co(II), Cu(II), Pb(II) and Ni(II) ions
58 from water onto Kaolinite: Influence of acid activation, *Adsorption Science and Technology*, **27**(1), 47-68.
59 819
60

- Boudreau, B.P., 1996. The diffusive tortuosity of fine-grained unlithified sediments, *Geochimica Et Cosmochimica Acta*, **60**(16), 3139–3142.
- Bruggeman, D.A.G., 1935. Berechnung verschiedener physikalischer Konstanten von heterogenen Substanzen. I. Dielektrizitätskonstanten und Leitfähigkeiten der Mischkörper aus isotropen Substanzen, *Ann. Phys*, **416** (7), 636–664.
- Butler, K. E., Russell, R. D., Kepic, A. W. & Maxwell, M., 1996. Measurement of the seismo-electric response from a shallow boundary, *Geophysics*, **61**(6), 1769–1778.
- Butler, K.E., Kepic, A.W. & Rosid, M.S., 2002. An experimental seismo-electric survey for groundwater exploration in the Australian outback, *SEG Technical Program Expanded Abstracts*, **21**(1), 1484.
- Charlaix, E., Kushnick, A.P. & Stokes, J.P., 1988. Experimental study of dynamic permeability in porous media, *Physical Review Letters*, **61**(14), 1595–1598.
- Chen, L., Zhang, L., Kang, Q., Yao, J. & Tao, W.Q., 2015. Nanoscale simulation of shale transport properties using the lattice Boltzmann method: permeability and diffusivity, *Sci Rep*, **5**, 8089.
- Chorom, M. & Rengasamy, P., 1995. Dispersion and zeta potential of pure clays as related to net particle charge under varying ph, electrolyte concentration and cation type, *European Journal of Soil Science*, **46**(4), 657–665.
- David, C., 1993. Geometry of flow paths for fluid transport in rocks, *Journal of Geophysical research*, **98**(B7), 12267–12278.
- Debye, P. & Hückel E., 1923. Theory of electrolytes: I. lowering of freezing point and related phenomena, *Phys. Z*, **24**, 185–206.
- Deng, J.X., Wang, H., Zhou, H, Liu, Z.H., Song, L.T. & Wang, X.B., 2015. Microtexture, seismic rock physical properties and modeling of Longmaxi Formation shale, *Chinese J. Geophy.*, **58**(6), 2123–2136.
- Dupuis, J.C., Butler, K.E., Kepic, A.W. & Harris, B.D., 2009. Anatomy of a seismo-electric conversion: measurements and conceptual modeling in boreholes penetrating a sandy aquifer, *Journal of Geophysical Research Solid Earth*, **114**(B10).
- Epstein, N., 1989. On tortuosity and the tortuosity factor in flow and diffusion through porous media, *Chemical Engineering Science*, **44** (3), 777–779.
- Fosu-Duah, E.L., Padmanabhan, E. & Gamez, V. J., 2016. Electrophoretic behavior of some shales from the Setap Formation, Sarawak Basin, Malaysia, *International Journal of Applied Engineering Research*, **11**, 337–341.
- Fosu-Duah, E. L., Padmanabhan, E. & Gamez, V. J., 2017. Characteristic zeta potential of selected oligocene-Miocene shale from the Setap Formation onshore Sarawak-Malaysia, *International Journal of Applied Engineering Research*, **12**, 6360–6368.
- Fu, C.Q., 2017. Study on Reservoir Characteristics and Shale Gas Enrichment of Wufeng-Longmaxi Formation Shale in Southeast Chongqing, Doctor's dissertation, China University of Mining and Technology, Beijing.
- Jougnot, D., Revil, A. & Leroy, P., 2009. Diffusion of ionic tracers in the Callovo-Oxfordian clay-rock using the Donnan equilibrium model and the formation factor, *Geochimica et Cosmochimica Acta*, **73**(10), 2712–2726.
- Jougnot, D., Revil, A., Lu, N. & Wayllace, A., 2011. Transport properties of the callovo-oxfordian clay rock under partially saturated conditions, *Water Resources Research*, **46**(8), 416–428.
- Glover, P. W.J., 2009. What is the cementation exponent? A new interpretation. *Leading Edge* (Tulsa, OK), **28**(1), 82–85.
- Glover, P.W.J., 2012. *Geophysical Properties of the Near Surface Earth: Electrical Properties*, Treatise on

- Geophysics, Second Edition, pp. 89–137. Elsevier.
- Glover, P.W.J. & Déry, N., 2010. Streaming potential coupling coefficient of quartz glass bead packs: Dependence on grain diameter, pore size, and pore throat radius, *Geophysics*, **75**(6), F225–F241.
- Glover, P.W.J. & Jackson, M.D. (2010). Borehole electrokinetics, *The Leading Edge*, **29**(6), 724–728.
- Glover, P.W.J., Walker, E., Ruel, J. & Tardif, E., 2012a. Frequency dependent streaming potential of porous media—Part 2: Experimental measurement of unconsolidated materials, *Int. J. Geophys.*, **2012**, 1–17.
- Glover, P.W.J., Walker, E. & Jackson, M.D., 2012b. Streaming-potential coefficient of reservoir rock: A theoretical model, *Geophysics*, **77**(2), D17–D43.
- Haartsen, M.W. & Pride, S.R., 1997. Electro seismic waves from point sources in layered media, *J. Geophys. Res.*, **102**(1022), 24745–24784.
- Haines, S.H. & Pride, S.R., 2006. Seismo-electric numerical modeling on a grid, *Geophysics*, **71**(6), 57–65.
- Haines, S.H., Pride, S.R., Klemperer, S.L. & Biondi, B., 2007. Seismo-electric imaging of shallow targets, *Geophysics*, **72**(2), G9–G20.
- Johnson, D., Koplik, J. & Dashen, R., 1987. Theory of dynamic permeability and tortuosity in fluid-saturated porous media, *Journal of Fluid Mechanics*, **176**, 379–402.
- Jouniaux, L. & Zyserman, F. I., 2016. A review on electrokinetically induced seismo-electrics, electro-seismics, and seismo-magnetics for Earth sciences, *Solid Earth*, **7**, 249–284.
- Kitagawa, Y., Yorozu, Y. & Itami, K., 2001. Zeta potentials of clay minerals estimated by an electrokinetic sonic amplitude method and relation to their dispersibility, *Clay Sci*, **11**(4), 329–336.
- Kozak, M.W. & Davis, E.J., 1989. Electrokinetics of concentrated suspensions and porous media: 2. moderately thick electrical double layers, *Journal of Colloid & Interface Science*, **129**(1), 166–174.
- Kröger, B., Yaramanci, U. & Kemna, A., 2009. Modelling of seismo-electric effects, *The COMSOL Conference 2008, Hannover*.
- Leroy, P. & Revil, A., 2004. A triple-layer model of the surface electrochemical properties of clay minerals, *Journal of Colloid and Interface Science*, 371–380.
- Leroy, P. & Revil, A., 2009. A mechanistic model for the spectral induced polarization, *Journal of Geophysical Research*, **114**, B10202.
- Leroy, P., Tournassat, C., Bernard, O., Devau, N. & Azaroual, M., 2015. The electrophoretic mobility of montmorillonite. Zeta potential and surface conductivity effects, *Journal of Colloid and Interface Science*, **451**, 21–39.
- Liu, Z. C., Zhang, F. & Li, X. Y., 2019, Elastic anisotropy and its influencing factors in organic-rich marine shale of southern China, *Science China Earth Sciences*, **62**(11), 1805–1818.
- Lu, S., Huang, W., Chen, F., Li, J., Wang, M., Xue, H., Wang, W. & Cai, X., 2012. Classification and evaluation criteria of shale oil and shale gas resources: Discussion and application, *Petrol. Explor. Dev.* **39**(2), 268–276.
- Macht, F., Eusterhues, K., Pronk, G.J. & Totsche, K.U., 2011. Specific surface area of clay minerals: Comparison between atomic force microscopy measurements and bulk-gas (N₂) and -liquid (EGME) adsorption methods, *Applied Clay Science*, **53**(1), 20–26.
- Müller, R.H., 1996. Zeta potential und partikeladung in der laborpraxis, Doctor's dissertation, Free University of Berlin, Berlin
- Peng, R., Wei, J.X., Di, B.R., Ding, P.B. & Liu, Z.C., 2016. Experimental research on seismo-electric effects in

- sandstone, *Applied Geophysics*, **13**(3), 425–436.
- Peng, R., Di, B.R., Wei, J.X., Ding, P.B., Zhao, J., Pan, X. & Liu, Z.C., 2017. Experimental study of the seismo-electric interface response in wedge and cavity models, *Geophysical Journal International*, **210**, 1703–1720.
- Peng, R., Glover, P.W.J., Di, B., Wei, J., Lorinczi, P., Ding, P. & Liu, Z., 2018a. The Effect of Rock Permeability and Porosity on Seismo-electric Conversion, *80th EAGE Conference and Exhibition 2018 Proceedings. Denmark*.
- Peng, R., Di, B., Wei, J., Glover, P.W.J., Lorinczi, P., Ding, P. & Liu, Z., 2018b. The Seismo-electric Coupling in Shale, *80th EAGE Conference and Exhibition 2018 Proceedings. Denmark*.
- Pride, S., 1994. Governing equations for the coupled electromagnetics and acoustics of porous media, *Physical Review B Condensed Matter*, **50**(21), 15678.
- Pride, S. & Morgan, F. D., 1991. Electrokinetic dissipation induced by seismic waves, *Geophysics*, **56**(7): 914–925.
- Revil, A. & Glover, P. W. J., 1997. Theory of ionic-surface electrical conduction in porous media, *Physical Review B - Condensed Matter and Materials Physics*, **55**(3), 1757–1773.
- Revil, A. & Glover, P. W. J., 1998. Nature of surface electrical conductivity in natural sands, sandstones, and clays. *Geophysical Research Letters*, **25**(5), 691–694.
- Revil, A., Pezard, P. A. & Glover, P. W. J., 1999a. Streaming potential in porous media: 1. theory of the zeta potential, *Journal of Geophysical Research Solid Earth*, **104**(B9), 20021–20031.
- Revil, A., Schwaeger, H., Iii, L. M. C. & Manhardt, P. D., 1999b. Streaming potential in porous media: 2. theory and application to geothermal systems, *Journal of Geophysical Research Solid Earth*, **104**(B9), 20033–20048.
- Revil, A., Leroy, P. & Titov, K., 2005. Characterization of transport properties of argillaceous sediments: Application to the Callovo-Oxfordian argillite, *Journal of Geophysical Research-Solid Earth*, **110**, B06202.
- Revil, A., Jardani, A., Sava, P. & Haas, A., 2015. *The Seismo-electric Method: Theory and Application*. Wiley-Blackwell.
- Rezaee, R. & Rothwell, M., 2015. Gas shale: Global significance, distribution and challenges, In *Fundamentals of Gas Shale Reservoirs*, First Edition, Ed. Reza Rezaee, John Wiley & Sons Inc.
- Samyukta, K., Pashin, J. C., Ramsey, J. D. & Clark, P.E., 2017. Quantitative characterization of polyacrylamide–shale interaction under various saline conditions, *Petroleum Science*, **14**(3), 586–596.
- Santiwong, S. R., Guan, J. & Waite, T. D., 2008. Effect of the ionic strength and pH on the hydraulic properties of accumulating solid assemblages during microfiltration of montmorillonite suspensions, *J. Colloid Interface Sci.*, **317**(1), 214–227.
- Saomoto, H. & Katagiri, J., 2015. Direct comparison of hydraulic tortuosity and electric tortuosity based on finite element analysis, *Theoretical and Applied Mechanics Letters*, **5**(5), 177–180.
- Schakel, M. D., Smeulders, D. M. J., Slob, E. C. & Heller, H. K. J., 2011a. Laboratory measurements and theoretical modeling of seismo-electric interface response and coseismic wave fields, *J. Appl. Phys.*, **109**(7), 074903.
- Schakel, M. D., Smeulders, D. M. J., Slob, E. C. & Heller, H. K. J., 2011b. Seismo-electric interface response: experimental results and forward model, *Geophysics*, **76**(4), N29–N36.
- Shaw, D. J., 1992. *The solid-liquid interface, in Introduction to Colloid and Surface Chemistry*. 4th edn (pp. 151–173). Butterworth-Heinemann.
- Thompson, A. H. & Gist, G. A., 1993. Geophysical applications of electrokinetic conversion, *The Leading Edge*,

- 12, 1169–1173.
- Thompson, A. H., Hornbostel, S., Burns, J., Murray, T., Raschke, R. & Wride, J., et al., 2007. Field tests of electroseismic hydrocarbon detection, *Geophysics*, **72**(72), 565.
- Tournassat, C., Chapron, Y., Leroy, P., Bizi, M. & Boulahya, F., 2009. Comparison of molecular dynamics simulations with triple layer and modified Gouy-Chapman models in a 0.1 M NaCl-montmorillonite system, *Journal of Colloid and Interface Science*, **339**(2), 533–541.
- Tournassat, C., Grangeon, S., Leroy, P. & Giffaut, E., 2013. Modeling specific pH dependent sorption of divalent metals on montmorillonite surfaces. a review of pitfalls, recent achievements and current challenges, *American Journal of Science*, **313**(5), 395–451.
- Walker, E. & Glover, P.W.J., 2010. Permeability models of porous media: Characteristic length scales, scaling constants and time-dependent electrokinetic coupling, *Geophysics*, **75**(6), E235–E246.
- Walker, E., Glover, P.W.J. & Ruel, J., 2014. A transient method for measuring the DC streaming potential coefficient of porous and fractured rocks, *Journal of Geophysical Research: Solid Earth*, **119**, 957–970.
- Walker, E. & Glover, P.W.J., 2018. Measurements of the relationship between microstructure, pH and the streaming and zeta potentials of sandstones, *Transport in Porous Media*, **121** (1), 183–206.
- Wang, L., Chen, Y. & Liu Y., 2014. Shale Porous Structural Characteristics of Longmaxi Formation of Southeast Sichuan Basin, *China Petroleum Exploration*, **19**(5), 80–88.
- Wang, J., Hu, H., Guan, W., 2015a, Experimental measurements of seismoelectric signals in borehole models, *Geophys. J. Int.*, **203**(3): 1937–1945
- Wang, J., Hu, H., Guan, W., Li, H., 2015b. Electrokinetic experimental study on saturated rock samples: zeta potential and surface conductance, *Geophys. J. Int.*, **201**(2): 869–877.
- Wang, J., Hu, H. & Guan, W., 2016. The evaluation of rock permeability with streaming current measurements, *Geophys. J. Int.*, **206** (3): 1563–1573.
- Wang, J., Zhu, Z., Gao, Y. et al., 2020. Measurements of the seismoelectric responses in a synthetic porous rock. *Geophys. J. Int.* (under reviewing).
- Xie, Q., Chen, Y., You, L., Hossain, M.M. & Saeedi, A., 2018. Drivers of wettability alteration for oil/brine/kaolinite system: implications for hydraulic fracturing fluids uptake in shale rocks, *Energies*, **11**(7), 1666.
- Zacur S.S., Naranjo P.M. & Farfán, E.M., 2011. Organoarcillas: Propiedades electrocinéticas y estructura, *Anales XVII Congreso Argentino de Físicoquímica*, Córdoba, Argentina.
- Zacur, S., Sham, E. & Torres, E.M.F., 2015. Measure of zeta potential of titanium pillared clays, *Procedia Materials Science*, **8**, 599–607.
- Zhang, T. S., Yang, Y., Gong, Q. S., Liang, X., Wei, X. F., 2014, Characteristics and Mechanisms of the Micro-pores in the Early Marine Shale, Southern Sichuan Basin. *Acta Geologica Sinica*, **88**(9), 1728–1740.
- Zhu, Z., Toksöz, M.N. & Burns, D.R., 2008. Electrostatic and seismo-electric measurements of rock samples in a water tank, *Geophysics*, **73**(5), E153–E164.
- Zhu, Z. & Toksöz, M.N., 2013. Experimental measurements of the streaming potential and seismo-electric conversion in Berea sandstone, *Geophysical Prospecting*, **61**(3), 688–700.
- Zhu, Z., Toksöz, M.N. & Zhan, X., 2015, Seismo-electric measurements in a porous quartz-sand sample with anisotropic permeability, *Geophysical Prospecting*, **64**(3), 700–713.

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Modified theoretical model for temperature-dependent electro-kinetic properties of porous media

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Abstract

I report theoretical modelling of streaming potential coefficient C_{sp} and zeta potential ζ in porous media with NaCl electrolytes as a function of temperature up to 150°C. The modelling captures the essential behaviour of both C_{sp} and ζ as a function of both temperature, salinity and pH, the latter of which also depends on temperature and salinity. The modelling also shows ζ which is negative, decreasing in magnitude with temperature, and as pH becomes smaller. It also shows that C_{sp} is also negative but has a complex temperature and pH dependence. These behaviours are controlled mainly by the temperature dependence of pH, but also depend on the temperature dependence of the density, viscosity, relative permittivity and electrical conductivity of the pore solution. The model has been compared to experimental data for sandstones up to 150°C, and describes the data reasonably well when using physically reasonable values for the electro-chemical parameters describing the electrical double layer. This modelling has implications for the calculation of electro-kinetic parameters in geothermal reservoirs, CCUS targets and radioactive storage underground.

1. INTRODUCTION

The temperature dependence of the steady-state streaming potential coefficient C_{sp} and zeta potential ζ of rock samples are important because their magnitude and frequency-dependence (Tardif et al., 2011; 2012; Glover, 2018; Glover et al., 2012; 2020; Walker and Glover, 2018) determine the size of coupling between fluid and electrical flow in techniques such as electro-seismics (Saunders et al., 2008; Dupuis et al., 2009). Applications of these electro-kinetic parameters are found in hydrocarbon, geothermal resources, aquifers and prospective sites for

carbon capture and storage by deep underground injection, and the storage of radioactive waste in the subsurface. All these applications involve elevated temperatures.

However, it is not clear whether C_{sp} and ζ of rock samples increases or decreases with temperature. Previous measurement and modelling have focused on the variation of C_{sp} and ζ with pH, frequency and microstructure (Walker et al., 2014; Glover, 2018; Walker and Glover, 2018; Peng et al., 2020). Vinogradov and Jackson (2015) have collated the existing data on the temperature dependence of these electro-kinetic phenomena and found data which showed apparently conflicting behaviour as temperature increases (Reppert and Morgan, 2003a; 2003b; Ishido and Mizutani, 1981; Alekhin et al., 1984; Dunstan, 1994). Vinogradov and Jackson (2015) attributed the observed contrasting behaviour to the pH, which was controlled in some studies and free to vary with temperature in others. This explanation not only explains the apparently conflicting behaviour but is also supported by their data and conclusions.

Vinogradov and Jackson (2015) made electro-kinetic measurements on four sandstones for two salinities (0.01 mol/dm³ and 0.5 mol/dm³) as a function of temperature up to about 150°C. They found that the ζ was negative, decreasing in magnitude with increasing temperature at low ionic strength (0.01 mol/dm³), and remaining independent of temperature at high ionic strength (0.5 mol/dm³). They measured electrolyte pH and observed its decrease with increasing temperature at low salinity with no verifiable change with temperature at high salinity, which led to their assertion that the temperature dependence of the ζ can be explained by the temperature dependence of the pH. Vinogradov and Jackson (2015) carried out some basic modelling of only the ζ , finding it consistent with the approximate formulation by Revil et al. (1999a), which was implemented by Saunders et al. (2012) and discussed in Glover (2018).

The first theoretical model that was capable of providing both the ζ and C_{sp} efficiently for rock samples with individual pore microstructural parameters was reported by Glover et al. (2012) who combined the Revil and Glover (1997; 1998) and Revil et al. (1999a) method for calculating the ζ from the electro-chemical relationships arising from the electrical double layer (EDL) with the modifications to the Helmholtz-Smoluchowski formula (e.g., Glover (2015)) to take account of the pore microstructure (Glover and Déry, 2010).

This paper modifies the Glover et al. (2012) model so that it can be used to model the variation of C_{sp} and ζ as a function of temperature, recognising that many of its input parameters are dependent on both temperature and salinity, and then attempts to model the Vinogradov and Jackson (2015) data.

2. THE MODEL

The model I use in this study is a modified and re-implemented version of that of Glover et al. (2012) incorporating all known temperature dependencies and implemented for the temperature range 0 – 150°C and for salinities between 10⁻⁵ to 1 mol/dm³. Details of the model can be found in the text below, and the supplementary information (Glover, 2026) includes the model implemented in Excel with active graphics.

2.1 Streaming potential and zeta potential

The streaming potential of capillary tubes is known to be governed by the Helmholtz-Smoluchowski (HS) equation (e.g., Overbeek, 1952; Hunter, 1981; Maineult et al., 2004; Saunders et al., 2008), which is given in its simplest form by

$$C_s = \frac{\Delta V}{\Delta P} = \frac{\varepsilon_r \varepsilon_o \zeta}{\eta_f \sigma_f^*}, \quad (1)$$

where the streaming potential C_s (in V/m) is the ratio of the measured streaming potential ΔV (in V) to the applied fluid pressure difference ΔP (in Pa) that drives the fluid through the capillary tube. This value depends upon the electric permittivity of the pore fluid $\varepsilon_f = \varepsilon_r \varepsilon_o$ (in F/m), ε_r is the relative permittivity of the pore fluid (unitless), ε_o is the electric permittivity of free space ($\approx 8.854 \times 10^{-12}$ F/m), the dynamic viscosity of the pore fluid η_f (in Pa.s), the pore fluid electrical conductivity σ_f^* (in S/m) and the zeta potential ζ (in V). The zeta potential is the electric potential on the shear plane when a part of the diffuse layer is transported by fluid flow. Equation 1 is valid for bundles of capillary tubes but is commonly applied to porous media including rocks even though it has never been validated for these media since there exists no independent measurement of the zeta potential for complex porous media.

The approach used in this work is a combination of fundamental theory and the use of empirical relationships. Theoretical models have been used for the heart of the model, i.e., to calculate the Debye screening length, Stern plane potential, zeta potential, surface conductance and streaming potential coefficient of the rock. Empirical relationships have been used to allow certain of the model's input parameters to vary as a function of temperature and fluid salinity instead of using input data from tables; these are the density, electrical conductivity, dynamic viscosity, the disassociation constant of water and the relative electric permittivity of the bulk fluid.

In parallel with the approach of other researchers, I have initially treated dielectric saturation (Pride and Morgan, 1991) and visco-electric effects (Lyklema and Overbeek, 1961; Hunter, 1966; Pride and Morgan, 1991) as negligible, the justification for which is given in Glover et al. (2012).

The fluid conductivity in Equation 1 has contributions from the bulk fluid and surface conduction. The latter depends both on the surface conductance and the microstructural properties of the rock (porosity, cementation exponent, grain size etc.) as discussed in detail in Glover and Déry (2010). Consequently, Equation 1 can be rewritten as (Morgan et al., 1989)

$$C_s = \frac{\Delta V}{\Delta P} = \frac{\varepsilon_r \varepsilon_o \zeta}{\eta_f \left(\sigma_f + \frac{2 \Sigma_s}{\Lambda} \right)}, \quad (2)$$

where Σ_s is the specific surface conductance (in S) and Λ is a length scale characteristic of the pore microstructure (in m) introduced by Johnson et al. (1987). In this paper I follow the approach of Glover and Déry (2010) by substituting the relationship of Revil and Cathless (1999) for the characteristic length scale ($\Lambda = d / (3(F - 1))$) into Equation 2 to obtain an equation for the streaming potential coefficient in terms of the mean grain diameter and formation factor of the rock giving Equation 4 of Glover and Déry (2010)

$$C_s = \frac{\Delta V}{\Delta P} = \frac{d \varepsilon_f \zeta}{\eta_f \left(d \sigma_f + 6 \Sigma_s (F - 1) \right)}, \quad (3)$$

where d is the mean grain diameter (in m) and $F = \phi^m$ is the formation factor of the rock. It is worthwhile noting that I have chosen to work with grain diameter in this modelling. However, other equations in Glover and Déry (2010) allow the modelling to be carried out as a function of pore radius or pore throat radius if one wishes, and these are also included in Figure 1, which shows the main functional dependencies of the whole model.

The zeta potential and surface conductance have been calculated according to the model of Revil and Glover (1997; 1998) and Revil et al. (1999a). While the modelling has been carried out as a function of bulk fluid pH , it should be noted that the pH of the solution decreases in the diffuse layer as one approaches the mineral surface. Since the position of the shear plane is very close to the surface (a distance of about 2.4×10^{-10} m according to Revil and Glover (1997)), it is possible that the zeta potential would be significantly altered by the perturbation of the pH close to the mineral surface. So, there is the question over whether the bulk fluid property is the most appropriate one to use, or whether an adjusted pH value should be used in the calculation of the zeta potential.

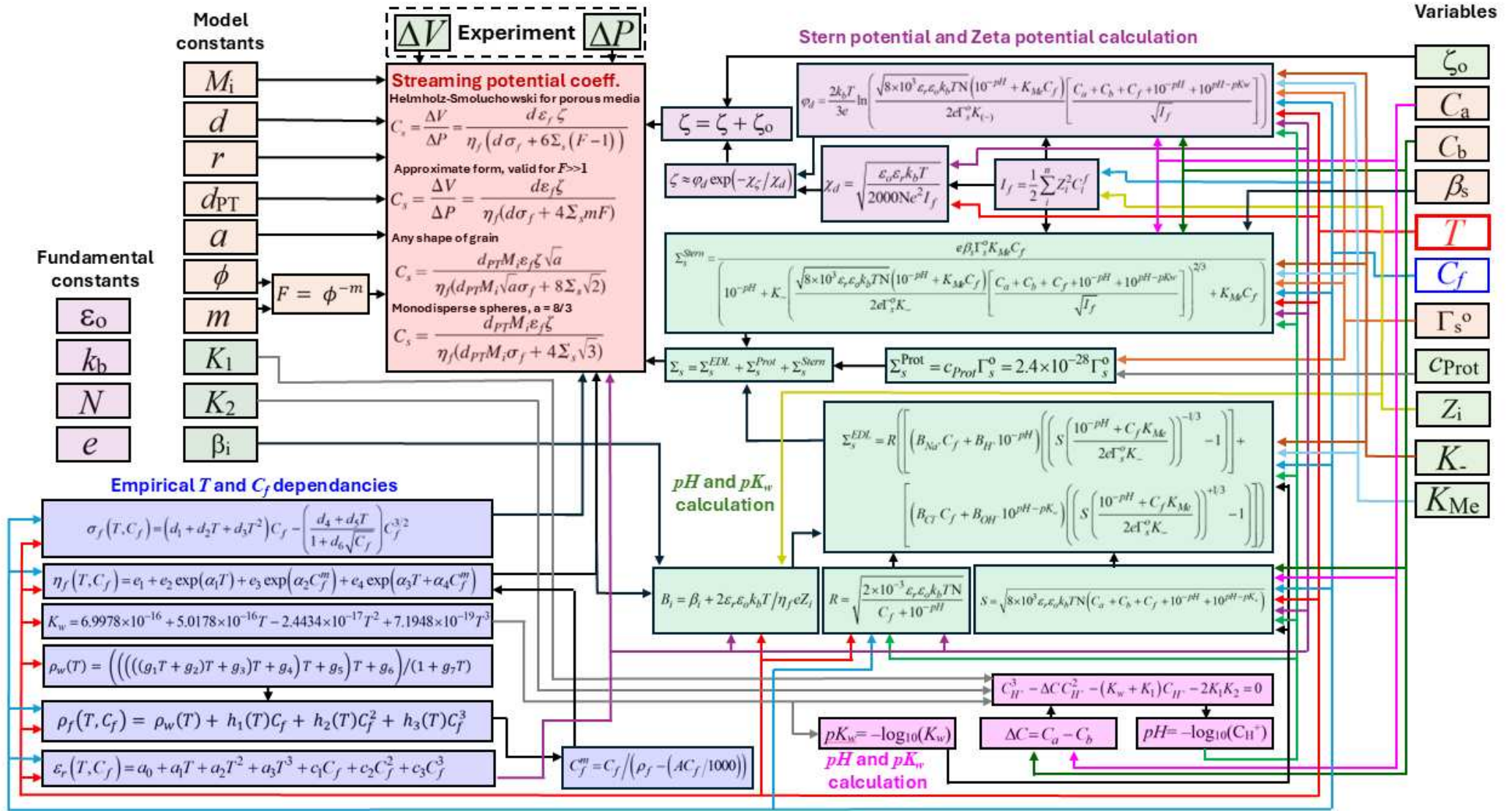


Figure 1. Please see over.

Figure 1. Illustration of the main known influences in the model, which are described in Section 2 of this work, C_s (in V/m) is the ratio of the measured streaming potential ΔV (in V) to the applied fluid pressure difference ΔP (in Pa), $\epsilon_f = \epsilon_r \epsilon_0$ (in F/m) is the electric permittivity of the pore fluid, ϵ_r is the relative permittivity of the pore fluid (unitless), ϵ_0 is the electric permittivity of free space ($\approx 8.854 \times 10^{-12}$ F/m), η_f (in Pa.s) is the the dynamic viscosity of the pore fluid, σ_f^* (in S/m) is the pore fluid electrical conductivity, and ζ (in V) is the zeta potential. The symbol Λ is a length scale characteristic of the pore microstructure (in m), d is the mean grain diameter (in m), r is the grain radius (in m), d_{PT} is the pore throat diameter (in m), m is the cementation exponent (no units), ϕ is the porosity or fluid fraction, a is a grain packing parameter whose value is 8/3 for spheres, M_i is the pore size to pore throat ratio for the i^{th} packing, and $F = \phi^m$ is the formation factor of the rock; α_i , a_i , c_i , e_i , h_i and d_i are various empirical parameters with different units, T is the operant temperature (in °C or K, depending in the equation (see text)), and C_f is the salinity of the bulk pore fluid (mol/dm³) and C_f^m is the molality of that solution. K_w is the disassociation constant of water, and K_1 and K_2 are reaction coefficients of the reaction of CO₂ with water. C_a and C_b are the acid and base concentrations (in mol/dm³) needed to be added to attain a given pH. $k_b = 1.38 \times 10^{-23}$ is Boltzmann's constant (J/K), $e = 1.602 \times 10^{-19}$ is the elementary charge (C), $N = 6.022 \times 10^{23}$ is Avogadro's constant (/mol), I_f is the ionic strength (mol/dm³), Z_i is the valence of the i^{th} species, K_{Me} , is the binding constant for cation (usually sodium) adsorption on quartz, (-), $K_{(-)}$ or K_{-} is the disassociation constant for dehydrogenisation of silanol surface sites, Γ_s^0 is the surface site density (/m²). ϕ_d is the Stern plane potential (V), χ_d is the Debye screening length (m), and $\chi_\zeta = 2.4 \times 10^{-10}$ is the shear plane distance, (m). The values of B_i are the equivalent ionic mobilities, and β_i are the ionic mobilities for each ion (both in m²/s.V), while β_s is the stern plane mobility. Σ_s is the surface conductivity, which is the sum of contributions from the EDL (Σ_s^{EDL}), from proton and electron conduction (Σ_s^{Prot}), and from the Stern layer (Σ_s^{Stern}), and where c_{Prot} is the proton surface conductance contribution rate.

However, I have no mechanism for making this adjustment because the surface conduction theory of [Revil and Glover \(1997\)](#) and [Revil et al. \(1999a\)](#) only accounts for pH variation in the bulk fluid. I am forced, therefore, to make the unsupported assumption that the

second-order pH effects close to the surface, including their effect on the zeta potential, are negligible for most pore fluid salinities.

The following subsections consider the development of the theoretical model, referencing those parts that have been drawn from previous work, followed by a subsection considering the main model parameters.

2.2 Theoretical Development

This subsection contains a full theoretical treatment of the temperature-dependent electrokinetic modelling of porous media presented in this work. Further supporting information can be found in [Glover et al. \(2012\)](#). It is possible to calculate the streaming potential coefficient of reservoir rocks thanks to theoretical developments that began in 1997 ([Revil and Glover, 1997; 1998; Revil et al, 1999a](#)) and have been supplemented and modified ([Glover and Déry, 2010, Walker and Glover, 2010](#)). Here I combine them to provide a model for calculating the streaming potential coefficient as a function of 18 fundamental parameters including temperature, pore fluid concentration and pH , as well as rock parameters such as porosity, grain size, pore size and formation factor.

Examination of [Equation 2](#) shows that in order to calculate the streaming potential coefficient I need to obtain (i) the zeta potential, (ii) the electrical permittivity of the pore fluid, (iii) the viscosity of the pore fluid, (iv) the bulk fluid conductivity, (v) the surface conductance, and (vi) some measure of the characteristic length scale of the pores as discussed in [Walker and Glover \(2010\)](#). Here I use the grain size and [Equation 4](#) in [Glover and Déry \(2010\)](#), but one might equally well use the pore radius or the pore throat radius together with the respective equation in [Glover and Déry \(2010\)](#) instead.

The calculation method follows the following steps:

1. Calculation of the electrical conductivity of the pore fluid as a function of temperature and pore fluid concentration using the empirical model of [Sen and Goode \(1992a; 1992b\)](#).
2. Calculation of the relative permittivity of the pore fluid as a function of temperature and pore fluid concentration using the empirical model of Olhoeft (unpublished note, 1980, as mentioned by [Revil et al., \(1999b\)](#)).
3. Calculation of the mass density of the pore fluid as a function of salinity and temperature using the approach of [Kell \(1975\)](#) and [Novotny and Söhnel \(1988\)](#).

4. Calculation of the molality of the pore fluid as a function of salinity and temperature, with consequent calculation of the pore fluid viscosity as a function of temperature and pore fluid molality using the empirical model of [Phillips et al. \(1978\)](#).
5. Definition of the physical chemistry of the double layer based on the substrate (here silica) and the fluid (here aqueous solutions of NaCl in equilibrium with dissolved CO₂).
6. Calculation or imposition of the *pH* of the solution.
7. Calculation of the Debye screening length and the shear plane distance ([Revil and Glover, 1997; 1998; Revil et al., 1999a](#)).
8. Calculation of the Stern plane potential ([Revil and Glover, 1997; 1998; Revil et al., 1999a](#)).
9. Calculation of the zeta potential ([Revil and Glover, 1997; 1998; Revil et al., 1999a](#)).
10. Calculation of the surface conductance ([Revil and Glover, 1997; 1998; Revil et al., 1999a](#)).
11. Use of one of the forms of the modified Helmholtz-Smoluchowski equation given in [Glover and Déry \(2010\)](#), to finally calculate the streaming potential coefficient as a function of either (i) grain size, (ii) pore size, or (iii) pore throat size.

These steps are also given in the large summary diagram shown as [Figure 1](#). It will be noted that the use of the Helmholtz-Smoluchowski equation or modified Helmholtz-Smoluchowski equation ([Equations 1 to 3](#)) is the last step in the streaming potential coefficient calculation, and has a temperature dependence solely because most of its input parameters have a temperature dependence.

It should be noted that this work considers how temperature change affects the physical properties of the fluid (viscosity, bulk conductivity, dielectric permittivity, *pH* etc). However, in many field applications, temperature can also enhance chemical reactions, which can also modify fluid *pH*, ionic strength, conductivity, and therefore surface properties ([Bernabé et al., 2003](#)). These second order effects can be very significant, but are specific to the particular applications and would need to be considered when modelling any specific field application.

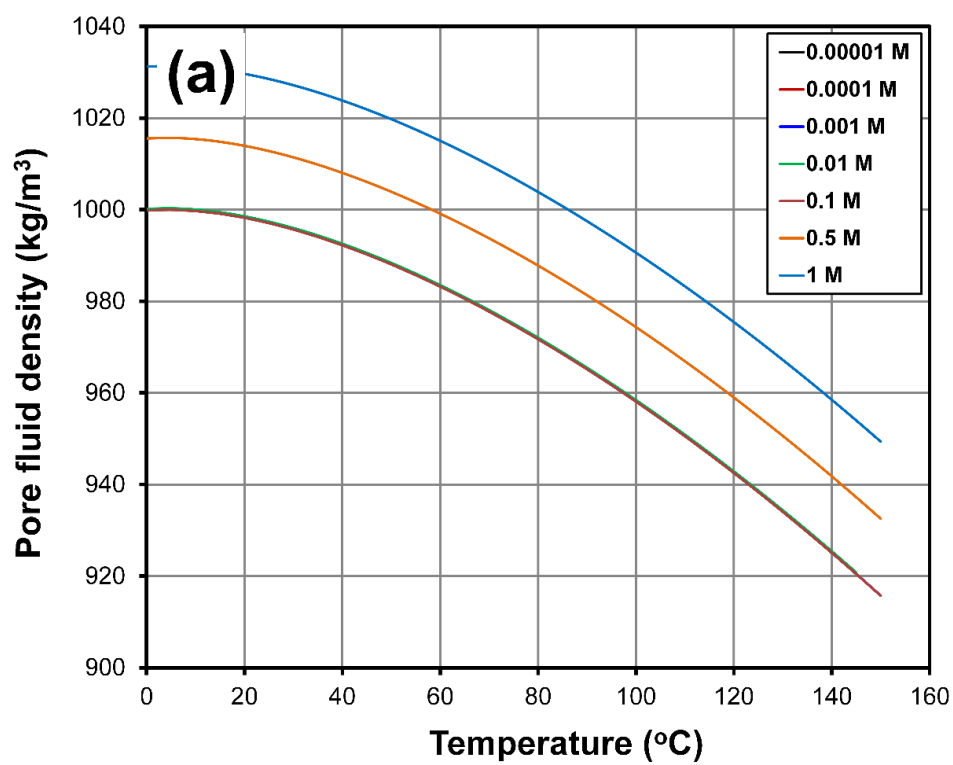
There are a large number of input parameters in the model. Not all of them have a large impact upon the final streaming potential coefficient values, and this will be described in more detail in a further scientific contribution.

The input parameters fall into 5 groups:

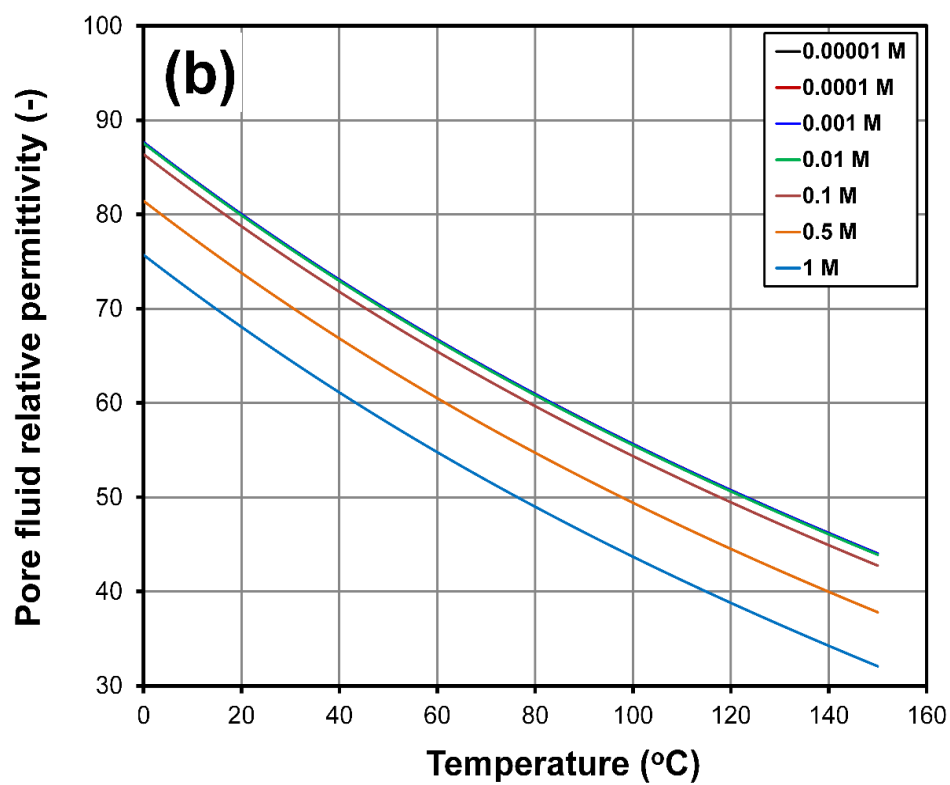
- (i) Fundamental constants (e.g., Boltzmann's constant and Avogadro's number).
- (ii) Environmental conditions (e.g., temperature).
- (iii) Fluid parameters (e.g., solute concentration, pH , pK_w , pK_1 and pK_2).
- (iv) Rock microstructure parameters (e.g., formation factor, cementation exponent, porosity, grain size etc.).
- (v) Rock-fluid interface parameters, i.e., the electro-chemical parameters associated with surface adsorption reactions (e.g., pK_{me} , pK_-).

All the following modelling has been carried out as a function of pore fluid concentration assuming that the pore fluid is aqueous NaCl and that all changes in pH can be accommodated by the addition of small amounts of HCl or NaOH. Computations were carried out in Microsoft Excel from 10^{-5} to 10 mol/dm³, with 5 points per decade, and in the range $4.5 \leq pH \leq 9$, where I note that for a silica surface the point of zero charge is about pH 3 ($pH_{pzc} \approx 3$), which the pH at which the EDL collapses. The methods for calculating the Stern plane potential and hence zeta potential used in this modelling become only approximately valid at $pH \leq 5$ (Revil et al., 1999a). Modelling has also been carried out as a function of temperature in the range $0^\circ\text{C} \leq T \leq 100^\circ\text{C}$. Extension beyond this range is only a matter of finding empirical relationships for the density, electrical conductivity, dynamic viscosity, dielectric permittivity and disassociation constant of the bulk fluid which are valid at the higher temperatures and also include pressure. There have been few studies on the effect of temperature upon those parameters entering the HS equation, the most important being those of Reppert and Morgan (2003a; 2003b).

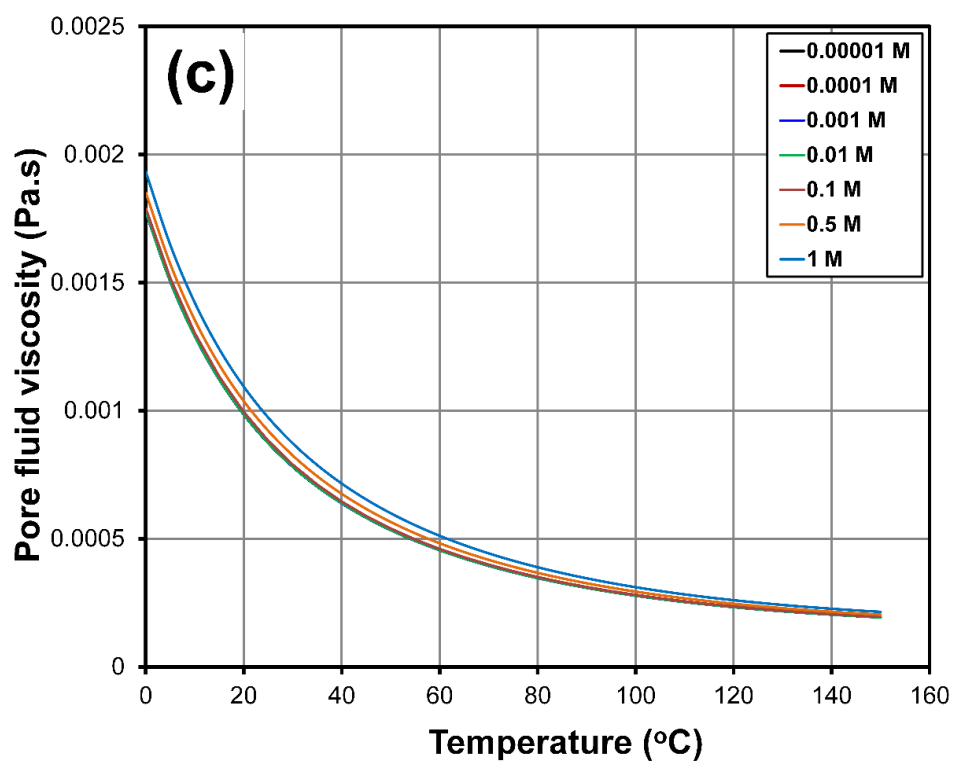
Following the Roman numeral numbered list above, I consider each step in the following sub-sections, with the temperature dependence of some of the main input parameters shown in [Figure 2](#).



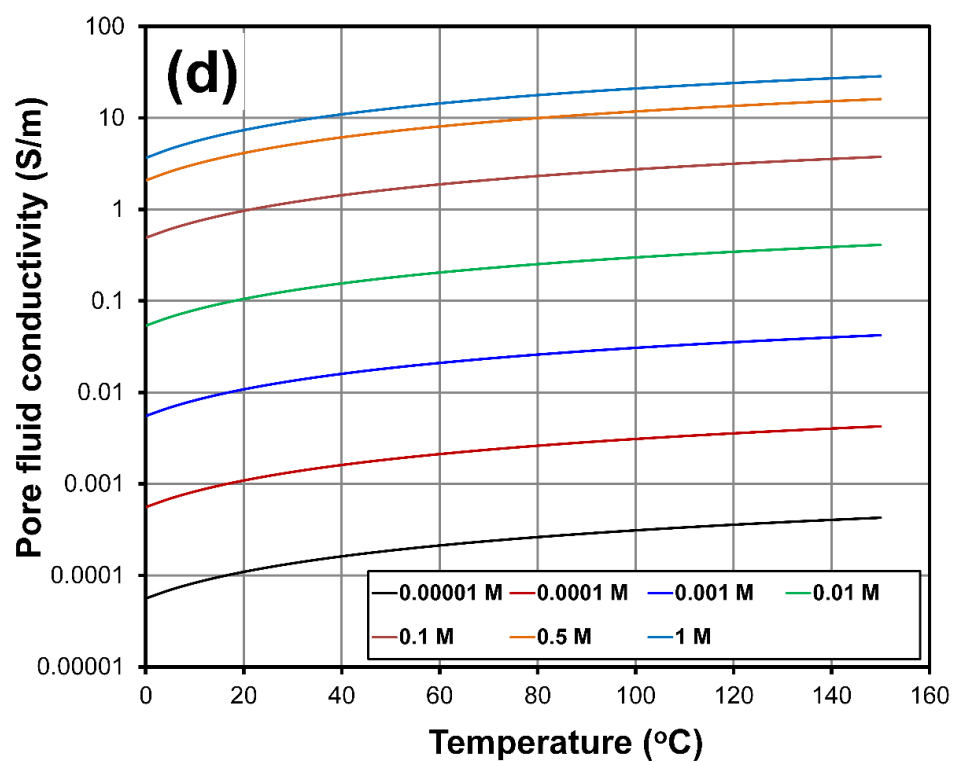
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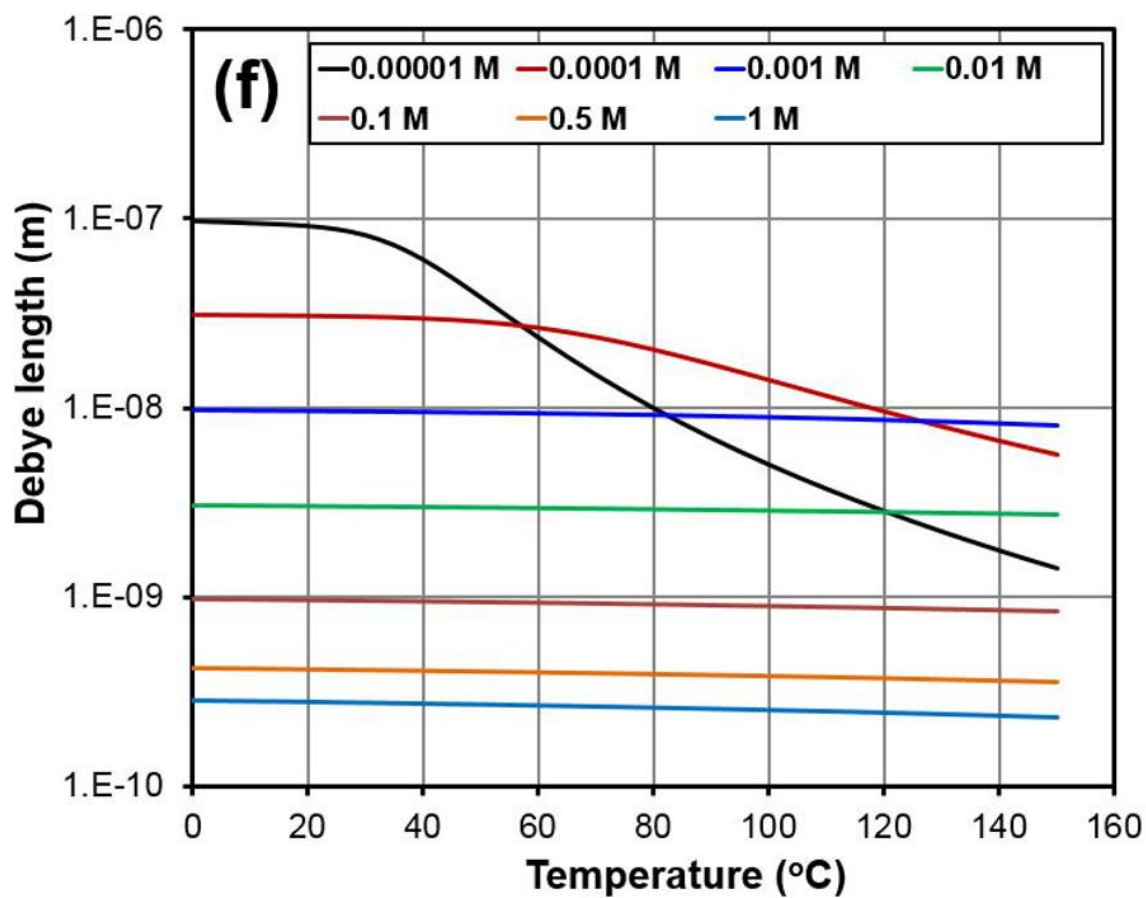
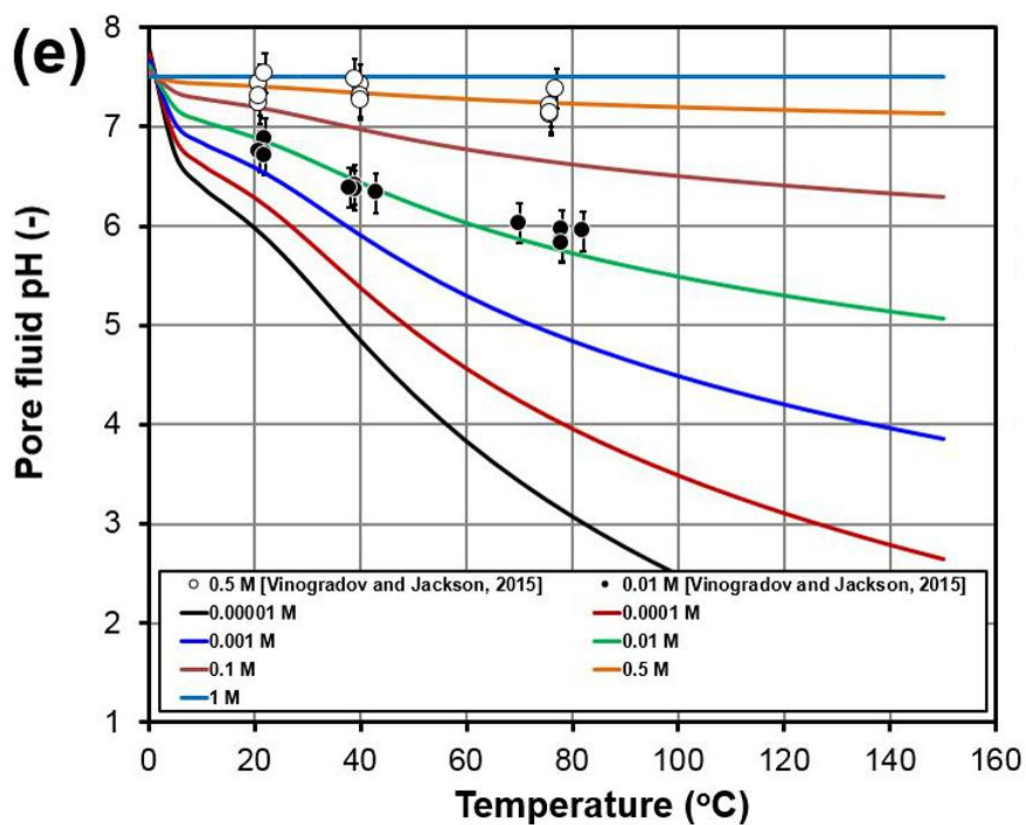
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Figure 2. The temperature and salinity dependencies of the main empirical parameters entering into the model: (a) pore fluid density, (b) pore fluid relative permittivity, (c) pore fluid viscosity, and (d) the pore fluid electrical conductivity, noting that the molality calculated in the model is only used to calculate the viscosity and is consequently not shown. In addition, I show two intermediate parameters calculated by the model and used to calculate the ζ and C_{sp} : (e) the pH as a function of temperature and salinity together with the measured values and uncertainties for $C_f = 0.01 \text{ mol/dm}^3$ and $C_f = 0.5 \text{ mol/dm}^3$, and (f) the Debye length.

2.2.1 Pore fluid conductivity

I have used the corrected [Sen and Goode \(1992a; 1992b\)](#) empirical equation to provide the electrical conductivity of the bulk solution (in S/m) (noting the later reference provides an important correction to the equation)

$$\sigma_f(T, C_f) = (d_1 + d_2 T + d_3 T^2) C_f - \left(\frac{d_4 + d_5 T}{1 + d_6 \sqrt{C_f}} \right) C_f^{3/2}, \quad (4)$$

where $d_1 = 5.6 \text{ (S} \cdot \text{dm}^3/\text{m} \cdot \text{mol)}$, $d_2 = 0.27 \text{ (S} \cdot \text{dm}^3/\text{m} \cdot \text{mol})/^\circ\text{C}$, $d_3 = -1.51 \times 10^{-4} \text{ (S} \cdot \text{dm}^3/\text{m} \cdot \text{mol})/^\circ\text{C}^2$, $d_4 = 2.36 \text{ (S/m}/(\text{mol/dm}^3)^{3/2})$, $d_5 = 0.099 \text{ (S/m}/(\text{mol/dm}^3)^{3/2}/^\circ\text{C})$, $d_6 = 0.214 \text{ ((mol/dm}^3)^{-1/2})$, T is in $^\circ\text{C}$ and C_f is the salinity of the bulk pore fluid (mol/dm^3). The variation of pore fluid conductivity with temperature is shown in [Figure 2d](#).

2.2.2 Pore fluid relative electric permittivity

Most researchers take a value of $\epsilon_f = \epsilon_r \epsilon_0 = 80 \times 8.854 \times 10^{-12} \text{ F/m}$ (e.g., [Revil et al., 1999a; 1999b](#)), which would correspond approximately to the bulk fluid being an aqueous solution at 25°C . However, I wish to model the streaming potential coefficient as a function of pore fluid salinity and temperature, both of which affect the value of the relative dielectric permittivity.

Consequently, I have determined the relative permittivity ϵ_r (no units) using Olhoeft's empirical equation (Olhoeft, unpublished note, 1980 mentioned in [Revil et al., 1999b](#)) as

$$\epsilon_r(T, C_f) = a_0 + a_1 T + a_2 T^2 + a_3 T^3 + c_1 C_f + c_2 C_f^2 + c_3 C_f^3, \quad (5)$$

where $a_0 = 295.68$, $a_1 = -1.2283 \text{ K}^{-1}$, $a_2 = 2.094 \times 10^{-3} \text{ K}^{-2}$, $a_3 = -1.41 \times 10^{-6} \text{ K}^{-3}$, $c_1 = -13.00 \text{ dm}^3/\text{mol}$, $c_2 = 1.065 \text{ (dm}^3/\text{mol})^2$, $c_3 = -0.03006 \text{ (dm}^3/\text{mol})^3$, T is in kelvin and the equation is valid in the

range from 273 K to 373 K and C_f is the salinity of the bulk pore fluid in mol/dm³. The permittivity *in vacuo* $\epsilon_0=8.854\times10^{-12}$ F/m (Lide, 2009).

The variation of pore fluid relative electric permittivity with temperature is shown in Figure 2b.

It is debatable whether the value of ϵ_r for the bulk fluid is the most appropriate in this modelling. This is because the value of ϵ_r diminishes within the diffuse layer as one approaches the mineral surface (within a few angstroms), taking values as low as 5 near the Stern Plane (Grahame, 1950). It would, perhaps, be more correct to use a smaller value of ϵ_r that would consider the reduction of the permittivity in the diffuse layer, which would give larger values of calculated zeta potential at high salinities, and which seem to be supported by experimental determinations (Jaafar, 2009; Jaafar et al., 2009). Jaafar et al. found that above 0.4 mol/dm³ (i.e., for Debye lengths less than 4.65×10^{-10} m) the zeta potential becomes constant and may even increase slightly, however they attributed the behaviour to surface charge density saturation not to dielectric saturation.

Unfortunately, at present there exists no theoretical model capable of providing us with a more appropriate effective permittivity value as such a value would depend on the microstructure of the rock. I have carried out some rudimentary CRIM based mixing calculations using the method of Iglesias and Peon Fernandez (2001) with a simplified rock microstructure and the results support the supposition that the dielectric saturation effect is negligible if the entire pore space contributes to the permittivity used by the modified H-S equation. Additionally, both Booth (1951) and Hunter (1966) provide a model of relative permittivity as a function of local field strength that indicates that the dielectric saturation effect is negligible for most bulk electrolyte solutions (Pride and Morgan, 1991), but again, it is assumed that the whole of the pore fluid contributes to the permittivity to be used in the H-S equation.

One might consider that the appropriate permittivity to be used in the H-S equation is that at the shear plane. In which case it would be some exponential mixture of that at the Stern plane, which presumably approaches 5 after Grahame (1950), and that at the Bjerrum length from the surface, where the fluid ions are more affected by thermal agitation than the presence of the Stern layer, and where the permittivity approaches that given by Equation 5 above. Under this hypothesis, permittivities of the order of 20 or less would be perfectly possible.

Finally, I note that if I replace the relative electric permittivity calculated by Equation 5 in the model by the value of 5 for all salinities, the result is that the streaming potential

coefficient is increased (becomes less negative) by an order of magnitude and zeta potential is decreased (becoming more negative) by approximately 15 mV, which is of the same order of magnitude as the experimental observations of zeta potential at high salinities made by Jaafar et al. (2009).

In this work I have chosen to use the values of fluid permittivity for the bulk fluid (i.e., Equation 5) because I considered the solution of the permittivity problem too large to include in this paper. However, I note that often the results of the model give positive zeta potentials at high salinities and low pHs that are not reproduced in experimental results. In these cases, I have added a constant zeta potential contribution, ζ_0 , to that calculated by the model for all salinities. This *ad hoc* parameter improves the behaviour of the model at high salinity data, and is fully described in the Section 3. I expect that this parameter will not be required once the model contains a method for including the appropriate electric permittivity.

2.2.3 Pore fluid density

Pore fluid density is also required by the model, primarily to help calculate the pore fluid viscosity. The calculation is a two-step process. First, I calculate the density of pure water using the Kell equation (Kell, 1975), which is given by

$$\rho_w(T) = \frac{\left(\left(\left(\left(g_1 T + g_2\right) T + g_3\right) T + g_4\right) T + g_5\right) T + g_6}{1 + g_7 T}, \quad (6)$$

where; ρ_w is the density of water in kg/m^3 , T is the temperature ($^{\circ}\text{C}$), and the empirical parameters are $g_1 = -2.8054253 \times 10^{-10}$ (in $\text{kg/m}^3/^{\circ}\text{C}^5$), $g_2 = 1.0556302 \times 10^{-7}$ (in $\text{kg/m}^3/^{\circ}\text{C}^4$), $g_3 = -4.6170461 \times 10^{-5}$ (in $\text{kg/m}^3/^{\circ}\text{C}^3$), $g_4 = -0.0079870401$ (in $\text{kg/m}^3/^{\circ}\text{C}^2$), $g_5 = 16.945176$ (in $\text{kg/m}^3/^{\circ}\text{C}$), $g_6 = 999.83952$ (in kg/m^3), and $g_7 = 0.01687985$ (in $/^{\circ}\text{C}$).

The density of the pore fluid containing a given amount of NaCl can be calculated using the Novotný & Söhnel (1988) equation

$$\rho_f(T, C_f) = \rho_w(T) + h_1(T)C_f + h_2(T)C_f^2 + h_3(T)C_f^3, \quad (7)$$

where; $\rho_f(T, C_f)$ is the solution density (g/dm^3), $\rho_w(T)$ is the density of pure water (g/dm^3) at a given temperature T , C_f is the salinity of the solution in mol/dm^3 , and the empirical parameters h_1 to h_3 are temperature-dependent functions, which can be found in Novotný & Söhnel (1988).

The variation of pore fluid density with temperature is shown in Figure 2a.

2.2.4 Pore fluid viscosity

The dynamic viscosity η_f of the pore fluid (in Pa.s) was calculated using the [Phillips et al. \(1978\)](#) empirical equation

$$\eta_f(T, C_f) = e_1 + e_2 \exp(\alpha_1 T) + e_3 \exp(\alpha_2 C_f^m) + e_4 \exp(\alpha_3 T + \alpha_4 C_f^m), \quad (8)$$

where $e_1=4.95166 \times 10^{-5}$ Pa.s, $e_2=6.034658 \times 10^{-4}$ Pa.s, $e_3=9.703832 \times 10^{-5}$ Pa.s, $e_4=1.025107 \times 10^{-3}$ Pa.s, $\alpha_1= -0.06653081$ /°C, $\alpha_2= 0.1447269$ /molal, $\alpha_3= -0.02062455$ /°C, $\alpha_4= 0.1301095$ /molal, T is in °C and C_f^m is the molality of the bulk pore fluid. For weak solutions it is possible to say that $C_f^m \approx C_f$, and consequently to use salinity in mol/dm³ in place of molality in [Equation 8](#). For stronger solutions it is necessary to calculate the density of the pore fluid and convert C_f into C_f^m using $C_f^m = C_f / \left(\rho_f - (AC_f/1000) \right)$, where ρ_f is the pore fluid density in g/cm³ and A is the atomic mass of the salt in g/mol (here for NaCl, $A=58.44$ g/mol). Although the difference is small even at high salinities, I have implemented the conversion in our model for all salinities. The variation of pore fluid viscosity with temperature is shown in [Figure 2c](#).

As discussed in [Glover et al. \(2012\)](#), visco-electric effects have been assumed to be negligible in accordance with the results of [Lyklema and Overbeek \(1961\)](#), [Hunter \(1966\)](#) and the interpretation of [Pride and Morgan \(1991\)](#). The separation at which the electrostatic interaction between two elementary charges is comparable in magnitude to the thermal energy scale is termed the Bjerrum length λ_b , and is given by $\lambda_b = e^2 / 4\pi\epsilon_r\epsilon_0 k_b T$. For an aqueous solution of NaCl and salinities less than about 0.4 mol/dm³, the Bjerrum length remains constant at $\lambda_b = 7.16 \times 10^{-10}$ m, increasing to a value of 13.5×10^{-10} m at 3.98 mol/dm³. The value of λ_b represents a threshold between a regime where fluid ions are more affected by thermal agitation than the presence of the Stern layer and the regime where the Stern layer dominates, and where the local field strength is sufficient to alter the viscosity of the fluid considerably. [Figure 3](#) in [Glover et al. \(2012\)](#) shows (i) the Bjerrum length λ_b , (ii) the Debye screening length χ_d , (iii) the thickness of the EDL (conventionally, twice the Debye screening length $2\chi_D$), and (iv) the shear plane distance $\chi_\zeta = 2.4 \times 10^{-10}$ m which has been taken from [Revil and Glover \(1997\)](#), all as a function of salinity. They showed that the Bjerrum length is greater than the EDL thickness for all salinities greater than about 0.5 mol/dm³. This is the same as saying that inter-ionic electrostatic interactions are significant for the entire EDL, including all of its mobile fraction ($\chi > \chi_\zeta = 2.4 \times 10^{-10}$ m) for salinities greater than 0.5 mol/dm³, and it is these electrostatic interactions that increase the fluid viscosity in the double layer. This viscosity perturbation is not taken account of in the Helmholtz-Smoluchowski equation which instead

uses the bulk fluid viscosity rather than the local viscosity in the EDL. The bulk viscosity is perturbed negligibly by the local viscosity changes in such a thin layer near the mineral surface, which justifies our approach and that of previous authors (e.g., [Pride and Morgan, 1991](#)).

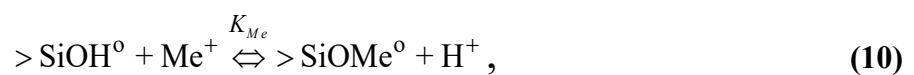
2.2.5 Physical chemistry of the EDL

The value of the Stern plane potential, and hence the zeta potential depends upon the surface chemistry which is different for different substrates. This paper confines itself to a silica surface. The reactions at a quartz surface in the presence of aqueous fluids has already been described well by many authors (e.g., [Iler, 1979](#); [Revil and Glover, 1997](#); [1998](#); [Revil et al., 1999a](#)). There are two types of neutral surface group. The first is doubly coordinated siloxal $>\text{Si}_2\text{O}^0$, which can be considered inert, and other is singly coordinated silanol $>\text{SiOH}^0$, which is reactive and gives $>\text{SiOH}_2^+$ when $pH < pH_{pzc}$, and $>\text{SiO}^-$ when $pH > pH_{pzc}$ (the symbol $>$ in the chemical formulae represents the silica crystal framework). The symbol pH_{pzc} represents the point of zero charge for the mineral, and is the pH at which $[>\text{SiOH}_2^+] = [>\text{SiO}^-]$, where square brackets indicate concentration. For silica $pH_{pzc} \approx 3$ ([Lorne et al., 1999](#)).

Following the approach of [Revil et al. \(1999a\)](#) this paper considers a 1:1 electrolyte with single valency anions and cations in the pH range 6-8 where the surface reactions can be written



and



where the metal cation is Me^+ , and for most of our modelling it represents Na^+ , though it might equally well represent K^+ or some other single valence cation. Note that the positive surface site $>\text{SiOH}_2^+$ does not occur in [Equation 9](#) and [10](#) because at $pH > 6$ I have assumed that $(>\text{SiOH}_2^+) = 0$. Hence, there results three sites types; two neutral ($>\text{SiOH}^0$ and $>\text{SiOMe}^0$) and one negative ($>\text{SiO}^-$).

The position of the equilibrium in [Equation 9](#) is controlled by the disassociation constant for dehydrogenization of silanol surface sites $K_{(-)}$. If $K_{(-)}$ is large, [Equation 9](#) has an equilibrium towards the left, giving very many fewer $>\text{SiO}^-$ sites than $>\text{SiOH}^0$ sites, and *vice versa*. The position of the equilibrium in [Equation 10](#) is controlled by the binding constant for cation (sodium) adsorption on quartz K_{Me} . If K_{Me} is small, [Equation 10](#) has an equilibrium towards the left, where there are very many fewer $>\text{SiOMe}^0$ sites than $>\text{SiOH}^0$ sites, and *vice*

versa. Hence, $K_{(-)}$ and K_{Me} , which are unitless, describe the relative concentrations of the three surface sites. The final parameter of interest is the total surface site density Γ_s° (in m^{-2}), which is the sum of the surface site densities for each of the three types of surface site.

2.2.6 Fluid pH

The last fluid property that is required is the fluid pH . Knowledge of the fluid pH is needed to calculate the surface conductance and Stern plane potential.

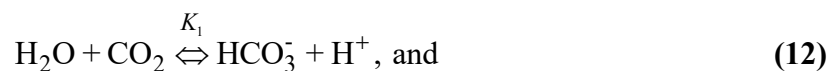
For an aqueous electrolyte in contact with the atmosphere the pH is defined by (a) the disassociation constant of water K_w , which varies with temperature, (b) the reaction of water with carbon dioxide from the atmosphere in a 2-step process, and (c) the addition of acid, and/or (d) the addition of a base.

The value of K_w varies with temperature. I have obtained an equation by fitting a polynomial to K_w data which was sourced from [Lide \(2009\)](#) and which is valid in the range $0^\circ\text{C} \leq T \leq 100^\circ\text{C}$

$$K_w = 6.9978 \times 10^{-16} + 5.0178 \times 10^{-16} T - 2.4434 \times 10^{-17} T^2 + 7.1948 \times 10^{-19} T^3 \quad (11)$$

It is worthwhile noting that this variation implies that the pH of pure water is 7.47 at 0°C , about 7 at 25°C and 6.14 at 100°C . The relationship is likely approximately valid in the range $100^\circ\text{C} \leq T \leq 150^\circ\text{C}$ which is the limit of the data space explored in this model, but it should be noted that the fit in this range is more approximate because it is an extrapolation outside the original [Lide \(2009\)](#) fitting data.

The reaction of water with atmospheric carbon dioxide occurs in two steps



I have used equilibrium constants $K_1 = 10^{-7.53}$ and $K_2 = 10^{-10.3}$ after [Wu et al. \(1991\)](#) and [Revil et al. \(1999a\)](#). These two constants may depend upon temperature. I have assumed them to be independent of temperature. Implementing temperature dependencies of K_1 and K_2 is one area where this model could be improved.

The result of [Equation 11](#) and [Processes 12](#) and [13](#) is that pure water in equilibrium with atmospheric carbon dioxide tends to be acidic. However, this initial acidity is reduced substantially when the fluid is equilibrated with the silica of the rock matrix. The process fluid can, of course, take any value of pH depending on the addition of an acid or a base (or both).

In this paper, I assume that the pH is imposed on the system. This choice is associated with the constraint that the model can be used with the data in [Vinogradov and Jackson \(2015\)](#) later in this paper as a case study. Modelling is only possible if the pH is measured during the experimental process and published with the other data. So many potentially useful datasets cannot be modelled because the experimental pH is either not measured, not recorded or not published. Consequently, the model assumes that a more acidic *pH* is obtained by adding HCl to the fluid in a concentration equal to C_a (in mol.L⁻¹), and a more basic *pH* is obtained by adding NaOH to the fluid in a concentration equal to C_b (in mol/dm³). Each addition of acid or base will perturb the concentration of Cl⁻ ions and Na⁺ ions in the solution, respectively. Both acid and base can be added together and their effect on *pH* will balance out, however, an augmentation of the concentration of Cl⁻ ions and Na⁺ ions in the solution will still occur. This effect is taken account of in the model by the terms 10^{-pH} for hydrogen ions, $10^{(pH-pK_w)}$ for hydroxyl ions, C_a for chloride ions that are added as HCl, and C_b for sodium ions that are added as NaOH. Hence, the model accounts for the addition of acid and base to the pore fluid solution to ensure that the ionic fluid concentrations in the model reflect those in a real pore fluid at a given *pH*.

The *pH* of the silica-H₂O-CO₂ system is defined by the cubic equation ([Revil et al., 1999a](#))

$$C_{H^+}^3 - \Delta C C_{H^+}^2 - (K_w + K_1) C_{H^+} - 2K_1 K_2 = 0 \quad (14)$$

where $\Delta C = C_a - C_b$ and $pH = -\log_{10}(C_{H^+})$.

Consequently, there are two ways that a link can be built between acid and base concentrations and *pH* into the model. The first is to define the *pH* and then calculate the respective concentrations of acid and base required, which will then give us the augmentation in Cl⁻ and Na⁺ ions in the pore fluid. If this approach is taken, it must assumed that either C_a or C_b is zero to ensure that the solution of [Equation 14](#) is unique. The second approach is to solve [Equation 14](#) for given values of C_a and C_b , and use the *pH* thus calculated in the model. I have implemented both approaches. The first is more convenient for fitting the model to experimental data where the *pH* of the pore fluid is known and has been used for the results presented in this paper, while the second is more convenient if the recipe for the pore fluid is defined, for example, in experiment planning.

[Figure 2e](#) shows an example of the modelled pH as a function of temperature showing the greater functional dependence of the lower salinity solutions, and the agreement of the model with experimental data from [Vinogradov and Jackson \(2015\)](#).

2.2.7 Debye screening length and shear plane distance

The Debye screening length is given by

$$\chi_d = \sqrt{\frac{\epsilon_o \epsilon_r k_b T}{2000 N e^2 I_f}} \quad (15)$$

where

$k_b = 1.38 \times 10^{-23}$	Boltzmann's constant	(J/K)	Lide (2009)
$e = 1.602 \times 10^{-19}$	elementary charge	(C)	Lide (2009)
$N = 6.022 \times 10^{23}$	Avogadro's constant	(/mol)	Lide (2009)
$\epsilon_o = 8.854 \times 10^{-12}$	electrical permittivity <i>in vacuo</i>	(F/m)	Lide (2009)
ϵ_r	relative electrical permittivity	(-)	From Equation 5
I_f	ionic strength	(mol/dm ³)	From Equation 16
T	temperature	(K)	Model variable

The factor of 2000 arises due to the units for ionic strength being here mol/dm³. This equation is often found cited with a 2 in place of the 2000, in which case the units for I_f would be mol/m³.

The ionic strength is given by

$$I_f = \frac{1}{2} \sum_i^n Z_i^2 C_i^f \quad (16)$$

In this work the acid and alkali contributions are usually small compared with that of the dissolved NaCl. Hence for the most common scenarios where $C_f > 10^{-5}$ mol/dm³ and $5 < pH < 9$, it may be said that $I_f \approx C_f$.

There currently does not exist a method for independently evaluating the shear plane distance. Instead, a commonly followed approach is to use the value obtained by [Revil and Glover \(1997\)](#) when they fitted their model to the data of [Scales et al. \(1990\)](#), which is $\chi_\zeta = 2.4 \times 10^{-10}$ m.

The value of the Debye screening length generated in this work varied from 1.75×10^{-11} m to 9.68×10^{-8} m for $C_f = 3.98$ mol/dm³ and $T = 100^\circ\text{C}$, and $C_f = 10^{-5}$ mol/dm³ and $T = 0^\circ\text{C}$, respectively, both at $pH = 7$. The thickness of the diffuse layer may be considered to be approximately twice the Debye length. For $C_f = 0.4$ mol/dm³ at $pH = 7$ and $T = 25^\circ\text{C}$ Equation 13 gives $2\chi_d = 9.31 \times 10^{-10}$ m. Since $2\chi_d \geq \chi_\zeta$, a significant amount of the double layer (74.2%) is moved during electro-kinetic coupling, and this amount increases swiftly to approach 100% of the double layer for lower pore fluid concentrations. The condition that $2\chi_d = \chi_\zeta$ is reached at a concentration of 3.395 mol/dm³, but streaming potentials are possible at higher salinities because the EDL decays away exponentially, ensuring that the arbitrary limit on its size given

by $2\chi_d$ is not a true reflection on the extent of the EDL. The EDL extends beyond $2\chi_d$, but very weakly, and it is this weak portion of the EDL that gives rise to the small streaming potential coefficients at concentrations greater than 3.395 mol/dm^3 .

Figure 2f shows the variation of the Debye length with salinity and temperature, capturing the smaller Debye lengths for the higher salinity fluids, and noting that there is only a small decrease in Debye length with temperature up to 150°C for most fluids (those with $C_f > 5 \times 10^{-4} \text{ mol/dm}^3$). However, there is a much more pronounced decrease in Debye length with increase in temperature for very low salinity solutions ($C_f = 10^{-4} \text{ mol/dm}^3$ and $C_f = 10^{-5} \text{ mol/dm}^3$) which we associate later with the cause for the apparently positive streaming potential coefficients for these low salinities.

2.2.8 Stern Plane potential

The Stern plane potential was calculated using the techniques described in [Revil and Glover \(1997\)](#) and in [Revil et al. \(1999a\)](#), and is given by

$$\phi_d = \frac{2k_b T}{3e} \ln \left(\frac{\sqrt{8 \times 10^3 \epsilon_r \epsilon_o k_b T N} (10^{-pH} + K_{Me} C_f)}{2e \Gamma_s^o K_{(-)}} \left[\frac{C_a + C_b + C_f + 10^{-pH} + 10^{pH - pK_w}}{\sqrt{I_f}} \right] \right) \quad (17)$$

where:

$k_b = 1.38 \times 10^{-23}$	Boltzmann's constant	(J/K)	Lide (2009)
$e = 1.602 \times 10^{-19}$	elementary charge	(C)	Lide (2009)
$N = 6.022 \times 10^{23}$	Avogadro's constant	(/mol)	Lide (2009)
$\epsilon_o = 8.854 \times 10^{-12}$	electrical permittivity <i>in vacuo</i>	(F/m)	Lide (2009)
$Z_i = 1$	valence of the ionic species <i>i</i>	(-)	Lide (2009)
ϵ_r	relative electrical permittivity	(-)	From Equation 5
I_f	ionic strength	(mol/dm ³)	From Equation 16
K_{Me}	binding constant for cation (usually sodium) adsorption on quartz	(-)	Model variable
$K_{(-)}$	disassociation constant for dehydrogenisation of silanol surface sites	(-)	Model variable
C_a	acid concentration	(mol/dm ³)	From Equation 14
C_b	base concentration	(mol/dm ³)	From Equation 14
Γ_s^o	surface site density	(/m ²)	Model variable
C_f	concentration of the solution	(mol/dm ³)	Model variable
pH	<i>pH</i>	(-)	Model variable
T	temperature	(K)	Model variable

This equation is the same as that developed in Glover et al. (2012) but is used here to explore the temperature dependence of the Stern potential which is both explicit and hidden in the temperature dependence of many of its other parameters.

While the fundamental constants are known to great accuracy, there is some considerable uncertainty over the exact value of other parameters such as K_{Me} , $K_{(-)}$, and Γ_s^o as discussed earlier, in Revil et al. (1999a), and in detail in Section 3 below.

It will be noted that Equation (17), though the same as that in Glover et al. (2012) is more comprehensive than the version given in Revil et al. (1999a). This is because the addition of the C_b and 10^{pH-pK_w} terms allow it to be applied for a full range of pH values, whereas strictly the equation in Revil et al., (1999a) is only valid for $pH < 7$.

This work chooses to obtain the temperature dependence of pH from the temperature dependence of K_w . This allows the temperature dependence of pH and pK_w to be calculated for a given set of C_a and C_b , K_1 and K_2 , which makes the calculation more generally applicable. Use of a direct empirical fit to the $pH(T)$ relationship would be tied to a fixed set of ΔC , K_1 and K_2 values, which would reduce its utility as a model. Such a general approach is needed when many of the input parameters to the model (and to Equation 17) are themselves temperature-dependent; not to do so risks not including latent functionalities.

The theoretical values of Stern plane potential arrived at in this work were between -181.6 mV and 15.1 mV for $C_f=10^{-5}$ mol/dm³ and $T=100^\circ\text{C}$, and $C_f=3.98$ mol/dm³ and $T=0^\circ\text{C}$, respectively, and both with $pH=7$, $pK_{Me}=7.5$, $pK_{(-)}=7$, and $\Gamma_s^o=10$ sites/nm³.

2.2.9 Zeta potential

The electrical potential φ in the EDL has, approximately, an exponential distribution given by $\varphi \approx \varphi_d \exp(-\chi/\chi_d)$, where φ_d is the Stern plane potential (V), χ_d is the Debye screening length (m) and χ is the distance from the mineral surface (m). It is, in fact, the solution of the linearised approximation of the Poisson-Boltzmann equation for the EDL (Equation 35 of Revil and Glover (1997)). The exponential form of Equation 18 is often called the Debye-Hückel approximation.

The zeta potential can then be calculated using (Revil and Glover, 1997; 1998; Revil et al., 1999a)

$$\zeta \approx \varphi_d \exp(-\chi_\zeta/\chi_d) \quad (18)$$

where

φ_d	Stern plane potential	(V)	From Equation 17
χ_d	Debye screening length	(m)	From Equation 15
$\chi_\zeta = 2.4 \times 10^{-10}$	shear plane distance	(m)	Revil and Glover (1997)

The theoretical values of the zeta potential arrived at in this work were between -201 mV and zero for $C_f = 10^{-5}$ mol/dm³ and $T = 100^\circ\text{C}$, and $C_f =$ mol/dm³ and $T = 0^\circ\text{C}$, respectively, and both with $pH = 7$, $pK_{Me} = 7.5$, $pK_{(-)} = 7$, and $\Gamma_s^0 = 10$ sites nm⁻³.

2.2.10 Surface conduction

The surface conductance was also calculated using the techniques described in Revil and Glover (1997; 1998) and in Revil et al. (1999a). The surface conductance is given by

$$\Sigma_s = \Sigma_s^{EDL} + \Sigma_s^{Prot} + \Sigma_s^{Stern} \quad (19)$$

where

$$\Sigma_s^{Stern} = \frac{e\beta_s \Gamma_s^0 K_{Me} C_f}{\left(10^{-pH} + K_- \left(\frac{\sqrt{8 \times 10^3 \varepsilon_r \varepsilon_o k_b T N} (10^{-pH} + K_{Me} C_f)}{2e\Gamma_s^0 K_-} \left[\frac{C_a + C_b + C_f + 10^{-pH} + 10^{pH-pK_w}}{\sqrt{I_f}} \right] \right)^{2/3} + K_{Me} C_f \right)} \quad (20)$$

and where

$$\Sigma_s^{EDL} = R \left[\left(B_{Na^+} C_f + B_{H^+} 10^{-pH} \right) \left(\left(S \left(\frac{10^{-pH} + C_f K_{Me}}{2e\Gamma_s^0 K_-} \right) \right)^{-1/3} - 1 \right) + \left(B_{Cl^-} C_f + B_{OH^-} 10^{pH-pK_w} \right) \left(\left(S \left(\frac{10^{-pH} + C_f K_{Me}}{2e\Gamma_s^0 K_-} \right) \right)^{+1/3} - 1 \right) \right] \quad (21)$$

where

$$R = \sqrt{\frac{2 \times 10^{-3} \varepsilon_r \varepsilon_o k_b T N}{C_f + 10^{-pH}}} \quad (22)$$

and where;

$$S = \sqrt{8 \times 10^3 \varepsilon_r \varepsilon_o k_b T N (C_a + C_b + C_f + 10^{-pH} + 10^{pH-pK_w})} \quad (23)$$

In Equation 20 to Equation 23 $pH = -\log_{10}(C_H^+)$, $pK_w = -\log_{10}(K_w)$, and K_w is the disassociation constant of water that is given by Equation 9. In Equation 21, The terms B_i are the equivalent ionic mobilities for each ion, where $B_i = \beta_i + 2\varepsilon_r\varepsilon_0k_bT/\eta_f eZ_i$, and where β_i is the ionic mobility of each ion.

$\beta_{Na^+} = 5.20 \times 10^{-8}$	Ionic mobility of sodium ions	($m^2/s \cdot V$)	Crow (1988)
$\beta_{H^+} = 3.63 \times 10^{-7}$	Ionic mobility of hydrogen ions	($m^2/s \cdot V$)	Crow (1988)
$\beta_{Cl^-} = 7.90 \times 10^{-8}$	Ionic mobility of chloride ions	($m^2/s \cdot V$)	Crow (1988)
$\beta_{OH^-} = 2.05 \times 10^{-7}$	Ionic mobility of hydroxyl ions	($m^2/s \cdot V$)	Crow (1988)

The ionic mobility controls the size of the electro-migration contribution to the diffuse layer conductance, while the term $2\varepsilon_r\varepsilon_0k_bT/\eta_f eZ_i$ controls the size of the electro-osmotic contribution to the diffuse layer conductance. The equivalent ionic mobilities B_i thus account for both processes.

Although the contribution from the diffuse layer Σ_s^{EDL} was shown to be negligible by Revil and Glover (1998) for a small range of parameters, I have implemented it in full in this model to be as complete as possible.

The contribution of the protons (and electrons) is thought to involve conduction along the surface only (O'Konski, 1960) and to be a function of surface site density but not of salinity (Revil and Glover, 1998). Watillon and de Backer (1970) have found that the surface conduction for a silica glass capillary full of pure water is $\Sigma_s = 2.4 \times 10^{-9}$ S. Given that Revil and Glover (1998) have found that $\Gamma_s^0 = (1.0 \pm 0.06) \times 10^{19}$ sites/ m^2 for the same data, I have assumed linearity and used the following formulation to calculate the proton contribution to the surface conductance

$$\Sigma_s^{Prot} = c_{Prot} \Gamma_s^0 = 2.4 \times 10^{-28} \Gamma_s^0 . \quad (24)$$

where $c_{Prot} = 2.4 \times 10^{-28}$ Sm²/site is the proton surface conductance contribution rate. All the other parameters in Equation 16 to Equation 24 have already been described except β_s , which is the ionic surface mobility (in $m^2/s \cdot V$). I have used $\beta_s = 5 \times 10^{-9}$ $m^2/s \cdot V$ from Revil et al. (1998).

The theoretical values of the surface conductance arrived at in this work were between 8.76×10^{-9} S and 2.64×10^{-9} S for $C_f = 3.98$ mol/ dm^3 and $T = 100^\circ C$, and $C_f = 10^{-5}$ mol/ dm^3 and $T = 0^\circ C$, respectively, and both with $pH = 7$, $pK_{Me} = 7.5$, $pK_{(-)} = 7$, and $\Gamma_s^0 = 10$ sites nm^{-3} .

2.2.11 Streaming potential coefficient

The streaming potential coefficient has been calculated from the values of the pore fluid dielectric permittivity $\epsilon_f = \epsilon_r \epsilon_o$, zeta potential ζ , bulk fluid conductivity σ_f , pore fluid viscosity η_f , and surface conductance Σ_s , all calculated previously, together with the grain diameter d , cementation exponent m , and formation factor $F = \phi^m$, where ϕ is the porosity, using the Equation 4 of [Glover and Déry \(2010\)](#)

$$C_s = \frac{\Delta V}{\Delta P} = \frac{d \epsilon_f \zeta}{\eta_f (d \sigma_f + 6 \Sigma_s (F - 1))}. \quad (25)$$

The grain diameter d , cementation exponent m , formation factor F , and porosity ϕ describe the microstructure of the rock matrix, and allow the previous modelling to be applied to real geological materials, such as rocks, sands and soils.

2.2.12 Influences

In reading the complex and interwoven functional relationships described in the last eleven subsections, it will be realised that the model is a web of influences that contain feedback loops, and where any single parameter (salinity, for example) is the input to many of the operating equations. The value of both C_f and the prime variable in this modelling, Temperature T are both inputs to 10 different operating equations, each of which often interact with each other. This makes the model extremely difficult to predict. This complexity is illustrated by the main summary diagram shown as [Figure 1](#).

3 THE PARAMETERS

Since the model presented in this paper is a temperature-dependent modification of that given by [Glover et al. \(2012\)](#), it shares a similar high number of independently adjustable parameters (22 *cf* [Glover et al.'s \(2012\)](#) 18), but as before many are either model variables or are fixed by independent measurements.

3.1 Model variables

Temperature T , salinity C_f , and pH are all considered to be model variables. These variables are either fixed, or the model is run as a function of one or more of them. For the modelling of the [Vinogradov and Jackson \(2015\)](#) both salinity and pH are known, and are treated as experiment (sample) fixed parameters.

3.2 Rock-specific parameters

Grain diameter d , porosity ϕ , grain radius r , pore throat diameter d_{PT} , grain packing parameter a , the pore size to pore throat ratio M_i , cementation exponent m , and formation factor F , are all rock-specific microstructural parameters which are fixed by whatever rock sample one wishes to model.

3.3 Fluid-specific parameters

The four effective ionic mobilities in a given aqueous fluid are independently available from electrochemical research (e.g., [Crow, 1988](#)). The equilibrium constants $K_1=10^{-7.53}$ and $K_2=10^{-10.3}$ are likewise known from studies made on the silica-H₂O-CO₂ system (e.g., [Wu et al., 1991](#)). The values of C_a and C_b that are added to attain a given pH are other fixed fluid-related parameters, while valence Z_i is also defined for the specific fluid being modelled.

3.4 Key parameters

There is more uncertainty over the value of the six remaining variables; each is discussed at greater length below.

3.4.1 Surface site density

The range of this parameter for silica surfaces varies considerably because silica surfaces can exist in a variety of forms, from amorphous silica, for which values of 4.6 sites/nm² ([Iler, 1979](#)), 5 sites/nm² ([Park and Regalbuto, 1995](#)) and 6 sites/nm² ([Kosmulski, 1996](#)) have been published, to crystalline silica for which both 9.6 sites/nm² and 6 sites/nm², for the (001) and (010) crystallographic planes, respectively, have been measured ([Hiemstra and van Riemsdijk, 1990](#)). For crushed quartz, a lower value of 2.6 sites/nm² was obtained by [Jørgensen and Jensen \(1967\)](#) which was attributed to the crushing process by [Revil et al. \(1999a\)](#), while at high pH values and during dissolution values as high as 25 sites /nm² are theoretically possible and confirmed by acid/base titration measurements at high pH values ([Tadros and Lykelma, 1969](#); [Yates and Healy, 1976](#)).

[Revil and Glover \(1998\)](#) found a value of 10.0 ± 0.6 sites/nm² while modelling the data of [Watillon and de Backer \(1970\)](#) which was made using an aqueous KNO₃ electrolyte at pH=6.8 in a silica glass capillary. Later, [Revil et al. \(1999a\)](#) refined the surface site density to

9.3 sites nm⁻² modelling the same data, and also stated that they consider that the value should generally lie in the range 5 to 10 sites/nm² providing $pH \leq 8$.

Glover et al. (2012) carried out modelling varying this parameter between 2.5 and 25 sites/nm² for a NaCl pore fluid with a pH between 5 and 8 while keeping all other parameters constant ($pK_{me}=7.5$, $pK_{(-)}=7.1$, $\chi_{\zeta} = 2.4 \times 10^{-10}$ m, $c_{Prot} = 2.4 \times 10^{-28}$ Sm²/site, $\beta_s = 5 \times 10^{-9}$ m²/s·V, $d=2 \times 10^{-5}$ m, $m=1.8$, $\phi=0.2$) and found that it has the effect of changing the streaming potential negligibly for $C_f > 0.01$ mol/dm³ NaCl, and diminishing it by up to one order of magnitude for $C_f < 0.01$ mol/dm³ NaCl. They noted that the strength of the low salinity effect is dependent upon the microstructural properties of the rock (grain size, formation factor, porosity etc.). In addition, they noted that the change from 2.5 to 25 sites nm⁻² reduced the sensitivity of the streaming potential coefficient to pH in the range 5 to 8.

In this work, a value of 10 sites/nm² has been used.

3.4.2 Binding constant for cation adsorption

The value of this parameter is not well known, partly because it is rarely measured and varies according to which cation is dominant. Existing data for silica give $pK_{me}(\text{Li}^+) = 7.8$ and $pK_{me}(\text{Na}^+) = 7.1$ (Dove and Rimstidt, 1994), $pK_{me}(\text{Li}^+) = 7.7$, $pK_{me}(\text{Na}^+) = 7.5$ and $pK_{me}(\text{Cs}^+) = 7.2$ (Kosmulski, 1996), and $pK_{me}(\text{Na}^+) = 3.25$ and $pK_{me}(\text{K}^+) = 2.8$ (Revil et al., 1999a).

Glover et al. (2012) carried out modelling varying this parameter between 3.25 and 7.5 for a NaCl pore fluid with a pH between 5 and 8 while keeping all other parameters constant ($\Gamma_s^0=10$ sites/nm², $pK_{(-)}=7.1$, $\chi_{\zeta} = 2.4 \times 10^{-10}$ m, $c_{Prot} = 2.4 \times 10^{-28}$ Sm²/site, $\beta_s = 5 \times 10^{-9}$ m²/s·V, $d=2 \times 10^{-5}$ m, $m=1.8$, $\phi=0.2$) and found that it has little effect $C_f < 0.01$ mol/dm³ NaCl, but seems to have a drastic effect in the range $0.01 > C_f > 2$ mol/dm³ NaCl where positive streaming potentials are predicted as $pK_{me} \rightarrow 3.25$. They concluded that it is clearly important to understand the sensitivity of this parameter.

In this work, a value of $pK_{me} = 7.5$ has been used.

3.4.3 Disassociation constant for dehydrogenization

Once again, the actual value of this parameter is not well known. Experimental values include $pK_{(-)} = 6.8$ (Dove and Rimstidt, 1994), $pK_{(-)} = 6.5$ (Kosmulski, 1996), and $pK_{(-)} = 7.5$ (Hiemstra and van Riensdijk, 1990). A modelling study by Rustad (1996) predicted that $pK_{(-)} = 8.5$. Revil et al. (1999a) have used $pK_{(-)} = 7.4$ to 7.5 in their modelling, while Glover et al. (2012) used $pK_{(-)} = 7.7$.

Glover et al. (2012) also carried out modelling varying $pK_{(-)}$ between 6.5 to 8.5 for a NaCl pore fluid with a pH between 5 and 8 while keeping other parameters constant ($\Gamma_s^0=10$ sites/nm², $pK_{me}=7.5$, $\chi_\zeta = 2.4 \times 10^{-10}$ m, $c_{Prot} = 2.4 \times 10^{-28}$ Sm²/site, $\beta_s = 5 \times 10^{-9}$ m²/s·V, $d=2 \times 10^{-5}$ m, $m=1.8$, $\phi=0.2$) and found that increasing $pK_{(-)}$ decreases the streaming potential coefficient for all salinities except those greater than 1 mol/dm³. The decrease was found to be small for solutions with pH 7 and pH 8 (about a twofold decrease), but is larger for pH 6 and leads to more than an order of magnitude for pH 5.

In this work, a value of $pK_{(-)} = 7.7$ has been used.

2.3.4 The shear plane distance

There exist few independent measurements of the shear plane distance χ_ζ . Smith (1976) found $\chi_\zeta = 5.0 \times 10^{-10}$ m, while Revil and Glover (1997) used χ_ζ as an adjustable parameter in their model to obtain $\chi_\zeta = 2.4 \times 10^{-10}$ m. A similar fitting procedure was used by Ishido and Mizutani (1981) to obtain $\chi_\zeta = 2.0 \times 10^{-9}$ m.

It should be noted that Revil et al. (1999a) used $\chi_\zeta = 0$ m, which corresponds to the shear plane and Stern plane being co-located (i.e., fluid flow transports the entire diffuse layer). While this is a useful tool to reduce the number of parameters in the model by one, Glover et al. (2012) considered it to be unduly restrictive because the modelled streaming potential coefficient at high salinities is sensitive to changes in this parameter.

Glover et al. (2012) also carried out modelling varying χ_ζ between 2.4×10^{-11} m and 2.4×10^{-9} m for a NaCl pore fluid with a pH between 5 and 8 while keeping the other parameters constant ($\Gamma_s^0=10$ sites/nm², $pK_{me}=7.5$, $pK_{(-)}=7.1$, $c_{Prot} = 2.4 \times 10^{-28}$ Sm²/site, $\beta_s = 5 \times 10^{-9}$ m²/s·V, $d=2 \times 10^{-5}$ m, $m=1.8$, $\phi=0.2$) and found that increasing χ_ζ reduces the streaming potential coefficient curves for all four pH s, but only negligibly for χ_ζ changing from 2.4×10^{-11} m to 2.4×10^{-10} m, but reduces the streaming potential by half an order of magnitude for all four pH s when augmenting the value further to 2.4×10^{-9} m, which provided model curves that were an extremely good fit to the data.

The modelling in this paper sets $\chi_\zeta = 5 \times 10^{-9}$ m.

3.4.5 Contributions to surface conduction

The surface conduction has contributions from so-called proton conduction, from the Stern plane and from the diffuse layer. The contributions from the Stern plane and diffuse layers are

not input parameters as they are calculated in the model. Although the contribution from the so-called proton conduction is also calculated in the model, it is directly controlled by the values of the proton surface conductance contribution rate c_{prot} and the density of surface sites (discussed in subsection 3.4.1. above). Consequently, c_{prot} may be considered to be an important input parameter.

Watillon and de Backer (1970) carried out experimental determinations using pure water on silica glass at 20-25°C, obtaining $c_{\text{Prot}} = 2.4 \times 10^{-28} \text{ Sm}^2/\text{site}$, which gives a value of $\Sigma_s^{\text{Prot}} = 2.4 \times 10^{-9} \text{ S}$ using Equation 24 and $\Gamma_o = 10 \text{ sites/nm}^2$, coinciding with the value used by Revil et al. (1999a) and Glover et al. (2012).

Glover et al. (2012) carried out a brief sensitivity of this parameter, increasing it from $2.4 \times 10^{-29} \text{ m}$ to $2.4 \times 10^{-27} \text{ m}$ for a NaCl pore fluid with a pH between 6 and 8 while keeping other parameters constant ($\Gamma_s^o = 10 \text{ sites/nm}^2$, $pK_{me} = 7.5$, $pK_{(-)} = 7.1$, $\chi_\zeta = 2.4 \times 10^{-10} \text{ m}$, $\beta_s = 5 \times 10^{-9} \text{ m}^2/\text{s} \cdot \text{V}$, $d = 2 \times 10^{-5} \text{ m}$, $m = 1.8$, $\phi = 0.2$). They found that increasing c_{prot} has negligible effect upon the streaming potential for fluids with $C_f > 0.01 \text{ mol/dm}^3 \text{ NaCl}$, pH 5-8. However, there is a significant reduction in the streaming potential for fluids where $C_f < 0.01 \text{ mol/dm}^3 \text{ NaCl}$, pH 5-8. Glover et al. (2012) concluded that increasing c_{Prot} has a similar effect as changes to rock microstructural properties which enhance the role of surface conduction, with both increasing the effect of surface conduction on the streaming potential coefficient.

The modelling in this paper uses $c_{\text{Prot}} = 2.4 \times 10^{-28} \text{ Sm}^2/\text{site}$.

3.4.6 The surface mobility

There is significant uncertainty over the value of the surface mobility (Glover et al., 2012). Watillon and de Backer (1970) has used value of $\beta_s = 4 \times 10^{-9} \text{ m}^2/\text{s} \cdot \text{V}$ in modelling their data, while Revil and Glover (1998) used a value of $5.14 \times 10^{-9} \text{ m}^2/\text{s} \cdot \text{V}$ that was derived by fitting their theory to shaly sand data. Later, both Revil et al. (1999a) and Glover et al. (2012) used a value of $5 \times 10^{-9} \text{ m}^2/\text{s} \cdot \text{V}$ in their modelling. These surface mobilities are approximately ten times less than the mobility of Na^+ in water ($\beta_f^{(+)} = 5.2 \times 10^{-8} \text{ m}^2/\text{s} \cdot \text{V}$) according to Crow (1988) and Waxman and Smits (1968).

It is known that the streaming potential coefficient is extremely insensitive to changes in the surface mobility in this restricted range. Unreasonably high surface mobilities, four orders of magnitude higher, are required any effect upon the modelled streaming potential coefficient can be noted (Glover et al., 2012).

In this modelling , a value of surface mobility $\beta_s = 5 \times 10^{-9} \text{ m}^2/\text{s} \cdot \text{V}$ has been used.

4. IMPLEMENTATION

The model begins by calculating important pore fluid parameters as a function of temperature using the well-understood empirical and theoretical relationships set out in the last section: (i) density ρ_f , (ii) molality b_f , (iii) relative permittivity ϵ_f , (iv) viscosity η_f , (v) electrical conductivity σ_f , (vi) Debye length χ_D , and (vii) pK_w . The values of pH and pOH can then be calculated. The value of pH is particularly important not only because the ultimate values of C_{sp} and ζ are extremely sensitive to this parameter, but the pH varies with both temperature and fluid salinity. The temperature dependence of most of these input parameters are shown in [Figure 2](#), while [Table 1](#) shows the source of the relationships used to obtain them.

Each of these input parameters undergoes significant variation over the range of salinities (10^{-5} to 1 mol/dm^3) and temperatures (0 to 150°C) considered in this paper. Pore fluid density, molality, relative permittivity, viscosity and electrical conductivity undergo a change of -8.4%, -9.2%, -49.8%, 89.1%, and +664%, respectively, over the temperature interval 0 to 150°C . In addition, the water product coefficient pK_w (and hence pH) also varies with temperature, leading to a change of +22% in pH over the same temperature range for a 0.01 mol/dm^3 NaCl solution, which becomes less for fluids with a larger salinity and greater for lower salinity fluids ([Figure 2e](#)).

The modelled temperature dependence of electrolyte pH accounts for water dissociation, addition of acid/base, and CO_2 dissolution in water due to fluid equilibration with atmospheric CO_2 . The [Vinogradov and Jackson \(2015\)](#) experimental data were obtained in a closed systems without any access to atmospheric CO_2 , thus no change in pH associated with CO_2 dissolution was expected during the experiment but the fluids would have been equilibrated with the atmosphere before the experiment started. [Vinogradov and Jackson \(2015\)](#) noted experimentally that the pH of brines of high salinity (0.5 mol/dm^3) did not change with temperature in the range 20°C to 80°C , while lower salinity brines reduced in pH , exemplified by that of 0.01 mol/dm^3 NaCl brine which decreased from about pH 6.5 to pH 5.5 in the range 20°C to 80°C . It is interesting to note that our model provides the same behaviour as found experimentally by [Vinogradov and Jackson \(2015\)](#).

The Stern potential can be calculated using the method of [Revil et al. \(1999a\)](#). This potential decays exponentially throughout the thickness of the EDL on a scale that is characterized by the Debye length χ_D . The ζ can be obtained by considering the value of the

electrical potential at the shear plane χ_ζ , which is an important input parameter to the model, and then adding the ζ offset (Glover et al., 2012).

The zeta potential offset is a parameter that was introduced by Glover and Déry (2010) in the initial formulation of the model because it was recognized that the zeta potential calculated using the Revil et al. (1999a) approach did not provide values which matched the measured values at high salinities.

In this work, modelling has been carried out while setting the zeta potential offset ζ_o to zero but it was not possible to find solutions of the overall model that would fit the measured zeta potential and streaming potential coefficient curves while retaining physically reasonable values of the other model parameters. Consequently, and in agreement with previous reports, it is concluded that a non-zero zeta potential offset is necessary for the operation of the model. The requirement for the zeta potential offset is probably due to a physical limit on the Debye length at high salinities caused by the size of the hydrated ions (Figure 2f) (Vinogradov and Jackson, 2015) or capacitor-like behaviour between excess ions on the Stern plane and the shear plane if the latter is inside the Stern layer as proposed by Eq. (1) of Leroy et al. (2022).

Consequently, the results given in this paper arise from use of the model with $\zeta_o = -9.5$ mV, kept constant irrespective of salinity or temperature. This assumption is forced by the lack of data allowing a more nuanced approach. In reality, it is likely that the zeta potential offset is both a function of temperature and salinity if we are correct in attributing it to a physical limit on the Debye length.

In the case of low and medium salinities the zeta potential is always much greater than the zeta potential offset, and the zeta potential offset will consequently have a small if not negligible effect on the final results. For high salinities, the zeta potential offset is needed for the model to provide physically realistic results, whether they match measurements or not. We implemented the model with a variable zeta potential offset such that at any salinity the total zeta potential was constrained to be greater than -9.5 mV, and found that the results were negligibly different from those presented in this paper.

This paper shows that the zeta potential generally decreases as temperature increases. Consequently, there will be a temperature where a zeta potential offset needs to be imposed if a physical limit needs to be introduced. This value is likely to be the same as that for salinity if it is caused by a physical limit.

The C_{sp} is calculated using one of the modified forms of the Helmholtz-Smoluchowski relationship (e.g., Glover, 2015) that were developed by Glover and Déry (2010). This model

takes account of the individual microstructural parameters which describe the pore network (grain size d_{grain} , porosity ϕ , cementation exponent m , and formation factor F), allowing modelling to be carried out for individual samples. In this study I have carried out C_{sp} modelling for individual samples using d_{grain} , ϕ , m data given in [Vinogradov and Jackson \(2015\)](#). However, since their data is restricted to the two relatively high salinities, 0.01 mol/dm³ and 0.5 mol/dm³, the effect if the microstructure of the pores upon the streaming potential is smaller than it would be for samples saturated with lower salinity pore fluids. This allows a single curve to represent all 4 samples. I have used the data from the St. Bees 1 sandstone to calculate the curve, with the data shown in [Table 1](#).

Table 1. General modelling parameters.

Parameter	Units	Value	Source
Temperature	°C, K	Modelling variable	Imposed
Salinity, C_f	M	Modelling variable	Imposed
pH	(-)	Modelling variable	Calculated in the model
Pore fluid density, ρ_f	kg/m ³	Empirical	Lide (2009)
Pore fluid molality, b_f	mol/kg	Empirical	Lide (2009)
Pore fluid viscosity, η_f	Pa.s	Empirical	Phillips et al. (1978)
Pore fluid relative permittivity, ϵ_f	(-)	Empirical	Revil et al. (1999b)
Pore fluid electrical conductivity, σ_f	S/m	Empirical	Sen and Goode (1992a, 1992b)
Disassociation constant of water pK_w	(-)	Empirical	Lide (2009)
Surface site density Γ_o	sites/nm ²	10	Revil and Glover (1998)
Binding constant for cation (Na ⁺) adsorption on quartz pK_{Me}	(-)	7.5	Kosmulski (1996)
Disassociation constant for dehydrogenization of silanol $pK_{(-)}$	(-)	7.7	see range in Glover et al. (2012)
Shear plane distance χ_ζ	m	5.0×10^{-9}	see range in Glover et al. (2012)
Protonic surface conduction Σ_s^{prot}	S	2.4×10^{-9}	Revil et al. (1999b)
Ionic Stern plane mobility β_s	m ² /s.V	5.0×10^{-9}	see range in Glover et al. (2012)
Ionic mobility of Na ⁺ ions β_{Na^+}	m ² /s.V	5.2×10^{-8}	Crow (1998)
Ionic mobility of H ⁺ ions β_{H^+}	m ² /s.V	36.3×10^{-8}	Crow (1998)
Ionic mobility of Cl ⁻ ions β_{Cl^-}	m ² /s.V	7.9×10^{-8}	Crow (1998)
Ionic mobility of OH ⁻ ions β_{OH^-}	m ² /s.V	20.5×10^{-8}	Crow (1998)
Zeta potential offset ζ_o	mV	-9.5	Vinogradov and Jackson (2015)
Modal grain size d_{grain}	m	1.3×10^{-4}	St. Bees Sandstone 1 Vinogradov and Jackson (2015)
Porosity ϕ	(-)	0.19	
Cementation exponent m	(-)	1.74	

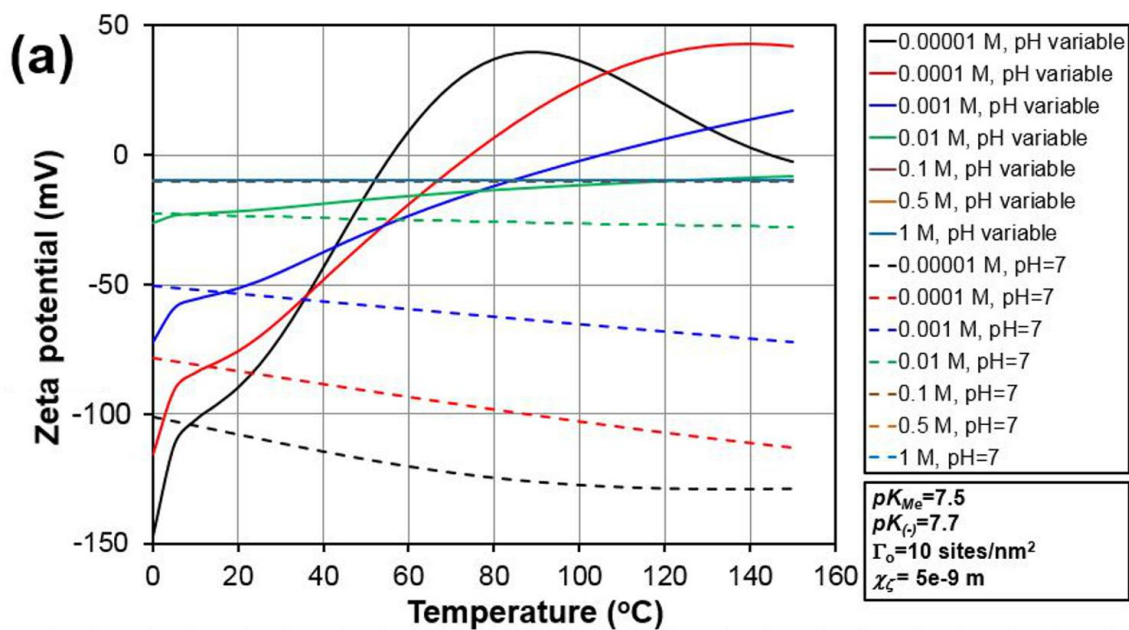
Other input parameters such as the pore fluid relative permittivity, viscosity and electrical conductivity, which were discussed earlier, are also temperature and salinity dependent. The final input parameter is the surface conduction. The surface conduction is made

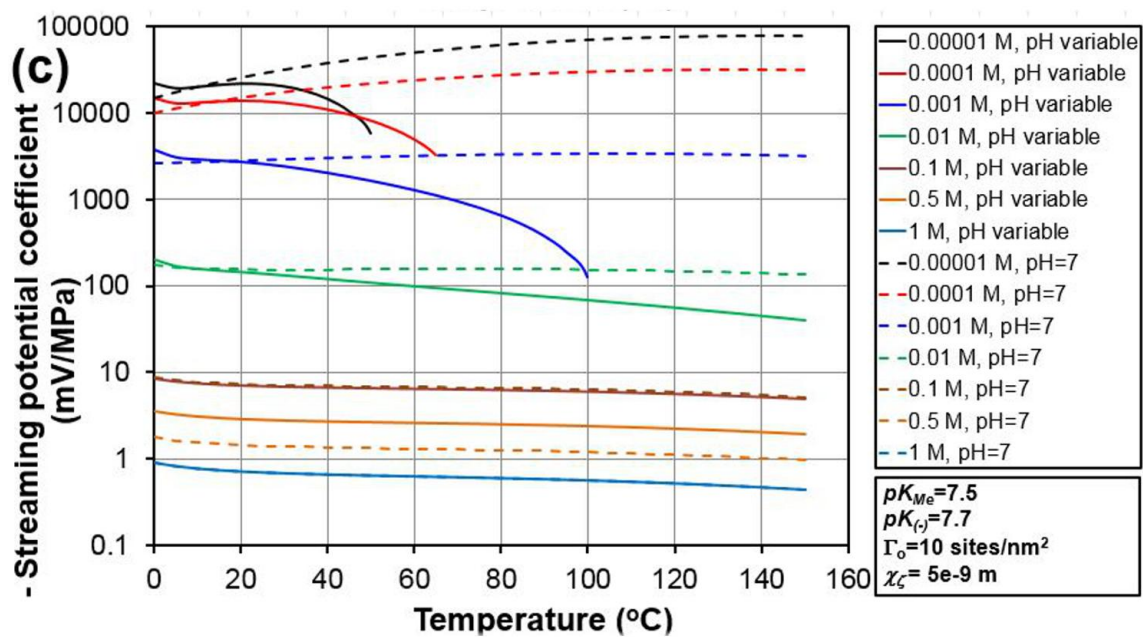
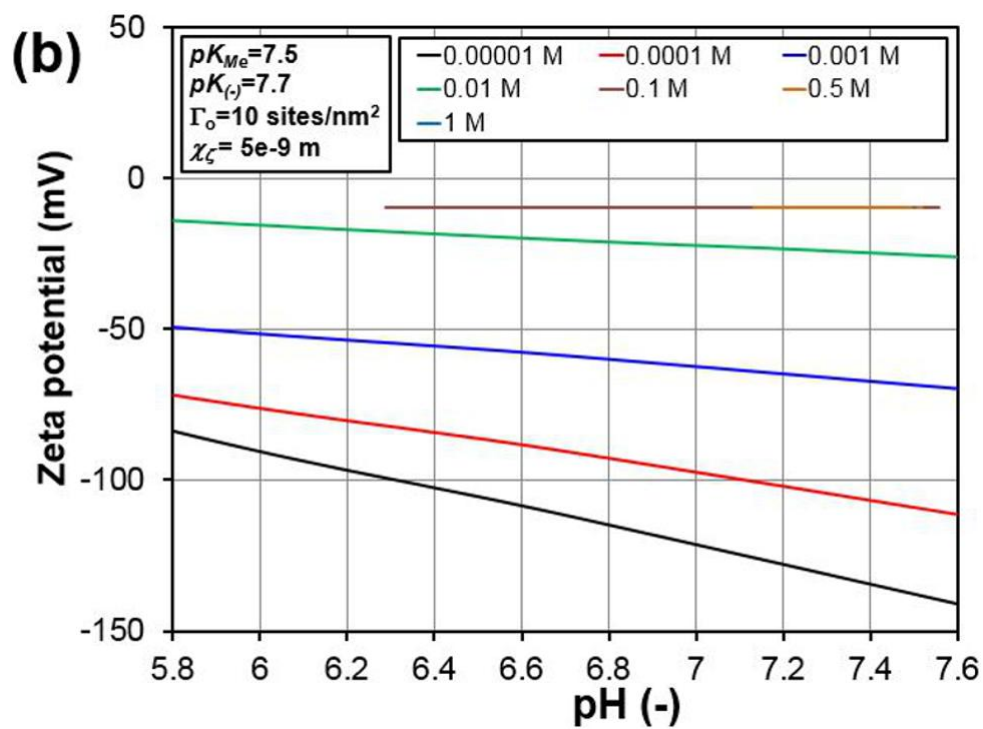
up of three contributions, each of which is calculated in the model using the methods in [Revil and Glover \(1997; 1998\)](#) and [Revil et al. \(1999a\)](#), and summed to provide an input to the final calculation, which is also temperature-dependent because it depends on a number of temperature-dependent parameters including pH . Unfortunately, there is no space to examine the temperature dependency of the surface conduction in this paper.

The sources and values of all the parameters used in the modelling are shown in [Table 1](#). A full description of all the parameters used in the model together with independent measurements and the ranges over which they vary is given in [Glover et al. \(2012\)](#) and in the supplementary material ([Glover, 2026](#)).

5. RESULTS

[Figure 3](#) shows the implementation of the model for the calculation of ζ and C_{sp} as a function of temperature from 0°C to 150°C and the consequential changes in pH for seven salinities from 10^{-5} mol/dm³ to 1 mol/dm³ and for the modelling parameters given in [Table 1](#).





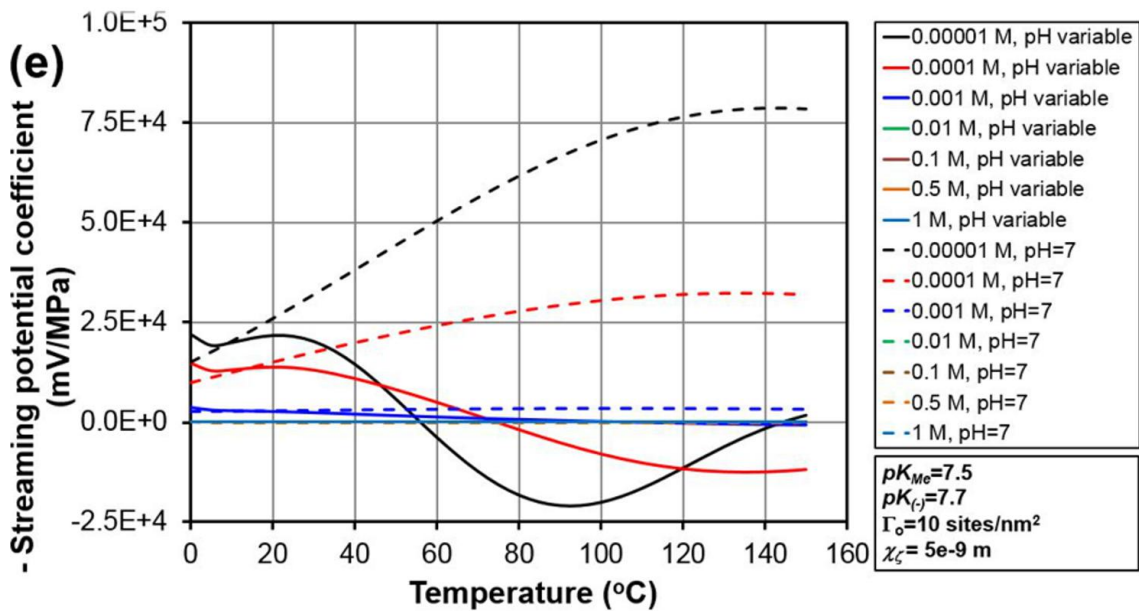
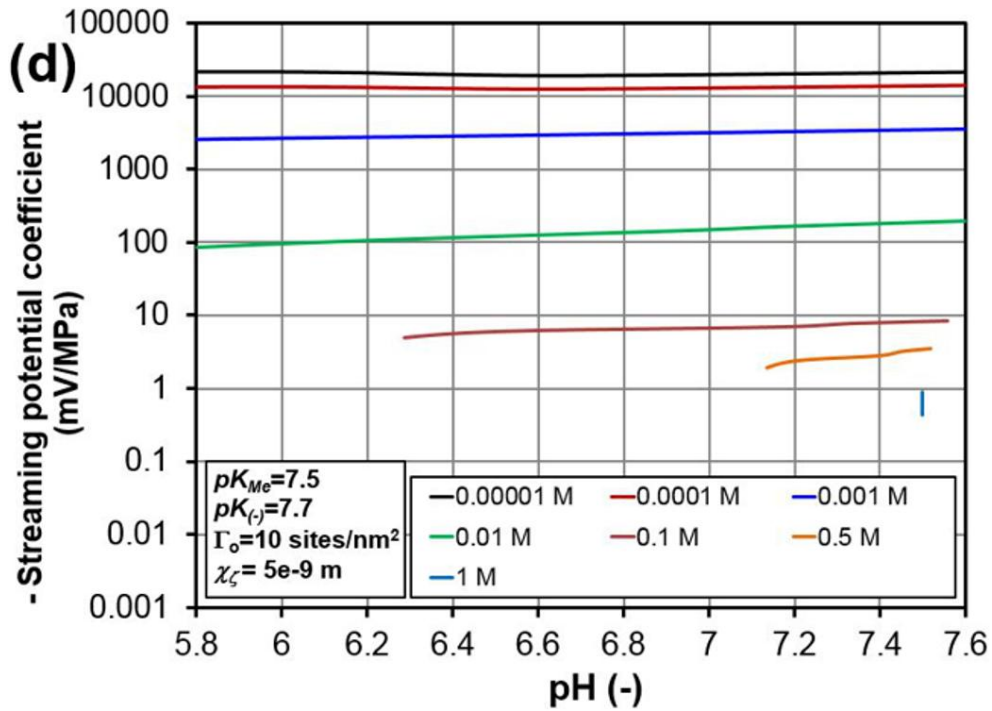


Figure 3. The results of the model for the parameters shown in Table 1. (a) the ζ as a function of temperature with variable pH (solid lines) and fixed $pH=7$ (dashed lines), (b) ζ as a function of pH (note the curves for 1, 0.5 and 0.1 mol/dm³ are coincident), (c) C_{sp} as a function of temperature with variable pH (solid lines) and fixed $pH=7$ (dashed lines), and (d) C_{sp} as a function of pH . Part (e) shows the same data as (c) but on linear axes in order to portray the full variation of the streaming potential without the curves being curtailed.

The ζ is negative at lower temperatures with larger magnitudes for lower salinities (Figure 3a). As temperature increases, ζ decreases in magnitude at a faster rate for low salinity fluids than with higher salinity fluids ultimately becoming positive. This behaviour is strongly influenced by the decrease in pH with temperature and its larger effect for low salinity solutions (see Fig. 2e). Comparison of the solid curves for variable pH with the dashed curves, which are calculated for a fixed pH=7, shows the degree to which ζ is highly influenced by changing pH, which is supported by Figure 3b showing the variation of ζ with pH. This agrees in all respects with the observations of Vinogradov and Jackson (2015). In each case the change is not linear, having a shape that is clearly influenced by the non-linearity in the pH/temperature curve (Figure 2e). The change in ζ over the temperature interval 0°C to 150°C varies with salinity, being -98.1% at 10⁻⁵ mol/dm³, reaching the greatest change of -136.4% at 0.01 M and tending to zero change for salinities greater than 1 mol/dm³ for the modelling parameters in Table 1. This is also consistent with the experimental measurements.

Varying the modelling parameters such as pK_{Me} , $pK_{(-)}$, Γ_o , and χ_ζ all change the sensitivity of the ζ to temperature, but the general pattern, with the greatest changes occurring for salinities between 0.001 mol/dm³ and 0.01 mol/dm³, and with a tendency to negligible temperature dependence at high salinities, is retained. There is a greater sensitivity to temperature as pK_{Me} , $pK_{(-)}$ and Γ_o all increase, though the effect is weak for pK_{Me} , and varying χ_ζ controls the salinity at which the greatest decrease of ζ per change in temperature occurs.

The behaviour of the C_{sp} is more complex as shown by Figure 3c and 3e. These two parts show the same data but on differently scaled axes. Figure 3c uses a logarithmic axis which represents the large range of variation but curtails the curves when they become positive for the low salinity fluids, while Figure 3e uses a linear scale which shows the full variation of the low salinity curves, but does not discriminate the high salinity curves whose small values become indistinguishable from zero when plotted this way.

Once again C_{sp} is always negative at low temperatures with much larger magnitudes for lower salinities. However, while C_{sp} increases slightly in magnitude as temperature increases for low salinities until about 20°C, it then decreases in magnitude significantly especially for the low salinity fluids. For the lowest salinities, it may be seen that the trend to become more positive stops after about 100°C, with the value of C_{sp} approaching zero as 1 mol/dm³ is approached. These effects can be seen to be controlled by the behaviour of ζ , which is ultimately controlled by the pH variation. For medium to higher salinity fluids C_{sp} decreases in magnitude as temperature increases to 150°C remaining negative. The pH variation is also

complex, decreasing with pH for low salinity pore fluids and increasing with pH for medium to high salinity pore fluids. Once again [Figure 3d](#) and comparison of the solid curves for variable pH with the dashed curves for a fixed $pH=7$ in [Figure 3c](#) and [3e](#) show the large influence of pH , approaching a factor of 10^4 in the case of 10^{-5} mol/dm³ at 150°C. This behaviour also agrees well with the limited data coverage provided by [Vinogradov and Jackson \(2015\)](#).

The temperature dependence in C_{sp} over the temperature interval 0°C to 150°C varies with salinity, being -92.3% at 10^{-5} mol/dm³, reaching the greatest negative change of -181.1% at about 10^{-4} mol/dm³, and tending towards a temperature dependence of about -52% for salinities greater than 1 mol/dm³.

Once again, varying pK_{Me} , $pK_{(-)}$, Γ_o , and χ_ζ all change sensitivity of the ζ to temperature, but the general pattern, with the greatest positive temperature dependence occurring for the lowest salinities and the greatest negative temperature dependence between 0.001 mol/dm³ and 0.01 mol/dm³ with a tendency to negligible temperature dependence at high salinities, is retained. There is a greater sensitivity to temperature as pK_{Me} , $pK_{(-)}$ and Γ_o all increase, especially at low salinities, though the effect is weak for pK_{Me} . Once again, varying χ_ζ controls the salinity at which the greatest decrease of ζ per change in temperature occurs, and increasing the protonic surface conduction (and hence the total surface conduction) only affects the C_{sp} , making the positive temperature dependence greater at low salinities and the negative temperature dependence weaker at high salinities.

[Vinogradov and Jackson \(2015\)](#) speculated that most of the temperature sensitivity of ζ was due to consequential changes in the pH of the pore solution, implementing a simplified model of the ζ for both variable and fixed mol/dm³. This modelling confirms that such a hypothesis is consistent with the latest understanding, but shows that modelling with a fixed pH is invalid. Indeed, [Figure 3](#) also shows our modelling with a fixed $pH=7$ (dashed lines), which is clearly in agreement with the limited ζ modelling at $pH=7$ carried out by [Vinogradov and Jackson \(2015\)](#), and shows how the modelling results of both ζ and C_{sp} must include variable pH for pore fluids less than about 0.01 mol/dm³.

We note particularly that the zeta potential becomes positive as temperature increases above about 82°C, 77°C and 52°C for 10^{-3} mol/dm³, 10^{-4} mol/dm³, and 10^{-5} mol/dm³, respectively ([Figure 3a](#)) and has a corresponding effect on the streaming potential coefficient ([Figure 3e](#)). The model seems to be consistent with all of the data presented in [Vinogradov and Jackson \(2015\)](#). We know of no data for the higher temperature behaviour of sufficiently low

salinity fluids, so cannot validate this behaviour, but note that it may not be realistic and may indicate that an important aspect of the model is lacking for low salinity fluids.

6. CASE STUDY: The Vinogradov and Jackson (2015) data

While the previous section considered the general modelling results as a function of temperature, pH and salinity, this section considers how well the full model can fit the data provided by [Vinogradov and Jackson \(2015\)](#). The large number of parameters in the model implies that a good fit might be possible. However, detailed examination reveals that many of these parameters are fixed by the conditions of the experiment or independent measurements, and where there is scope for varying the parameters, independent electro-chemical measurements restrict the range over which the remainder may be varied. Consequently, of the 22 input parameters listed in [Table 1](#), 8 of them are defined by the temperature and pressure conditions of the measurements (T , C_f and the 6 empirical relationships), 4 of them are rock-specific parameters or other parameters provided by [Vinogradov and Jackson \(2015\)](#) (ϕ , m , d_{grain} and ζ_0), 7 are electro-chemical parameters for which the values are well-known or to which the model is relatively insensitive (pK_{Me} , Σ_s^{prot} , β_s , $\beta_{\text{Na}+}$, $\beta_{\text{H}+}$, $\beta_{\text{Cl}-}$, $\beta_{\text{OH}-}$), leaving three parameters ($pK_{(-)}$, Γ_0 , and χ_ζ) which exert the greatest control over the shape of the ζ and C_{sp} curves. These three parameters are constrained by independent measurements to specific ranges ($6.5 < pK_{(-)} < 8.5$; $2.6 < \Gamma_0 < 25$; $2 \times 10^{-10} \text{ m} < \chi_\zeta < 5 \times 10^{-10} \text{ m}$), which are sourced and discussed in full in Section 3 above and in [Glover et al. \(2012\)](#).

The modelling values used in this study ([Table 1](#)) provide the closest fit of the ζ and C_{sp} curves to the data while varying these three parameters within the experimental ranges. The fit was carried out by manual exploration rather than any curve fitting process and consequently a slightly better fit might be obtained by using slightly different input parameter values.

[Figure 4](#) shows the results of modelling the [Vinogradov and Jackson \(2015\)](#) ζ and C_{sp} data as a function of temperature and pH for both of the experimental salinities. The models perform creditably in all cases but cannot be considered to be an exact match to the data. This is likely to be due to a combination of experimental errors and inadequacies in the model.

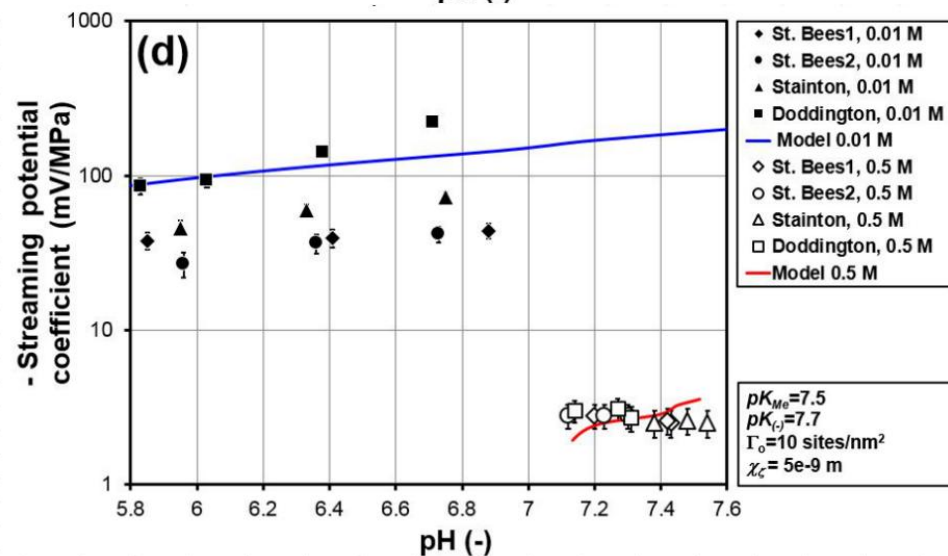
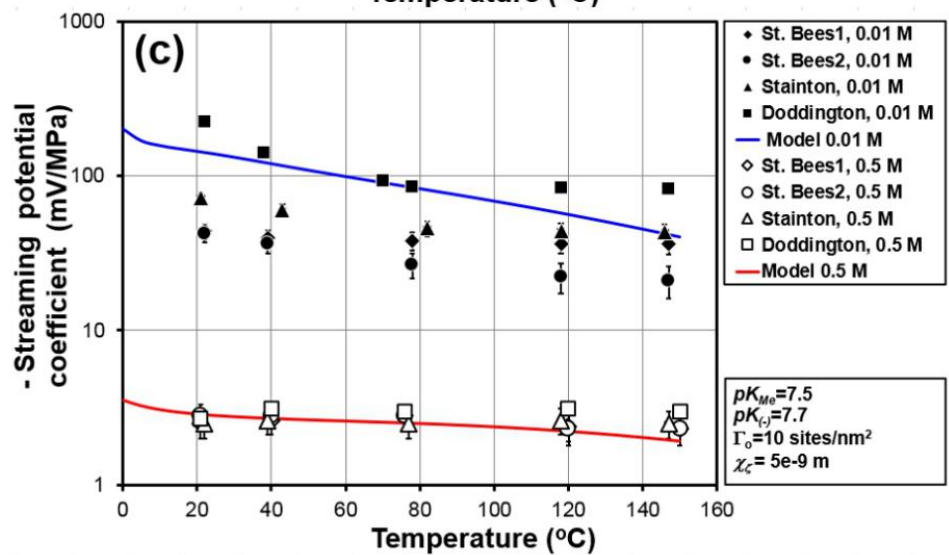
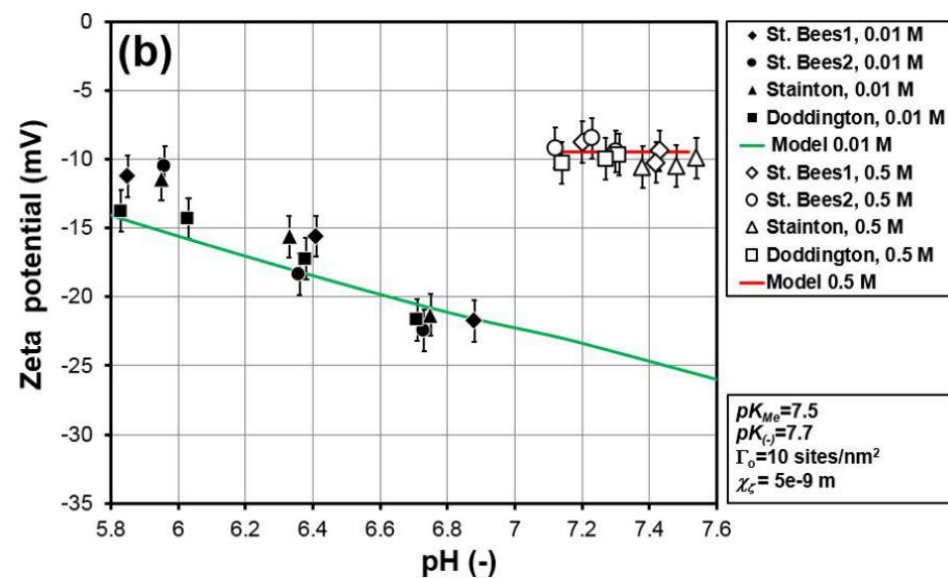
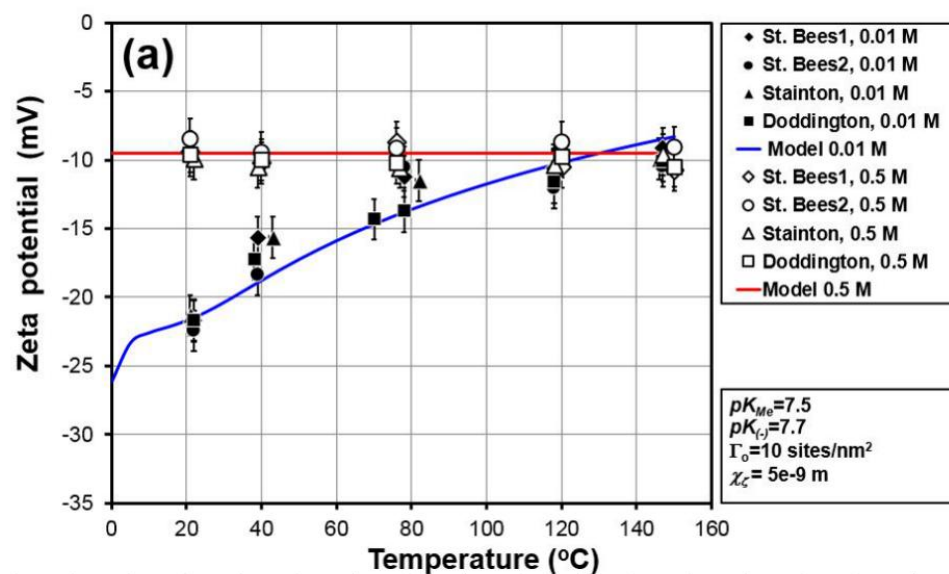
For the ζ the high salinity data (0.5 mol/dm^3) as a function of both temperature and pH is modelled well mainly because its value is controlled by the imposed value of the ζ offset (-9.5 mV).

The model for the lower salinity (0.01 mol/dm^3) as a function of pH is also modelled very well, while that as a function of temperature reproduces the sign, approximate magnitude and shape of the data well, but overestimate the magnitude in the middle salinities by about 3 mV, which I regard as a creditable performance.

The fit to the temperature dependence can be improved by small changes to the three main parameters (*e.g.*, any one of $pK_{(-)}$ $7.7 \rightarrow 8.0$; Γ_o $10 \rightarrow 7$; χ_ζ 3.5×10^{-9}), all of which remain within the acceptable range, but in each case the fit to the pH dependency is degraded for temperatures above 150°C . The reason for this small difference in the model efficacy is not easy to ascertain. It might be that there was a systematic error in either the temperature or pH measurement, but this is very unlikely as an error of either 30°C , or 0.3 pH points would be necessary. Another reason could be a failure of the relationship between temperature and pH used in the model, but this is a well-known and well-tested empirical relationship. A third, and more plausible reason might be that the pH in the rock is different from that calculated in the model because of reactions between the pore fluid and the rock matrix. It was recognised (Revil and Glover, 1997) that the pore fluid pH changes within the EDL, and that there is some uncertainty as to the value of pH that ought to be used in the modelling of process operating at these scales, and which define the ζ and C_{sp} .

For the C_{sp} modelling of both the high salinity data (0.5 mol/dm^3) and the lower salinity (0.01 mol/dm^3) as a function of both temperature and pH reproduces the sign, approximate magnitude and trend of the data well, with the model underestimating the high salinity data by a factor of 2 and falling within the spread of the low salinity data.

It is tempting to ascribe the spread of the C_{sp} data to variations in the pore microstructure between the samples measured by Vinogradov and Jackson (2015), which includes samples with porosities ranging from 0.13 to 0.19, cementation exponents ranging from 1.34 to 1.86, formation factors between 10 and 45 and grain sizes from $60 \times 10^{-5} \text{ m}$ to $200 \times 10^{-5} \text{ m}$. Analysis of the magnitudes of the C_{sp} measurements shows no obvious relationship to the microstructural parameters. In any case, the microstructural parameters have almost no effect on the modelled C_{sp} , for these high salinities. The curves shown in Figure 4 are for St. Bees1 parameters. Implementation of the model for any of the other of the Vinogradov and Jackson (2015) rocks moves the curve by a very small amount, insufficient to allow the curve to match the relevant data points.



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Figure 3. Implementation of the model for the parameters pertaining to the data from [Vinogradov and Jackson \(2015\)](#). (a) the ζ as a function of temperature, (b) ζ as a function of pH, (c) C_{sp} as a function of temperature, and (d) C_{sp} as a function of pH. Modelling parameters are shown in [Table 1](#)

5. Conclusions

[Vinogradov and Jackson \(2015\)](#) have reported experimental measurements of C_{sp} and derived ζ . The [Glover et al. \(2012\)](#) approach for calculating the ζ and C_{sp} for porous silica-based rocks filled with aqueous NaCl pore fluids has been implemented as a function of temperature, pH and salinity. The model curves reproduce the sign, magnitude and shape of the ζ and C_{sp} as a function of both temperature and pH dependence, fitting the [Vinogradov and Jackson \(2015\)](#) data reasonably well. However, small discrepancies between the model and the data show that there is scope for improvement of the model, and new experimental methodologies ([Walker et al., 2014](#)).

The modelling confirms [Vinogradov and Jackson's \(2015\)](#) assertion that the temperature effect on both the ζ and the C_{sp} is mainly through its effect on the pH of the pore fluid. While the temperature dependencies of the ζ and the C_{sp} have long been ambiguous, the combination of the [Vinogradov and Jackson \(2015\)](#) data with the modelling carried out in this work has confirmed the sign, magnitude and trend of these parameters as temperature, pH and salinity varies, opening the door for more accurate electro-kinetic predictions in hydrocarbon exploration and production, geothermal resources, aquifers, prospective sites for carbon capture and storage by deep underground injection, and in characterizing subsurface sites for the storage of radioactive waste.

I have been made aware of further datasets that might be treated by this model ([Hidayat et al., 2022; Vinogradov, 2024](#)). While such data cannot be included in this paper due to time constraints, this modelling will be prepared for a further publication

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Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Full supplementary information are available at ([Glover, 2026](#)). This information includes the model implemented in Excel with active graphics and a large summary diagram which shows the main functional associations within the model.

References

- Alekhin, Y.V., M.P. Sidorova, L.I. Ivanova, and L.Z. Lakshtanov (1984), Investigation of the electrokinetic potential and adsorption of ions on Al_2O_3 and SiO_2 at high temperature and pressures, *Kolloidn. Zh.*, 46, 1195– 1198.
- Bernabé, Y., U. Mok, A. Maineult and B. Evans (2003) Laboratory measurements of electrical potential in rock during high-temperature water flow and chemical reactions, *Geothermics*, 32(3), 297-310, doi: 10.1016/S0375-6505(03)00020-8
- Booth, F., 1951, The dielectric constant of water and the saturation effect: *J. Chem. Phys.*, 19, 391–394, doi: 10.1063/1.1748233.
- Crow, D.R., 1988, *Principles and applications of electrochemistry*: Chapman and Hall, London.
- Dove, P.M., and J.D. Rimstidt, 1994, Silica-Water Interactions: in *Reviews in Mineralogical Series: The Silica Polymorphs*, edited by P.J. Heaney, C.T. Prewitt, and G.V. Gibbs, 30, 210-260, Mineral. Soc. Am., Washington, D.C.
- Dunstan, D.E. (1994), Temperature-dependence of the electrokinetic properties of two disparate surfaces, *J. Colloid Interface Sci.*, 166, 472– 475, doi:10.1006/jcis.1994.1319.
- Dupuis, J., K. Butler, A. Kepic, and B. Harris (2009), Anatomy of a seismoelectric conversion: Measurements and conceptual modeling in boreholes penetrating a sandy aquifer, *Journal of Geophysical Research*, doi:10.1029/2008JB005939.
- Glover, P.W.J. (2015), *Geophysical Properties of the Near Surface Earth: Electrical Properties*. In: Gerald Schubert (editor-in-chief) *Treatise on Geophysics*, 2nd edition, Vol 11. Oxford: Elsevier; 2015. p. 89-137.

- Glover, P.W.J. (2018) Modelling pH-Dependent and Microstructure-Dependent Streaming Potential Coefficient and Zeta Potential of Porous Sandstones. *Transp. Porous Med.* 124, 31–56. <https://doi.org/10.1007/s11242-018-1036-z>
- Glover, P. (2026). Modified theoretical model for temperature-dependent electro-kinetic properties of porous media - Supplementary material - Model. In Not yet published (1.1). Zenodo. <https://doi.org/10.5281/zenodo.19133221>
- Glover, P.W.J. and N. Déry (2010), Dependence of streaming potential on grain diameter and pore radius for quartz glass beads, *Geophysics*, 75(6), F225-241, doi: 10.1190/1.3509465.
- Glover, P.W.J., E. Walker and M.D. Jackson (2012), Streaming-potential coefficient of reservoir rock, *Geophysics*, 77(2), D17-D43, doi:10.1190/GEO2011-0364.1.
- Glover, P.W.J., Walker, E., Ruel, J. & Tardif, E. (2012), Frequency-Dependent Streaming Potential of Porous Media—Part 2: Experimental Measurement of Unconsolidated Materials, *International Journal of Geophysics*, 2012, 728495, doi: 10.1155/2012/728495.
- Glover, P.W.J., Peng, R., Lorinczi, P. et al. (2020) Experimental Measurement of Frequency-Dependent Permeability and Streaming Potential of Sandstones. *Transp. Porous Med.* 131, 333–361. <https://doi.org/10.1007/s11242-019-01344-5>
- Grahame, D.C., 1950, Effects of dielectric saturation upon the diffuse double layer and free energy of hydration of ions: *J. Chem. Phys.*, 18, 903-909.
- Hidayat, M., Sarmadivaleh, M., Derksen, J.J., Vega-Maza, D., Iglauer, S. & Vinogradov, J. (2022) Zeta potential of CO₂-rich aqueous solutions in contact with intact sandstone sample at temperatures of 23 °C and 40 °C and pressures up to 10.0 MPa. *Journal of Colloid and Interface Science*, 607, 1226–1238. <https://doi.org/10.1016/j.jcis.2021.09.076>
- Hiemstra, T., and W. H. Van Riemsdijk, 1990, Multiple activated complex dissolution of metal (hydr)oxyde: A thermodynamic approach applied to quartz: *Journal of Colloid and Interface Science*, 136, 132–150, doi: 10.1016/0021-9797(90)90084-2.
- Hunter, R.J., 1966, The interpretation of electrokinetic potentials: *J. Colloid Interface Sci.*, 22, 231-239, doi:10.1016/0021-9797(66)90028-2.
- Hunter, R.J., 1981, *Zeta Potential in Colloid Science*: Academic Press, New York.
- Iglesias, T. P., and J. Péon Fernandez, 2001, A mixing rule for the permittivity of binary mixtures in a framework of a weak-fluctuations model: *J. Mater. Sci. Lett.*, 20, 1333-1324, doi:10.1023/A:1010962920679.

1099 Iler, R.K., 1979, *The Chemistry of Silica*: Wiley, New York.
 1100 Ishido, T., and H. Mizutani (1981), Experimental and theoretical basis of electro-kinetic
 1101 phenomena in rock-water systems and its applications to geophysics, *J. Geophys. Res.*,
 1102 86(B3), 1763–1775, doi:10.1029/JB086iB03p01763.
 1103 Jaafar, M.Z., 2009, *Measurement of Streaming Potential for Oilfield Monitoring in Intelligent*
 1104 *Wells*: PhD Thesis, Imperial College London, UK.
 1105 Jaafar, M.Z., J. Vinogradov and M.D. Jackson, 2009, Measurement of streaming potential
 1106 coupling coefficient in sandstones saturated with high salinity NaCl brine: *Geophys. Res.*
 1107 *Lett.*, 36, L21306, doi: 10.1029/2009GL040549.
 1108 Johnson, D. L., J. Koplik, and R. Dashen, 1987, Theory of dynamic permeability in fluid
 1109 saturated porous media: *J. Fluid. Mech.*, 176, 379–402. Jørgensen, S.S., and A.T. Jensen,
 1110 1967, Acid-base properties of quartz suspensions: *J. Phys. Chem.*, 71, 745-750,
 1111 doi:10.1021/j100862a040.
 1112 Jørgensen, S. S., and A. T. Jensen, 1967, Acid-base properties of quartz sus-pensions: *Journal*
 1113 *of Physical Chemistry*, 71, 745–750, doi:10.1021/j100862a040.
 1114 Kell, G. S. (1975). Density, thermal expansivity, and compressibility of liquid water from 0°
 1115 to 150°C: Correlations and tables for atmospheric pressure and saturation reviewed and
 1116 expressed on the 1968 temperature scale. *Journal of Chemical & Engineering Data*,
 1117 20(1), 97–105. <https://doi.org/10.1021/je60064a005>
 1118 Kosmulski, M. (1996), Adsorption of cadmium on alumina and silica: Analysis of the values of stability constants of surface
 1119 complexes calculated for different parameters of triple layer model: *Colloids and*
 1120 *Surfaces, A: Physicochemical and Engineering Aspects*, 117, 201–214, doi:
 1121 10.1016/0927-7757(96)03706-5.
 1122 Kosmulski, M., Adsorption of cadmium on alumina and silica: Analysis of the values of
 1123 stability constants of surface complexes calculated for different parameters of triple layer
 1124 model: *Colloids Surf. A*, 117, 201-214, 1996.
 1125 Leroy, P., Mainault, A., Li, S. & Vinogradov, J. (2022) The zeta potential of quartz: Surface
 1126 complexation modelling to elucidate high salinity measurements. *Colloids and Surfaces*
 1127 *A: Physicochemical and Engineering Aspects*, 650, 129507.
 1128 <https://doi.org/10.1016/j.colsurfa.2022.129507>
 1129 Lide, D.R. (2009), *Handbook Of Chemistry And Physics*, 90th ed.: CRC Press.
 1130 Lorne, B., F. Perrier, and J.-P. Avouac, 1999, Streaming potential measurements 1. Properties
 1131 of the electrical double layer from crushed rock samples: *J. Geophys. Res.*, 104, 17857-
 1132 17877, doi:10.1029/1999JB900156.

1133 Lyklema, J., and J.T.G. Overbeek, 1961, On the interpretation of electrokinetic potentials: J.
 1134 Colloid Interface Sci., 16, 501–12, doi:10.1016/0095-8522(61)90029-0.
 1135 Maineult, A., Y. Bernabé, and P. Ackerer, 2004, Electrical response of flow, diffusion, and
 1136 advection in a laboratory sand box: Vadose Zone J., 3, 1180-1192.
 1137 Morgan, F.D., E.R. Williams, and T.R. Madden, 1989, Streaming potential properties of
 1138 Westerly Granite with applications: J. Geophys. Res., 94, 12449-12461,
 1139 doi:10.1029/JB094iB09p12449.
 1140 Novotný, P. & Söhnel, O. (1988). Densities of aqueous solutions of inorganic substances: A
 1141 comprehensive correlation. Journal of Chemical & Engineering Data, 33(1), 49–55.
 1142 O’Konski, C.T., 1960, Electric properties of macromolecules, V: Theory of ionic polarization
 1143 in polyelectrolytes: J. Chem. Phys., 64, 605-619.
 1144 Overbeek, J.Th.G., Electrochemistry of the double layer: Colloid Science 1, Irreversible
 1145 Systems, edited by H.R. Kruyt, 115- 193, Elsevier, New York, 1952.
 1146 Park, J., and J.R. Regalbuto, A simple, accurate determination of oxide PZC and the strong
 1147 buffering effect of oxide surfaces at incipient wetness; J. Colloid Interface Sci., 175, 239-
 1148 252, 1995.
 1149 Phillips, S.L., H. Ozbek, and R.J. Otto (1978), Basic energy properties of electrolytic solutions
 1150 database: paper presented at Sixth International CO-DATA Conference, 2–25, accessed
 1151 10 June 2010, [http://www.etde.org/etdeweb/purl.cover.jsp?purl=/6628103-
 1152 xaRVpW/](http://www.etde.org/etdeweb/purl.cover.jsp?purl=/6628103-xaRVpW/)
 1153 Pride, S.R., and F.D. Morgan, 1991, Electrokinetic dissipation induced by seismic waves:
 1154 Geophysics, 56, 914– 925, doi:10.1190/1.1443125.
 1155 Reppert, P. & F. Morgan, 2003a, Temperature dependent streaming potentials: 1. Theory: J.
 1156 Geophys. Res., 108, 2546, doi:10.1029/2002JB001754.
 1157 Reppert, P. & F. Morgan, 2003b, Temperature dependent streaming potentials: 2. Laboratory,
 1158 J. Geophys. Res., 108, 2547, doi:10.1029/2002JB001755.
 1159 Revil, A. and P.W.J. Glover (1997), Theory of ionic surface electrical conduction in porous
 1160 media, Phys. Rev. B, 55(3), 1757-1773
 1161 Revil, A. and P.W.J. Glover (1998), Nature of surface electrical conductivity in natural sands,
 1162 sandstones, and clays, Geophys. Res. Lett., 25(5), 691-694.
 1163 Revil, A., P.A. Pezard, and P.W.J. Glover (1999a), Streaming potential in porous media: 1.
 1164 Theory of the zeta potential, J. Geophys. Res., 104(B9), 20021–20031,
 1165 doi:10.1029/1999JB900089.

- Revil, A., H. Schwaeger, L. M. Cathless, and P. D. Manhardt (1999b), Streaming potential in porous media 2. Theory and application to geothermal systems: *Journal of Geophysical Research*, 104, 20033–20048, doi: 10.1029/1999JB900090.
- Revil, A., and L.M. Cathles, 1999, Permeability of shaly sands: *Water Resources Research*, 35, 651–662, doi: <https://10.1029/98WR02700>
- Rustad, J.R., E. Wasserman, A.R. Felmy, and C. Wilke, 1998, Molecular dynamics study of proton binding to silica surfaces: *J. Colloid Interface Sci.*, 198, 119-129.
- Saunders, J., M. Jackson, and C. Pain (2008), Fluid flow monitoring in oil fields using downhole measurements of electrokinetic potential, *Geophysics*, 73, no. 5, E165–E180.
- Saunders, J.H., M.D. Jackson, M.Y. Gulamali, J. Vinogradov, and C.C. Pain (2012), Streaming potentials at hydrocarbon reservoir conditions, *Geophysics*, 77, E77–E90, doi: 10.1190/geo2011-0068.1.
- Scales, P.J., F. Grieser, and T.W. Healy, Electrokinetics of the muscovite mica-aqueous interface: *Langmuir*, 6, 582-589, 1990.
- Sen, P., and P. Goode (1992a), Influence of temperature on electrical conductivity on shaly sands: *Geophysics*, 57, 89–96, doi: 10.1190/1.1443191.
- Sen, P., and P. Goode (1992b), Errata, to “Influence of temperature on electrical conductivity of shaly sands”: *Geophysics*, 57, 89, doi: 10.1190/1.1443191.
- Smith, A.L., 1976, Electrokinetics of the oxide - solution interface: *J. Colloid Interface Sci.*, 55(3), 525-530, doi:10.1016/0021-9797(76)90062-X.
- Tadros, Th.F., and J. Lyklema, 1969, The electrical double layer on silica in the presence of bivalent counter-ions, *J. Electroanal. Chem. Interfacial Electrochem.*, 22, 1-7.
- Tardif, E., Glover. P.W.J. & Ruel, J. (2011), Frequency-dependent streaming potential of Ottawa sand, *J. Geophys. Res.*, 116, B04206, doi: 10.1029/2010JB008053.
- Tardif, E., Glover. P.W.J. & Ruel, J. (2012), Frequency-Dependent Streaming Potential of Porous Media—Part 1: Experimental Approaches and Apparatus Design, *International Journal of Geophysics*, 846204, doi: 10.1155/2012/846204.
- Vinogradov, J. and Jackson, M.D. (2015), Zeta potential in intact natural sandstones at elevated temperatures, *Geophysical Research Letters*, 42(15), 6287-6294, doi: 10.1002/2015GL064795.
- Vinogradov, J., Hidayat, M., Sarmadivaleh, M., Vega-Maza, D., Iglauer, S., Zhang, L., Mei, D. & Derksen, J. (2024) Zeta potential of supercritical CO₂-water-sandstone systems and its correlation with wettability and residual subsurface trapping of CO₂. *Water Resources Research*, 60(11), e2023WR036698. <https://doi.org/10.1029/2023WR036698>

1200 Walker, E. & Glover, P.W.J., (2010) Permeability models of porous media: Characteristic
 1201 length scales, scaling constants and time-dependent electrokinetic coupling. *Geophysics*,
 1202 75(6), pp. E235–E246. <https://doi.org/10.1190/1.3506561>

1203 Walker, E., Glover, P.W.J. and Ruel, J. (2014), A transient method for measuring the DC
 1204 streaming potential coefficient of porous and fractured rocks, *Journal of Geophysical*
 1205 *Research: Solid Earth*, 119(2), 957-970, doi: 10.1002/2013JB010579.

1206 Walker, E., Glover, P.W.J. (2018) Measurements of the Relationship Between Microstructure,
 1207 pH, and the Streaming and Zeta Potentials of Sandstones. *Transp. Porous Med.* 121, 183–
 1208 206. <https://doi.org/10.1007/s11242-017-0954-5>

1209 Watillon, A., and R. de Backer, 1970, Potentiel d'écoulement, courant d'écoulement et
 1210 conductance de surface à l'interface eau-verre: *J. Electroanal. Chem.*, 25, 181-196,
 1211 doi:10.1016/S0022-0728(70)80336-9.

1212 Waxman, M.H. and Smits, L.J.M., 1968, Electrical conductivities in oil-bearing shaly sands,
 1213 *Soc. Pet. Eng. J.*, 8, 107-122.

1214 Wu, L., W. Forsling, and P.W. Schlindler, 1991, Surface complexation of calcium minerals in
 1215 aqueous solution, *J. Colloid Interface Sci.*, 147, 178-185.

1216 Yates, D.E., and T.W. Healy, The structure of the silica/electrolyte interface, *J. Colloid*
 1217 *Interface Sci.*, 55, 9-19, 1976.